CHAPTER TWO

A secure base script perspective on attachment: progress, promise, and prospects

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Abstract

Bowlby's attachment theory has long been a dominant framework for understanding the development of intimate relationships, particularly the role of early caregiving in shaping later socio-emotional functioning. While Bowlby's proposed cognitive mechanism allowing for early experience to guide later adaptation, the Internal Working Model, has been central to the theory, the specific psychological constructs and processes at play remain underexplored. Waters and Waters (2006) advanced our understanding of the Internal Working Model by introducing the "secure base script," a cognitive representation that summarizes secure base experiences and guides behavior in attachment-related contexts. Twenty years on, this review summarizes current progress in terms of: a) assessment; b) latent structure; c) antecedents; d) sequalae; and e) stability/change. Further, it evaluates the secure base script's potential to advance attachment theory and our understanding of attachment processes by

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integrating both basic and clinically oriented perspectives from cognitive psychology with attachment theory.

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1. Introduction

Bowlby's (e.g., 1969/1982) attachment theory has served as a dominant perspective for understanding the development of intimate relationships across the lifespan for more than a half century. Attachment theory not only serves as a framework for understanding the development of parent-child relationships but also how those relationships inform both our later attachments and our socio-emotional adjustment into the years of maturity (Roisman & Fraley, 2012; Waters & Waters, 2024). For Bowlby, the critical dynamic between parent and child was the provision and use of secure base support. The infant/child's use of their primary attachment figures as a source of support during distress, as well as a secure base from which to explore, is thought to organize the attachment system and build competency at utilizing this secure base relationship.

Those who receive consistent and supportive care during their continuous cycles of exploration and proximity seeking form what Bowlby termed a "secure" attachment, marked by clear signals of distress and need for proximity followed by the acceptance of comfort and an expeditious return to exploration (Bowlby, 1969). However, not all caregivers provide the type of consistent, responsive, and sensitive care required to build a secure attachment. In cases where care is aloof, inconsistent, or even harsh and rejecting the attachment that forms is "insecure" (or as Ainsworth et al., 1978, termed it, "anxious") and marked by tentative or limited proximity seeking, difficulty accepting or rejection of comfort, and a protracted period of distress. Once established, these individual differences in secure base dynamics are internalized as an Internal Working Model (IWM) of attachment and carried forward across development, serving as a guide for behavior and expectations in intimate relationships (Bowlby, 1988). Thus, for attachment scholars, the IWM serves as a primary mechanism by which early caregiving experiences come to exert enduring effects across the lifespan.

Over the decades, attachment researchers have applied rigorous observational, longitudinal, and experimental methods to test Bowlby's predictions. The data largely supports the assumptions contained within attachment theory regarding to origins of attachment being in the early caregiving environment (e.g., Madigan et al., 2024), attachment being carried forward as a (modestly) stable individual difference across development (e.g., Pinquart et al., 2013; Waters et al., 2021a), and its developmental sequelae (e.g., Dagan et al., 2021; 2024a; Groh et al., 2014). Despite the promising body of research supporting attachment theory, the primary mechanism that gives the theory its explanatory power is comparatively less well understood. Said another way, the specific contents of the IWM or the psychological processes that account for its origins, stability, and sequalae have long been underdeveloped in attachment theory and a point of reliable criticism (e.g., Hinde, 1982; 1991). The IWM lacked a degree of specificity due, in part, to the limitations of the cognitive psychology of Bowlby's day which was still developing its theories of mental representations of experience.

Building from Bretherton's seminal work on the IWM (e.g., Bretherton, 1990; see also Main et al., 1985) and the cognitive and developmental literatures on event representations, Waters and Waters (2006) argued that the attachment IWM contains (among other things) a cognitive script summarizing the history of secure base use and support, termed the *secure base script*. This theorizing represented an important step toward specificity and created a meaningful bridge between attachment theory and the extensive literature on event representations (e.g., Nelson, 1986; Schank & Abelson, 1977). Now, nearly twenty years since Waters and Waters' (2006) proposal, we provide a review of the literature on the secure base script. We focus on three key assumptions underlying both attachment and cognitive scripts, mainly that they are (a) tolerably veridical with experiences in the parent–child relationship, (b) a heuristic or guide for behavior/adaptation, and (c) relatively stable once established. Finally, we evaluate the literature on the secure base script in light of the initial promise of this construct for advancing attachment theory and offer several prospects for moving the field forward.

2. Assessment

Waters and Waters (2006; see also Waters et al., 1998) defined the secure base script as consisting of eight elements of secure base support linked in a temporal–causal sequence. From the secure base script perspective, individual differences in attachment security and the IWM are operationalized as varying degrees of knowledge and elaboration of the script. Assessment of the secure base script has generally taken two parallel approaches: 1) fictional storytelling tasks and 2) autobiographical narrative assessments.

2.1 Prompt-word methods

The attachment script assessment (ASA; Waters & Waters, 2021) is the most widely used assessment tool for secure base script knowledge. Versions of the task cover a broad range of development (i.e., middle childhood, adolescence, emerging adulthood, and adulthood) and cultural contexts, including versions in English, Spanish, Dutch, Portuguese, Chinese, and Japanese, with varying degrees of validity data (Waters & Roisman, 2019). Common to all is the utilization of the prompt-word methodology (Waters & Hou, 1987) to elicit fictional attachment narratives which are then rated on a scale from 1 to 7 for the extent to which the narrative follows the secure base script and provides elaboration/detail around the various script elements (e.g., offering explicit detail about comfort, including physical touch and verbal encouragement). The prompt-word method provides participants with a list of words organized in columns that form a beginning, middle, end structure with the instruction being for the participant to tell a story using the outline as a guide and to attempt to fill in as much detail around the outline as they can. Each story in the ASA focuses on a distress scenario in which it would be appropriate to seek support from a primary attachment figure such as a child seeking support after falling off a bike, an adolescent seeking support from a parent after a conflict with peers, or a romantic partner seeking support from a spouse during a health scare. Individual differences in secure base script knowledge are then assessed from transcripts of each story and the average score across a set of three to four attachment narratives is taken to maximize the reliability and validity of the assessment (Waters & Waters, 2021). Neutral/control stories about routine activities with peers are often include for practice or to limit hypothesis guessing.

2.2 Story-stems

The prompt-word method's reliance on literacy and understanding of narrative structure makes it challenging to use with children under the age eight or nine. As such, researchers interested in studying the secure base script in young children have turned to the story stem procedure (e.g., Bretherton & Oppenheim, 2003; Oppenheim & Waters, 1995) to help scaffold children's emerging narrative skills and examine individual differences in secure base script knowledge (Ruiz et al., 2019; Vaughn et al., 2019; Waters et al., 1998). The story stem procedure provides children with various attachment/relationship-relevant scenarios using a set of dolls or toys. The first part of the story is provided by the experimenter and the child is then asked to show and tell what will happen next. These narratives are then coded for the extent to which they reflect the secure base script and averaged across multiple story stems, once again in the interest of maximizing reliability and validity of the overall assessment. Again, this procedure for assessing secure base scripts has been applied in a variety of cultural contexts (see Vaughn et al., 2019).

2.3 Autobiographical interview approaches

Attachment research has a long and well validated tradition of assessing attachment security through autobiographical interview which started with the introduction of the Adult Attachment Interview (AAI; Main et al., 1985). To date, more than 26,000 AAIs (Bakermans-Kranenburg, Dagan, Cárcamo and van IJzendoorn, 2024) have been collected and parallel interviews focused on other developmental periods or relationships (romantic partnership; Current Relationship Interview (CRI); Crowell & Owens, 1996) have been developed as well. Following the initial validation and success of the fictional narrative approaches to assessing secure base script knowledge, researchers began applying similar frameworks for evaluating script knowledge reflected in the autobiographical narratives produced during the AAI (AAI_{sbs}; Waters & Facompré, 2021) and the CRI (CRI_{sbs}; Nivison, et al., 2022). Unlike the traditional coding systems for these interviews that focus on narrative coherence across the entire interview, the secure base script coding focuses on the first portion of the interview that emphasizes more generalized relationship expectations and prompts for specific narrative examples of these expectations. Despite significant differences in the tasks, convergent validity between the ASA and the AAI suggests that both approaches assess the same underlying construct (Waters et al., 2021b).

3. Latent structure

Issues of latent structure are of both theoretical and methodological import. Traditionally, attachment research has approached the assessment of individual differences under two key assumptions that: a) attachment is generalized in that it reflects a quality of the individual rather than that of a specific relationship and b) individual differences should be thought of as differences in kind rather than of degree (e.g., Fraley & Speaker, 2003). The majority of the evidence on the latent structure of attachment indicates that it becomes generalized after infancy and that, in contrast to expectations, individual differences are continuously distributed (e.g., Fraley et al., 2013). The emergence of the secure base script construct naturally renewed interest in these questions, especially in the context of cognitive script theory, which shares similar assumptions about generalization and all-or-nothing individual differences (Waters et al., 2015).

The ASA lends itself well to examining the generalization of secure base script knowledge because both the adolescent and adult versions include separate items for mother/father and parent/romantic partner scenarios, respectively. As such, confirmatory factor analysis can be used to examine the relative fit of a generalization model (covariance between latent variables for mother/father or parent/romantic partner ASA scores is freely estimated) versus a discrete representations model (covariance between latent variables is fixed to zero). Waters et al. (2015) compared the fit of these two models using both the adolescent and the adult version of the ASA. In both cases, evidence strongly supported the generalization model. Waters et al. (n.d.) performed a downward and cross-cultural extension of this work in China. They examined the same models using a new adaptation of the middle childhood ASA in a sample of 10-year-old Chinese children included two mother-child and two father-child items. Again, results supported the generalization model.

Building on work by Fraley and colleagues (e.g., Fraley et al., 2014; Roisman et al., 2007) which utilized taxometric analysis (e.g., Meehl, 1992; 1995; Ruscio et al., 2007) to distinguish between continuous and categorical latent structure, Waters et al. (2015) also tested for categorical individual differences in the same adolescent and adult samples which supported the generalization model. They found that, in contrast to assumptions often made by attachment theorists, secure base script knowledge was best thought of as continuously distributed. Waters et al. (2019a) followed up this work with a downward extension into middle childhood in a sample of 10-year-olds from Western Europe. In contrast to what was observed at later ages, they found that individual differences in secure base script knowledge in childhood were best understood as categorically (i.e., secure v. insecure) rather than continuously distributed. This led the authors, informed by both cognitive and developmental theories on event representations (e.g., Fivush, 1984; Hudson, 1992; Schank & Abelson, 1977), to propose that secure base script knowledge emerges in an all-or-nothing fashion. The later expansion of their social world and the need for secure base support in more diverse contexts during the transition from middle childhood to early adolescence leads to a process of elaboration, nuance, and refinement that results in continuously distributed individual differences.

To test this hypothesis, Houbrechts et al. (2024) conducted a large-scale longitudinal study of the latent structure of secure base script knowledge across the middle childhood period. Specifically, they collected the middle childhood version of the ASA in three separate waves when the participants were 10 years, 11 years, and 12 years old. Replicating Waters et al. (2019a), they found categorical latent structure at age 10 years. At age 11 years, however, the taxometric analyses were inconclusive. By age 12 years, a continuous latent structure had emerged and evidence for the developmental shift in the latent structure of secure base script knowledge was observed. Taken together, the work on the latent structure of secure base script knowledge suggests that across development, we construct a generalized script-like mental representation of our caregiving experiences that is initially learned in an all-or-nothing fashion but shifts toward individual differences in terms of degree (more or less secure) and away from differences in kind (secure v. insecure).

4. Antecedents

Data over the past 50 years support the predicted association between early caregiving quality and the development of attachment (Madigan et al., 2024). The link between any measure of attachment and its theoretical antecedents is a cornerstone of validation. This is especially critical when operationalizing attachment as a cognitive script, as cognitive script theory *also* clearly specifies that scripts are generalized representations based on lived experiences (Schank & Abelson, 1977). As such, any new assessment of script-like attachment representations must replicate this general association between early caregiving and later attachment.

Researchers have produced substantial evidence for the association between higher levels of secure base script knowledge in childhood, adolescence, and adulthood and early maternal and paternal sensitive caregiving as well as abuse and neglect. Steele et al. (2014) reported on the association between the adolescent version of the ASA at age 18 years and maternal (9 assessments) and paternal sensitivity (5 assessments) from infancy through age 15 years in the normative-risk Study of Early Child Care and Youth Development (SECCYD). They found that both maternal and paternal sensitivity positively predicted ASA scores in the target adolescents (see also Vaughn et al., 2016). In addition to ASAs (n = 673), the

SECCYD also collected AAIs during the same age 18 years assessment wave. These AAIs ($N = 857$) were later coded using the AAI_{sbs} system and Nivison et al. (2024) replicated the findings from Steele et al. (2014) in the SECCYD, with both maternal and paternal sensitivity positively correlating with AAI_{sbs}, although the findings for maternal sensitivity were more robust to the inclusion of covariates than were those observed with paternal sensitivity. Looking at secure base scripts in middle childhood, Runze et al. (2024) found that parental sensitivity at roughly age 6 years was positively associated with secure base script scores three years later.

To extend the work on the developmental antecedents of secure base script knowledge to higher risk contexts, Waters et al. (2017) recoded the existing AAIs collected with the Minnesota Longitudinal Study of Risk and Adaptation (MLSRA) cohort. AAIs were collected at both age 19 years and 26 years and observations of maternal sensitivity were collected across seven waves through the first 13 years of life. A composite of the maternal sensitivity assessments was significantly and positively associated with AAI_{sbs} at both ages. Schoenmaker et al. (2015) found similar results using the ASA at age 23 years in a sample of adoptees, thus accounting for genetic confounds, with maternal sensitivity in early and middle childhood significantly positively predicting script knowledge in young adulthood. In the same MLSRA sample as Waters et al. (2017), Eller et al. (2022) examined the role of consistency in sensitive caregiving in addition to mean levels, as both cognitive script theory and attachment theory also make arguments that the construction and organization of mental representations benefit from consistency across experiences. Supporting this view, they found that individuals who experienced greater fluctuations in maternal sensitivity during childhood showed significantly lower levels of secure base script knowledge in adulthood, even after accounting for mean levels of sensitivity. Finally, Waters et al. (2025) tested for cross-cultural replication of the sensitivity to secure base script knowledge association in a prospective longitudinal study of Chinese families. They found that maternal sensitivity observed at 14 and 24 months positively predicted secure base script knowledge in children at age 10 years. Taken together, the literature demonstrates a significant association between the development of the secure base script and the level and consistency of sensitive caregiving experienced during development in normative risk, high-risk, genetically unrelated, and cross-cultural samples.

Nivison et al. (2021) looked beyond sensitive caregiving as a positive predictor and examined the role of abuse and neglect in secure base script development. They explored the role of perpetrator, timing, and type of abuse in the MLSRA cohort. They found that total abuse/neglect experienced was negatively associated with later secure base script knowledge even after accounting for maternal sensitivity. In terms of timing and type, the effects were present for abuse/neglect experienced during middle childhood and adolescence and specifically for physical abuse. They also found that these effects were unique to maternal and paternal abuse/neglect and did not extend to non-caregiver relationships. The parental caregiver finding parallels work by Waters et al. (2021c) with the SECCYD which found that early child care/daycare experiences (quality, quantity, and type) were unrelated to later secure base script scores. These results provide some discriminant validity to the secure base approach to studying attachment representations in that the effects seem confined to primary attachment figures.

Given the strong emphasis placed on the caregiving environment by attachment theory it is not surprising that the vast majority of research on the origins of individual differences in attachment (secure base script knowledge included) has focused on environmental influences within biologically-related families. As a result, the potentially confounding role of genetics remains largely unaccounted for (but see Schoenmaker et al., 2015). The only study to date looking at the relative contributions of genetic and environmental factors to secure base script knowledge found that genetic influences accounted for 38 percent of the variation in secure script scores in middle childhood (Runze et al., 2024). This result stands in contrast to the sensitivity hypothesis and prior genetically-informed attachment work in infancy, which found minimal evidence for genetic influences on attachment (Bakermans-Kranenburg & Van IJzendoorn, 2016; Bokhorst et al., 2003; Fearon et al., 2006; Roisman & Fraley, 2008). This result, however, was consistent with those reported by Fearon et al. (2014) in sample of adolescents using the traditional coding approach with the Childhood Attachment Interview. Though the genetically informed literature on secure base representations is scant, initial evidence suggests the presence of potential gene-environment correlations. As children develop, their increased ability to engage with environments that align with their genetic predispositions could increase the heritability of secure base script knowledge. As increasing evidence suggests attachment, and the secure base script, is subject to substantial influence from genetics, more research is needed to replicate these initial findings. It will be especially critical to leverage high-powered genetically-informed samples in testing the origins of secure base representations where possible, to control for genetic confounding as well as to look at the genetic and environmental contributions to secure base script knowledge at various stages across the lifespan, particularly given that behavioral geneticists suggest that the heritability of a trait increases over time (Turkheimer, 2000).

5. Sequalae

Over the past two decades, researchers have reported associations between secure base script knowledge and a myriad of theory consistent sequelae including caregiving behavior, social competence, and psychological adjustment. Bowlby suggested that a child who is "much-loved" and sensitively cared for develops a sense of self-confidence that others will see them as worthy of love also; whereas when a child is treated neglectfully or as unwanted by his caregivers, the child comes to internalize this feeling to a more general level (Bowlby 1973, p. 204). As such, a strong secure base script allows an individual to form not only a set of expectations regarding attachment figures and their support but also an adaptive understanding of oneself. In doing so, this affords a sense of confidence in exploring the environment as well as dealing with potentially distressing situations. Conversely, individuals who have low expectations of security and safety from their caregivers, that is, relatively weak secure base script knowledge or an internal working model that is antithetical to the secure base script, may have difficulties regulating their emotions or be at increased risk of maladaptive psychosocial pathways.

5.1 Psychological functioning

Secure base script knowledge can provide a protective effect against problematic psychological development. A secure base script reflects internally-held themes about oneself and one's relationships; when an individual repeatedly experiences harsh or neglectful interactions with attachment figures., their attachment expectations can take on anomalous qualities, such as the child feeling responsible for relieving their caregiver's distress or believing that they are unwanted or flawed. Evidence from middle childhood through late adolescence indicates that higher secure base script knowledge is associated with fewer early maladaptive schemas when concurrently assessed (Li et al., 2024). Research on young adults similarly indicated that higher secure base script knowledge, derived from free recall of close relationships, was inversely associated with four out of five maladaptive schemas related to emotional disconnection with others, such as social isolation and defectiveness (McLean et al., 2014). Thus, the development of script-like representations of early caregiving experiences may serve as a cognitive framework through which children internalize positive and adverse childhood experiences, shaping their psychosocial adjustment.

As children internalize their interactions with caregivers, they may also develop interpersonal skills and an implicit understanding of the self in relation to others that guides their behavior toward or away from psychopathology. Across multiple cultural contexts, researchers have produced converging evidence indicating the negative association between secure base script knowledge and externalizing behaviors (Portugal: Fernandes et al., 2019; Dutch-speaking Belgium: Houbrechts et al., 2023; Waters et al., 2015; Singapore: Cheung et al., 2024).

Longitudinal evidence, from multiple studies utilizing the NICHD SECCYD, indicates that secure base script knowledge is negatively associated with depressive symptoms such that stronger secure base script content at age 18 was associated with fewer depressive symptoms when assessed in adulthood (Dagan et al., 2021; Dagan et al., 2024b). In one such study, secure base script knowledge, assessed via the ASA, partially mediated the relation between parental sensitivity, assessed at multiple time points and developmental stages within the first 15 years of life, and depressive symptoms at age 26. However, it should be noted that the effect size was modest.

Multiple studies highlight the potential indirect role that secure base script knowledge may play in the relationship between environmental stressors and psychological development. Instability within an environment undermines not only a child's ability to successfully regulate emotions and behavior (Ackerman et al., 1999; Bobbitt & Gershoff, 2016), but also their ability to abstract a cognitive script from their early experiences with caregivers. Ruiz and colleagues (2019) found that development of secure base script knowledge in six-year-olds was inversely associated with parent-reported stress exposure, such as death, divorce, and residential moving, and partially mediated the pathway between early life stress and increases externalizing behavior problems at age eight in a high-risk sample. In this study, higher levels of secure base script knowledge predicted smaller increases in externalizing problems relative to children with lower levels of script knowledge. As such, development of the secure base script provides a potential mechanism for explaining the trajectory of psychosocial adjustment, and by this logic, secure base script knowledge may also provide a buffer from psychological symptoms. In a study conducted in Belgium, secure base script knowledge moderated the relationship between daily stress and depressive symptoms in middle childhood (van Aswegen et al., 2023). In addition, script knowledge during middle childhood served as a protective factor/buffer against the detrimental effects of cumulative family stress in relative change of externalizing problems over a two-year period in a normative risk sample (Houbrechts et al., 2023).

The secure base script may buffer responses to stress as it allows an individual to effectively utilize their attachment figures in the face of challenging environments. However, the secure base expectations that are imbued in an individual with a strong secure base script may reveal a vulnerability towards less than favorable conditions later in life. In a sample of primiparous mothers, higher secure base script knowledge moderated the relation between perceived social support and postpartum psychosocial functioning (Gu et al., 2024). Gu and colleagues found that mothers with higher ASA scores but lower perceived social support during the perinatal period reported higher postpartum maternal maladjustment, a composite of postpartum depression, emotion dysregulation, and internalizing and externalizing behavior metrics. The authors argued that the secure base script reflects a set of expectations and when those expectations are contradicted by the environment (in this case, low social support during the transition to motherhood), it results is psychological distress. As such, scripted representations of attachment serve as a set of expectations by which we evaluate and adapt to our current and immediate social environments, not as an inoculation against negative outcomes.

5.2 Social competence

Throughout development, the opportunities for novel social environments inevitably increase. Attending school, socializing with peers of a similar age, and later, exploring romantic relationships all present opportunities to engage with new social partners. As suggested by Bowlby, attachment representations can be generalized to social interactions outside of the immediate caregiving environment (1969/1982, 1973). The internal working model of attachment allows individuals to organize their expectations and behavior within such novel social contexts in accordance with their secure base script, and as such, secure base script knowledge is activated during such interactions and serves to guide or shape how the individual processes and behaves in unexplored social environments (Bowlby, 1973).

5.2.1 With peers

As children shift from an early caregiving environment into a school environment, the dramatic shifts in context allow for ripe opportunities to utilize their script-like cognitive representations to interact with peers and teachers. Secure base script knowledge has been empirically associated with multiple indices of peer and school-based social competencies across multiple cultural contexts. Fernandes et al. (2019) found significant associations between script-like content elicited from the Attachment Story Completion Task (ASCT) and social competence metrics assessed through direct classroom observation and sociometric interviews in a sample of Portuguese and American four- to six-year-old children. Posada and colleagues (2019) replicated these significant positive associations between secure base script knowledge, assessed via the Attachment Story Completion Task, and teacher reports of social competence in two independent samples of US children drawn from Midwestern and Southern contexts with a moderately strong effect size. Similar associations in preschool-aged children have also been observed in South Korean (Shin, 2019), Portuguese (Fernandes et al., 2019), Mexican, and Peruvian samples (Nóblega et al., 2019).

Researchers have also sought to understand how the secure base script can bolster positive prosocial competencies in children and buffer the effects of negative school-based experiences. Secure base script knowledge, assessed in middle childhood, has been shown to moderate the relation between Reactive Attachment Disorder (RAD) symptoms and parent and teacher-reported prosocial behavior such that in children with more RAD symptoms, those who had higher secure base script scores exhibited higher prosocial behavior (Cuyvers et al., 2020). Similarly, research in Singapore suggests that higher secure base script knowledge is negatively associated with self-reported peer victimization and bullying and positively associated with academic performance (Cheung et al., 2024).

Within an urban Chinese sample, researchers have found that secure base script knowledge in middle childhood partially mediates the relation between early caregiving sensitivity, measured during the first two years, and later friendship conflict at age 15. Yang et al. (2024) found that higher maternal sensitivity was longitudinally associated with lower adolescent conflict within friendships and that ~19 percent of this effect was mediated by secure base script knowledge. Further, the same study found that adolescents with lower secure base script reported greater loneliness in high friend/peer conflict environments . This potentially highlights how the secure base script may guide expectations and interpretations of peer conflict so as to buffer from perceptions of loneliness.

5.2.2 With romantic partners

Attachment representations of early caregiving experiences have been shown to reliably predict variation in multiple indices of romantic functioning (e.g., Dagan et al., 2021; Dagan et al., 2024a; Nivison et al., 2022; Roisman et al., 2001; Waters et al., 2018). Utilizing the SECCYD ample, researchers found that participants' scripted representations of attachment at age 18 years, indexed by the Attachment Script Assessment, mediated the relationship between parental sensitivity assessed in multiple waves across the first 15 years of life and perceptions of romantic partner's extreme hostility at age 26 (Dagan et al., 2021). Further, secure base script content extracted from the AAI at age 18 predicted high romantic dyad adjustment scores at age 30 years (Dagan et al., 2024a). In a high-risk sample, secure base script knowledge in early adulthood was found to mediate the effect of maternal sensitivity quality and consistency on romantic relationship effectiveness at age 32 years such that higher mean levels of early sensitivity predicted higher reports of relationship competency via higher script knowledge (Eller et al., 2022).

5.3 Caregiving outcomes

A script-like representation of attachment and secure-base interactions provides caregivers with both expectations for how secure base interactions *should* occur as well as a cognitive script from which to draw upon when caregivers are faced with attachment scenarios in real time. When a caregiver develops a secure attachment representation (i.e., there is strong evidence of understanding the secure base script), this caregiver is hypothesized to behave in accordance with their understanding of secure base interactions and as such, provide appropriate care in accordance with the secure base script. Additionally, the caregiver's secure base script knowledge would also inform their expectations for the child's actions, affect, and responses, thus contributing to the dyadic quality of secure base interactions. Parenting sensitivity has been widely studied as an associated outcome of attachment representations in both normative and high-risk populations (e.g., Coppola et al., 2006; Raby et al., 2021; Van Ee et al., 2017; Witte et al., 2023). Raby and colleagues found that in parents that were Child Protective Services-referred as well as low-risk parents, higher SBSK was associated with more sensitive caregiving during interactions with middle childhood-aged children

(Raby et al., 2021). In a highly stressful caregiving environment, secure base script knowledge can modulate the types of care that caregivers provide. For example, mothers with a history of childhood maltreatment with higher secure base script knowledge, using both the ASA aggregate score as well as only parent-child stories, were observed to have more positive parenting behaviors with their infants (Huth-Bocks et al., 2014). Evidence from asylum-seeking and refugee families indicates that when caregivers have less access to the secure base script, post-traumatic stress disorder symptoms are strongly associated with an increase in insensitive parenting behaviors (Van Ee et al., 2017). Further, utilizing a genetically-informed twin design, Witte and colleagues found that parent secure base script knowledge was significantly associated with both parental sensitivity and sensitive discipline and that these associations were not associated with child-based genetic factors (2023).

Taken together, this research has established a link between parents' secure base script knowledge and caregiving sensitivity, suggesting that in both normative and high-risk caregiving contexts, parents possessing a well-developed secure base script are more likely to engage in sensitive and responsive parenting. Research now highlights the potential neural mechanisms underlying this association. For example, there is evidence that mothers with lower secure base script knowledge exhibit significantly elevated P3b event-related potential (ERP) responses when attending to infant distress, as well as a disproportionate allocation of cognitive resources toward distress cues from their infants rather than positive cues. This pattern suggests that low SBSK mothers engage more cognitive resources in processing attachment-related signals compared to mothers with higher SBSK as well as have a bias in their attachment-signal related processing (Groh & Haydon, 2017). Further, there is early evidence that maternal secure base script knowledge may affect neural development in the next generation, particularly in child brain structure (Fitter et al., 2022).

Researchers have established strong support for the intergenerational transmission of attachment security (e.g., Van IJzendoorn, 1995; Vaughn et al., 2007; Verhage et al., 2016). Specifically for scripted attachment representations, research indicates that parental secure base script knowledge may predict child attachment security (Waters et al., 2018). Similarly, Huth-Bocks et al. (2022) found a longitudinal association between maternal SBS knowledge and child attachment behavior during infancy in a sample of caregivers at risk of parenting difficulties. One potential explanation in this intergenerational attachment representation transmission may be that the secure base script may play a role in how mothers jointly discuss emotional content with their children (Bost et al., 2007). Although secure base script knowledge has been shown to increase parenting sensitivity, there is also evidence that a strong command of the secure base script increases parents' likelihood and encouragement to openly engage with their children in emotion-based speech (Apetroaia & Waters, 2018; Waters et al., 2018). Relatedly, in understanding potential mechanisms underlying how secure base script knowledge affects parenting, research has also indicated indirect associations between secure base script knowledge, self-reported attachment styles in parents and their reflective functioning (Borelli et al., 2017). In this cross-sectional study, Borelli and colleagues found that lower secure base script knowledge was associated with higher self-reported attachment avoidance in parents as well as less positive/more negative emotionality after reflecting on positive connections with their child (Borelli et al., 2017).

Parent secure base script knowledge has also demonstrated long-term predictive validity in intergenerational outcomes for academic success, particularly in at-risk populations. For example, McLear et al. (2016) found that parents' SBSK uniquely predicted their kindergarteners' academic achievement at the end of the school year, above and beyond the effects of effortful control and child verbal ability. This suggests that, beyond cognitive and self-regulatory skills, a parent's internalized attachment representations contribute significantly to their child's educational trajectory. Similarly, parent ASA scores have been linked to child social competence, an important predictor of academic engagement. Sparks (2018) demonstrated that parental attachment script knowledge (ASA) was positively associated with children's social competence, which in turn mediated the relationship between family cumulative risk and social development.

Beyond caregiving and outcomes in the next generation, research also suggest that higher secure base script knowledge predicts the quality of interactions between adult caregivers and their elderly parents (i.e., less criticism and hostility), when there was a high level of perceived caregiving stress (Chen et al., 2013). This work points not only to how secure base script knowledge influences caregiving across different life stages but also to the cognitive script's role in shaping relational sequelae beyond the parent-child dyad. By providing an abstract framework for navigating attachment-related interactions, SBSK may serve as a protective factor in stressful and emotionally demanding caregiving contexts, such as supporting aging parents (Waters & Waters, 2024).

Though a large body of research has established the predictive validity of secure base script knowledge for a range of psychological and social outcomes, there is growing interest in understanding how attachment representations shape physiological development. Initial empirical evidence is inconclusive given the relatively small number of studies that specify relations between secure base script representations and physical health indices. For example, research from the Minnesota Longitudinal Study of Risk and Adaptation indicates that early caregiving sensitivity during infancy in an at-risk sample predicted an composite index of cardiometabolic health (i.e., mean arterial pressure (MAP), body mass index (BMI), C-reactive protein (CRP), and blood pressure) at ages 37 and 39 with secure base script knowledge in early adulthood (ages 19 and 26) partially mediated this relationship (Farrell et al., 2018). However, research utilizing a normative-risk prospective longitudinal sample did not find evidence of secure base script knowledge mediating the relationship between observed parental sensitivity from infancy to age 18 and BMI at age 26 (Dagan et al., 2021). As we continue to consider the multifinality of attachment representations, there is an increasing need for further research investigating the role of secure base script knowledge on markers of physical well-being and psychophysiological development.

6. Stability and change

Beyond typical methodological concerns related to reliability, stability is a critical assumption of attachment theory. Attachment theory does allow for lawful change but is also assumes a degree of stability which supports the mechanistic role attachment representations play in the association between early caregiving experiences and later attachment relevant sequalae. Cognitive script theory also predicts that there should be some degree of stability in scripts and that, assuming experiences in the script domain continue to share common features, this stability may increase over time (Waters et al., 2021d).

Several studies have examined the stability of secure base script knowledge across moderate and long-term intervals, at various periods of development, and across risk status. Vaughn et al. (2006) found that ASA scores were moderately stable over roughly a one-year interval in a small sample of normative risk mothers. This provided the first piece of evidence that the secure base script demonstrated theory-consistent levels of stability over a significant interval. At the same time, the normative risk nature of the sample along with modest sample size prompted several replications and extensions. Waters et al. (2019b) examined ASA stability across the transition from middle childhood to adolescence, a period where greater instability might be expected given socio-emotional and cognitive changes in the child, as well as a shift in the parent-child relationship dynamic toward a supervisory partnership (Koehn & Kerns, 2015; Waters & Waters, 2024). Similar to Vaughn and colleagues (2006), they found moderate stability in ASA scores from age 11 years to 14 years (4-waves with 1-year intervals) of age in a study of normative risk children (see also Houbrechts et al., 2024). Moderate stability was again observed with the same sample following a final 2-year follow-up at age 16 years (Waters et al., 2021d).

Greater instability in attachment may also be expected in higher-risk contexts due to the greater potential for significant changes to relationship dynamics and increased stress combined with less flexibility/availability of attachment figures due to socioeconomic circumstances (e.g., Pinquart et al., 2013). However, secure base script stability has generally produced equivalent stability data irrespective of context. In the MLSRA, the high-risk longitudinal cohort, AAI_{sbs} scores were stable from age 19 years to age 26 years with a comparable effect size to the findings discussed above. One potential confound in stability estimates provided for the secure base script comes from shared method variance. The use of the same measure across multiple timepoints may artificially inflate stability due to consistency driven by the task itself rather than the underlying construct. To address this concern, Waters et al. (2021) examined the stability of the secure base script using two different assessment tools, the ASA and AAI. In a pooled sample of both high and low-risk cohorts ($N = 113$), they found that the AAI_{sbs} assessed at age 20 years significantly predicted the ASA at age 40 years ($r = 0.43$) to a roughly equivalent degree as in prior tests (medium and longer term) of stability. Further, they found that the association was not moderated by risk status.

Despite the notable stability in the secure base script, plenty of room for change exists and attachment and cognitive script theories make several predictions of what governs such changes. To date, however, very little work has examined changes in secure base script knowledge. One study examined change in the secur ebase script over a three-year period and found that the number of self-reported daily hassles (e.g., conflicts with peers) but not big stressors (e.g., parental divorce; death of a loved one) predicted *increases* in SBSK during middle childhood and early adolescence (Waters et al., 2019b). The authors argue that frequent opportunities to practice secure base use with manageable stressors allow for the reinforcement and elaboration of the secure base script. Big events, which are more likely to cause large shifts in family dynamics or be challenging to resolve through secure base support, were unrelated to changes in script scores but this may have resulted from low base rates given the normative risk sample examined. In a somewhat higher-risk context, Finet et al. (2021) found that a sample of internationally adopted children did not differ from their non-adopted comparison group in terms of secure base script knowledge in middle childhood, although they were more likely to have been insecurely attached when first observed a few months after adoption had taken place. Further, concurrent maternal sensitivity was positively associated with children's secure base script scores. Together, these results suggest that with time, supportive care can shift attachment security and promote secure base script knowledge even after significant life changes like international adoption. There is also emerging evidence that secure base script knowledge can be changed through clinical intervention. Raby et al. (2021) evaluated the effectiveness of the Attachment and Biobehavioral Catch-up (ABC) intervention on secure base script knowledge. They examined the impact of ABC on ASA scores in a sample of mothers who had been referred to Child Protective Services due to risk for child maltreatment when their children were infants. Seven years after completing ABC, mothers were tested using the ASA. Mothers who had been randomized to receive the ABC intervention had higher ASA scores than those who had received a control intervention and were equivalent to a sample of low-risk comparative control mothers. Although not specifically designed to target the secure base script, ABC did lead to improvements. Interestingly, one of the main targets of ABC is maternal sensitivity and mediational analyses suggested that ABCs effects on ASA scores was through increasing sensitive caregiving via intervention.

The encouraging theory-consistent stability and change results for the secure base script construct offer numerous opportunities to extend the cognitive script perspective on attachment, especially in the context of studying lawful change. Although the data suggests increasing stability in secure base script knowledge from middle childhood and beyond, early intervention does seem promising. The only intervention work to examine change in SBSK followed up with participants several years after intervention. It is still unclear if the ABC intervention directly alters script knowledge or whether this effect is more an indirect effect resulting from changes in caregiver sensitivity and improvement of parent-relationship quality. Interventions that directly target the secure base script are still in development and represent another promising research avenue as how best to affect change in the secure base script is still very much an open question.

7. Conclusion: Promise and prospect

The initial introduction of the secure base script construct was not met with immediate uptake, as is to be expected in a mature field such as attachment. Any new measure or perspective is going to be confronted with the burden of overcoming the "old wine in a new bottle" problem when entering into a research area with a rich tradition of well-validated assessments across the lifespan (Waters, 2021; Waters & Roisman, 2019). The evidence reviewed above speaks to the validity of the secure base script construct, but less so to the novelty and added value of adopting a cognitive script perspective on attachment. The data support the predictions, stemming from both attachment and cognitive script theory, that the secure base script is learned from early attachment relevant experiences, stable and generalized, and applied to novel attachment relevant contexts.

However critical to validity they are, these findings do not necessarily speak to the promise the cognitive script approach offers. Waters et al. (2013) argued that the standard approach to assessing attachment representations in adolescence and adulthood through the AAI coherence coding was troublingly distant from the constructs thought to give rise to, or result from, attachment representations. Theory and evidence are sparse when it comes to accounting for how early caregiving experiences result in coherent autobiographical narratives in the AAI. It is perhaps even more difficult to account for how coherent autobiographical narratives produce behavior in novel contexts. The rich tradition of cognitive script research does provide a straightforward account for how scripts are acquired, generalized, and guide behavior in novel contexts given their high heuristic value and integration with goal structures (Markman, 1999). However, relying on making inferences from the cognitive literature to account for attachment processes falls short of a true cognitive account of attachment. Attachment researchers need to demonstrate that the secure base script has similar impacts on basic cognitive processes like perception, memory, and decision making as has been seen in the cognitive literature in order implicate specific cognitive mechanisms underlying the associations between the secure base script and various attachment processes.

The clinical promise and potential of the cognitive perspective on attachment has also yet to be realized. The work by Raby et al. (2021) demonstrated that the secure base script may be impacted by intervention efforts, even though the intervention examined was not designed specifically with the secure base script in mind. Instead, the ABC intervention program studied focuses on parental coaching with live feedback targeting sensitive caregiving behaviors (e.g., Dozier & Bernard, 2017). There is, however, a substantial clinical literature focused on intervening with cognitive scripts and schemas which becomes relevant once attachment scholars take a cognitive script approach. Cognitive Behavior Therapy (CBT; Hollon & Beck, 1994; Seligman & Ollendick, 2011; see also Schema Therapy, Young, et al., 2003) is a problem focused approach which attempts to alter both maladaptive behaviors and cognitions through cognitive restructuring and reinforcement and has generally produced modest but consistent positive outcomes (Bosmans, 2016; Wenzel, 2017). However, CBT and related approaches have largely not targeted attachment related clinical concerns, in part because attachment theorists had not fully articulated the cognitive underpinnings of attachment beyond the IWM construct. Bosmans (2016) has argued that the cognitive script perspective has opened the door for integration of some of the well-worn paradigms and approaches from CBT with the parenting and sensitivity focused approaches already being used in the attachment field. The development of new integrative interventions that leverage the strengths of both the cognitive and attachment traditions to better understand and effect change represents a great opportunity for future research.

In short, the cognitive script perspective has made tremendous progress in its first twenty years. This has been facilitated by a highly adaptive set of methods and benefited from the wealth of longitudinal studies available in the attachment field (e.g., MLSRA; SECCYD). Waters and Waters (2006) argued for the benefits of the explanatory power of cognitive scripts for the attachment literature, noting that:

> *"Given their impact on social perception, expectations, and memory, not to mention self-definition and goal structures (e.g., Carver & Scheier, 1990; Epstein, 1994; Horowitz, 1988; Lazarus, 1991), studying script-like representations of secure base experience should have far-ranging and fundamental implications for our understanding of attachment related cognitions, emotions, and behavior. (p. 188)"*

This promise has yet to be fully realized, but with a set of well-validated tools and clear developmental, cognitive, and clinical theory as a guide, the future of the cognitive script perspective on attachment has great potential.

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CHAPTER EIGHT

Assessing children's spatial thinking: Insights, challenges, and implications

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Contents

Abstract

In the past few decades, interest in children's spatial thinking has increased substantially, and consequently, interest in spatial assessments for children has also increased. However, there are not many reliable, validated, and widely accessible spatial assessments

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for this segment of the population, which affects researchers' ability to conduct and interpret spatial thinking research. While some limitations of these tests relate to broader issues with spatial assessments in general (see Uttal et al., 2024), creating assessments that are appropriate for children presents unique challenges. In this chapter, we review the current state of tests of children's spatial thinking, including mental rotation and perspective-taking. We draw on insights from psychometrics, open science, and cognitive development research. Furthermore, we examine how spatial assessments affect research on the relation between spatial and STEM abilities, particularly research aimed at leveraging spatial thinking through interventions and training that improve children's spatial skills and, in turn, their STEM performance (i.e., Bruce & Hawes, 2015; Cheng & Mix, 2014; Hawes et al., 2022; Judd & Klingberg, 2021; Mix et al., 2021). Lastly, we outline recommendations for improving these assessments to ultimately improve research and theory creation on the development of spatial thinking.

Keywords: Assessment and measurement, Development, Psychometrics, Spatial thinking

Spatial thinking is an important part of everyday life and learning. In addition to being essential for tasks like navigation or arranging objects in a room, spatial thinking is particularly important for STEM disciplines. This relationship has been explored broadly (Uttal & Cohen, 2012) and within specific STEM domains, including, but not limited to, mathematics (Atit et al., 2022), chemistry (Harle & Towns, 2011), anatomy (Roach et al., 2021), and geoscience (Nazareth et al., 2019). Importantly, spatial thinking has been shown to be a significant and unique contributor to STEM achievement and participation, even when controlling for verbal and mathematical abilities (Shea et al., 2001; Wai et al., 2009). More recently, research has focused on training and intervention studies aimed at improving students' STEM abilities directly and indirectly by enhancing their spatial skills (Cheng & Mix, 2014; Langlois et al., 2020; Sorby et al., 2013). This research on spatial training is supported by robust findings regarding the malleability of spatial skills (Uttal et al., 2013). Despite this emphasis on leveraging spatial thinking for STEM education, many unanswered questions remain about the structure and framework of the skills that are in this domain (Uttal et al., 2024). Moreover, efforts to delineate the various factors that comprise spatial thinking have yielded different results (Hegarty & Waller, 2005).

1. The need for developmental spatial thinking research

In addition to unanswered questions about the structure of spatial thinking, many questions remain about the development of spatial skills. For example, different studies that have examined the developmental trajectory of spatial transformation abilities have found different answers regarding when children demonstrate such abilities (Frick et al., 2014a). Typically, spatial skills assessed in adults have been evaluated in children with minimal changes to the assessments. Thus, the lack of consensus about how different spatial skills relate to each other in research with adults is echoed in research with children. However, focusing on developmental research can be key in unraveling the structure of spatial thinking; studying and comparing the development of various spatial abilities may provide insight into how different spatial skills are related. Nevertheless, research with children presents unique challenges. Creating and administering assessments for children can be challenging, and there are fewer spatial assessments for children available compared to adults.

1.1 The importance of assessment and construct validity

To understand how the lack of rigorous assessments for children may affect research on spatial thinking, we must consider the importance of assessment in training and intervention research, as well as the current state of spatial thinking assessment. Operationalizing constructs is essential for quality psychological research. In cognitive psychology, this operationalization typically occurs through the creation and use of assessments targeting specific cognitive constructs. However, in the case of spatial thinking, the development of constructs has often followed from assessments (Hegarty & Waller, 2005); many spatial assessments were designed to meet specific needs rather than to provide rigorous measures of well-defined constructs. Furthermore, many studies measuring spatial skills rely on only one or two specific types of spatial skills, yet they often refer to these as general measures of spatial thinking.

Given that assessments play a critical role in measuring spatial thinking and determining which spatial skills are examined, it is essential to understand what the assessments we use actually measure. Generally, researchers separate spatial thinking into small-scale spatial thinking, which deals with smaller objects and spaces, and large-scale spatial thinking, which focuses on navigation and wayfinding. There are more small-scale spatial assessments than large-scale ones. Uttal et al., (2024) recently published a review addressing current challenges with spatial thinking tests. Among these challenges are difficulties in understanding what different spatial tests measure and how they relate to one another. This can leave researchers uncertain about which tests are most appropriate for contexts such as investigating spatial training for STEM improvement. While these issues have been addressed in assessments designed for adults (Uttal et al., 2024), spatial assessments for children are similarly affected. Additionally, assessments developed for children, particularly young children, present unique challenges that further complicate understanding and interpreting these tests.

2. What spatial assessments are there for young children?

To understand the current state of spatial assessment in children, we need to examine which tests are being used. While there are too many existing tests to review all of them (Schenck & Nathan, 2024), some assessments are more commonly used than others. Here, we will review some of the most common spatial assessments for children and use these exemplar tests to highlight and discuss broader issues related to children's spatial testing. To identify the tests available for young children, we must define the age range that "young children" encompasses. On the lower end, children younger than 2 or 3 years old are typically given assessments that resemble infant assessments more than adult assessments. Additionally, while some judgment-based spatial assessments are administered to three-year-olds, there are far fewer tests for younger children compared to older ones (Verdine et al., 2017). On the upper end of the age range, researchers have administered adult spatial tests to children as young as 7–8 years old (Hoyek et al., 2012). Thus, in this chapter, we will generally review assessments designed for children between the ages of 3- and 8-years-old. Tables 1, 2, 3 and 4 provide examples of different types of spatial assessments for children.

2.1 Mental rotation

As mentioned above, mental rotation is the most used type of spatial thinking test. Mental rotation is often defined as the cognitive operation of transforming a mental image by imagining its rotation through different orientations in space. This definition stems from the original mental rotation test by Shepard and Metzler (1971), in which participants determined whether two 3D cube figures presented at different orientations were the same or different. In addition to measuring accuracy, participants' reaction times were also recorded. Reaction time was strongly and linearly related

Table 1 Examples of tests used to assess children's mental rotation.

Test Name	Source	Original age range (years old)
Picture Rotation Test	Quaiser-Pohl, 2003	4–6
Children's Mental Transformation Test[a]	Levine et al., 1999	4–7
Ghost Mental Rotation Test	Frick et al., 2013	3–5 (though 3-year-olds were near chance level)
MRT-Animal	Titze et al., 2010 (based on Vandenburg & Kuse, 1978)	8–10
Gilligan et al., 2019	Gilligan et al., 2019 (modified from Broadbent et al., 2014)	6–10

[a]The original CMTT was not solely a mental rotation test but a general mental transformation test. However, it has been used as a substitute for mental rotation tests several times and also modified to include only rotation items.

to the angular disparity between the two cube figures. Some researchers suggest that this linear relationship indicates that participants were imagining the rotation of the cube figures. in an analog process; the more they had to rotate the cube figures., the longer it took. Thus, many researchers have assumed that both in the original mental rotation task and in the numerous subsequent variations of the test individuals were mentally rotating the test stimuli (i.e., Just & Carpenter, 1985; Khooshabeh et al., 2013). However, evidence suggests that individuals do not always imagine rotation on these tests. Several studies have shown that individuals can instead use alternative strategies, such as analytic strategies that are not spatial in nature (Boone & Hegarty, 2017; Hegarty, 2018; Just & Carpenter, 1985).

The study of children's mental rotation dates back to the 1970s. Marmor (1975, 1977) investigated when children first represent movement in mental images, a concept known as kinetic mental imagery at the time. Piaget and Inhelder (1971) proposed that kinetic imagery, or mental imagery that represents movement, emerges during the concrete operational stage, which occurs around ages 7–8. Marmor (1975, 1977) used Shepard and Metzler's mental rotation paradigm but found the original 3D cube figure stimuli too difficult for young children and redesigned the task to be appropriate for 4- and 5-year-olds (see Fig. 1). Marmor found that both 4- and 5-year-olds seemed able to perform mental imagery; while they were much slower than adults, they did show the linear rate of rotation reported by Shepard and Metzler. However, Dean and Harvey (1979), using a similar paradigm with

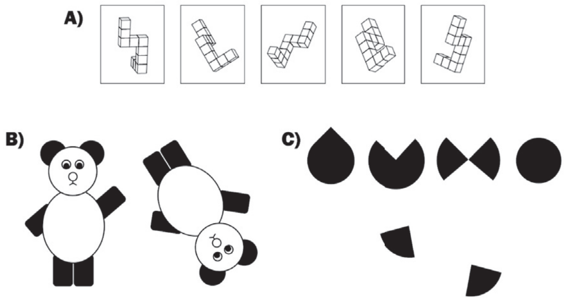

Fig. 1 Author renditions of examples of mental rotation tests that use different questions, dimensionality, and stimuli. Note. *A) Example of an adult mental rotation test based on Vandenburg and Kuse, 1978 (as described in Wilbur et al., 2024). B) Marmor (1975, 1977) bears mental rotation. C) Levine et al. (1999) Children's Mental Transformation Test.*

different stimuli, failed to replicate these findings, ultimately suggesting that children could not successfully perform mental rotation until age 6. Since then, numerous studies on children's mental rotation have often yielded conflicting results, particularly regarding the age at which mental rotation skills develop. For example, some studies suggest that Marmor's original findings were accurate and that children develop mental rotation skills around 4.5 years old (i.e., Frick et al., 2013), while more recent studies indicate that even younger children exhibit mental rotation skills (i.e., Krüger, 2018). These findings are not only contradictory within this age range, but are also contradictory with infant studies of mental rotation that show infants as young as three-months-old show precursors of mental rotation, such as detecting mirror reflections at various orientations (Enge et al., 2023; Moore & Johnson, 2008; Quinn & Liben, 2008). When considering the entire literature onmental rotation from birth to early childhood, there are unanswered questions about the developmental trajectory of mental rotation (Frick et al., 2014a). Further, the apparent dip in mental rotation ability of pre-school aged children may further suggest potential issues with the mental rotation tests used with this age range; these tests may be too confusing or have too high of cognitive demands.

2.2 Perspective taking

Perspective taking is another common category of spatial thinking tests; however, there is a significant gap between the use of mental rotation and

Table 2 Examples of tests used to assess children's perspective taking.

Test name	Source	Original age range (years old)
Short Object Perspective-Taking Task (sOPT)	Kozhevnikov and Hegarty, 2001	Undergraduates (also used with children 8- to 16-years-old, e.g. (Orefice et al., 2024))
Perspective-Taking Test for Children (PTT-C)	Frick et al., 2014b	4–8
Photographic Perspective Taking Test	Liben and Downs, 1993	5.5–11.5

perspective taking (Uttal et al., 2024). Spatial perspective taking is generally defined as mentally adopting a viewpoint different from one's own. The literature on spatial perspective taking is extensive, as "perspective taking" can refer to various abilities, not all of which are spatial. For example, perspective taking can involve understanding another person's beliefs or feelings, as well as viewing an environment or object from a different angle (Healey & Grossman, 2018; Michelon & Zacks, 2006; Tarampi et al., 2016). However, only the latter interpretation of perspective taking is typically considered a spatial skill (Brucato, 2022). Although some hypotheses suggest that all types of perspective taking share common cognitive processes (Parkinson & Wheatley, 2013), recent work has supported the claim that spatial perspective taking is a distinct process (Brucato, 2022).

Unlike mental rotation, assessments of spatial perspective taking have historically focused more on young children. One of the first spatial perspective taking assessments is the Three Mountains Test developed by Piaget and Inhelder (1956). In this test, a child is seated next to a model of three distinct mountains, while a doll is positioned differently around the Table. The child is then asked to identify the view of the mountains that the doll would see. The cognitive process involved in perspective taking requires imagining a different frame of reference. Research has shown that perspective taking is distinct from mental rotation in both adults and children (Frick & Pichelmann, 2023; Hegarty & Waller, 2004). Furthermore, like mental rotation tests, there are various versions of perspective taking tests for children, though there are fewer of them and the differences between these tests are not as pronounced.

2.3 Block design test

While mental rotation and perspective-taking tests are designed to target a specific type of spatial thinking skills, some tests are used as more general spatial thinking assessments. One such general spatial thinking test is the Block Design Test (Groth-Marnat & Teal, 2000; Jirout & Newcombe, 2015). In this test, participants must recreate a design using colored blocks within a certain time limit. The Block Design Test was not originally intended to be a spatial thinking test but rather a general intelligence test to be used primarily for clinical purposes (Hutt, 1932; Kohs, 1920). Since it was first developed, researchers have used the block design test in other settings, particularly as a spatial assessment in research and educational settings (McKee et al., 2024). Despite not being originally designed as a spatial thinking assessment, research indicates that performance on the Block Design Test is correlated with other spatial assessments and experiences (Caldera et al., 1999; Casey et al., 2008). Because the Block Design Test was not designed to target a specific spatial skill, its use in spatial thinking research is varied; researchers in different contexts have referred to the Block Design Test as an assessment of spatial visualization, visual processing, spatial relations, and general visuo-spatial ability (Casey et al., 2008; Flanagan & Kaufman, 2004; Jirout & Newcombe, 2015). Further, the Block Design Test also exemplifies spatial construction assessments, a type of spatial assessment primarily used with young children (Verdine et al., 2014). These spatial construction or block construction assessments typically aim to mimic naturalistic block-building play.

2.4 Other small-scale spatial tests

Outside of mental rotation and perspective taking, there are very few categories of spatial thinking with multiple assessments available for young children. For instance, several spatial skills have multiple assessments for adults, such as cross-sectioning (Cohen & Hegarty, 2007; Ormand et al., 2014, 2017; Titus & Horsman, 2009), but there are far fewer cross-sectioning assessments specifically designed for children (Ratliff et al., 2010). This lack of established assessments does not indicate a complete absence of research on these spatial skills; rather, these skills have often been measured in children using one-off, ad hoc tests. Additionally, some assessments originally developed for adults have been adapted for use with children (Aguilar Ramirez et al., 2021; Gori et al., 2024; Hoyek et al., 2012; Morawietz et al., 2024).

Table 3 Examples of other small-scale spatial assessments for young children.

Test name	Source	Original age range (years old)
Crossing Sectioning for Children	Ratliff et al., 2010	5–9
Spatial Scaling Task	Frick and Newcombe 2012	3–6
Diagrammatic Representations Task	Frick and Newcombe, 2015	4–8
Proportional Reasoning	Möhring et al., 2015	4–5
Mental Folding Task for Children	Harris et al., 2013	4–7
Test of Spatial Ability (TOSA) 2-D & 3-D	Verdine, Golinkoff et al., 2014, 2014	3–5
Block Design Test	WISC Version: Weschler, 2014	6–16

Furthermore, the scarcity of tests for certain spatial skills can be beneficial in research if it leads to multiple studies using the same test. This reduction in variation can enhance the comparability of results across different studies, as performance in various samples on the same assessment can be directly compared. For example, the Children's Embedded Figures Task (Witkin et al., 1971, 1971) has been utilized by various researchers in studies of children's spatial thinking over many decades, including those that also examined children's STEM abilities (Hodgkiss et al., 2018; Tracy, 1987; Quaiser-Pohl et al., 2004). However, many of the spatial assessments currently in use were developed only in the last decade.

2.5 Large scale spatial assessments

There are more spatial assessments for small-scale than for large-scale spatial thinking, both for adults and children (Uttal et al., 2024). The relative lack of large-scale assessments is at least partly due to the challenges of testing large-scale spatial skills. Navigation is often assessed using virtual environments, sometimes presented through virtual reality (Nguyen et al., 2023). To run these simulations effectively, researchers need suitable computers, screens, and headsets (if using virtual reality). Furthermore, creating and sharing virtual environments is more challenging than sharing a paper-and-pencil assessment.

Table 4 Examples of test used to assess children's navigation.

Test name	Source	Original age range (years old)
Silcton	Schinazi et al., 2013	Various; 8–16 (Nazareth et al., 2018)
Virtual Path Mazes	Jansen-Osmann and Wiedenbauer, 2004	7–8

3. Challenges of children's spatial assessments

With the increased interest in spatial thinking, many researchers in education and related fields have begun using spatial assessments in their research. However, it can be challenging for those with limited knowledge of spatial cognition to find, select, and administer these assessments (Schneck & Nathan, 2024). Additionally, there are broader issues related to the absence of a comprehensive framework for spatial thinking as a whole (see Uttal et al., 2024 for a review). These issues also apply to assessments for children. However, creating assessments for children, especially young ones, presents unique challenges that can compound the general difficulties associated with spatial assessments, resulting in children's spatial assessments being significantly less effective than those for adults. In this section, we review these unique challenges.

3.1 Challenges of designing assessments understandable for children

Perhaps the main challenge in creating assessments for children is balancing the need to capture the construct of interest while keeping the design simple enough for children to understand the task. Despite numerous cognitive and attentional differences between adults and children, many assessments for children have been developed by modifying adult tests, changing only a few features, such as the stimuli or total number of items. For example, in one of the first mental rotation tests for children, the Shepard & Metzler (1971) mental rotation paradigm was adapted, replacing the three-dimensional cube figures. with child-friendly drawings of bears and ice cream cones (Marmor, 1975, 1977). Similarly, the Animal-MRT maintained the general structure of the Vandenburg & Kuse (1978) test but substituted the cube figures. with pictures of animals. Although these changes make tests more child-friendly compared to the adult tests, researchers have more recently rethought how mental rotation questions can be asked in more child-friendly ways in new test paradigms (i.e. Beckner et al., 2023; Frick et al., 2013; Pedrett et al., 2023).

Effects of Training and Instruction. Researchers have sought to ensure that children understand assessments by including more practice and training questions. However, it is widely accepted that children's spatial skills are malleable (Yang et al., 2020). A meta-analysis of spatial training found that improvement can occur within a single session (Uttal et al., 2013). Thus, the question arises at what point additional practice and training questions shift from demonstrating what children should do on a task to training the targeted skill. Platt and Cohen (1981) demonstrated significant performance differences between children who received training and those who did not on a mental rotation test. Moreover, the specific kind of instruction and practice children receive can also affect their performance. Studies have shown that physical experience with manipulatable stimuli can enhance children's spatial thinking and can be used as instructional material for assessments (Byrne et al., 2023; Frick et al., 2013a; Frick et al., 2013b; Ha & Fang, 2018). Further, the language used in the instructions of a task may influence how children approach or understand the task. For example, in a study done with adults, the language used in a navigation task affected how participants completed the task (Boone, Maghen, & Hegarty, 2019). Thus, if the instructions of a task give explicit instruction related to how participants should solve a task, the participants might use strategies they otherwise would not. Although these tools can be used to explain a task to children, they might also give children too much information about how to solve a particular task, leading to artificially improved performance.

Attentional differences between children and adults. Another challenge in designing tests for children, compared to adults, is their lower attentional capacities (Betts et al., 2006). Tests for children are typically shorter than those for adults, and those for younger children are generally shorter than those for older children. Differences in the number of items in a test can affect its reliability, a point discussed later. Additionally, tests for children often use engaging designs to maintain their interest throughout the assessment. However, design choices intended to enhance engagement may interfere with or distract children from completing the task. An example of tests with conflicting designs is the Picture Rotation Test and the Ghost Test (Frick et al., 2013; Quaiser-Pohl, 2003). In the Picture Rotation Test, the stimuli consist of colorful pictures of animals, selected after pilot testing indicated that children were more motivated by colored images than by black-and-white stimuli.

In contrast, the ghost stimuli in the Ghost Test are grayscale and lack many details, which reduces potential distractions for children.

Individual differences among children. Like any population, children exhibit individual variation. Differences among children can lead to some successfully completing an assessment while others of the same age perform at chance level (i.e., Frick et al., 2013). A more important variation to consider is the differences in children's approaches to assessments. For example, consider research on adults' varying strategy use on mental rotation tests. Studies showing that adults employ non-rotational strategies on these tests have existed for decades (Boone & Hegarty, 2017; Hegarty, 2018; Just & Carpenter, 1985). Although research on adult strategy use in these tests is extensive, relatively little has focused on how children solve mental rotation problems, even though children have demonstrated the use of strategies other than rotation (Estes, 1998). Furthermore, researchers have suggested that children's inability to complete mental rotation tests successfully at young ages can stem from using inefficient strategies. For instance, Quaiser-Pohl et al. (2010) found that young children employed various strategies on the Picture Rotation Test, some more appropriate than others. In this test, children view a target picture of an animal or human and are asked to choose which of three answer choices matches the target. The two incorrect choices are mirror-reflected versions of the target image, and all three choices are rotated by varying amounts. One semi-appropriate strategy children used was selecting the answer choice with the smallest angular disparity. In other words, children who employed this strategy would choose the answer with the picture closest to upright, regardless of whether it matched the target image.

This difference in strategy use among children also indicates that children may perform differently on various tests. The least angular disparity strategy is not applicable to all types of mental rotation tests, as not all tests provide answer choices that vary in degrees of rotation. Additionally, without understanding how a child solved a particular assessment, it is challenging to determine whether poor performance was due to an inability to demonstrate the targeted skill or a lack of understanding of the task. Young children may not grasp what is being asked of them on a given assessment, which could contribute to their failure on mental transformation tests, despite infants showing precursors to mental transformation (Frick et al., 2014a).

Many children around the ages of 3–5 likely have not encountered judgment-based multiple-choice tests before. In educational settings, policy recommendations suggest that more formal test scores should be accompanied by informal observations due to children's unpredictability on tests (Epstein et al., 2004). Furthermore, there is even less research on children's strategy use or response patterns on non-mental rotation spatial assessments.

3.2 Test availability

One major limitation in children's spatial assessment is the availability of tests. Although this issue is also a concern for adult spatial tests (Uttal et al., 2024), children's assessments face a unique challenge: there are both too many and too few tests. Many tests have been created, but only afew are widely accessible to other researchers.

Overabundance of tests. One characteristic of spatial assessment is that tests are often developed for use in one or two studies due to reasons such as lack of access to or funding for existing tests (Uttal et al., 2024). This results not only in an overabundance of tests but also in less comparable results across studies. Furthermore, when spatial tests are developed for a study, they are typically mental rotation tests. A notable feature of the literature on children's mental rotation is the existence of numerous different mental rotation tests. These tests can vary in several ways, including their stimuli, the number of practice questions, and the number of answer choices. Additional differences among these tests include the type of questions, which means many children's mental rotation tests do not resemble the original mental rotation tests (Frick & Pichelmann, 2023). The presence and nature of alternative strategies used by adults have been shown to differ depending on specific task characteristics of the mental rotation test (Boone & Hegarty, 2017; Jost & Jansen, 2024). Thus, specific test characteristics that vary across tests, such as stimuli or question type, may influence how individuals approach the problem. Different mental rotation tests for children may measure different constructs due to these design differences, leading individuals to solve the tests in various ways, possibly using non-spatial strategies. Like the potential use of alternative strategies by children, the effects of different test characteristics on children's performance have not been systematically evaluated.

Lack of availability of children's tests. While many tests have been developed to measure children's spatial thinking, most are not widely accessible. Currently, there are no online resources that specifically provide free spatial assessments for children or information on where to find various tests. The Spatial Intelligence Learning Center (SILC; spatiallearning.org) maintains a repository of spatial assessments that includes some for children. However, out of the 41 resources listed, only seven assessments are explicitly labeled as children's assessments in their name or description (excluding surveys and coding manuals). Of these seven, SILC has complete test materials for only two: the Children's Mental Transformation Test and the Mental Folding Test for Children. Consequently, the Children's Mental Transformation Test is one of the most used assessments for children.

Furthermore, not all children's tests can be easily shared through online resources because they require physical manipulatives, either in the instructions or as part of the test itself (i.e., Casey et al., 2008; Frick et al., 2013). One example of a physical assessment is the Test of Spatial Assembly (TOSA), which asks children to recreate a target design using either flat geometric pieces (2D) or interlocking blocks (3D) (Verdine et al., 2014). To administer this test, a researcher must obtain the target stimuli designs, scoring manual, and purchase the physical manipulatives needed for the test. These physical assessments create additional barriers to administration, and researchers cannot use them for online studies, which have become increasingly common post-COVID-19 (Lapidow et al., 2021). The lack of widely available tests fuels ad-hoc test creation, leading to a cycle where the unavailability of existing tests results in the over-creation of new ones.

Modifications of tests. Sometimes researchers do not create entirely new tests but modify existing ones. These changes can range from seemingly minor adjustments, such as reducing the total number of items or digitizing a previous paper-and-pencil test, to more significant alterations, like developing a new set of stimuli based on an existing test. Similar to our uncertainty about how many tests of the same spatial skill relate to each other, we do not know how these modifications influence children's understanding, performance, and strategy use on different tests. Furthermore, altering one test characteristic may affect other characteristics. For example, Bambha and Casasola (2021) report on digitizing the Picture Rotation Test, a mental rotation test traditionally administered in person, for use in an online study. This change not only alters the modality of the test but also requires updating the instructions to reflect that the experimenter is not in the same room as the child.

Missing Spatial Assessments. The types of spatial thinking assessments used with children tend to mirror those used with adults. In other words, the spatial constructs measured in adults are also assessed in children. While this overlap can be useful for researching the development of specific spatial skills, the focus on a limited set of abilities often leads to the oversight or complete neglect of important spatial skills (Uttal et al., 2024). For example, non-rigid spatial skills, such as those used when mentally bending or breaking an object, are considered particularly important for STEM capability, yet they are rarely, if ever, measured in young children (Atit et al., 2013, 2020; Resnick & Shipley, 2013). Additionally, there may be areas of spatial thinking that have not yet been identified in either adults or children.

3.3 Lack of psychometric rigor

Even in cases where established children's spatial assessments exist, these tests can lack adequate standardization and psychometric evaluation. Many researchers do not report the psychometric properties of their tests. For example, when Levine et al. (1999) introduced the Children's Mental Transformation Task, they did not provide reliability measures or discuss validation processes for the test. Furthermore, the reported reliability of some children's tests has varied substantially across studies; a test may demonstrate acceptable reliability in one study but low reliability in another. For example, the reported Cronbach's alpha for different forms of the Children's Mental Transformation Test has ranged from .24 to .82 (Casasola et al., 2020; Ramírez, 2018). Some of these problems may arise in part because children's spatial assessments may be inherently more challenging to evaluate than adult tests.

Reliability. The reliability of an assessment is a fundamental aspect of determining its quality (Schmidt & Hunter, 1996). The most used measure of test reliability among psychologists is Cronbach's α, although some psychometricians have pointed out its limitations (Revelle & Condon, 2019). However, calculating traditional reliability measures, such as α or Guttman's λ_2, which rely on inter-item correlations, can be misleading or challenging to compute for some children's tests. For example, tests with items that vary in difficulty can result in decreased reliability under classical test theory (Gulliksen, 1945; Hambleton & Jones, 1993). Many spatial tests exhibit variation in item difficulty, such as differing angular disparities in mental rotation tests or varying amounts of folds in mental folding tests (Frick, 2019; Harris et al., 2013). Additionally, some children's tests are not paper-and-pencil based but instead require a child to perform a physical task (e.g., Casey et al., 2008; Verdine, Golinkoff, et al., 2014). These tests can resonate better with children because they mimic natural activities, like block-building, rather than relying on multiple-choice questions. Although the reliability of these tests can be calculated by simplifying a child's response to correct or incorrect, this scoring method overlooks the richness that can be captured through more qualitative approaches. Moreover, even with quantitative scoring systems, these physical tests have shown low reliability in some studies (Dearing et al., 2012). Thus, a significant challenge in designing children's spatial assessments lies in balancing the creation of open-ended, qualitative assessments that children may understand better with the need for assessments rooted in traditional psychometric rigor.

Another factor that can influence test reliability calculations is sample size (Charter, 1999). Studies involving children tend to have fewer participants than those involving adults for various reasons, such as the difficulty of recruitment (Hurwitz et al., 2017). Consequently, if measures like Cronbach's α or Guttman's λ_2 are calculated for a children's assessment using a small sample size, the resulting reliability may not be accurate. Collecting large samples of children can also be quite expensive, as tests that do collect a large sample to create testing norms, such as the Block Design Test, tend to be costly to acquire. Furthermore, more complex psychometric models, such as Item Response Theory (IRT), also require large sample sizes, typically in the hundreds, to be effective (Edelen & Reeve, 2007). For example, Munns et al. (2023) recently conducted IRT analysis on four different adult mental rotation tests. For this analysis, they obtained a large, representative sample of 500 adults recruited online through Qualtrics. Collecting a similarly sized, nationally representative sample of children would be much more challenging, as most widely used online data collection tools are primarily suited for gathering data from adult participants. Despite recent pushes for more rigorous psychometric evaluations of spatial assessments, children's assessments have so far not been included in this work (Brucato et al., 2023; Munns et al., 2023; Uttal et al., 2024).

Validity. In addition to reliability, proper validation of tests is a core aspect of psychometric work. As mentioned earlier in the discussion of challenges in designing spatial assessments for children, creating a test that is both age-appropriate and accurately measures the target construct is challenging. Therefore, validating children's assessments is crucial to ensure the target construct is being assessed. Many children's assessments are either not validated, or the validation procedures are not reported. Furthermore, validating assessments for children is challenging since one of the most common measures researchers use to validate assessments is convergent validity. Convergent validity suggests that an assessment is valid if it relates to an existing assessment of the same or similar ability (DeVon et al., 2007). For example, if a researcher creates a new assessment for children on mental folding, they might administer both the new test and a mental rotation test. If the children's performance on the two spatial tests is correlated, this suggests the new test is measuring a facet of spatial thinking. However, conducting this kind of validation depends on the validity of the comparison test; if there are no gold-standard assessments to compare new tests to, validating new tests can be difficult. Thus, the lack of psychometrically validated spatial assessments for children can result in more assessments not being properly validated.

4. Interim summary

Through the previous sections, we have established the importance of researching the development of spatial thinking. However, despite the attention that spatial thinking has received in the last few decades as a potential tool to improve students' STEM learning, the current state of spatial assessments for young children has room for improvement. Compared to tests for adults, several factors make it more challenging to create, share, and validate assessments for children. Although these challenges present hurdles to improving children's spatial assessments, they are not insurmountable. Additionally, researchers have been working on addressing these challenges and are making progress towards valid, age-appropriate spatial measures for children (i.e. Frick & Pichelmann, 2023). Nevertheless, the limitations of these assessments should be considered when evaluating research that has used them, especially in studies with mixed results or many unknowns.

5. Implications of children's spatial assessments on spatial-stem research

One area of research that relies on spatial assessments and has many open questions is the relationship between spatial thinking and STEM performance. The evidence linking spatial thinking to STEM outcomes is continuously increasing; however, there is still much unexplored potential in studying the *development* of this relationship (Newcombe & Frick, 2010).

Spatial thinking plays an important role in early STEM education. Evidence of its importance includes the integration of the skill set in elementary science and math education standards. For example, according to the Next Generation Science Standards, elementary school science education should emphasize skills such as interpreting and creating diagrams and developing physical models (NGSS, 2013). Similarly, Common Core math standards state that early math should include concepts like measurement, understanding a number line, and reasoning about shapes (NGACB & CCSSO, 2010). Although the specific topics of these standards differ across content areas (i.e., math versus science), the prevalence of spatial thinking underlines its substantive and foundational role in early STEM learning.

Furthermore, due to the importance of spatial thinking in early STEM mastery, spatial training interventions may be particularly effective for young children. In Uttal et al. (2013) meta-analysis of the malleability of spatial thinking, training with children under the age of 13 was shown to be effective compared to a control group, with a Hedge's $g = .61$. Years later, Yang et al., (2020) conducted a meta-analysis on spatial training specifically with children between the ages of 3 and 8, finding an overall Hedge's $g = .96$. While full interpretation of these results depends on understanding the heterogeneity within the included studies, their general positive findings suggest that spatial training with young children is effective. Experimental studies have also provided evidence that improvements in spatial ability in young children can transfer to STEM domains (Cheng & Mix, 2014; Mix et al., 2020); however, other studies have shown no transfer to STEM (Hawes et al., 2015).

Yet, as stated earlier, the results of spatial-STEM research with young children depend on the assessments that were used in the research. If an un-validated assessment was used, the results of the study should be interpreted with that in mind. Furthermore, assessments can play a larger role in research than merely a dependent variable. Thus, the state of spatial assessments can have consequences that extend beyond laboratory research. In this section, we discuss the potential implications of the current state of children's spatial assessment on both spatial-STEM research and educational contexts.

5.1 The role of assessment in spatial-stem research

While the quality of assessment is important in any research, it is particularly critical in spatial training and intervention studies. In most spatial training studies, one or two specific spatial skills are targeted, almost exclusively defined by spatial assessments. For example, multiple studies have examined the effects of mental rotation training on both spatial thinking and STEM skills in children (Cheng & Mix, 2014; Cheung et al., 2020; Fernández-Méndez et al., 2018; Hawes et al., 2015; Rodán et al., 2019). Although there is promising evidence that mental rotation-focused training can enhance mathematical abilities (Cheng & Mix, 2014), it is worth questioning whether the extensive focus on mental rotation stems from its true importance for STEM or from an overabundance and reliance on mental rotation assessments.

In a recent meta-analysis of the relationship between spatial skills and mathematics, of the 93 tests or test batteries used in the 45 included studies, approximately one-third were mental rotation tests or included mental rotation questions (Atit et al., 2022). No other spatial skill was as prominently represented as mental rotation among these tests. In a different meta-analysis specific to training spatial thinking in young children, mental rotation accounted for about 40 percent of the spatial outcome measures reported in the included studies (Yang et al., 2020). Thus, current theories regarding how spatial skills contribute to STEM abilities depend on a limited number of spatial thinking tests. Furthermore, without a wide range of spatial assessments for young children, researchers are unable to determine which spatial skills are related to specific STEM abilities.

One of the main motivators for researching spatial thinking and STEM is the potential to leverage the malleable nature of spatial thinking to improve students' spatial skills, with the hope that this improvement transfers to STEM ability. Spatial assessments also play an important role in spatial training studies. Not only do they serve as outcome measures in intervention studies, but they can also inform the training conducted during the study. For example, in Cheng & Mix (2014)'s spatial intervention study, the training children received was mental rotation training based on the Children's Mental Transformation Test. Bower et al., (2020)'s spatial training was based on the Test of Spatial Ability. This type of assessment-targeted training can lead to near transfer; children trained on a specific assessment can show improvement on that assessment. However, this kind of assessment-based training also limits the potential for transfer to various STEM domains; far transfer from spatial training to STEM would depend on the strength of the relation between the assessment informing the training and the specific STEM measure of interest.

5.2 Educational implications and outcomes

The state of spatial assessments for children can also impact education. Spatial thinking has historically been overlooked in traditional educational settings (Gagnier & Fisher, 2016; Mathewson, 1999). Unvalidated spatial assessments for children used in spatial-STEM research may obscure study results, making it difficult to present compelling arguments to education stakeholders about the educational importance of spatial thinking. Mixed research results also impede progress in evaluating best practices for leveraging spatial thinking in STEM education. Furthermore, the lack of widely available and validated spatial thinking assessments for children affects the diversity of students pursuing STEM in the future. The current limitations associated with spatial assessments make it challenging to identify students who could benefit from spatial training to enhance their STEM performance. Additionally, it restricts the identification of students who excel in spatial thinking and may be more likely to succeed in STEM (Lakin & Wai, 2020; Veurink & Sorby, 2019; Wai & Kell, 2017; Wai & Lakin, 2020). Identifying students from disadvantaged backgrounds who excel in spatial thinking can foster their potential and increase representation in STEM from historically underrepresented groups (Wai & Lakin, 2020).

6. Recommendations for future research

In this chapter, we reviewed the current state of children's spatial assessments, the challenges in designing these assessments, and their implications for broader research. While recognizing the existing problems with children's spatial assessments is the first step toward improvement, more dedicated efforts are necessary to achieve significant enhancements. This work is also essential for advancing research in spatial thinking, including the reevaluation of conceptual frameworks of spatial thinking (Uttal et al., 2024). Therefore, in this section, we offer recommendations to improve the state of children's spatial assessments.

6.1 Research focused on children's tests

One of the primary recommendations for improving children's assessments is to conduct research aimed at understanding these assessments and their properties. While there is significant momentum in evaluating adult spatial assessments through item response theory-based studies (Brucato et al., 2023; Munns et al., 2023), we recommend that research on children's assessments should begin by understanding existing assessments before attempting to refine them.

Systematically review children's assessments. To understand these assessments, researchers must know what assessments exist and have been used with young children. This work could help clarify how to locate different spatial assessments and lead to understanding differences in how children perform on various tests, particularly of the same construct. For example, Nguyen et al. (2023) recently reviewed the use of virtual environments to assess the development of navigation in children and adolescents. Their review, while not a full systematic review, compares the paradigms of different types of virtual environment studies, which is useful for understanding which aspects of navigation are and are not captured by these assessments. Furthermore, these kinds of reviews also highlight age ranges that are lacking assessments for certain types of spatial skills.

Effects of test design. Researchers should examine how various test characteristics affect children's performance and strategy use. By comparing different tests and versions of the same test, researchers can determine the significance of specific test characteristics. For instance, studies have explored the effects of modality differences on children's performance between paper-and-pencil and computerized versions of the same test (Bambha & Casasola, 2021; Frick & Pichelmann, 2023), as well as between paper-and-pencil tests and physically manipulable versions of the same test (Frick et al., 2013). Additionally, research has investigated the impact of different types of stimuli, such as variations between animals and cubes (Neuburger et al., 2011). However, many test characteristics remain unstudied, such as the spatial arrangement of questions and answer choices. Moreover, most of the research on test characteristics conducted thus far has focused on mental rotation tests.

Children's strategy use. To determine if a particular test is measuring the desired construct, it is important to examine how children approach the test. For example, understanding children's erroneous strategy use (or in other words, children failing to solve a task as intended) can provide insight into the test's validity and indicate whether changes are necessary to limit unwanted strategies. Additionally, analyzing children's strategy use over time can shed light on the development of spatial thinking skills (Quaiser-Pohl et al., 2010).

6.2 Make assessments accessible

To limit the number of tests created ad-hoc or modified for individual studies, reliable, validated assessments should be easily accessible to researchers. This accessibility includes not only making assessments available but also ensuring their formats of assessments are user-friendly by researchers.

Synthesize test resources and information. In addition to systematically reviewing existing assessments, efforts should be made to create a synthesized collection of test information. This collection should detail what each test measures, how to obtain it, how to administer it, its psychometric properties, and the appropriate age ranges for its use.

Adapt assessments for online studies. Many researchers rely on online studies to recruit participants and conduct research with children. Online child testing platforms, such as LookIt, have facilitated this research possible for both synchronous and asynchronous studies (Scott & Schulz, 2017). Although promising research indicates that digitized versions of assessments perform similarly to in-person assessments, more standardization is needed for online spatial assessments (Bambha & Casasola, 2021). These platforms enable researchers to engage families they might otherwise be unable to reach in a lab setting and provide a shared database of families among researchers.

6.3 Diversify which spatial skills are assessed

A significant portion of research on spatial thinking is focused on mental rotation. This emphasis is prevalent in both spatial cognition research and in the operationalization of spatial thinking within various STEM domains. While some studies suggest that mental rotation assessments are uniquely related to certain STEM fields (Bates et al., 2021), the focus on mental rotation has been cyclical; promising results from these assessments lead to increased interest, generating more results related to mental rotation. Importantly, this focus does not imply that other spatial skills lack promising results. Children's perspective-taking and spatial assembly skills have both been shown to be malleable and related to specific areas of mathematics and science (Bower et al., 2020; Bower & Liben, 2021; Gilligan-Lee et al., 2023; Plummer et al., 2016; Verdine, Irwin, et al., 2014). However, the full potential of non-mental-rotation spatial skills has yet to be explored in children. Future research should diversify the spatial assessments used to inform interventions in spatial-STEM studies. Additionally, incorporating multiple spatial assessments within a study examining spatial and STEM abilities can help researchers understand whether the relationship between these skills is more general or specific. For example, what spatial skills, besides mental rotation, are related to specific mathematical concepts (e.g., number line comprehension)? To answer these questions, researchers must have access to valid spatial thinking assessments across a range of spatial skills.

6.4 Incorporate existing research from other cognitive assessments

Spatial cognition is not the only field that has developed cognitive assessments for children. However, research across different fields tends to be siloed. Therefore, researchers aiming to improve or create spatial assessments for children should not only examine existing spatial assessments but also learn from successful assessments in other domains. Various assessments have been developed within psychology to evaluate young children's executive functioning, fluid reasoning, analogical thinking, and other cognitive abilities (Anderson, 2002; Shivaram et al., 2023; Weschler, 2003). Additionally, cognitive skills related to spatial thinking, such as patterning skills, have been effectively assessed in children (Rittle-Johnson et al., 2013). Researchers in these cognitive areas are also advocating for improved assessments for young children. For instance, a recent study explored the gamification of an executive function assessment for preschool-aged children (Eng et al., 2024). By evaluating which types of assessments have been shown to be valid and reliable in other cognitive domains, researchers can apply similar design practices to create and refine spatial assessments for children.

7. Conclusions

Spatial thinking research is an important and growing field, with increasing evidence suggesting that spatial thinking interventions can benefit students across various educational contexts (Gilligan-Lee et al., 2022; Newcombe, 2017). Although this research has been limited by unreliable, unvalidated, inaccessible, or nonexistent spatial assessments for children, these challenges can be addressed. Some researchers have begun focusing on validating children's spatial assessments and providing psychometric information (Frick & Pichelmann, 2023). Furthermore, this work parallels similar studies on adult spatial assessments (Brucato et al., 2023; Munns et al., 2023; Uttal et al., 2024). Given the crucial role of assessment in research, this assessment-focused work is foundational for improving theories and frameworks of spatial cognition, understanding the relationship between spatial thinking and STEM proficiency, and developing effective spatial interventions.

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CHAPTER FOUR

Autonomic and attentional pathways in the emergence of autism: bridging mechanisms and real-world contexts in infancy

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Abstract

Autism spectrum disorder (ASD) unfolds over the first two years of life through complex interactions among developmental systems. Attention and autonomic nervous system (ANS) regulation represent foundational processes critical for adaptive engagement with the environment. Disruptions in these systems during early infancy may initiate developmental cascades that contribute to core ASD features, including social-communication challenges and restricted and repetitive behaviors, as well as the vast heterogeneity found within ASD. This chapter reviews our approach to integrating mechanistic research and real-world methodologies to explore autonomic and attentional pathways in the early development of ASD.

We conduct prospective, longitudinal studies to track how ANS regulation and attention unfold in infancy in infants at elevated and low familial likelihood for ASD. By integrating multimethod approaches to study these functions in naturalistic contexts, we demonstrate that autonomically regulated attention mechanisms support key developmental milestones and may play an important role in the emergence of ASD. Findings presented provide preliminary support for the theory that infants later diagnosed with ASD exhibit early disruptions in parasympathetic modulation of arousal and attention across responses to social and nonsocial contexts.

This work emphasizes the value of leveraging portable, non-invasive technologies to study early autonomic and attention processes in naturalistic contexts. By using this approach to uncover formative mechanisms underlying ASD, we enhance the translational potential of this research while addressing critical gaps in inclusivity and representation. Understanding autonomic and attentional systems in infancy provides an opportunity to identify cascading developmental divergences and promote adaptive developmental trajectories in ASD.

Keywords: Autism, Autism spectrum disorder (ASD), Infancy, Dynamic systems, Developmental cascades, Respiratory sinus arrhythmia (RSA), Parasympathetic, Autonomic nervous system, Attention, sustained attention, social attention, Head-mounted eye tracking, Naturalistic

Autism spectrum disorder (ASD) emerges from the complex interplay of developmental systems, where early differences cascade across developmental domains to shape growth and adaptation over time. Understanding these dynamic processes in the early development of ASD requires bridging theory with application to address critical questions: How do foundational developmental systems unfold in ASD? How do these early developing systems initiate cascading pathways that shape the emergence of ASD? How can our science most directly translate to actionable early identification and intervention strategies that can be implemented during critical developmental windows?

The developmental emergence of core features of ASD occurs gradually over the first three years of life, with diagnostic certainty often achieved by age 2–3 years. Despite most diagnoses occurring between 2–5 years of age, robust evidence suggests that neurological processes underlying ASD are already underway shortly after birth (see (Dawson et al., 2023) for a review). Core features of ASD include delays and differences in social communication and interaction, alongside repetitive behaviors and restricted interests (American Psychiatric Association, 2013). Prospective longitudinal designs have been used to track the development of infant siblings of children with ASD, with the goal to identify early markers of ASD and advance early detection and intervention methods (Szatmari et al., 2016). These baby sibling research designs comprise cohorts of elevated likelihood infant siblings of children with ASD (EL) and low likelihood infants without a family history of ASD (LL). Approximately 20% of ASD siblings are diagnosed with ASD at age 2–3 years and an additional 30% exhibit atypical development and/or subclinical features of ASD (Ozonoff et al., 2011; Ozonoff et al., 2024). These research designs have been instrumental in advancing the field and bringing us closer to identifying mechanisms and pathways in ASD (Messinger et al., 2013; Ozonoff & Iosif, 2019; Szatmari et al., 2016; Zwaigenbaum et al., 2005; Zwaigenbaum et al., 2015). For example, as a result of infant sibling research, we have identified social-communication delays and repetitive behaviors as early as 9–12 months, revealing early iterative processes that shape and are shaped by subsequent skills and developmental outcomes (Bradshaw et al., 2021; Pileggi et al., 2021). Outside of core features of ASD, findings from such studies have reported even earlier differences in foundational developmental processes, like attention and autonomic nervous system (ANS) functions.

Atypical attention is among the earliest of these developmental processes to characterize individuals with ASD and persists as a lifelong feature of ASD (Ness et al., 2019; Murias et al., 2018; Bradshaw et al., 2019; Kaliukhovich et al., 2020; Frazier et al., 2017). Research has demonstrated that atypical attention is not only detectable in early infancy but is also predictive of later developmental outcomes (Bradshaw et al., 2020; Di Giorgio et al., 2016; Jones & Klin, 2013). However, the biological mechanisms underlying these early disruptions remain poorly understood. One promising avenue of investigation lies in another foundational system—the autonomic nervous system. ANS is an early-emerging bioregulatory system essential for regulating neonatal arousal and attention (Brazelton & Nugent, 2011; Feldman, 2006; Porges & Furman, 2011). Disruptions in the ANS's regulation of attention, or "autonomic regulation of attention," may initiate a series of cascading effects that influence social interaction, learning, and broader development. While general ANS dysfunction has been documented in older children diagnosed with ASD (Cheng et al., 2020), less research has focused on whether and when such dysfunction contributes to atypical attention in infancy and how early autonomic regulation predicts the emergence of ASD symptoms.

Baby sibling research to date relies heavily on laboratory-based methods that prioritize experimental control. These approaches, while valuable, simplify the complexities of infants' natural environments, which are rich with experiential and adaptive factors critical to development. Naturalistic data collection—capturing behavior or specific neurodevelopmental processes in everyday environments—represents a conceptual shift in our scientific approach, embracing the dynamic, contextual factors that shape development, rather than removing them in lab-based experiments or controlling for them in statistical analyses (Tamis-LeMonda et al., 2024). By observing infants in their lived environments, researchers can access the very influences that drive developmental trajectories. Studying attention and ANS in these contexts, while leveraging portable, noninvasive technologies, not only advances knowledge of developmental mechanisms, but also bridges the gap between research and practice, facilitating earlier detection and intervention strategies that can be seamlessly implemented in real-world settings.

Naturalistic methods also offer a pathway to address longstanding disparities in participant representation by enabling research in a variety of home settings. Participants in baby sibling studies predominantly comprise white, higher-resource families. Descriptive data from the past three decades of autism intervention research show that white participants have the highest rate of participation, comprising 64.8% of sample cohorts (Steinbrenner et al., 2022). This underrepresentation limits understanding of ASD across diverse populations and addressing these disparities is essential, not only for equity, but also to fully capture the heterogeneity of developmental contributing to the phenotypic diversity of ASD.

Building on emerging findings in attention, ANS regulation, and the dynamic interactions among these systems in the early emergence of ASD, we propose a unified framework that integrates existing literature and our findings to better understand how foundational systems in real-world settings initiate cascading effects in the emergence of ASD. First, we explore how cascading developmental processes—specifically, disruptions in attention and autonomic regulation—shape typically developing and ASD trajectories during infancy. Next, we highlight the promise of portable, noninvasive technologies like head-mounted eye tracking (HMET) and synchronized home-based systems to capture these processes in real-world settings. Finally, we address critical challenges in participant inclusion and representation, underscoring the importance of studying diverse populations to ensure findings are relevant and translatable to broader contexts. By bridging mechanistic research with innovative methodologies and inclusive approaches to research, we aim to contribute to the current understanding of ASD development in infancy and create more direct pathways toward early detection and intervention.

1. Developmental cascades and the emergence of ASD

A developmental cascades framework views development as continuous, cumulative interactions and transactions across various developmental domains, biological systems, and environmental contexts (Masten & Cicchetti, 2010; Oakes & Rakison, 2019). It provides a lens for investigating how lower- or higher-level processes produce cross-domain, reciprocal effects that influence outcomes later in development. Importantly, cascades highlight both specific developmental mechanisms and the dynamic interactions among multiple mechanisms that unfold over time to shape development. Individual experiences contribute to the divergence of developmental trajectories, creating opportunities for individualized input driven by the infant, which, in turn, influences future developmental pathways.

Developmental cascades offer a useful framework for understanding the early development of ASD, comprised of cascading processes, where early differences in foundational systems, including autonomic and attentional systems, set off a ripple effect that can enrich or disrupt downstream developmental pathways (Bradshaw et al., 2022a; Iverson et al., 2023; Karmiloff-Smith, 1998; Thelen & Smith, 1995). Among these foundational systems, attention stands out as a particularly intriguing process that may be altered in individuals with ASD. Variability in early attentional processes can significantly influence later social, cognitive, and communicative development. In ASD, early attentional differences may contribute to the atypical development of social communication abilities (Bradshaw et al., 2022a). Early attentional processes undergo rapid development over the first few months of life (e.g., by 6–10 weeks), driven by a shift from subcortical to cortical control of behavior. Critical elements of early-developing attention processes—such as disengaging from stimuli, shifting focus, and filtering irrelevant information—are foundational for cognitive and behavioral development (Colombo, 2001; Xie, 2017; Conte & Richards, 2021; Hunnius, 2007). Achievements of these early attentional milestones coupled with co-occurring achievements in other domains (e.g., motor and cognitive) form a critical foundation for continued growth in social, language, and cognitive domains, making early attention a key focus for understanding the mechanisms underlying atypical developmental trajectories, including those observed in ASD.

In ASD, atypical patterns of visual attention represent the result of, and contributor to, influential developmental cascades that begin in infancy. Reduced attention to socially relevant stimuli, such as eyes and faces, as well as more generalized difficulty disengaging from focal stimuli, emerge within the first year of life (Chawarska et al., 2013; Fu et al., 2024; Jones et al., 2016). These subtle variations in attention may not constitute diagnostic criteria for ASD, but have broad implications for social development. For example, reduced visual saliency of the eyes of a face in early infancy could lead to diminished attention to eyes in later infancy, which may reduce the frequency and quality of interactions with caregivers, resulting in missed opportunities for social learning, and may reinforce nonsocial attentional preferences over time, as is observed in ASD (Pierce et al., 2016).

Atypical visual attention is also evident in young EL infants, irrespective of their eventual diagnostic outcome (Di Giorgio et al., 2021; Di Giorgio et al., 2016). While these attentional differences may be transient (Bradshaw et al., 2020), they point to an underlying genetic basis for atypical social attention in ASD. Genetics strongly influence visual attention patterns throughout childhood (Kennedy et al., 2017; Constantino et al., 2017) which further influence information processing and social interactions. Together, these findings highlight the role that biological predispositions for social or nonsocial stimuli play in the developmental cascades that lead to ASD.ASD-specific differences in lower-level attention processes, such as attention orienting and disengagement, may stem from atypical neural and neurochemical development. Disruptions in arousal regulation and visual processing circuits have been proposed as potential early contributors to attentional variability in ASD (Artoni et al., 2020; Hazlett et al., 2017). Arousal regulation is driven by the sympathetic-parasympathetic balance of the autonomic nervous system. The parasympathetic branch of the ANS supports attention and readiness for cognitive engagement, pointing to the ANS as a critical biological factor in the development of attention, attentional control, and visual processing. Cascading effects of early ANS dysfunction are less well studied, but early differences in ANS and neural circuitry could initiate developmental cascades that affect both regulatory behavior and subsequent brain development, which may shape later patterns of social engagement and learning. For instance, difficulties with both arousal modulation and attention shifting may compound, negatively affecting infants' ability to engage in triadic interactions with objects and caregivers. These interactions require a calm arousal state and effective attention shifting to facilitate entry into a shared space of joint attention and engagement. Such states of joint engagement with caregivers and objects are critical for language acquisition and higher-order cognitive integration (Adamson et al., 2019; Adamson et al., 2009; Thorp, 2006; Conway et al., 2018).

Overall, the developmental cascades model frames development as a dynamic system, where interactions between biological predispositions, developmental processes, and environmental inputs continuously shape unfolding trajectories. Early disruptions in the foundational systems discussed here, including autonomic regulation and visual attention, can cascade to affect social communication, motor development, and broader neurocognitive functions. Mechanistic frameworks for ASD therefore highlight the convergence of multiple subsystems and underscore the need for integrative, multimodal approaches to studying its emergence.

2. The role of autonomic nervous system function as an early regulatory mechanism in ASD

Systems-level biological factors known to distinguish those with vs. without ASD include genetics, brain development, and physiological arousal. Among these biological systems, the ANS plays a central role in regulating behaviors essential for social and cognitive development. Understanding parasympathetic function as a component of ANS provides insight into its contribution to developmental cascades in ASD. There is robust evidence that ANS-driven physiological systems support behavioral regulation, social engagement and interaction, and attention and cognitive processing (Porges & Furman, 2011; Heilman et al., 2007; Porges, 2003; Richards, 2010; Thayer et al., 2009; Whedon et al., 2018). This body of work lays the foundation for examining how developmental *trajectories* of autonomic and attentional mechanisms in infancy shape development, initiate cascades, and contribute to outcomes.

ANS regulation is governed by the balance of sympathetic and parasympathetic activation, with the sympathetic nervous system activating during times of stress ("fight or flight") and the parasympathetic nervous system activating during times of rest ("rest and digest"). This delicate balance enacts physiological changes in basic functions, such as decreasing heart rate or increasing body temperature to support infants' healthy responsiveness and adaptation to environmental demands.

In early prenatal development, physiological functions are driven primarily by sympathetic influence. Higher cortical processes begin to integrate autonomic control in the late prenatal and early postnatal period leading to increasing influence of the parasympathetic nervous system. As ANS matures, parasympathetic influences support calm states of alertness and sustained attention that are essential and provide countless opportunities for development across domains, including social and language learning. In this way, ANS is considered a "developmental resource" in the neonatal and early infant period that mediates infant-environment interactions.

Physiological measures used to index parasympathetic function include heart rate variability (HRV) and respiratory sinus arrhythmia (RSA). Both HRV and RSA reflect variability in the influence of the parasympathetic system on the heart, with HRV reflecting variability in heart rate and RSA reflecting variability in heart rate at the frequency of respiration. An adaptive, regulated ANS involves continuous changes to heart rate, making increased variability (higher HRV and RSA) indices of parasympathetic influence over the heart. HRV and RSA undergo significant maturation in infancy and early childhood, and, when measured during periods of rest, our work demonstrates that the first six months of life is a period of particularly rapid growth (Fig. 1). Throughout development, higher HRV and RSA at rest reflect an infant's readiness and capacity to regulate arousal and maintain engagement with their surroundings. During times of stress, RSA decreases due to the suppression of the parasympathetic system to make way for sympathetic influence. RSA suppression supports mobilization of the fight or flight system, which includes heightened arousal and cognitive focus. Evidence indicates that more RSA suppression in response to stress reflects a healthy, responsive neurophysiological system (Bazhenova et al., 2001; Porges et al., 1996). Studies that demonstrate links between RSA suppression and developmental outcomes, for example social interaction and language, strengthen these claims.

To evaluate the link between RSA suppression and positive developmental outcomes, we conducted a longitudinal study of infants at elevated and low likelihood for ASD (Bradshaw & Abney, 2021). RSA was measured in 3-4-month-old infants during a semi-structured infant-caregiver interaction. The structure of the interaction was modeled after the still-face procedure, but infants and parents were not constrained in terms of positioning or movement. The infant was placed in supine on the Table while the caregiver stood in front of them interacting as they normally would at home for 5 minutes. At that time, the caregiver was cued to sit down and ignore

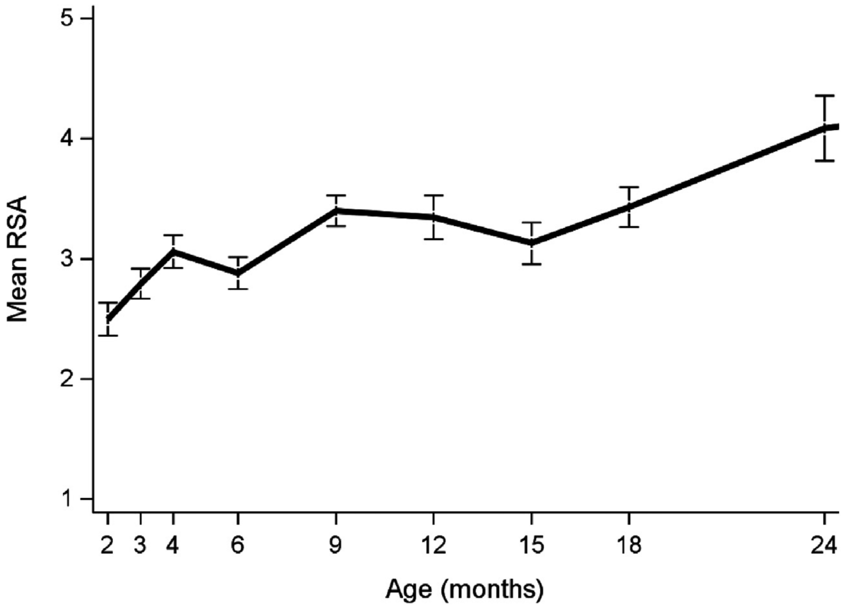

Fig. 1 Trajectory of mean, resting RSA in typically developing infants from 2 to 24 months of age.

their infant for 2 minutes while maintaining a neutral expression. Because they were no longer face-to-face during this time, both the infant and caregiver were free to visually explore their environment, and the infant was able to move as they liked. After this 2-min period, the caregiver interacted with their infant again for 2 min. RSA was measured during each phase of this interaction, and RSA suppression was indexed as the difference between RSA during the "play" phase and RSA during the "ignore" phase. We found that RSA suppression during this task at 3–4 months positively associated with parent-reported social communication skills at 9 months. These findings suggest that appropriate parasympathetic responsivity to environmental demands, particularly those that are social in nature, may initiate developmental cascade that supports early social learning, which in turn fosters early social-communication development.

2.1 Autonomic dysfunction and autism in preterm infants

Decreased parasympathetic influence on the heart, reflected in lower HRV and RSA, can signal underlying ANS dysfunction (Thayer et al., 2009; Mathewson et al., 2014; Propper, 2012). Consistent with studies of children and adults, and theoretical frameworks that describe ANS as a foundational regulatory function that is disrupted in ASD (Cheng et al., 2020; Patriquin et al., 2014; Sheinkopf et al., 2019), we have found differences in early ANS function and a high prevalence of ASD in infants born preterm. Preterm birth presents a unique context for studying ANS dysfunction, as these infants experience major challenges in early autonomic development. Substantial ANS maturation occurs in the second half of pregnancy, and preterm birth (born <37w GA) significantly disrupts the development of ANS, along with the overall functioning of the PT newborn (Mulkey & Plessis, 2018). The drastic transition to the extrauterine environment that occurs amidst ongoing ANS maturation is a clash that is compounded by exposure to medical stressors. As a result, autonomic maturation is impaired throughout the PT period, fails to normalize at term age equivalent (Fyfe et al., 2015), and may alter long-term temporal programming of ANS (Haraldsdottir et al., 2018). This early process initiates a developmental cascade that disrupts initial and subsequent physiological adaptations to the environment, and may affect the integration of autonomic control into later-developing, higher-level cortical processes important for social learning and cognitive development, such as attention. These early disruptions may be a critical factor in the emergence of ASD features, leading to heightened likelihood of ASD in very PT infants compared to the general population.

Building on the theoretical and empirical evidence linking ANS dysfunction to PT infants and to ASD, we conducted a prospective, longitudinal pilot study of very preterm infants (born < 32 weeks gestational age) in which we examined the predictive link between neonatal ANS dysfunction and later-developing ASD symptoms (Bradshaw et al., 2024). In this study, we collected continuous, multimodal measures of ANS dysfunction. ANS measures comprised minute-by-minute thermal gradients, defined as the difference in neonatal peripheral temperature and core temperature (Knobel-Dail et al., 2017), and hour-by-hour abnormal heart rate indices, measured using Heart Rate Characteristic (HRC) scores derived from HeRO Monitors, (Fairchild & Aschner, 2012). Heart Rate Characteristic Scores reflect an algorithm that considers beat-to-beat heart rate, heart rate variability, interbeat interval (IBI) standard deviation, and IBI skewness (Fairchild & Aschner, 2012). Thermal gradients and HRCs were collected for 28 days after birth (>40,000 samples/infant) in the neonatal intensive care unit (NICU). Throughout the 28 days, ANS dysfunction was characterized by an average HRC score >1 and abnormal negative thermal gradients for about 20%

of the time (6 of 28 days). Following NICU discharge, comprehensive neurodevelopmental follow-up consisting of standardized measures of cognition, language, and motor skills were collected at adjusted ages 6, 9, and 12 months. Adjusted ages equate the neurological and physical development of preterm infants with that of full-term infants, whereas chronological age equates duration of time in the extrauterine environment with that of full-term infants. At 12 months adjusted age, assessments of social communication and early ASD symptoms were administered using the Systematic Observation Rating of Features of ASD (SORF; Dow et al., 2017).

In this small pilot study of very preterm infants who were retained in the study and followed to 12 months, we observed that the neonatal abnormal heart rate characteristics were strongly correlated with ASD symptoms at 12 months ($r = 0.81$, $p < .01$), as was birth gestational age, birth weight, and abnormal negative thermal gradients. A regression model including GA, BW, and HRC scores revealed a significant, unique predictive effect of abnormal heart rate indices on ASD outcomes (model $r = 0.86$, $p = .017$; HRC Score: $r = .72$, $p = .029$). Overall, these results show a surprisingly high association between ANS dysfunction in the neonatal period and ASD features at 1 year of age in very PT infants. This study provides compelling evidence for the role of ANS dysregulation and its cascading effects in the development of ASD. Whether this points to ANS as a catalyst for an ASD-specific developmental cascade, or reflects a developmental endpoint in an earlier cascade that serves as a more direct mechanistic pathway to ASD remains unknown, but highlights the need for further research into early autonomic mechanisms as predictors of ASD.

3. The role of attention as an early regulatory mechanism in ASD

Although atypical attention does not constitute a core feature of ASD, mounting evidence suggests impairments in attentional networks are pervasive and lifelong, constituting a central feature of ASD. Attentional disengagement is among the earliest reported features of ASD, particularly in EL infants (Zwaigenbaum et al., 2005; Elsabbagh et al., 2009). Attention disengagement is the process of shifting attention from one stimulus to another and serves as a regulatory mechanism that modulates arousal. A bidirectional relationship between arousal and disengagement would involve effective arousal modulation facilitating attention disengagement, and failure to disengage leading to further dysregulation.

Our research demonstrates that atypical attention can be observed as early as three months of age in EL infants (Bradshaw et al., 2020) and infants who go on to develop ASD (Bradshaw et al., 2021). In these studies, we used a neonatal neurobehavioral exam to evaluate attention throughout the first five months of life. In the first study, we examined trajectories of visual and auditory attention to animate (faces and voices) and inanimate (objects) from 1 week to 5 months of age in EL infants and compared them to LL infants. We also tested associations between early attention and 12-month social-communication and cognitive outcomes.

Trajectories of attention across the first five months of life illustrated significant growth over time for both EL and LL infants. However, EL infants exhibited lower performance on the attention tasks at 2 and 3 months, particularly in visual attention tasks, compared to their LL peers. However, this difference was transient and by 4 and 5 months, their performance aligned with that of LL infants. Contrary to our expectations that social attention (i.e., animate attention tasks) would diverge most from that of LL infants, we found that attention to both animate and inanimate visual stimuli was lower during this time period, highlighting generalized vulnerabilities in attention during early development. At 12 months, early attention at 3 months was significantly associated with social-communication skills and, to a lesser extent, nonverbal cognitive skills.

In a follow-up study, we compared the same attention measures in infants who went on to develop ASD with those of typically developing infants (Bradshaw et al., 2021). Based on our earlier findings (Bradshaw et al., 2020), we focused on attention at 1, 2, and 3 months. Results showed that attention development differentiated infants with ASD from their TD peers. At 1 month, attention scores were comparable between groups, but by 2 and 3 months, infants with ASD exhibited significantly diminished attention. Further exploration revealed that this finding was most robust for visual, inanimate, attention (attention to objects). The difference in inanimate attention at 3 months in infants later diagnosed with ASD may reflect disruptions in the transition from reflexive to volitional control of attention during a critical period of neurodevelopment.

At birth, attention to visual stimuli is primarily subserved by subcortical mechanisms that enable reflexive orienting. This allows the youngest of neonates to orient to and track objects and faces. By 2 to 3 months, however, cortical systems, particularly those in the parietal and prefrontal regions, begin to mediate volitional attention and attentional shifting. This transition supports the development of more complex behaviors, such as joint engagement with objects and caregivers, which are foundational for social and communication skills. Infants with ASD may experience delays or disruptions in this cortical-driven process, resulting in diminished ability to orient to and sustain attention on inanimate stimuli. Moreover, orienting requires visual disengagement from other salient stimuli and visual tracking requires motor control and movement of head and neck that aligns with eye movements. Both attention disengagement and motor challenges, particularly head, neck, and trunk control (Bradshaw et al., 2023), are observed in very early development in ASD. These difficulties with attention disengagement from social or environmental stimuli, as well as impaired coordination between visual and motor systems, may also explain the attention differences observed in these studies. Together, these results emphasize the importance of attentional processes during the 2- to 3-month window, possibly the result of underlying neurodevelopmental or regulatory vulnerabilities, which may shape developmental pathways later observed in ASD.

4. The coupling of ANS and attention: autonomic regulation of attention

Attention and parasympathetic activation both serve as foundational regulatory functions throughout the lifespan. In infancy, the coupling of autonomic regulation and attention is a cornerstone of an adaptive engagement system, The ANS modulates attention by regulating heart rate responses during information processing, particularly through parasympathetic activity. A well-functioning ANS enables neonates to override metabolic demands and generate physiological responses, such as heart rate deceleration, which support sustained attention and enhance stimulus processing (Lester et al., 1996; Richards, 2004; Richards & Turner, 2001).

Parasympathetic-driven attention responses underpin three distinct phases of attention—attention orienting, sustained attention, and attention termination—each linked to cognitive and learning processes (Richards, 2010; Richards, 2004). Among these, sustained attention, plays a particularly crucial role in cognitive engagement. Characterized by prolonged look duration and heart rate deceleration, sustained attention facilitates prioritization of relevant stimuli while reducing distractibility, resulting in selective information processing and recognition memory (Richards & Hunter, 1998; Reynolds & Richards, 2005).

Heart rate measures of sustained attention represent a noninvasive method for understanding cognitive and neurological processes that underly sustained attention. During sustained attention, the heart rate slows and remains decelerated, signaling heightened cognitive engagement and information processing. As sustained attention shifts to the third phase, attention termination, heart rate returns to baseline, reflecting diminished arousal and increased distractibility (Richards, 2004). Sustained heart rate deceleration reflects the activation of arousal systems, crucial for maintaining attention, while the return to baseline during termination marks disengagement (Richards & Casey, 1992). Heart rate-defined sustained attention (HRDSA) can be indexed with duration of the heart rate deceleration, and the magnitude of the heart rate change, or the depth of deceleration. HRDSA is said to offer a more precise measure of information processing than looking behavior alone (Colombo, 2001; Reynolds et al., 2011; Xie & Richards, 2016). This sequence highlights the dynamic role of parasympathetic activity in regulating attention and underscores the importance of HRDSA as a measure of meaningful attention and engagement during infancy.

4.1 Developmental course of autonomic regulation of attention

Attention undergoes significant changes during the first two years of life, shaped by the maturation of attention networks. Early in infancy, posterior attention systems support visual orienting and disengagement, while anterior attention systems involved in executive control emerge after six months. These neurodevelopmental milestones contribute to varied trajectories in attention metrics. For instance, look duration, which indicates visual learning and encoding, follows a triphasic trajectory: increasing during the first two months, decreasing between three and six months, and rising again after six months (Colombo, 2001; Richards & Anderson, 2004).

To further investigate the dynamic unfolding of these regulatory processes, and to comprehensively document the development of attention in context, we examined age- and context-dependent differences in looking behavior and physiologically-regulated sustained attention. In a two-year longitudinal study, we mapped trajectories of attention in over 50 infants from 1 to 24 months of age. To understand the effect of stimulus on attention patterns, we included four distinct stimulus conditions—social videos, nonsocial videos, social interactions, and nonsocial interactions—across multiple attention metrics that included overall looking time, mean look duration, percent time in HRDSA, and heart rate deceleration during HRDSA episodes.

Results of this study showed that during the first six months, overall looking time increased across all conditions, with screen-based tasks eliciting more looking than live interactions. However, contrary to expectations, a drastic decline in attention to the caregiver's face during the social interaction was observed between 4-6 months and was lower than all other stimuli by 6 months. The decline in face-looking coincides with developmental gains in motor skills and executive control of attention, which facilitates a broader, more dynamic exploration of the environment. The emergence of these skills is also closely tied to the maturation of cortical control of attention, particularly the development of the anterior attention network. This network underpins voluntary attention shifting and disengagement, processes that are crucial for adapting to and learning from a complex, multimodal environment. Importantly, our data showed that HRDSA during face-looking remained stable or increased slightly during this period, suggesting that while infants may look at caregivers' faces less frequently between four and six months, these looks reflect stable cognitive engagement and efficient processing. This highlights the value of HRDSA as a measure of meaningful attention.

In the second year of life (9–24 months), attention measures become more stable across all task conditions, with videos (both social and nonsocial) eliciting more sustained attention than live toy interactions. This may be due to the perceptual saliency and cognitively engaging content of videos compared to the real-world demands of interactive tasks, which require the infant to initiate the interaction and engage motor and cognitive resources for sustained engagement. Social videos elicited longer look durations compared to nonsocial videos during this period, suggesting an enhanced attention to social stimuli as infants' communication and language skills develop.

Findings from this study highlight the importance of considering both context and measurement method in studies of infant attention. Our results suggest that behavioral measures—percent looking and look duration—capture general behavioral trends, while HRDSA and heart rate deceleration provide deeper insights into cognitive engagement. The divergence between these attention measures demonstrates that reduced looking time does not necessarily indicate diminished attentional engagement, especially in dynamic, interactive contexts. This underscores the need for multi-method approaches to studying infant attention, particularly for identifying early markers of atypical development. Overall, this study establishes a multimethod-informed trajectory of typical infant attention over the first two years of life, providing a reference point for comparisons with other populations, such as EL infants or those with ASD.

4.2 Autonomic regulation of attention and ASD: a developmental framework

The evidence is clear that physiological measures of attention, reflecting ANS's regulation of attention, are related to cognition and learning. Early disruptions in autonomic regulation of attention then may have cascading consequences for interaction, learning, and development (Feldman, 2006; Porges & Furman, 2011; Bradshaw et al., 2022). Based on current literature and our preliminary data, we propose a developmental framework that highlights the role of ANS dysregulation in attention very early in development and has cascading effects on learning and development. In this framework, global ANS function, which supports neonatal arousal and reactivity, appears intact at birth. Arousal and reactivity during this period are primarily driven by sympathetic activation, allowing newborns to respond to environmental demands, such as crying in response to internal states like hunger and pain. However, as parasympathetic regulation matures during infancy, it has increasing influence over the heart and becomes critical for maintaining a calm state and modulating attention. Adaptive modulation of arousal and attention then supports sustained cognitive engagement and readiness for social interaction. In infants who develop ASD, this transition to effective parasympathetic regulation may be disrupted.

This disruption may underly observed attention differences in the 2-3 month period (Bradshaw et al., 2020; Bradshaw et al., 2021), when parasympathetic modulation of attention becomes increasingly necessary for adaptive functioning. Ineffective parasympathetic modulation interferes with the ability to sustain attention to relevant stimuli and flexibly shift attention to novel, salient stimuli. This impairment may diminish the quality and frequency of learning experiences, particularly those that involve social and communicative interactions. Over time, the resulting cascade may contribute to the emergence of core ASD features, including difficulties in social communication and behavioral flexibility.

Preliminary data that support this framework include trajectories of heart-defined sustained attention during naturalistic parent-infant interactions from 1-4 months across EL, LL, and PT infants. These trajectories are characterized by a significant increase in heart rate (HR) deceleration during sustained attention and point to a dampened HR deceleration that

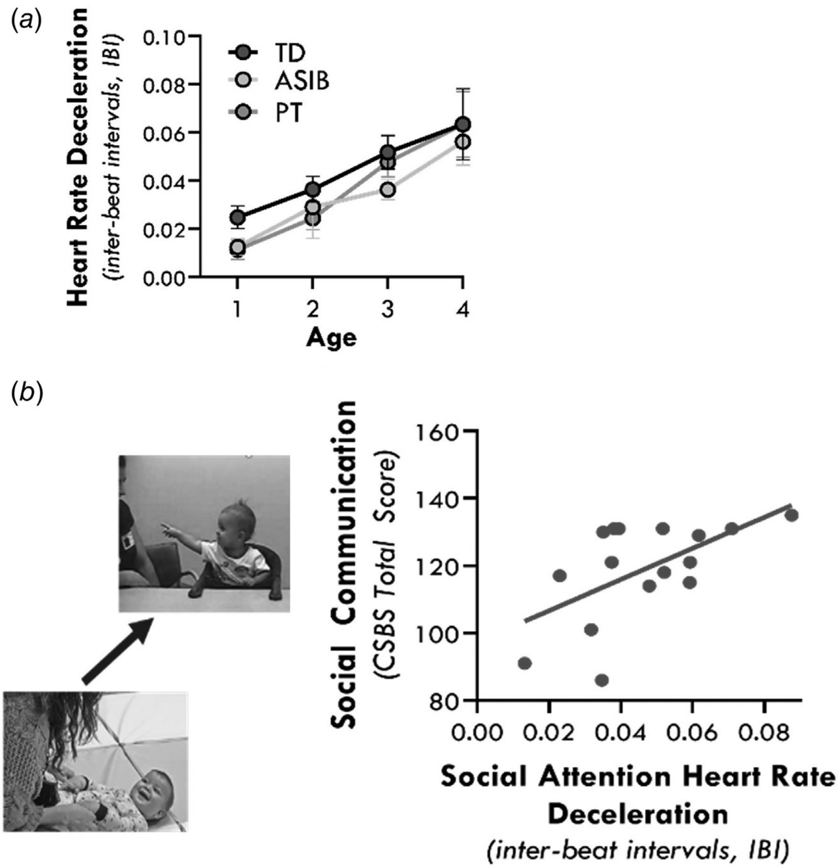

Fig. 2 **a**. Developmental trajectory of heart rate deceleration within periods of heart rate-defined sustained attention (HRDSA) during infant-caregiver interactions across ASD, TD, and PT infants. **b**. Association between deceleration within periods of HRDSA measured during infant-caregiver interactions at 3-4 months and social-communication skills measured at 9 months.

is developmentally transient for PTs, but persistent for EL infants (Fig 2A; (Bradshaw et al., 2022b). Moreover, we have found that HR deceleration during sustained attention to caregivers at 3 months is associated with early-emerging symptoms of ASD (Bradshaw and Abney, 2021) (*Fig. 2*B).

In a preliminary investigation and an extension of our previous work (Bradshaw et al., 2024), we examined physiological measures of attention to social and nonsocial videos across EL and LL infants from 1 to 24 months of age. While LL infants demonstrated an emerging distinction between social and nonsocial attention based on both behavioral and physiological measures in the second year of life, with greater attention to social stimuli, this differentiation was absent in EL infants. Specifically, for EL infants, the proportion of time spent looking to the screen and the proportion of looking classified as HRDSA were equivalent across both types of stimuli. This was not driven by diminished attention to both social and nonsocial stimuli, but rather high levels of both social and nonsocial attention. Trajectories of social attention between EL and LL infants did not differ. These findings suggest that from 12 to 24 months, LL infants exhibit enhanced attention to social stimuli or attenuated attention to nonsocial stimuli. In contrast, EL infants engage in equally high levels of attention to both social and nonsocial stimuli, reflecting a lack of differentiation in attentional prioritization that may underlie or reflect early disruptions in social engagement.

5. ANS and attention: cascading consequences in ASD

In ASD, the transition to effective parasympathetic modulation of ANS in the first year of life emerges as a critical developmental inflection point. ANS dysregulation during this period may limit the capacity for sustained attention and adaptive engagement. This framework has yet to be fully tested, but there are at least two alternative consequences to a dysregulated autonomic-attention system. First, instead of ANS dysregulation cascading into a less efficient attention control system, compensatory mechanisms may lead to the need for increased parasympathetic activity, evidenced by deeper heart rate decelerations and higher RSA, needed to achieve similar levels of regulation and attention as neurotypical infants. And second, there may be a decoupling of ANS regulation and attention mechanisms in ASD, meaning that other systems are employed to regulate attention and that parasympathetic influence over the heart may not necessarily serve to regulate attention or other processes. This may lead to the deployment of other systems and mechanisms to facilitate regulation. These other systems may not only be less effective, but may take away from resources allocated to other neurodevelopmental processes.

Taken together, these findings emphasize the importance of examining parasympathetic function and autonomically regulated attention as both a developmental resource and a potential early marker of ASD. By integrating behavioral and physiological measures of attention, such as HRDSA, future research can refine models of neurodevelopment and identify early pathways leading to ASD. This approach offers promise for earlier detection and intervention strategies that target attention and autonomic regulation to mitigate developmental disruptions and promote adaptive learning and interaction.

6. Multimodal methods in naturalistic contexts: an approach to understanding heterogeneity in ASD

The phenotypic manifestation of ASD is strikingly variable across individuals. This heterogeneity is apparent in early developmental pathways as well. While some infants exhibit clear trajectories toward an ASD diagnosis, others show more variable patterns, including transient symptoms, or delayed onset of early ASD markers and behaviors (Landa et al., 2007). Through a developmental cascades framework, we understand environmental contexts play an important role in developmental trajectories and outcomes. The effect of environmental contexts on developmental trajectories of ASD is understudied, but interactions between biological and experiential factors likely influence the observed heterogeneity in outcomes. While statistical models can, and often do, control for context variables like socioeconomic status (SES) or early intervention history, these adjustments may not fully account for the nuanced variability observed in naturalistic social interactions or home environments. As an example, differences in motor actions and possibly motor skill development in infancy can be influenced by the home environment, even subtle features like furniture arrangement or flooring textures. For instance, a bedroom with limited furniture at the height level of an infant might lead to fewer opportunities for cruising behavior, an important bridge milestone to independent walking. Alternatively, an infant in this environment may benefit from enriched social interactions as caregivers provide support for the infant to cruise while they practice walking. The child's behavior is therefore a result of the interaction between the child's own abilities and the environmental opportunity to develop and express these abilities. Similarly, infant behavior and physiological regulation, measured with RSA, have been found to be modulated by maternal sensitivity (Girod et al., 2024), highlighting how physiological and behavioral differences emerge only under specific contextual conditions. Together, these examples highlight the need to study early markers situated within an individual's environment, familial context, behavior, and biology. These nuanced, interconnected dynamics set the stage for examining how specific physiological and attentional mechanisms interact with environmental contexts to uncover specific developmental pathways of ASD.

6.1 Naturalistic studies of attention and ANS

Thus far, we have discussed studies of physiological regulation and attention that occur in largely controlled, lab-based environments. While these studies provide valuable insights into the mechanisms underlying early differences in infants who develop ASD, they often fail to capture the dynamic and multifaceted nature of real-world infant experiences. In everyday life, infants are immersed in rich, complex environments where attention is constantly influenced by interactions with caregivers, fluctuating contextual demands, and evolving developmental capacities. These complex factors both shape and are shaped by the infant's ongoing development. Naturalistic settings present challenges for researchers aiming to measure attention and other neurodevelopmental processes in a way that reflects their real-world functionality while maintaining methodological rigor.

To address this gap, we have leveraged specialized approaches that bridge two highly valued features of developmental science: experimental control and naturalistic contexts. Advanced tools like head-mounted eye tracking (HMET) capture infants' moment-to-moment visual engagement within everyday social environments. This method allows us to observe how infants allocate their attention to social and nonsocial features of their environment as they interact with their caregivers and surroundings. By applying this approach in infant sibling studies of ASD, we aim to uncover nuanced patterns of attention that are both influenced by and predictive of early developmental trajectories. These efforts provide a more comprehensive picture of how attention unfolds in the dynamic, real-world contexts shaped by social, motor, and cognitive development.

6.2 Feasibility of head-mounted eye tracking in very young infants

Infant social attention develops within an intricate, embodied context shaped by dynamic interactions with caregivers, objects, and the broader environment. Traditional, lab-based approaches fail to capture the real-time, bidirectional processes through which caregivers scaffold infant behavior and infants explore their surroundings. HMET technology offers an innovative lens to study infant social attention within naturalistic and even home-based contexts. Unlike traditional screen-based eye-tracking methods that constrain participants to fixed positions and controlled environments, HMET provides the unique ability to capture real-time gaze patterns as individuals move, interact, and engage with their surroundings. By integrating lightweight cameras mounted on a headband or cap, researchers record both the wearer's gaze and their field of view, enabling precise mapping of attention within three-dimensional, everyday environments. This approach is particularly valuable for studying infants, whose behaviors and interactions unfold in dynamic and unpredictable ways that cannot be fully captured in static laboratory settings (Bradshaw et al., 2023). Recording gaze behavior during naturalistic interactions facilitates the study of how attention is dynamically shaped by factors like body posture, movement, and environmental affordances. Moreover, HMET enables researchers to capture moment-to-moment coordination between individuals, such as how a caregiver's gaze and actions influence an infant's attention or how joint attention is established during shared activities. These capabilities make HMET an essential tool for understanding the embodied and interactive nature of attention, particularly in populations with atypical development, such as children with ASD, where traditional paradigms may not fully capture the nuances of their real-world social behavior.

The first study to employ HMET with children diagnosed with ASD offered surprising results that challenge longstanding models of social attention deficits in ASD (Yurkovic-Harding et al., 2022). Contrary to the extensive literature emphasizing social attention deficits, in particular gaze patterns to faces, and impaired joint attention skills in children with ASD, this research revealed that children with ASD show no significant differences in real-world attention to faces during free play with a caregiver compared to their typically developing peers. Both TD children and those with ASD spent minimal time looking at their caregiver's face during naturalistic play, with face-looking accounting for less than 2% of the interaction time. These results are particularly striking considering the vast research landscape illustrating social attention deficits in ASD that has guided our understanding and core models of ASD etiology.

In addition, the study examined joint attention, or coordinated visual attention (CVA), between children and caregivers. CVA is defined by moments of child and caregiver simultaneous gaze to the same object. Again, both groups demonstrated comparable rates of CVA, predominantly via alternative pathways, such as manual interactions with objects. This observation also runs counter to prevailing notions that gaze to face is the primary conduit for establishing shared attention, or CVA. In the early ASD literature, joint attention, which involves CVA with a social partner for the explicit purpose of sharing attention to the same object or event, is predominantly characterized by discrete instances of child behavior, including point- and gaze-following (child response to joint attention), or distal pointing and/or two-point and three-point gaze shifts between faces and objects (child initiation of joint attention). These cues represent the primary pathways to joint attention. While deficits in these pathways are well-documented and clinically significant (Adamson et al., 2009; Dawson et al., 2004; Leekam & Ramsden, 2006; Naber et al., 2007; Nyström et al., 2019; Warreyn et al., 2007), Yurkovic-Harding et al. (2022) suggests they may represent reliable markers of ASD rather than causal mechanisms underlying the disorder.

The study's context—a naturalistic, free-flowing play environment—exemplifies the critical importance of context and advanced tools that effectively capture the breadth and impact of this context. Unlike the constrained settings of traditional laboratory-based paradigms, the use of HMET in this study captures the complexity of real-world interactions, during which infants and caregivers co-construct shared attention through dynamic, multimodal exchanges. By highlighting the potential for alternative pathways to support social and communicative development, this research opens new avenues for understanding developmental trajectories in ASD.

Leveraging the strengths of HMET in naturalistic contexts, we have explored how this tool can be applied in the earliest stages of infancy to examine embodied attention in everyday environments. Existing HMET research has largely focused on toddlers and children, leaving a gap in how infants, particularly those as young as four months old, coordinate attention with their caregivers and surroundings. In a longitudinal study of infants at low and elevated familial likelihood for ASD, we evaluated the feasibility, acceptability, and data quality of using HMET with four- and eight-month-old infants (Bradshaw et al., 2023). These age points represent developmental periods of motoric transitions and achievement of key milestones, that is, reaching and crawling. The eight-month study age point also represents a formative developmental period just preceding the beginning emergence of social and communicative behaviors. By situating these investigations in the home environment, where infants engage in the vast majority of interactions with their caregivers, this study pushes the boundaries of what can be achieved with wearable eye-tracking technology. As such, a key goal of the study was to evaluate whether HMET could be successfully implemented in the home with younger infants who lack independent head control—a population previously understudied due to these technical challenges—and to assess the acceptability of these procedures for caregivers. Additionally, the study aimed to explore early developmental changes in infant gaze behavior, particularly shifts in attention between caregivers and objects, and how these patterns might vary by age or ASD likelihood.

The application of HMET to infants as young as 4 months in the home environment proved to be highly feasible. A custom headband was designed for 4-month-olds that prioritized comfort and camera stability, as even slight shifts can have significant impacts on data quality. All infants tolerated the placement of the headset, and the data collected met established criteria for accuracy and precision, with only minimal data loss across sessions. Importantly, caregivers reported high levels of acceptability, with most indicating that the experimental procedures did not disrupt their infant's natural behavior. The retention rate from the 4-month to 8-month visit was also high, reflecting strong engagement from participating families. Together, these findings suggest that HMET, when paired with appropriate equipment and methodological adjustments, can be successfully adapted for use in a home setting and with young, possibly neurodivergent, infants—even those as young as four months of age.

6.3 Gaze patterns in naturalistic contexts in young infants at elevated likelihood for ASD

Preliminary analyses of gaze behavior reveal intriguing developmental trends. At 4 months, infants spent significantly more time looking at their caregiver's face compared to 8-month-olds, whose gaze was more frequently directed toward objects in the play environment. This pattern reflects developmental shifts in social and attentional priorities, as younger infants rely heavily on face-to-face interactions while older infants begin to explore their surroundings more independently. Notably, these trends align with existing literature demonstrating a decline in face-looking and an increase in object-focused attention across the first year of life (e.g., Bradshaw et al., 2024).

In addition, these results extend those of Yurkovic-Harding et al. (2022) showing that despite the availability of caregiver faces for the vast majority of the time during the 4-month-old infant-caregiver interactions (76%), their gaze to face was strikingly low (16%). At 8 months, availability of the caregiver's face was dropped significantly (only 17% of the interaction) and proportion of time infant's gazed to the caregiver's face was only 5%. These results seem to challenge long-standing assumptions driven by evidence of infants' attentional bias toward faces in lab-based settings (e.g., Leppänen, 2016; Peltola et al., 2018; Reynolds & Roth, 2018). While faces are often emphasized as highly salient in controlled settings, naturalistic data reveal more nuanced patterns of social attention, shaped by real-world complexities. The home environment offers a rich, dynamic backdrop for studying how infants and caregivers co-construct interactions, with caregivers often scaffolding attention through behaviors like naming, touching, and positioning objects. By capturing these behaviors *in situ*, HMET allows researchers to examine not only what infants attend to but also how their attention is guided and supported by their social partners.

We have also explored CVA and the pathways into CVA among 8-month-old EL and LL infants. Analyses identified three key differences between the groups. We found that look durations to toys were greater for EL infants and bouts of CVA were fewer in EL infants compared to LL infants. This aligns with deficits in attention shifting and attention disengagement that is observed in infants and toddlers who later develop ASD. Additional analyses are needed to thoroughly interpret this preliminary finding, but it is likely that EL infants attend to the same toy for longer periods, and their caregiver follows their attention, leading to fewer CVA bouts. In addition, we see that EL infants' CVA episodes are more often infant-initiated, with 66% of CVA bouts in the EL group being infant-led. That is, the CVA bout began with the infant directing their gaze toward a toy, followed by the parent coordinating their attention to the same toy. This proportion was significantly higher than that observed in LL infants, whose CVA episodes were more balanced between parent-led (47%) and infant-led (53%) pathways.

Together, these studies offer new perspectives on how lab-based mechanistic studies of infant attention may translate to and inform studies of attention allocation in real-world settings—the very contexts that shape and support development and learning. Leveraging HMET and other technologies to study embodied attention and natural behavior, we can further understand attention, not only as a set of discrete behaviors, (e.g., attention disengagement, attentional social preference) but as an emergent property of dynamic interactions, offering new pathways to understanding how developmental trajectories take shape in both typical and atypical populations.

6.4 A synchronized system for studying physiological regulation of attention in naturalistic settings

HMET is not the only advanced tool that can be made portable and implemented in the home to study developmental trends in infant development. In our longitudinal studies of EL and LL autonomic regulation of attention, we have used home-based data collection to study behavior in naturalistic contexts and to address barriers to access for research participants. Our team has developed the Lab2Home (L2H) system – a tailored, portable, and scalable apparatus designed to bring rigorous laboratory-quality data collection directly into participants' homes. The L2H system is tailored for studies requiring synchronized data streams from multiple video angles and concurrent heart rate monitoring for both parent and child. Synchronization, a cornerstone of L2H, is achieved using a combination of custom-written and open-source software with redundant methods to ensure millisecond precision. Devices utilize a shared computer clock for alignment of time-stamped data, while a custom-built microcontroller triggers an LED array recorded by all cameras and logged into the experimental system for added accuracy. Additionally, visual time displays are embedded into recorded video streams to align seamlessly with heart rate data. These measures ensure that all data modalities remain synchronized throughout the procedure, even in diverse and variable home environments.

L2H is compact, lightweight, and designed for rapid setup and breakdown, reducing logistical overhead during testing. The system's hardware includes three high-definition (1080p, 60fps) USB-C cameras mounted on adjustable tripods to accommodate a range of home layouts, from spacious living rooms to confined apartments. High-capacity portable power banks power the system for over 10 hours, allowing uninterrupted operation even in homes without access to reliable power. Heart rate data collection is facilitated by two CamnTech Actiheart-5 wireless monitors—one for the parent and one for the child—while a small-profile condenser microphone with preamp ensures high-quality audio recording and strategic placement. The central microcontroller, an Arduino UNO WiFi REV2 outfitted with a multicolor LED array, coordinates the system's operations seamlessly.

The L2H apparatus not only facilitates the collection of rich, high-quality data but also reduces the systemic barriers that often limit the participation of historically underrepresented populations in research. By enabling data collection in familiar home environments, L2H supports studies that aim to understand developmental processes in the very contexts in which they unfold.

7. In-Home studies as a representative and inclusive approach to science

Naturalistic data collection, particularly in the home, offers a unique window into infant behavior and development in real-world contexts, affording the possibility of more ecologically valid data while maintaining high standards of rigor. Moreover, the cost of travel for participants, in terms of time, vehicle mileage, and gas expenses, decreases accessibility of research participation for families from diverse sociodemographic backgrounds, even when monetary compensation is provided. Inclusivity and representation are incredibly important to confirm our science to be representative of and generalizable to the broader population at large.

To evaluate the experimental and ecological validity of in-home data collection, we administered parallel experimental and naturalistic tasks longitudinally to two groups of infants who participated in in-lab study visits (n = 29, 63%) or in-home study visits (n = 17, 37%). Infants were seen from 2-24 months of age, resulting in a total of 167 lab visits and 106 home visits. Following data collection, caregivers rated their child's observed emotional and interactive states. Caregiver report of emotional reactivity during experimental and clinical procedures significantly differed by context ($\chi 2(2) = 12.64$, $p = 0.002$) with lab-based sessions yielding more parent-reported atypically negative emotional reactivity (n = 23; 13.8%) compared to home-based sessions (n = 4; 3.8%). Ratings of interest and attention during the tasks did not significantly differ by setting ($\chi 2(2) = 2.76$, $p = 0.252$). These findings suggest that the controlled lab environment may contribute to more atypical behavior, that is, negative emotional reactivity, suggesting more typical behavior in the home environment and possibly more representative and valid data. Experimental data from variables of interest was also comparable across home vs. lab contexts. The percentage of valid, usable behavioral and electrocardiogram (ECG) data was not significantly different ($p = 0.73$, $p = 0.89$, respectively).

A home-based approach may also increase inclusivity and accessibility of participant cohorts. Traditional lab-based studies often fail to engage families living in rural areas or from underrepresented socioeconomic groups, such as Black, Indigenous, and people of color (BIPOC). By bringing research into the home, we expand the demographic reach of participants, paving the way for a representative, enriched dataset that includes diverse populations. In our analysis of a separate cohort of participants who opted to complete home vs. lab study visits (see Table 1), household income significantly differed ($\chi 2(1) = 8.99$, $p = 0.003$), with 40.7% of home-enrolled participants earning $60,000 or less annually compared to 11.1% of lab-enrolled participants—well below the state average household income (South Carolina; $92,833). As expected, the average distance from the participant's home to the lab was significantly higher for the home-enrolled participants (t (131.5) =

Table 1 Participant demographics of participants across lab vs. home-based data collection.

	Lab enrolled, N = 40	**Home enrolled, N = 103**	**Test statistic (*p* value)[b]**
Child race			
White-non-Hispanic	26 (65%)	51 (49.5%)	0.257
Black or African American	9 (22.5%)	33 (32%)	
More than one race	3 (7.5%)	12 (11.7%)	
White-Hispanic	1 (2.5%)	4 (3.9%)	
Asian	0 (0%)	3 (2.9%)	
Other/Unknown	1 (2.5%)	0 (0%)	
Child race[a]			
Black, indigenous, and persons of color (BIPOC)	14 (35%)	52 (50.5%)	0.139
White	26 (65%)	51 (49.5%)	
Reported Primary Caregiver			
Mother	37 (92.5%)	102 (99%)	0.119
Father	3 (7.5%)	1 (1%)	
Primary Caregiver Education[a]			
High School/GED/some college	12 (30%)	47 (45.6%)	0.129
Bachelor/Graduate/Professional degree	28 (70%)	56 (54.4%)	
Not reported	2	0	
Annual Household Income*			
<US$20,00–$60,000	4 (11.1%)	37 (40.7%)	**0.003**
US$60,001 or higher	32 (88.9%)	54 (59.3%)	
Not sure or prefer not to answer	4	12	
Distance from the Lab			
Average roundtrip mileage (SD)	23 (22)	73.5 (76.6)	**<0.001**

[a] Dichotomized groups and chi squared analyses.
[b] Bold values indicate statistical significance at the $p < .05$ level.

-6.0, $p < 0.001$). Results from these analyses suggest that distance to the lab stands as one of the barriers to lab-based participation and that offering in-home study visits increases accessibility of research participation for lower-resourced families.

Confirming that findings from controlled, lab-based experiments completed in the home are comparable to those completed in the lab, while also demonstrating differences in infant behavior and sociodemographic factors across the two contexts, underscores the feasibility and impact of in-home study visits. To continue this work, we have developed a sophisticated method using portable, synchronized technologies that can effectively capture synchronous behavioral and physiological data without disrupting the familial environment. By situating research in participants' homes, with advanced technology explicitly designed for our study procedures, we aim to gain a deeper understanding of how early ASD markers manifest in naturalistic settings and reduce barriers for families from diverse demographic backgrounds.

8. Conclusions

The science of ASD in infancy is at an exciting stage, offering new opportunities to elucidate the mechanisms and developmental cascades underlying atypical neurodevelopment in real-word contexts. The naturalistic, behavioral, and non-invasive nature of this work elevates its potential for rapid translation to novel and, most importantly, actionable early detection and intervention strategies. By integrating a developmental cascades approach with prospective, longitudinal designs, innovative methodologies, and home-based data collection, we can illuminate new pathways that characterize early development of ASD. Grounded in both mechanistic insights and real-world applicability, we can understand how autonomic regulation and attentional processes shape developmental outcomes.

Prioritizing inclusivity and ecological validity through home-based data collection not only broadens the scope of ASD research but also ensures its relevance to the lived experiences of diverse families. This multifaceted approach not only advances scientific understanding but also lays the groundwork for interventions that resonate with the complexities of infant development and with the cultural context of families traditionally underrepresented in research. As ASD research continues to evolve, bridging theory, empirical findings, and methodological innovation will remain critical to transform outcomes for children and their families.

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Child Development Perspectives

Capturing the complexity of autism: Applying a developmental cascades framework

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Abstract

Developmental change emerges from dynamic interactions among networks of neural activity, behavior systems, and experience-dependent processes. A developmental cascades framework captures the sequential, multilevel, cross-domain nature of human development and is ideal for demonstrating how interconnected systems have far-reaching effects in typical and atypical development. Neurodevelopmental disorders represent an intriguing application of this framework. Autism spectrum disorder (ASD) is complex and heterogeneous, with biological and behavioral features that cut across multiple developmental domains, including those that are motor, cognitive, sensory, and bioregulatory. Mapping developmental cascades in ASD can be transformational in elucidating how seemingly unrelated behaviors (e.g., those emerging at different points in development and occurring in multiple domains) are part of an interconnected neurodevelopmental pathway. In this article, we review evidence for specific developmental cascades implicated in ASD and suggest that theoretical and empirical advances in etiology and change mechanisms can be accelerated using a developmental cascades framework.

KEYWORDS
autism spectrum disorder, developmental cascades, neurodevelopment

INTRODUCTION

Developmental research has historically focused on documenting the timing and acquisition of specific skills and behaviors within established developmental domains. Infants reach by 5 months, crawl at 6 months, and walk at 12 months. Babbling emerges by 7 months and the first words can be heard by 1 year. Yet the onset of these skills varies significantly and their emergence does not occur in isolation. The onset of reaching and crawling generate new opportunities for interactions with the world. The emergence of communication quickly establishes a new motivation for locomotion. *Developmental cascades* describe how behavior and development in one domain (e.g., motor) have cascading, far-reaching effects on other, seemingly unrelated domains (e.g., language).

The idea of developmental cascades stems from the work of key developmental scholars and theoretical models, including Gottlieb's model of developmental behavioral genetics (Gottlieb, 2003), Smith and Thelen's work on developmental dynamics (Smith & Thelen, 2003), Sameroff's transactional model of development (Sameroff, 2009), and Karmiloff-Smith's neuroconstructivist approach to neurodevelopmental disorders (NDDs; Karmiloff-Smith, 1998). A developmental cascades framework considers development as the ongoing, cumulative consequence of transactions across developmental domains, biological systems,

Abbreviations: ASD, autism spectrum disorder; CSF, cerebral spinal fluid; NDD, neurodevelopmental disorder.

and environmental contexts (Masten & Cicchetti, 2010). In this way, it allows researchers to examine how low-level (or high-level) features have cross-domain, far-reaching, bidirectional effects on processes later in development. Cascades enable the examination not only of specific developmental mechanisms, but also of how multiple mechanisms interact in the moment (and across time) to *shape* individual development and interactive contexts. Individual developmental trajectories diverge in response to unique experiences and events, providing countless opportunities for rich input that is driven by the individual child and consequently shapes future input.

A developmental cascades framework applied to the study of autism

Recently, a surge of interest in developmental cascades has produced several robust models for explaining cross-domain effects on overall development (e.g., Oakes & Rakison, 2019). From a developmental psychopathology perspective, these models have typically focused on older populations (e.g., Masten et al., 2005). The cascades framework provides an opportunity to extend this work to infancy, a period of rapid neurodevelopment and peak brain plasticity. In this context, such an extension may advance the discovery of etiological mechanisms and understanding of heterogenic variability in atypical developmental pathways, such as those observed in NDDs.

Autism spectrum disorder (ASD) is one such NDD, with behavioral and biological differences that persist throughout the lifespan. The promise of early behavioral interventions to support infants and toddlers with ASD has motivated mechanistic research to advance early detection. However, the heterogeneity of the disorder is vast. Sensitive and specific biological or behavioral indicators of ASD prior to 12 months remain elusive. This is largely due to our current conceptualization of ASD wherein core features (i.e., social communication differences and restricted interests/repetitive behaviors) are inherently developmental and do not fully manifest in a clearly observable way until after the first year. From a developmental cascades perspective, these core features can be conceptualized as points on a cascade that represent the cumulative result of unique input and experience, driven by the interaction between individual and contextual differences and pre- and postnatal neurological variation. Accordingly, the search for etiological mechanisms in ASD must extend beyond core features.

Etiological theories of ASD, such as the social motivation hypothesis (Chevallier et al., 2012; Dawson, 2008) elegantly describe a core impairment (i.e., social motivation) that explains the secondary (defining) characteristics of ASD. The recently developed anterior modifiers in the emergence of neurodevelopmental disorders (AMEND) framework (Johnson et al., 2021) take a systems biology approach to understanding NDDs. A developmental cascades framework builds on these theories and allows us to understand ASD within the context of broad, cross-system, normative developmental processes, leading to rich descriptions of phenotypic heterogeneity within ASD across development. By considering the cross-domain effects of multiple experiences and behaviors that set the stage for specific developmental pathways, we can capture the complexity of neurodiversity, explain the variability and heterogeneity within ASD, and pinpoint how the timing and quality of behaviors and bioregulatory processes carry to downstream effects in development. Delineation of these developmental pathways, including biological, behavioral, and contextual contributors, can drive early detection and intervention efforts.

In some domains, infants with ASD may follow a unique developmental pathway that leads to a neurotypical outcome. For example, communicative gestures are hypothesized to be a critical point on a developmental cascade that culminates in the acquisition of first words for typically developing infants, but the use of gestures may not be part of the language acquisition cascade in ASD. Conversely, infants with ASD may show similar developmental skills early on that, because of cross-domain effects of other emerging behaviors, lead to vastly different outcomes. For example, a low-level attentional preference for high-contrast features may lead to an attentional preference for eyes in infants without ASD, but to sensory-seeking behaviors and attentional preference for objects in infants with ASD. Developmental cascades emphasize a cumulative model with additive components rather than a one-to-one mapping of single deficits that lead to maladaptive outcomes. This framework provides a unique opportunity for interventions that target a variety of domains at precise moments.

In summary, assuming that all infants experience the same developmental pathways narrows the potential for mechanism discovery. Rather than asking *when* observable features of ASD emerge, we can ask *how* they emerge. A developmental cascades framework expands our view of how cross-domain systems interact at multiple levels to influence how individual infants both experience and drive environmental input to shape their own development. Contextualizing ASD as the result of, and a contributor to, unique cascades of development that drive specific input and learning can transform and enrich our approach to research on ASD. Next, we illustrate potential developmental cascades in ASD that represent unique biobehavioral processes across three domains of development—infant sitting, visual attention, and sleep—which may be avenues for early intervention. We close with an overview of methods for using the developmental cascades framework in research on ASD and other NDDs.

DEVELOPMENTAL CASCADES

Infant sitting: A watershed moment in the first year of life

In the early months of the first year, infants spend most of their time lying supine on their backs or prone on their bellies. Between 5 and 7 months, they spend more time seated in an upright position, first with support from objects or their own hands. Over time, infants gain strength and improve their ability to balance in the sitting position, allowing them to remain upright without support for progressively longer periods.

The transition to upright, independent sitting is a watershed moment in the first year of life: It represents a significant advance in motor skill, but more importantly, it unleashes a cascade of effects in other behavioral domains (see Figure 1). This cascade provides infants with access to new experiences and opportunities for interaction with the objects and people in their environments. While infants lying supine or prone can see only the areas directly above or below the head, sitting infants have an expanded 180-degree panoramic view of their surroundings (Luo & Franchak, 2020). Arm movements and the ability to manipulate objects are also more constrained in supine and prone positions. Supine infants must extend their arms and work against gravity to hold an object in view. In a prone position, one arm is often used for balance and support. But in the sitting position, hands are free to move and they fall naturally within infants' visual fields, making it possible to coordinate looking at objects with acting on them (e.g., Soska & Adolph, 2014), an activity that supports the extraction of rich, multimodal perceptual information.

In addition to these cascading effects, the emergence of sitting also has downstream effects on the infant-caregiver dyad. Sitting infants and their caregivers spend more time with their bodies positioned at right angles to one another, effectively enlarging the space between them and providing new opportunities for caregivers to introduce objects into shared play (Schneider et al., 2021). The construction of a large shared dyadic play space creates ideal conditions for joint attention (e.g., caregiver and infant looking at the same toy at the same time). Indeed,

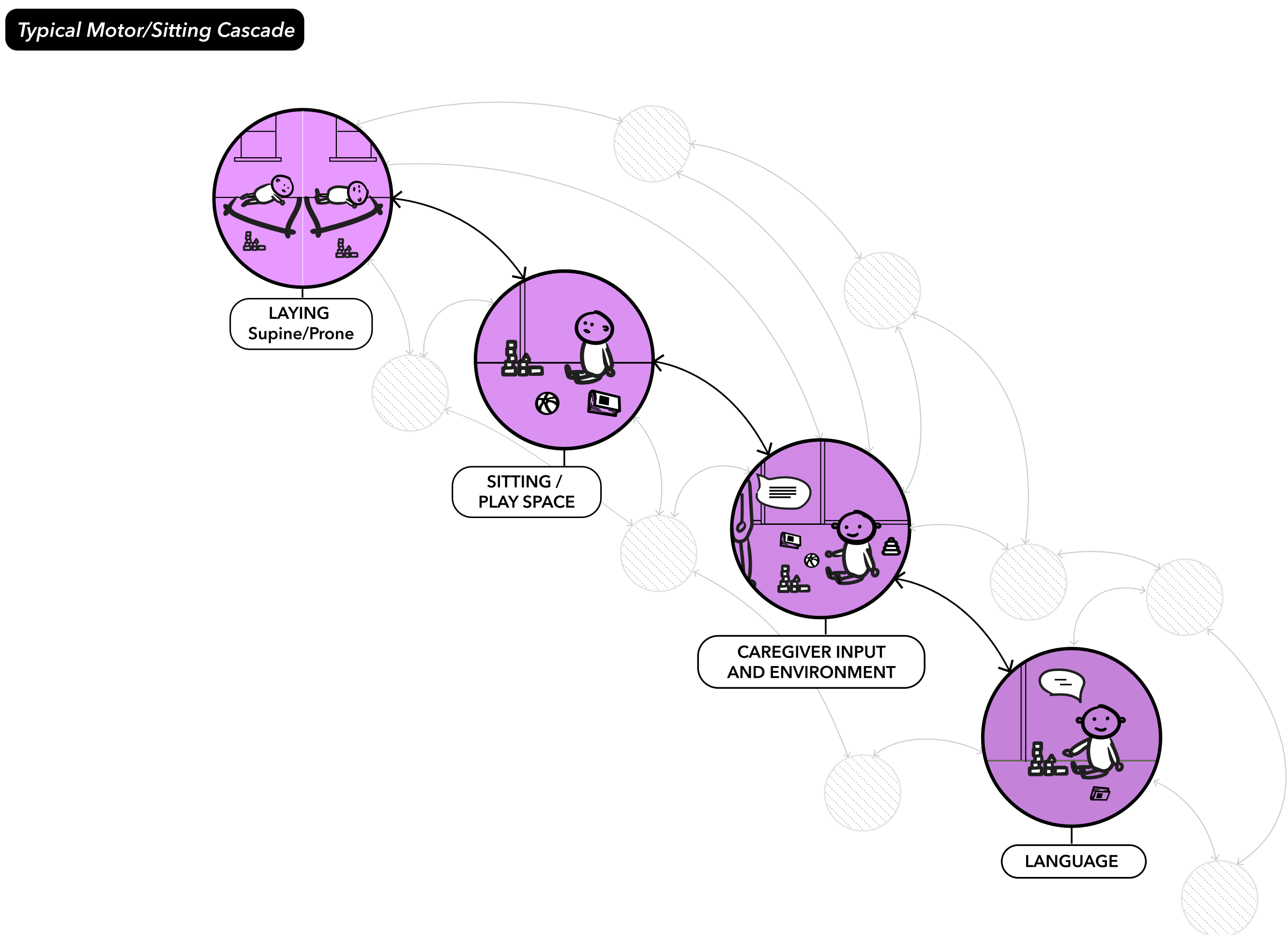

FIGURE 1 The transition from lying prone or supine to upright sitting transforms infants' experiences with objects and caregivers. Infants' ability to coordinate reaching and grasping objects with looking at a caregiver provides new opportunities for rich language input from the caregiver

dyads spend significantly more time in joint attention when infants are positioned in sitting compared to prone positions (Franchak et al., 2018). In addition, when caregivers provide rich language input about an object of interest, they create powerful opportunities for language learning.

Developmental differences in gross and fine motor skills are becoming a recognized feature of emerging ASD (e.g., Licari et al., 2021). Infants with ASD experience atypical head control compared to infants without ASD (Flanagan et al., 2012). On average, infants with ASD begin to sit independently later than their neurotypically developing peers (Nickel et al., 2013). In addition, even when they are able to sit independently, infants with ASD spend significantly more time in lying postures (prone, supine) than infants without ASD, likely because the sitting posture requires more protracted development of balance and control (Leezenbaum & Iverson, 2019). Less time spent sitting upright and slowed consolidation of sitting skills constrain opportunities for more sophisticated object manipulation and play. Moreover, because caregivers use infants' manual actions as a pathway to joint attention (Yu & Smith, 2017) and are more likely to label objects their infant is holding, looking at, and actively manipulating (West & Iverson, 2017), less object manipulation in infants with ASD may translate into reduced opportunities for joint attention and alterations in caregivers' language input. Thus, while the development of sitting may appear to be unrelated to the development of social communication and language, evidence suggests that sitting has cascading developmental effects on behaviors within and beyond the motor domain, and within and beyond the infant. For infants with ASD, differences in the timing and consolidation of sitting and its integration with other behaviors (e.g., object manipulation) may affect the unfolding of the developmental cascade in ways that fundamentally alter their experiences with objects and people.

Visual attention: A complex skill with cascading effects

Endogenous attention develops rapidly over the first months of life. The emergence of volitional, selective attention by 6–10 weeks is subserved by a transition from subcortical to cortical control of behavior (Johnson et al., 1991). Efficient disengagement from salient stimuli, attention shifting, and filtering of extraneous information are all critical components of selective attention (Atkinson et al., 1992). Developing attention in infancy is both a product of and a contributor to multiple cross-domain developmental cascades (see Figure 2). As described earlier, motoric achievements create new access to visual input. The coupling of motor and sensory systems, including coordination of eyes, head, hands, and body, is inherently linked to selective attention. Attention is affected by cognition and experience, and cognition is shaped in part by attention. The ubiquitous role of attention, sensorimotor coupling, and experience on higher- and lower-order processes throughout development makes it an intriguing access point for understanding etiological mechanisms and unique developmental pathways in ASD.

Atypical visual attention patterns, particularly to socially relevant features of the environment, constitute an early-emerging and enduring feature of ASD across the lifespan (Frazier et al., 2017). Many mechanistic theories of autism incorporate attention differences as a core feature of the ASD phenotype: social motivation (decreased social attention; Chevallier et al., 2012), central coherence (global vs. local visual processing; Happé & Frith, 2006), executive function (attention disengagement; Pellicano, 2012). Differences in lower-level attention in ASD begin very early in development, evidenced by decreased orienting to and tracking of faces and objects (Bradshaw et al., 2020, 2021), decreasing attention to the eyes and face (Jones & Klin, 2013), and difficulties with attentional disengagement and shifting (Elsabbagh et al., 2013). Atypical attention is not in itself a core diagnostic feature of ASD, but instead represents a subtle, early-emerging variation that has proximal and distal effects on individuals' perception of and interaction with the environment.

For example, eyes are visually salient and newborns demonstrate an attentional preference for eyes from the first hours of life. At a higher level, eyes are laden with rich social and communicative information, suggesting that decreased attentional preference for eyes in infancy can result in missed critical social learning opportunities. Moreover, because familiarity is an important component of selective attention, an early reduction in eye-looking may lead to increasingly diminished eye-looking over time, cumulatively resulting in reduced social attention and a nonsocial attentional preference (e.g., Pierce et al., 2016).

What causes this early variability in such low-level attentional behaviors? Diminished attention disengagement has been associated with atypical neural circuitry and suggested as a domain-general starting point for a developmental cascade that leads to social-specific deficits in ASD (Baranek et al., 2018). Recent work suggests atypical neural and neurochemical circuitry specific to visual processing and arousal modulation, which may constitute an even earlier point in a developmental cascade related to attention differences in ASD (Artoni et al., 2020; Hazlett et al., 2017).

Together, this research points to a potential cascade that crosses biology and behavior and begins shortly after birth. Differences in neurotransmitter levels and neural connectivity, particularly as they relate to arousal, attention, and sensory systems, may lead to behavioral variability in lower-order attentional processes, including attention orienting, disengagement, and shifting.

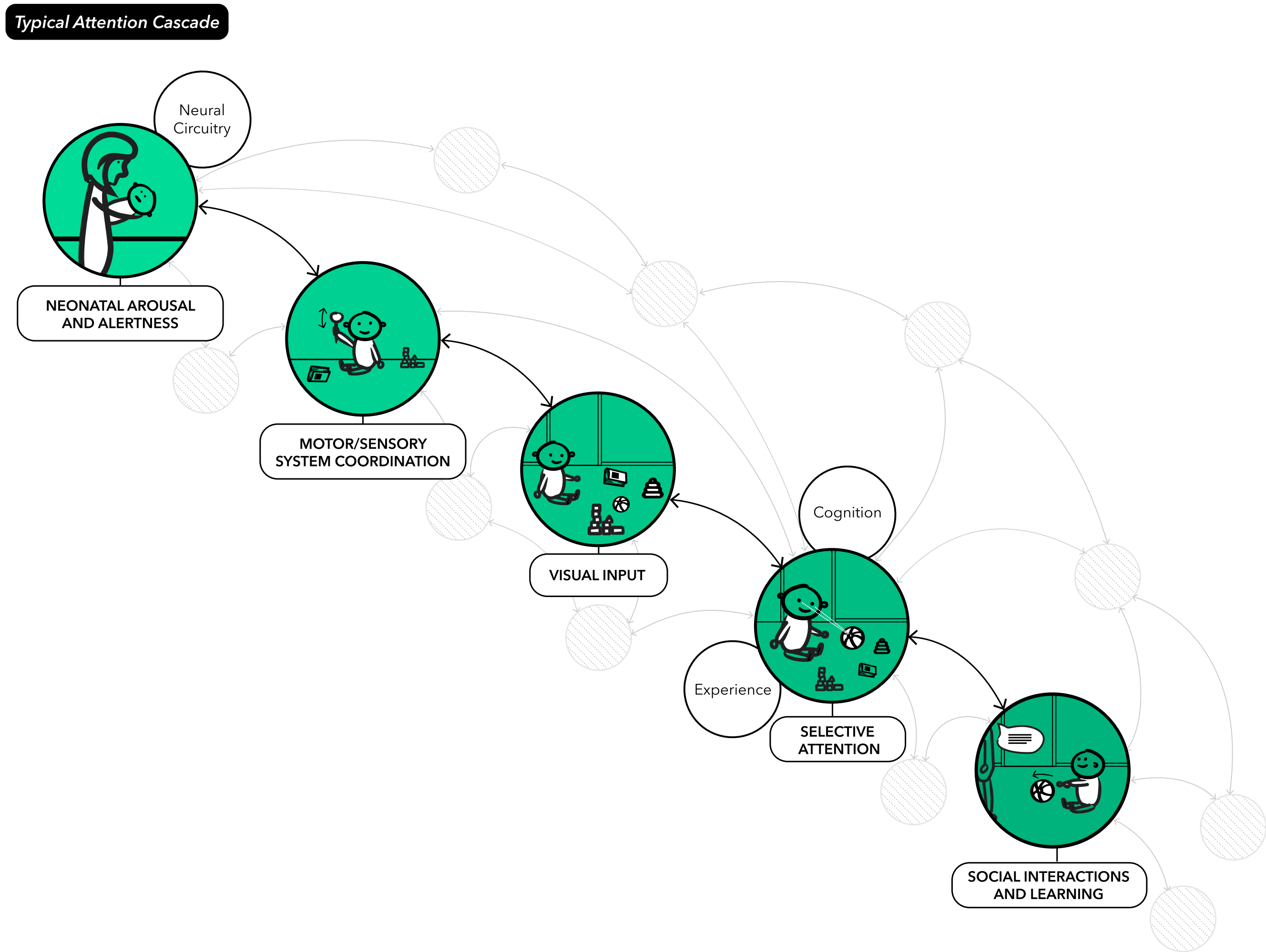

FIGURE 2 The development of attention begins in the neonatal period with modulated levels of arousal and periods of quiet alertness. Subsequent motor, sensory, and cognitive development allow for triadic (infant-object-caregiver) interactions, which provide increased opportunities for social interaction and language

These early, subtle attentional differences can have cascading effects on social interactions and learning. A reduced capacity for attention shifting in early infancy may decrease opportunities for triadic (caregiver-object-infant) interactions. This may interfere with integrating higher-order cognitive concepts, which requires jointly attending to both caregiver and object, and combining objects together in play—skills that are critical for language development. Infants drive caregiver-provided opportunities for learning and social interaction and, in ASD, individual differences in attention and the neural structures that support it may have cascading effects on social experience and learning.

Sleep: A critical bidirectional, bioregulatory behavior in infancy

Sleep is one of the first bioregulatory behaviors infants must master to support neurologic, digestive, and immune health. Infants with consistent sleep problems early in life are at elevated risk for problems across several developmental domains, including attention, learning, memory, language, elevated adiposity, emotion regulation, and social skills. Sleep problems are well-documented across life for individuals with ASD (Cohen et al., 2014) but the potential and affiliated developmental cascades are understudied. For example, the synaptic homeostasis hypothesis of sleep (Tononi & Cirelli, 2014) posits that a core function of sleep is to selectively prune synaptic connections to reinforce those used during the day, thus making established pathways more efficient. For children with ASD, applying a developmental cascade that starts with early sleep deficiency could help explain the mechanistic link between ASD and synaptic overgrowth (Courchesne et al., 2003), large brain volume and head size (Hazlett et al., 2017), and reduced synaptic efficiency (Just et al., 2012).

For example, excess extra-axial cerebral spinal fluid (CSF) has been documented in three groups of infants who later received a diagnosis of ASD (Shen, 2018). This could reflect a cascade that starts with sleep dysfunction.

Circulation of CSF in the brain occurs predominately during sleep, specifically during slow-wave sleep or non-rapid eye movement stage 3 sleep (Fultz et al., 2019). Disrupted sleep may compromise efficient CSF circulation, possibly resulting in the atypical CSF disbursement observed in ASD. Indeed, recent studies of ASD in early infancy show altered sleep patterns and direct links between early sleep problems and atypical brain development (MacDuffie et al., 2020).

A developmental cascades approach to the connection between sleep and behavior in ASD could also inform our mechanistic understanding. Deficient sleep in ASD is associated with the severity of core symptoms as well as co-occurring features (e.g., challenging behaviors, attention problems), with notable links to arousal and repetitive behaviors (Cohen et al., 2014). Sleep deprivation affects arousal modulation profoundly. This association is characterized by an initial increase in arousal with the onset of sleep deprivation, followed by a chronic state of underarousal with extended periods of deprivation (Tobaldini et al., 2017). In ASD, underarousal is associated with sleep dysregulation and challenging behaviors (Cohen et al., 2011).

The field's current practice of assessing behavior within domains (e.g., types of sleep problems, repetitive behavior profiles) can lead to missed opportunities to recognize developmental mechanistic pathways of influence. Applying a developmental cascades framework explicitly draws us toward cross-domain mechanisms and helps us understand how these relations change and grow across development (see Figure 3). In this example, a chronic sleep deficit that emerges in infancy may feed into a dampened arousal system. Consequently, in an attempt to modulate arousal, individuals with ASD may exhibit elevated repetitive or challenging behaviors. Therefore, repetitive behaviors may be one behavioral manifestation of underlying disrupted arousal modulation, a bioregulatory process that is inherently linked to atypical sleep and brain development.

RESEARCH METHODS AND ANALYTIC MODELS

In developmental science, most agree that applying theory is a best practice when establishing questions of interest or testable models. However, analytic models (and their limitations) often drive the types of questions posed in research and how they are tested. To help in the application of a developmental cascades framework in ASD research, in the next section, we outline several analytic approaches that marry well with this framework. The list is not exhaustive, prescriptive, or specific to the cascades framework. Rather, it exemplifies how developmental cascades and analytic models can be symbiotic.

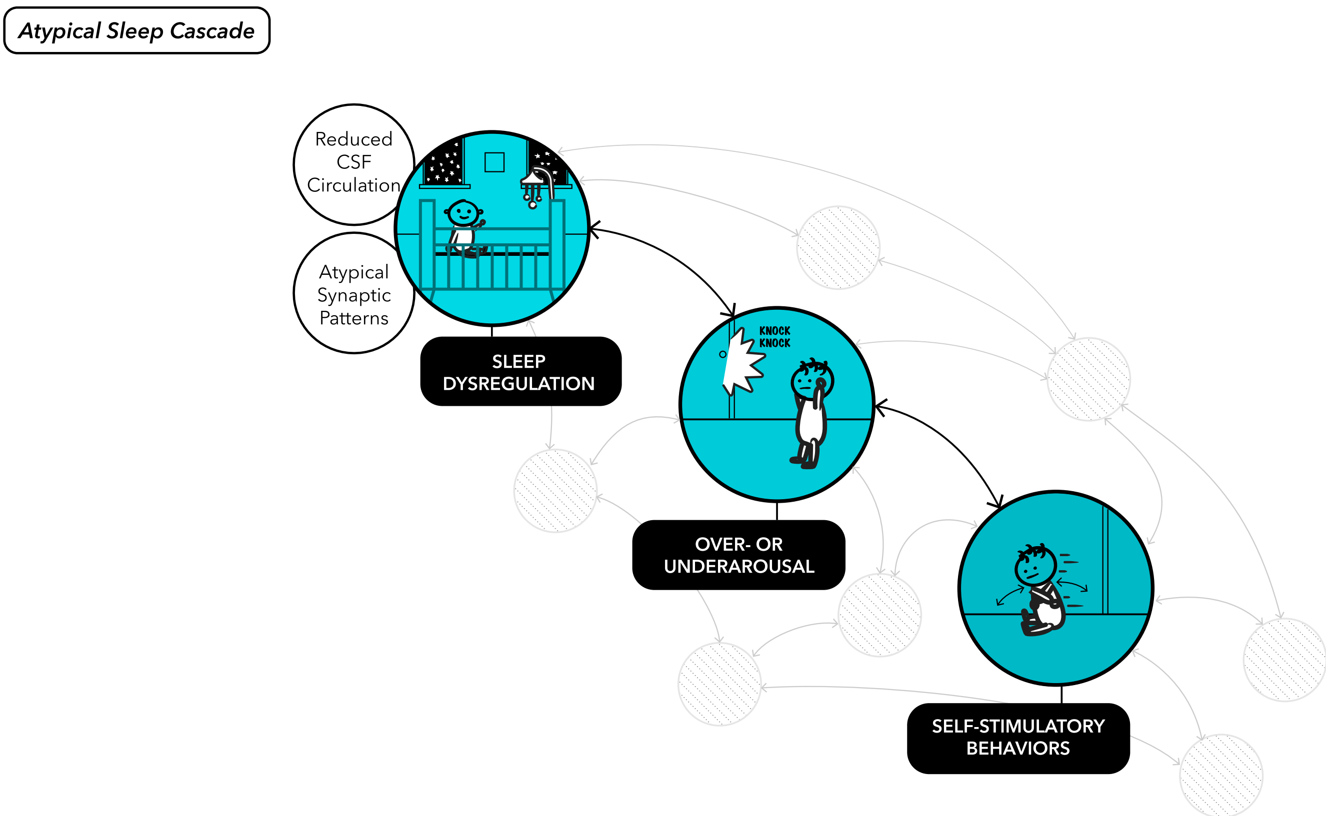

FIGURE 3 Atypical sleep early in life can influence several elements of brain development (e.g., synaptic pattern formation and pruning, cerebrospinal fluid [CSF] circulation) and biosocial processes (e.g., arousal). Over- or underarousal, in turn, is associated with self-stimulatory behavior as individuals aim to modulate their arousal using behavioral/physical actions

The initial application of a theoretical approach often includes *mapping* or specifying key factors or the pathway of interest. The accessible cause-outcome representation and notation system (Moore & George, 2011) is a tool for visualizing and modeling developmental pathways, including bidirectional effects, within- and cross-domain developmental change, and specifications for variance patterns and relative developmental timing. When testing a developmental cascade, researchers often use longitudinal designs and the most common analytic strategy is path model testing, which allows for simultaneously estimating several paths or associations with possible adjustments for model complexity, nesting, and autocorrelated features (e.g., Lynne-Landsman et al., 2010; Masten et al., 2005). For example, in one testable pathway model, sleep problems affected emotion-regulation difficulties persistently, which in turn affected attention regulation (Williams et al., 2015). Not only did this model support a sleep-emotion-attention cascade, but it also revealed transactional processes, providing a clear example of how one analytic approach can be used to combine theoretical frameworks.

When several cascades are possible, a comparative model approach can assess which model accounts for most of the variability in an outcome of interest. A supplemental approach that is sometimes used is a likelihood assessment to index the likelihood of a target outcome (e.g., ASD diagnosis) with the presence, absence, or varying degrees of other independent variables. In one example, researchers assessed the percentage of children classified as being at elevated risk for a particular outcome (in this case, substance abuse) at each stage of the proposed cascade (Dodge et al., 2009). Ultimately, few children followed the entire specified cascade, but inflection points of risk were identified and later used to inform interventions. Finally, researchers have used regression-based models of mediation and moderation, but this approach is less common and typically tests relatively short cascades.

CONCLUSION

The search for etiological mechanisms of ASD has proven to be a challenge, largely driven by the heterogeneous, cross-domain, developmental nature of the disorder. Core features of ASD change with development and manifest differently as a result of unique individual experiences and skills. In other words, the ASD phenotype is rich with variation across biological and behavioral domains. Studies of infants with ASD have expanded our understanding of the phenotype and highlighted nuanced, subtle variations in domains that do not necessarily constitute core features of the disorder. Novel discoveries can be made and the field advanced when the complexity of ASD is harnessed, rather than controlled for or reduced.

In this article, we described three developmental cascades that can help transform our understanding of ASD and social development. The development of postural control and the emergence of sitting influences attention, play, and communicative opportunities profoundly. Subtle differences in visual attention and underlying neurological systems can have cascading effects on sensorimotor behavior, social attention, and communication. Sleep is one of the most important bioregulatory processes that supports arousal modulation and may underlie some of the most challenging behaviors associated with ASD. Pathways within these cascades can occur bidirectionally in which higher-level systems (e.g., play skills) can affect lower-level systems (e.g., muscle development), and while not all domains at all levels can be measured in one study, this framework can be used to develop new studies that examine new mechanisms. These examples are intended to catalyze the use of developmental cascades in how we conceptualize studies, analyze data, and interpret findings. While the cascades we have outlined were inspired by potential areas of disruption in ASD, they are not specific to ASD. A developmental cascades framework can inform the sequential, multilevel, cross-domain nature of autism and demonstrate how interconnected systems have far-reaching effects in typical and atypical development across the lifespan.

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CHAPTER ONE

Childhood essentialism

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Contents

Abstract

Essentialism is the intuitive belief that certain categories, such as "tiger," "boy," or "gold," have an underlying reality that goes beyond surface appearances. Childhood essentialism provides insights regarding the nature, origins, and development of human cognition. This chapter reviews the current state of the art regarding research on childhood essentialism, addressing five key issues: (1) what is essentialism and why is it important?; (2) the role of experience (including context, culture, and identity); (3) language as a uniquely powerful mode of transmission; (4) developmental origins; and (5) consequences for social issues and education. Throughout, the chapter considers how children's essentialism works in concert with, and alongside, other cognitive, linguistic, social, and societal processes. Ultimately, childhood essentialism can be considered a "double-edged sword," contributing to human propensities that are both impressive

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(an early-developing ability to look beyond the obvious) and problematic (stereotyping, prejudice, and intergroup bias).

Keywords: Children, Cognitive development, Culture, Essentialism, Generics, Language, Social cognition

1. Childhood essentialism

I have been researching and writing about childhood essentialism for over 40 years—my entire career. I keep returning to this topic because I believe it speaks to enduring and foundational questions regarding cognitive development, including: the origins of human thought, the nature of conceptual change, the mechanisms by which ideas are transmitted from one generation to the next, and the consequences of childhood thought for education, social policy, and combating misconceptions that underlie social ills. In addition, the research on childhood essentialism keeps growing and getting more interesting. Every week, new findings and theoretical proposals are published—often thought-provoking, broad-ranging, and revealing. Essentialism is a story that extends back thousands of years, and will likely continue as long as there are human minds attempting to organize the world into categories, and making sense of their biological and social surroundings.

The goal of this chapter is to review the current state of the art regarding childhood essentialism. The chapter has five main sections: (1) What is essentialism and why is it important? (2) The role of experience (including context, culture, and identity). (3) Language as a uniquely powerful mode of transmission. (4) Developmental origins. (5) Consequences for social issues and education. These are followed by conclusions and open questions.

2. What is essentialism and why is it important?

Psychological essentialism is the intuitive belief that certain categories, such as "lion," "gold," or "girl," have an underlying reality that goes beyond surface appearances. On this view, there is some essential, inherent substance or quality that all lions possess, and that gives a lion its identity.

Psychological essentialism is important because it guides our inferences—for good and for ill. On the plus side, essentialism encourages overlooking misleading appearances and instead searching for hidden structure, which underlies the scientific impulse and knowledge creation. On the negative side, essentialism encourages treating categories like "woman" and "Black" as deeply predictive of the limits and properties of the individuals in those categories. Essentialism also predicts a host of problematic consequences for adults, including stereotyping and prejudice (Bastian & Haslam, 2006; Chen & Ratliff, 2018; Meyer & Gelman, 2016; Williams & Eberhardt, 2008; see Roth et al., 2023, for a review regarding race essentialism).

Childhood essentialism is important for challenging classic views about the nature of development. Essentialist beliefs in young children run counter to the standard view that children are concrete thinkers who build up their knowledge of the world from what they observe. Instead, research reveals an expectation that category members have hidden, non-obvious features (as detailed below).

Because the term "essentialism" has been used variously over the centuries with a host of different meanings, I wish to clarify what I do and don't mean when I refer to essentialism (see also Gelman, 2003, 2004, for more detail). It is a *psychological* construct (about people's beliefs and representations)—*not* a metaphysical claim about how the world actually is. Indeed, essentialism is a *cognitive bias* (Gelman & Marchak, 2020; Leslie, 2013; Medin, 1989). For example, essentialism may involve treating the Indian caste system as natural and immutable, rather than recognizing that it is a social construction with a deep historical basis. Essentialism may also involve treating racial categories as biologically based and having sharp boundaries (e.g., leading most Americans to agree with the false statement, "Two people from the same race will always be more genetically similar to each other than two people from different races"; Christensen et al., 2010), when in actuality they do not have a biological reality and correspond to as much within-category as between-category variability. Essentialism even leads to misconceptions of biological species. For example, individual members of a species do not have inherent properties that are identical for all members; rather, species differences are determined at a different level of analysis, by interbreeding populations (Sober, 1994).

Essentialism is a *placeholder*—a belief that members of a category have some essential quality or substance that make them what they are—but not necessarily any knowledge of what that essence might be (Medin, 1989). Thus, the essence in the sense used here is not a definition, nor does it specify which features are necessary for category membership (but for an example of definitional essentialism, see Machery et al., 2023). This is not to say that people have no beliefs about the essence placeholder, as it is understood to be internal, inherent, and causally powerful—but beyond that, the specifics are unknown (Ahn et al., 2001).

It is also important to distinguish between psychological essentialism and other constructs with which it may be confused. For example, essentialism is not the same as stereotyping (see Ritchie & Knobe, 2020), it is not the "pursuit of less" (McKeown, 2014), and it is not Plato's notion of an idealized, unrealized perfect form (Kraut, 2022). Psychological essentialism as discussed here is related to, but different from, value-based essentialism (Bailey et al., 2021), teleological essentialism (Rose & Nichols, 2019), and essentialism of personal identity, as reflected in belief in a true self (Horne & Cimpian, 2019; Umscheid et al., 2023).

2.1 Tests of childhood essentialism

Because the essence itself is a placeholder rather than an identifiable entity (property, part, or substance), it can be detected only indirectly, analogous to the detection of exoplanets: not visible via telescope but inferrable by alterations in the gravitational pull of nearby stars. Psychological essentialism in children likewise cannot be detected by naming or pinpointing the essence, but rather by tasks that reflect the assumption that a category possesses a deeper, inherent, non-obvious reality. Some of these tasks are provided in Table 1.

Altogether, these tasks reveal a host of essentialist intuitions by 4 or 5 years of age. These include judgments that category membership is stable over transformations (e.g., a cactus cannot be transformed into a porcupine), inductive inferences in the face of dissimilarity (e.g., a blackbird shares characteristics with a flamingo, not a bat), reliance on nature over nurture (e.g., a baby rabbit raised with monkeys will nonetheless prefer carrots over bananas), judgments that category boundaries are objective and discrete (e.g., it is wrong to call a cat and a dog the same kind of animal), and belief that innards are causally powerful (e.g., a dog needs its insides to bark and eat dog food). By 7 or 8 years of age, these intuitions are sharpened further (e.g., a raccoon cannot transform into a skunk; Keil, 1989; receiving a heart transplant from a mean person will alter someone's personality, making them somewhat meaner; Meyer et al., 2017).

Children's essentialism is sometimes expressed in the comments they spontaneously make when participating in the studies described above. Examples include: "Snakes are a little bit the same and a little bit different. Inside they're the same." "That's the way rabbits are, because that's how

Table 1 A sample of essentialism tasks used with young children.

Essentialism Task	Description	Citation
Transformation	A porcupine is superficially transformed to look like a cactus. What is it now: a porcupine or a cactus? Essentialist answer: a porcupine.	Keil (1989)
Induction	Triad task: Child learns a distinct property about a flamingo ["bird"] vs. a bat ["bat"]. Which property is true of a blackbird ["bird"] that looks like the bat? Essentialist answer: the same as the flamingo.	Gelman and Markman (1986)
Switched-at-birth	A newborn rabbit is raised by monkeys and never sees another rabbit. When it grows up, would it rather have carrots or a banana? Is it good at hopping or climbing? Essentialist answer: carrots, hopping	Gelman and Wellman (1991)
Category boundaries	Feppy (a character from far away), and all his friends, say that two items [a cat, a dog] are the same kind of animal. Could they maybe be right? Essentialist answer: no	Rhodes and Gelman (2009)
Causal innards	What if you take out all the stuff inside of a dog. Is it still a dog? Can it still bark and eat dog food? Essentialist answer: no.	Gelman and Wellman (1991)

they are when they're born." "Every dog has the same stuff [inside]… Just because [they] have different colors doesn't mean they have different stuff."

It is also worth noting that there is an asymmetry between the sophisticated inferences that young children make when told an item's category membership (as in the essentialist assumptions summarized above) versus the difficulties they encounter when asked to form a categorization on their own (where more typically they rely on superficial appearance cues, such as shape; e.g., Imai et al., 1994). Deciding how to categorize an ambiguous item requires weighing competing cues and making use of expertise that a child may lack, whereas in their essentialist inferences, children are provided with labels from a trusted adult, and their responses indicate what expectations

Table 2 Some proposed components of essentialism.

Author(s)	**Components**
Gelman (2003)	Inductive potential; non-obvious properties; stability over transformations; boundary intensification; innateness; causal power.
Haslam et al. (2000)	Discreteness; uniformity; informativeness; naturalness; immutability; stability; inherence; necessity; exclusivity.
Rhodes and Mandalaywala (2017)	Natural kinds; strict boundaries; homogeneity; stability; causal.
Neufeld (2022)	Natural kinds; causal power; large inductive potential; sharp category boundaries; homogeneity; immutability; intrinsicality; innate potential; hidden and unobservable; distinctness and sameness; objectivity.
Fine et al. (2024)	Biological basis; discreteness; informativeness; immutability.

they hold about these categories that are transmitted via language (Gelman et al., 1986).

2.2 One construal or many?

At its most fundamental level, I suggest that essentialism reflects a single coherent assumption: category realism: Just as individuals are real (e.g., Fido exists as an independent entity), so too categories are real (e.g., "dogs" exists independently of us as humans). On this assumption, certain categories are not subjective or arbitrary human constructions but are out there in the world, waiting to be discovered; they "carve up nature at its joints." At the same time, essentialism has been analyzed as having multiple components or manifestations, as shown in Table 2. There is no consensus as to how many components or what they are, although there is much overlap across the lists.

There is reason to treat different components as meaningfully distinct (see also Rhodes & Moty, 2020, for discussion). First, there are categories for which only some aspects of essentialism apply and not others. For example, age categories, such as "adolescent," are not stable over time because people move in and out of such groupings; kinship categories such as "cousin" have graded rather than absolute boundaries; and natural substances such as "gold" cannot be said to have innate potential—yet age, kinship, and natural substance categories are essentialized. Second, factor analyses with adults indicate at least two or three factors are needed to account for the full range of essentialist beliefs (Haslam et al., 2000; Haslam & Levy, 2006; Schudson & Gelman, 2023). Third, different aspects of essentialism are differentially predictive of outcomes such as stereotyping (e.g., Haslam & Levy, 2006) and consumer judgments (Gomez et al., 2024). Developmental data provide further evidence that essentialism has separable conceptual components. For children, different tasks or dimensions do not necessarily cohere (Diesendruck & Haber, 2009; Gelman et al., 2007). For example, learning that a characteristic is inborn does not lead first-grade children (6–8 years of age) to infer that it cannot change. More work is needed to map out this conceptual space: when different components of essentialism emerge in development, and the extent to which they cohere. Rhodes and Mandalaywala (2017) suggest that components will be more distinct when reasoning about social kinds.

3. The role of experience (including culture, context, and identity)

A serious turn over the past 20 years has been an appreciation for the important role of culture, context, and identity in essentialist beliefs. Whereas studies consistently find broad essentialism of animal kinds across the world despite widely varying kinds of input (Astuti et al., 2004; Atran et al., 2001), there is substantial variation in which social categories (i.e., categories of people) are essentialized and to what degree (e.g., Coley et al., 2019; Pauker et al., 2020; Rhodes, 2020; Xu et al., 2021). Whether a community essentializes race, ethnicity, religion, nationality, or occupation depends on the category, the perceiver, and the cultural context. Examining this variation in the construal of social kinds is thus a valuable—indeed, necessary—opportunity to learn about the mechanisms that can foster or inhibit essentialism.

3.1 Culture and context

Here I provide a few illustrative examples of cultural and contextual variation in children's essentialist beliefs; see also the papers in volume 59 of this journal (Rhodes, 2020) for more extensive and detailed overviews. An overarching theme is that children's essentialist beliefs are embedded within a cultural context, but at the same time do not simply or passively mirror the adult belief systems:

- An extensive investigation of the Vezo of Madagascar found that children tended to essentialize group identity (whether a person is Vezo or Karany), despite the adult understanding that these groups are defined in terms of individuals' occupational activities and religious beliefs (not an inborn essence) (Astuti et al., 2004). This result indicates an early-emerging essentialist construal in the absence of overt cultural support. At the same time, Vezo children typically did *not* essentialize the distinction between Vezo and a different group, the Masikoro (Astuti et al., 2004), suggesting that early theory-building mechanisms interact with a child's cultural environment.
- Children in the US and Israel essentialized gender substantially more than occupation, but children in the two societies differed in how they essentialized race: race essentialism increased with age in the US, but decreased with age in Israel (Diesendruck et al., 2013).
- In Northern Ireland—as opposed to the US—children 8–10 years of age essentialized the culturally salient category of religion (Catholic vs. Protestant), though this pattern was found predominantly among those attending religiously segregated schools (Smyth et al., 2017).

Systematic variation in essentialist beliefs is found not only when comparing different peoples in different parts of the world, but also when comparing communities within a country, region, or city. For example, gender and race essentialism show different developmental trajectories for urban vs. rural communities in the US, even when comparing neighboring communities within the same state (Rhodes & Gelman, 2009). Whereas gender essentialism was high in early elementary school in both settings, it decreased with age in the urban communities but remained high over age in the rural communities (Fine et al., 2024; Gross et al., 2024; Rhodes & Gelman, 2009). In contrast, race essentialism was low in early elementary school in both settings, then increased with age in the rural community but remained low in the urban community (Rhodes & Gelman, 2009). In order to understand the relevant mechanisms, "urban" and "rural" would need to be unpacked, as they correspond to a cluster of differences, including diversity, income, education, and political leanings.

The opportunity to engage with people from a different social group than oneself (contact effects) may play an important role in social essentialism. For example, race essentialism increased with age in children ages 4–11 in Massachusetts, but not those in Hawai'i, perhaps reflecting the notably greater community diversity in Hawai'i (Pauker et al., 2016). As noted earlier, essentialism of religion in Northern Ireland was lessened in integrated vs.

segregated schools (Smyth et al., 2017). Similarly, Israeli and Arab children in integrated schools were less essentialist over age (from kindergarten through sixth grade)—not by ignoring ethnicity, but by being more aware of it (Deeb et al., 2011).

In additional to differential rates of essentialism for different categories in different cultural contexts, there may also be cultural variation in the content or nature of children's essentialist beliefs. For example, the idea that an essence is inherent leaves open the causal mechanism—is it inborn (as assessed in the switched-at-birth task) or is it acquired through early experience, such as mother's milk or contact with the land (see Gelman & Hirschfeld, 1999, for discussion)? And if it is inborn, how is it transmitted? Initial findings indicate the important role of community context. For example, Waxman et al. (2007) found that Native American (Menominee) children (ages 4–10) were more likely than children in the three non-Native communities to endorse blood as a biological essence. And when comparing children ages 3–6 years in China and the US, Xu et al. (2025) found different developmental patterns in the two countries, with US children increasing in their endorsement of features as inborn, stable, and inflexible, and Chinese children increasing in their judgments of category homogeneity. The question of how children in different cultural settings explain the transmission of essentialist identities remains largely unexplored, and is ripe for future research.

3.2 Identity

Diesendruck (2021) argued for the need to consider motivational factors—not just cognitive factors—to explain patterns of essentialist attributions. He proposed that several factors, including dominance, in-group identification, and need to belong, all contribute to biased intergroup attitudes and behaviors, and these in turn lead to increased essentialism regarding certain social groups. Consistent with this view, Mahalingam (2007) argued that "essentialist thinking serves the needs of those in power to justify existing social and economic hierarchies." Conversely, those not in power may also engage in "strategic essentialism," whereby they reclaim and embrace an essentialized identity, as a means of achieving certain political goals (Eide, 2016), or to defend one's identity as real (Bartels et al., 2024).

Altogether, this theoretical perspective may help explain the pattern that a person's own identity may influence their essentialist beliefs—in children as well as adults. For example, among adults in India, members of a high-status caste (Brahmins) held more essentialist beliefs about caste than members of a low-status caste (Dalits) (Mahalingam, 1998, 2003). Similarly, Srinivasan et al. (2016) examined views of caste in children from grades 3–11 and adults. They found that, beginning in middle school (though not earlier), the more that a participant viewed caste as central to their identity, the more they displayed determinist views of caste (i.e., that biology determines success or failure, that the kind of person someone is can't change, that they are born a certain way, etc.—in other words, essentialist theories). They note that the causal direction here is unclear (e.g., caste attitudes may affect beliefs, the reverse may be true, or a third variable may be operative).

A study of Chilean kindergarteners examined their essentialist beliefs about poverty. Although all children endorsed certain essentialist beliefs (e.g., non-obvious basis and inductive potential), only the high-SES children (not the low-SES children) consistently reported that poverty is inherited and unchanging with age (del Río & Strasser, 2011).

Although the studies described above found that those in a higher-status position were more likely to endorse essentialist beliefs, this is not consistently the case. For example, gender essentialism is typically comparable for boys and girls (e.g., Taylor et al., 2009), despite the arguably higher status of boys, and race essentialism has been found to be higher in Black than White children ages 5–6 or 7 (Kinzler & Dautel, 2012; Mandalaywala et al., 2019; Roberts & Gelman, 2016), despite the arguably higher status of White children in the US (Mandalaywala et al., 2020). Pauker (2020) suggests that this latter pattern may reflect a tendency for many White families in the US to avoid discussion of race, though more evidence is needed (e.g., Scott et al., 2020).

It is of special interest to examine individuals whose own lived experiences may run counter to certain essentialist assumptions, in order to examine more closely the effects of identity and experience on essentialist beliefs. For example, transgender children's own experience of gender is that it is not rooted in biological sex, and they may have experienced their gender as mutable over time. Similarly, the siblings of transgender children live in close contact with people holding such identities. Yet despite these differences in experience, recent research has found relatively few differences in gender essentialism among cisgender children, transgender children, and cisgender siblings of transgender children. For example, in a study of children ages 3–11 years, all three groups were above-chance in predicting that a child's sex at birth would predict their clothing and play preferences (e.g., that a girl raised in an all-male environment would prefer to wear dresses and play with dolls, despite the lack of a female role model) (Gülgöz et al., 2019). Similarly, in a study of children ages 6–11 years, using a range of essentialist questions, all three groups essentialized both sex [body parts] and gender identity [whether a character felt like a boy or a girl], treating them as how they were born—although some subtle group differences did emerge (e.g., transgender children were less likely than the other two groups of children to essentialize gender/sex—i.e., when the meaning of the word "boy" or "girl" was unspecified) (Gülgöz et al., 2021). It will be valuable to examine as well those children whose identity does not fit into a gender binary.

Byers-Heinlein and Garcia (2015) studied 5- to 6-year-old children who experienced different conditions of language learning: either monolingual (spoke only one language), simultaneous bilingual (spoke two languages from birth), or sequential bilingual (learned their second language after age 3). They hypothesized that sequential bilingual children would have a greater appreciation that the language one speaks is not innately determined (in contrast to young monolingual children, who believe that one's language is determined by one's birth parents rather than the family of upbringing; Hirschfeld & Gelman, 1997). As predicted, sequential bilingual children were less essentialist about language learning, when compared to monolingual and simultaneous bilingual children. Surprisingly, however, the sequential bilingual children were also less essentialist about the vocalizations and physical traits of animals. Whereas the monolingual and simultaneous bilingual children were consistently essentialist in their attributions, the sequential bilingual children showed a bimodal distribution—roughly half consistently providing a "birth-parent" response, and the others consistently providing an "adoptive-parent" response. These provocative findings illustrate the influence of early experiences on children's essentialist theories.

Peretz-Lange and Kaebnick (2025) examine essentialist reasoning in children ages 4–8 years of age, comparing traditionally conceived children to those who were either adopted or donor-conceived. Peretz-Lange and Kaebnick made use of a switched-at-birth task, in which children were asked to predict whether a child's attributes (hair color, spoken language, personality, interests, or intelligence) more closely match those of their birth vs. adoptive parents. The findings indicated substantial group differences: traditionally conceived children were three times as likely to provide an essentialist response, as compared to adopted and donor-conceived children. There were also qualitatively distinct developmental patterns across the two groups: increasing essentialism with age in traditionally conceived children, but decreasing essentialism with age in adopted and donor-conceived children. The authors note that the mechanism(s) underlying these changes (e.g., motivated cognition, messages from parents and others in child's life, and/or observations of their own different environments) will require further research.

3.3 Future directions

Given the evidence reviewed in this section, documenting variation in children's essentialist reasoning as a function of culture, context, and identity, more systematic cross-cultural studies hold promise for providing important comparative evidence. Weisman et al. (2024) have published a Phase 1 registered report that will assess children's (ages 4–10) essentialism of religious groups, in comparison to gender and wealth groups, across 17 countries and 13 languages. The hope, of course, is not only to determine when these differences arise, but also why. A focus on carefully selected identities and cultural contexts will also be valuable in teasing apart different causal mechanisms that underlie the variation described in this section. For example, Amemiya et al. (2023) suggested that Indonesia would provide a compelling test case of different factors that may underlie early essentialist reasoning. Generally speaking, Native Indonesians have relatively greater political influence, whereas Chinese Indonesians have more wealth, and children are sensitive to these associations by age 6.5 years. This cultural context would therefore allow testing the importance of groups holding distinct societal roles (which is the case for both these ethnic groups) vs. one group having a more dominant status (which is the case for neither).

4. Language as a uniquely powerful mode of transmission

The variation reviewed above suggests that children are sensitive to cues from their environment to determine which categories to essentialize. There are likely a rich variety of cues that enhance the psychological salience of a category. Take the case of gender, for example. The salience of gender in the US is strengthened for children by "blue" and "pink" aisles in toy stores, the use of gender as an organizing principle in elementary-school classrooms (e.g., separate bathrooms; forming separate lines before going out to recess), and gender-specific dress codes (Arthur et al., 2008). Moreover, historical and structural forces support the inference that gender has vast inductive potential, and that gender-linked attributes are immutable (e.g., that most firefighters and all US presidents have been men) and have inductive potential.

With recognition that the nature and reach of these cues are likely important and deserving of study, here I focus on language as a starting point, given its uniquely powerful capacity for cultural transmission in the human species (Maynard Smith & Szathmáry, 1997). My focus is not on direct or explicit statements about categories, as essentialist beliefs can flourish even in the absence of such (e.g., Astuti et al., 2004; McIntosh, 2009). Indeed, explicit articulations of essentialism appear to be rare in parent-child speech (Gelman et al., 1998, 2004), even when parents are prompted to talk about appearance-reality distinctions (e.g., a bat that looks like a bird). Rather, I will review evidence suggesting that essentialist messages are *implicitly* conveyed, through labels and generic language (Gelman & Roberts, 2017). An important caveat to keep in mind throughout this section: the claim is not that labels and generics *create* childhood essentialism, but rather that they inform children as to *when to apply* this construal—that is, which categories to essentialize.

4.1 Labels

Labels are a simple yet powerful means of transmitting our hard-won adult knowledge about the structure of the world to young children. For children, a category label may signal that an animal or person shares hidden, non-obvious properties with others of the same kind. For example, upon hearing a pterodactyl labeled as a "dinosaur," 2-year-olds infer that it does not live in a nest even though it looks like a bird (Gelman & Coley, 1990); upon learning that a newborn baby is a "girl," 4-year-olds infer that she will later enjoy playing with dolls, regardless of her upbringing (Taylor, 1996), and upon hearing that a child received a heart from a "pig," 6- to 7-year-olds infer that the child would be slightly more likely to enjoy rolling in the mud (Meyer et al., 2017). The form of the label packs particular punch. For example, hearing that a person is a "carrot-eater" (a novel noun label), as opposed to "eats carrots whenever she can" (verb phrase), leads children 5–7 years of age to treat her behavior as more stable over time and contexts (Gelman & Heyman, 1999; but see Reynaert & Gelman, 2007, for how conventional patterns of use also affect interpretation of familiar expressions).

The use of the category label "scientist" can have downstream negative consequences for children's engagement and interest in science, by implying that the category is stable and distinct from other kinds of people—in other words, that scientific activities are appropriate only for this special kind of person (Rhodes et al., 2020). Describing science activities to children with a non-label (asking children to "do science") resulted in children's higher levels of interest and persistence in science activities, as compared to describing science activities with a label (asking children to "be scientists").

Although numerous labs have found that shared labels foster inductive inferences even in the face of dissimilarity (e.g., Booth, 2014; Davidson & Gelman, 1990; Graham et al., 2004; Jaswal & Markman, 2007), Sloutsky and colleagues (2001) suggest that this does not reflect a conceptual expectation about categories, but rather a reliance on perceptual similarity between items, which is increased when children hear the same auditory feature (e.g., "bird") for two different items. In a series of papers, they employed a paradigm designed to tease apart these two theoretical accounts (see Sloutsky, 2018 for review). The basic set-up was as follows: (1) Children were taught two novel categories in which labels (ziblet and flurp) were orthogonally crossed with item similarity; (2) the rule for applying the labels was a relational property (i.e., ratio of buttons to fingers); and (3) no labels were provided during the inference task. Under these conditions, children were much more likely to make inductive inferences on the basis of item similarity than category membership (indicated by the relational property). However, the artificial categories in these experiments differed markedly from basic-level natural kinds (Gelman & Waxman, 2007), and the relational property "rule" for identifying category membership was unfamiliar and cognitively demanding. In contrast, when the categories are clearly conceptually distinct, the category indicator is a feature rather than a relational rule, and information-processing demands are reduced, children consistently make use of category membership to draw their inferences (Gelman & Davidson, 2013).

I think there are two key conclusions from this debate. First, not all categories are alike, for children as well as adults, and children's assessment and use of labels for reasoning reflects these differences (see also Waxman & Gelman, 2009). And second, concepts and perceptual similarity are not in opposition to one another but are mutually informative (e.g., perceptual features provide information that helps us identify category membership; see also Medin, 1989). This is true for children as well; children are both "theorists" and "data-analysts," making use of both similarity and conceptual information (Waxman & Gelman, 2009).

A final important point about labels is that they engender what Hacking (1995) calls "looping effects." Building on Hacking's idea, Steven Roberts and I (Gelman & Roberts, 2017) suggest:

"Concepts of human kinds may lead to a cyclical pattern in which cultural practices lead groups to appear more distinct from one another, which confirms the categorizations, leading to more differentiating practices, and so forth. Viewing social kinds as having deep differences has cycling effects on behaviors that contribute to the reality of that social kind."

For example, differences between social groups (e.g., men/women; Jews/non-Jews, Whites/Blacks) have been exaggerated at different points in history via clothing, hairstyles, gait, bodily deformations, styles of speech, physical separation, and so forth—both chosen and imposed. These differences then further support the notion that these labels refer to natural groups that support indefinitely many non-obvious inferences, thereby reifying the categories they name. In essence, labels create a self-reinforcing loop. The developmental origins of such looping effects deserve more direct study.

4.2 Generics

Generics are the most common linguistic device that we use to refer to kinds directly (Gelman et al., 1998, 2000, 2004; Gelman, 2003; Rhodes et al., 2025): *Birds lay eggs; Girls wear pink; Italians eat pasta*. Generics are universal across the world's languages, frequent in child-directed speech, and both understood and produced in early childhood, by 2 or 3 years of age (Gelman, 2021; Gelman et al., 2008; Gelman & Raman, 2003; Hollander et al., 2002). Generics are notable for referring to a kind as a whole (e.g., birds, girls, or Italians, in the examples above)—not any particular instance or set of instances. As such, they have two semantic features that support essentialist inferences: (1) they express features that are not tied to any particular time, place, or context, but rather are assumed to be enduring and non-accidental (e.g., "Birds have hollow bones"), and (2) they gloss over exceptions, thereby minimizing within-category variation (e.g., "Birds lay eggs," even though male birds and baby birds do not).

Generics imply that the property predicated of the kind (e.g., wearing pink) is generally true (Cella et al., 2022; Cimpian et al., 2010), distinctive (Moty & Rhodes, 2021; Novoa et al., 2023), inherent (Cimpian & Markman, 2009, 2011), and normative (Leshin et al., 2021; Roberts et al., 2017). Moreover, above and beyond the specific properties that are predicated of the kind, generics imply that the category itself is a real, objective, natural kind. For children, just by virtue of a speaker choosing to use a generic about a given category (e.g., "girls" rather than "this girl" or "children"), this communicates that the speaker thinks the category is a richly structured, objective natural kind (e.g., that "girls" is a category about which a host of inferences can be made)—and this assumption holds even when the features expressed are innocuous, counter-stereotypical, or externally caused (see Rhodes et al., 2025, for review).

Studies of parent-child conversation indicate that the rate of parental generics corresponds to parents' own essentialist beliefs (Gelman et al., 2014) and racial attitudes (Britton et al., 2024) and contribute to children's ethnic essentialism (Segall et al., 2015). Experimental studies indicate that generics boost essentialism about novel animal kinds (Gelman et al., 2010) and novel social kinds (Pronovost & Scott, 2022; Rhodes et al., 2012, 2018). Additionally, hearing generics about a novel social group (non-citizens) led children to require less evidence to label individuals as members of that group, to ignore negative consequences (e.g., jail or deportation) to those so identified, and to express higher certainty about their choices (Goldfarb et al., 2017).

In summary, generics are ubiquitous in the speech that children hear, and they implicitly convey essentialist messages—even when the explicit messages are the opposite (Gelman et al., 2004). Messages conveyed generically may be particularly "sticky," in that they are difficult to refute. Whereas a single counterexample undermines a universal claim, it does not undermine a generic claim, as these allow for exceptions (Cimpian et al., 2010; Simmons & Gelman, 2025). However, more research is needed to examine generics in other languages and cultures (e.g., Shahbazi et al., 2024), and to examine the contexts that may undermine their negative consequences—a point to which I return in the final section of this chapter).

5. Developmental origins

The question of the developmental origins of essentialism is an important one. It is revealing not only about early cognition, but also why humans essentialize, and how we may intervene to curb its most problematic consequences. Theoretical proposals have been various and wide-ranging, and are summarized in Table 3. They include (but are not limited to) claims that essentialism is: (a) a late-emerging, historically contingent set of beliefs piggy-backed on learning about the work and tools of scientists (Historically Contingent), (b) an innate, domain-specific, modular capacity to reason about biological kinds (Biological Module), and (c) a broad assumption rooted in the logic of nouns—due to the use of single labels for varied sets (e.g., "mountain" for all mountains, suggesting some common essence) (Inherent Consequence of Naming).

Table 3 Some different theoretical accounts of the origins of essentialism.

Theoretical position	Key proponent(s)	Age of emergence	Scope of early essentialism
Historically contingent	Fodor (1998)	Late (when exposed to modern Western philosophy and technology)	[not specified]
Biological module	Atran (1998); Gil-White (2001)	Innate	Biological; later extended to social kinds
Inherent consequence of naming	Hallett (1991); Mayr (1991)	With emergence of language (10–12 months)	All named categories
Domain-general capacities, selectively applied	Gelman (2003)	Early (by 4–5 years of age, perhaps earlier)	Broad but limited, depending on fit with multiple perceptual, causal, linguistic, and social cues

Each of these first three views seems to fall short when considering the developmental data regarding when essentialism emerges in development, as well as the scope of early essentialism. In the remainder of this section, I review the theoretical implications of these issues, and then turn to my own view, "Domain-General Capacities, Selectively Applied." This view posits that essentialism emerges from a set of domain-general capacities that are beneficial for human development, capacities that are applied selectively to categories for which there is a perceived "fit." The convergence of multiple such capacities fosters psychological essentialism.

5.1 When essentialism emerges in development

As summarized in earlier sections of this chapter, numerous signatures of childhood essentialism are present early in development—prior to formal schooling, and in a range of societies, both industrialized and non-industrialized. This refutes the suggestion that essentialism is historically contingent on knowledge of Western philosophical traditions and modern scientific tools and discoveries, such as microscopes and genes (Fodor, 1998). Formal schooling and exposure to Western philosophy are not required for essentialism.

Although it is an open question as to when in development essentialism first appears, there is provocative evidence for essence-like construals in infancy. Croteau et al. (2024) suggest that two key aspects of infants' sortal-kind concepts display evidence for psychological essentialism: infants place greater reliance on deeper, causal features than surface-level features, selectively attending more to the former than the latter, when reasoning about objects (see evidence for this capacity by 13–14 months of age; e.g., Cacchione et al., 2013; Taborda-Osorio & Cheries, 2018). And by 9 months of age, infants draw rich inductive inferences from their kind concepts (e.g., Baldwin et al., 1993). This is consistent with the possibility that essentialism is part of core cognition, and inherent in the logic of basic-level kinds (Carey, 2015). Croteau et al. (2024) also note that infants treat ontological boundaries as immutable (e.g., an inanimate object cannot change into an agent or an animate object). Although the authors are careful to point to the need for further research to examine how these findings relate to psychological essentialism, certainly these results indicate an appreciation that categories extend beyond superficial features and have inductive potential.

Indeed, similar findings have been documented in some non-human species as well, although this evidence is more speculative and limited. Impressively, certain non-human primates can categorize based on non-obvious features—for example, baboons make use of sophisticated and subtle dominance hierarchies (Seyfarth & Cheney, 2009), and great apes can prioritize kind identity over superficial features (Cacchione et al., 2016; Lurz et al., 2022). However, it is also notable that without language, these concepts differ in several important ways from essentialism in language-using children (see Gelman & Roberts, 2017, for discussion): they are less efficient in transmitting category information, limited in inducing conceptual innovation and conceptual change, and narrow in scope of application (e.g., centered primarily on food and within-species social relations, such as mating and dominance, as opposed to the broad range of animal and social kinds to which young children apply essentialism). Thus the nature and extent of early essentialist-like behaviors—in infants and non-human primates—remains an important issue for future research (see also Rakoczy & Caccione, 2019).

The early emergence of essentialism would seem consistent with two theoretical positions: the Biological Module view and the Inherent Consequence of Naming view. However, both face problems when considering the scope of early essentialism, to which I turn next.

5.2 Scope of early essentialism

An important constraint on theories of developmental origins is the scope of early essentialism. Certainly for adults, essentialism does not apply broadly to all categories or domains. For example, people can form categories of simple artifacts (e.g., cups or Tables; Rosch et al., 1976), random dot patterns (Posner & Keele, 1968), and arbitrary groupings (e.g., items on the even-numbered pages of a google search for animals; Ahn et al., 2013)—yet none of those categories are presumed to support rich inductive inferences or have an innate basis. The domain-specific nature of essentialism is evident in young children as well. By age 3–4 years, children treat animals as having stricter category boundaries than artifacts (Rhodes et al., 2014); and by second grade, children distinguish between animals and artifacts in their judgments of immutability (Keil, 1989) and inductive potential (Gelman, 1988). Altogether, essentialism seems to be a better fit for natural kinds than artifacts, by preschool age.

Importantly, this result argues against the proposal that essentialism is a domain-general capacity that follows from how nouns function (Inherent Consequence of Naming).

By contrast, the domain-specificity of essentialism has been argued to support the Biological Module view. This account proposes that essentialism has its origins in reasoning about biological kinds, as part of an innate, universal biological module (Atran, 1998; Atran et al., 2001). On this view, we essentialize because we have special reasoning systems dedicated to processing the biological domain, and essentialism is extended to social kinds by analogy (see also Rothbart & Taylor, 1990)—especially those social categories that are most similar to biological kinds (Gil-White, 2001). Potentially consistent with this view, Davoodi et al. (2020) and Shahbazi et al. (2024) suggest that children's patterns of essentializing social categories across different cultural contexts (US, Iran, and Turkey) are strongest for categories that have a biological or quasi-biological basis (e.g., with gender the most strongly essentialized and sports teams least strongly essentialized). Thus, biological essentialist perceptions of social categories (e.g., that a member of a category was born that way; whether category membership could be detected in the blood) were associated with higher levels of essentialism on a switched-at-birth task (Shahbazi et al., 2024). This conclusion is limited, however, in that the measure of essentialism was itself a biological outcome (inheritance), and did not include other dimensions of essentialism, such as immutability or inductive potential.

However, there are two primary arguments against essentialism as initially a domain-specific, biological construal. First, young children also essentialize nonbiological natural kinds such as gold and salt, expecting dissimilar members to share non-obvious properties (Gelman & Markman, 1986) and expecting that a substance can be invisible yet still have causal effects (e.g., when sugar is dissolved in water, it still makes the water sweet; Au et al., 1993). Natural substances have none of the biological features of animal kinds. Second, there is no evidence that children essentialize biological kinds (e.g., animals) prior to their essentialism of social kinds (e.g., especially gender, but other social kinds as well, such as language). Not only are both social and biological kinds essentialized across a range of tasks in the preschool years, but children's essentialism of certain social kinds (again, most notably gender, but also language and personality) may be stronger for children than adults (Heyman & Gelman, 2000; Hirschfeld & Gelman, 1997; Taylor et al., 2009).

5.3 Domain-general capacities, selectively applied

I now turn to my own account of the developmental origins of essentialism, which I call "Domain-General Capacities, Selectively Applied." I propose that we essentialize not because we have special reasoning systems dedicated to processing the biological domain, not due to the logic of nouns, and not the consequence of a particular historical moment. Rather, I propose that essentialism is the application of multiple, domain-general psychological capacities or tendencies, each evolved to solve a different set of foundational problems in development (see Gelman, 2003, for more detail). Each of these capacities can apply to a broad range of concepts or problems that a child will face—including artifacts as well as animals. However, they also sensibly apply to some categories better than others. Which capacities are invoked will depend on cues from the real-world causal structure, as well as from context and language. When the full suite of capacities converge, they are more than the sum of their parts, and result in what we would call "essentialism."

The precise set of these capacities is up for debate, but some candidate tendencies include: causal determinism, induction from property clusters, the distinction between appearance and reality, deference to experts (but see Noyes & Keil, 2017, for an example of how essentialism can in certain contexts instead lead to rejection of expertise), attention to history, and an inherence heuristic. None is sufficient by itself for essentialism, but when they converge, they result in a powerful set of essentialist expectations.

Each is present early in development, and each applies broadly, not just to essentialized kinds.

- *Causal determinism* (the assumption that all events have a cause and do not occur randomly) may contribute to the essentialist belief that there is some unknown essence that accounts for why birds all have wings, feathers, and the same basic body structure. But causal determinism also applies to how infants and young children reason about simple mechanical devices, as when they search for a button or internal mechanism to explain why an object behaves in an unexpected way, or when they attribute a transparent artifact's apparently self-generated motion to "invisible batteries" (Bullock et al., 1982; Chandler & Lalonde, 1994; Gelman & Gottfried, 1996; Muentener & Schulz, 2014).
- *Induction from property clusters* can account for the rich inductive potential of natural kinds (where shared features serve as "magnets" for even more shared features), but it also can account for children's sophisticated tracking of property correlations to make inferences about the causal affordances of artifacts (e.g., Gopnik et al., 2001; Sobel et al., 2007; Walker et al., 2014) and overhypotheses regarding object sets (Dewar & Xu, 2010).
- The *appearance/reality distinction* and *deference to experts* can account for how children accept counterintuitive labels for natural kinds (e.g., accepting that a bat is not a bird, and a legless lizard is not a snake). But they can also account for young children's broad reliance on the labels that adults provide for artifacts—as when a sponge-rock is called "a sponge," or a crayon with a wick is called "a candle" (Lane et al., 2014).
- *Attention to history* may underlie the essentialist belief that once a raccoon, always a raccoon (immutability) and that a baby's gender at birth will determine her later preferences and abilities, regardless of the environment in which she's raised (innate potential). But it also accounts for how children apply labels to artifacts (Gelman & Bloom, 2000), decide who owns what (Friedman et al., 2011; Gelman et al., 2012, 2016), determine an object's value (Hood & Bloom, 2008; Gelman & Davidson, 2016; Tasimi & Gelman, 2021), and extract social information from traces of objects' history (Jara-Ettinger & Schachner, 2024).
- An *inherence heuristic* can explain that the causal essence is assumed to be internal to each member of a kind and immutable. But it also accounts for a range of cases in which children (and adults) appeal to inherent or intrinsic factors more than extrinsic or external factors to explain

the attributes of artifacts—as when explaining that people drink orange juice for breakfast because of inherent qualities it possesses (Cimpian & Salomon, 2014).

These underlying capacities do not themselves constitute essentialism, and do not emerge "for" essentialism. Rather, each has independent adaptive value for young humans, when they are taking in information and building knowledge structures. However, when these capacities converge, they may collectively foster a host of essentialist consequences, including a realist assumption about categories and names, boundary intensification, attention to internal features, and an expectation that a category is immutable over time and over transformations.

6. Consequences for social issues and education

To this point I have focused on basic research on childhood essentialism. In this section I address two broad sets of consequences that essentialism has for children beyond the laboratory: for social issues, and for education.

6.1 Implications for social issues

One of the more important and problematic aspects of childhood essentialism is its association with social ills, including stereotyping, prejudice, stigma, and negative intergroup relations (see Rhodes & Mandalaywala, 2017, for review). For example, children's race essentialism was predictive of reduced liking of Black children (Mandalaywala et al., 2019) and out-group stereotyping (Pauker et al., 2016), and children's gender essentialism was predictive of prejudice toward gender-non-conforming children; Fine et al., 2024; Gross et al., 2024). Inducing essentialism resulted in negative inter-group attitudes (Diesendruck & Menahem, 2015), and led children to share fewer resources with outgroup members (Rhodes et al., 2018).

At the same time, research with both adults and children has found that essentialism can at times have the opposite effect, for example reducing prejudice based on body weight, same-sex attraction, and incarceration (see Peretz-Lange, 2021, for review). Peretz-Lange (2021) proposes a causal discounting theory to explain these contrasting effects:

> *"Essentialism may promote prejudice by leading children to discount structural explanations (i.e., to reason that a group is low-status because of its personal deficiencies rather than its structural disadvantages), but it may mitigate prejudice by leading children to discount agentic explanations (i.e., to reason that a group was "born that way" rather than choosing to be that way). Thus, the consequences of essentialism may reflect both the explanations children endorse as well as those they discount."*

This important point indicates the double-edged sword of essentialist beliefs, as well as the need to probe more fully children's (and adults') causal explanations in order to understand their consequences. Consider, for example, essentialist beliefs about criminal behavior. There are striking developmental changes with age: children were substantially more likely than adults to attribute law-breaking to internal, unchanging behaviors, and to attribute imprisonment to internal moral qualities (e.g., "Prison is a place where bad people go"; Dunlea & Heiphetz, 2020). However, whether essentialist explanations for criminal behavior are linked to greater or lesser endorsement of punishment, depend on the component of essentialism that is assessed (e.g., biology vs. immutability), the target of essentialism (e.g., behaviors vs. moral character), and whether the actor is believed to have control of their actions (Martin & Heiphetz, 2021; Martin et al., 2022; Meyer et al., 2022; Xu et al., 2022).

This framework can be applied to a host of children's concepts of disability. Jaswal & Robertson (2024) note that children's essentialist views of disability could increase prejudice (by decreasing awareness of structural explanations), and/or could reduce prejudice by highlighting a person's lack of control (see Lebrón-Cruz & Orvell, 2023, for protective effects of essentialist beliefs about neurodivergence, in an adult sample). Menendez & Gelman (2024) likewise suggested that essentialist models of disability may foster viewing the disability as residing in the disabled individual, and as highly informative and predictive of other features (treating all with that disability as alike), and may lead to under-appreciation for the role of non-biological factors. At the same time, essentializing a disability might lead people to place less blame on the disabled individual, but also contribute to a fatalistic assumption that a disability cannot be improved. Both Jaswal and Robertson (2024) and Menendez and Gelman (2024) point to the need for future research in this area. For example, Menendez and Gelman argue for more systematically examining different components of essentialism, each of which may have different implications for children's attitudes and beliefs.

Given the potential for biased reasoning about essentialized groups, researchers have proposed methods for reducing such bias. One such means is via intergroup contact, which has been theorized to lower bias and discrimination against stigmatized outgroup members (e.g., Allport, 1954). Evidence for reduced essentialism resulting from intergroup contact has been documented in children (e.g., Deeb et al., 2011; Pauker et al., 2016). Intergroup contact can be either direct or indirect (i.e., via children's media). For example, in one study, cisgender children learned about a transgender child via a child-friendly animated video, resulting in their having a better understanding of transgender identities and lower gender essentialism on a few specific measures (Fine et al., 2025).

Another strategy may be to introduce alternative, non-essentialist explanations for group differences. For example, recent research indicates that even young children are capable of appreciating structural explanations, according to which group differences are attributable to the external circumstances in which the groups find themselves (e.g., Peretz-Lange et al., 2021; Vasilyeva et al., 2018; Zhang et al., 2024). This may be a promising approach for countering bias resulting from essentialism. At the same time, it should be noted that increasing structural explanations does not necessarily reduce children's reliance on internal explanations (Yang et al., 2022). Furthermore, children may ignore structural factors when they are consistent with an existing stereotype (Amemiya et al., 2022). To reduce prejudiced beliefs, it may be necessary not only to draw children's attention to external factors, but also to highlight individuals' potential for change (Vasilyeva et al., 2018).

To this point I have focused on children's own attitudes toward members of different social groups, but adults' own essentialist beliefs toward social policies also have consequences for children. For example, Roberts et al. (2017) found that adults who endorsed higher levels of essentialism were also more likely to endorse boundary-enhancing legislation, policies, and social services—both ones that were designed to disadvantage vulnerable social groups (e.g., mandating that transgender individuals use bathrooms for their biological sex), as well as ones that were design to help them (e.g., support for single-sex classrooms). This result was obtained even after controlling for participants' conservatism, education, religiosity, and anti-gay attitudes. Adults' essentialist views regarding language learning (that language is innate and biologically based) were associated with endorsing language-related educational neuromyths and rejecting educational policies that promote multilingual education (Sun et al., 2023). Adults' belief in the myth of "learning styles" corresponded to their beliefs regarding the efficacy of multimodal instruction for children (Nancekivell et al., 2020, 2021), as well as the beliefs of both parents and teachers about children's academic potential (Sun, Norton, et al., 2023). Adults' essentialist belief that success in certain fields requires raw, innate, natural talent or "brilliance" (Leslie al., 2015) is one that emerges early in childhood and may have negative consequences for children's own goals and aspirations (Bian et al., 2017; Muradoglu et al., 2025).

6.2 Implications for education

A second set of consequences are in the field of education, where lay essentialist beliefs may be imported into people's construal of scientific biological concepts, thereby introducing distortions and misunderstandings. I focus on two such examples: genetics and evolution.

Genetic essentialism. Dar-Nimrod and Heine (2011) introduced the notion of "genetic essentialism," which they define as "the tendency to infer a person's characteristics and behaviors from his or her perceived genetic makeup." They provide evidence that learning that an attribute has a genetic cause has several potentially negative consequences, including treating it as (1) immutable and determined, (2) having a specific etiology, (3) homogeneous and discrete, and (4) natural (and therefore more morally acceptable). Essentialist misconceptions about genetics are found in nationally representative samples of adults in the US (Christensen et al., 2010; Jayaratne et al., 2009). More problematically, essentialist misconceptions about genetics are also found in widely used biology textbooks aimed for high school students—for example, portraying sex and gender differences as discrete rather than continuously variable and overlapping, and overemphasizing the role of sex chromosomes vs. environmental factors and gene-environment interactions (Donovan et al., 2024). Curricular changes have been proposed with the goal of reducing essentialist views of race and gender (e.g., Blake, 2004; Donovan et al., 2024; Pérez, 2024).

The extent to which children likewise view genetics in an essentialist manner, and at what ages, is still a largely open question. Recent research does suggest elements of essentialist and non-essentialist views of genes in childhood, as well as age-related changes in children's reasoning (e.g., Menendez et al., 2024; Meyer et al., 2020)—but more research is needed.

Evolutionary theory. Marjorie Rhodes and I proposed that essentialist concepts contribute to two sorts of obstacles to evolutionary theory and natural selection: those that stand in the way of understanding, and those that stand in the way of acceptance (Gelman & Rhodes, 2012; see Table 4).

Shtulman and Schulz (2008) found that children (ages 4–9 years) and adults greatly underestimated the degree of variation within a species (needed for appreciating natural selection), and that this essentialist construal correlated with misunderstanding evolution (e.g., believing that change takes place over time within individuals, rather than at the population level). Ware and Gelman (2014) found that adults also endorsed a Lamarckian view of evolution: if an individual animal needs a feature to survive, then it will develop that feature and even pass it down to its offspring. Importantly,

Table 4 Essentialist components that undermine acceptance and understanding of evolutionary principles.

Essentialist components	Beliefs undermining understanding of evolution	Beliefs undermining acceptance of evolution
Stability over transformations	–	Species cannot change.
Strict boundaries	There are sharp, absolute boundaries between species.	There are no intermediate forms in biology.
Uniformity	Members of a species vary in only superficial respects; they are alike in their essence.	There is minimal variation across members of a species.
Inherence	Evolution entails individual-level change, not population-level change.	–

however, simple and direct instruction can successfully inform even young children (5–8 years old) of the principles of natural selection (Kelemen et al., 2014).

7. Conclusions

Childhood essentialism is a double-edged sword: it is a source of great achievements and serious misconceptions. On the one hand, it is impressive that preschool children are capable of thinking about entities in their everyday world in terms of non-obvious features and hidden potential. Essentialism reveals a precocious ability to look beyond the obvious, and contrasts with the classical view of children as captivated by appearances and fixated on the "here-and-now" (Gelman, 2023). Furthermore, the placeholder nature of essentialism, along with the expectation that natural kind categories have rich inductive potential, encourages children to assume that there is always more to learn about the world around them. It is a perspective with the potential to engender additional knowledge growth. In this regard, I view essentialism as a starting-point for learning—not merely its outcome (see also Gelman & Wellman, 1991). On the other hand, essentialism is a source of stereotyping, prejudice, and intergroup bias. It has contributed to some of the most terrible conflicts in human history. It also fosters serious and persistent misconceptions about science. Altogether, the study of childhood essentialism is a stark reminder that the cognitive mechanisms that make us so smart and sophisticated are simultaneously sources of distortion and bias (Gelman, 2023; Noyes & Keil, 2017).

Although essentialism can be examined as a construct unto itself, I have tried to emphasize, throughout this chapter, that children's essentialism always works in concert with, and alongside, other cognitive, linguistic, social, and societal processes. For example, essentialism does not entail the absence of similarity-based reasoning. As Medin (1989) points out, most of the time, things that look alike can be assumed to share deeper properties; appearances are therefore a useful heuristic for children as well as adults. Likewise, a full understanding of essentialism and its consequences will require continued study of the complex array of messages that children receive, and their sensitivity to such messages—in different communities, cultures, contexts, and languages.

Although childhood essentialism emerges early in development, it is also important to note that much of the research I have reviewed has also obtained age-related changes. A broad theme appears to be that childhood essentialism becomes increasingly differentiated over time, in terms of the categories and properties to which an essentialist framework does (and does not) apply (e.g., Diesendruck et al., 2013; Peretz-Lange & Kaebnick, 2025; Davoodi et al., 2020; Shahbazi et al., 2024; Waxman et al., 2007). Essentialism may also change as children gain experience with a scientific knowledge base (e.g., see Gottfried et al., 1999, for an example in children's reasoning about the brain) and with cultural or community beliefs (Rhodes & Gelman, 2009). Rhodes and Moty (2020) also propose an interesting developmental change, in which children are initially essentialist about specific categories (e.g., tigers, wolves) but then over time construct over-hypotheses such that essentialism applies also to entire domains (e.g., animals). They note that such a shift would be especially powerful, as it would guide children's reasoning about even novel categories they have not previously experienced.

An open question is the extent to which we (as adults) restructure our concepts away from essentialism as we learn about phenomena that conflict with some of its foundational assumptions (phenomena such as natural selection; epigenetics; and structural bases for disparities associated with socioeconomic status, race, ethnicity, or gender). Formal scientific knowledge about species, an appreciation for statistical variation within kinds, and exposure to historical, geographic, and epidemiological sources of social group differences may all contribute to an appreciation for within-category variability, the limits of genetic causation, the importance of structural factors and complex environmental influences, and so forth. At the same time, it is certainly not the case that essentialism disappears altogether, at least for many adults. Extensive evidence over the past few decades documents that many adults hold essentialist beliefs regarding a range of social and biological categories (e.g., Berent et al., 2020; Haslam & Whelan, 2008; Sun et al., 2021). Moreover, essentialist and anti-essentialist beliefs may co-exist within an individual, for example with the reappearance of essentialism under speeded testing conditions (e.g., Eidson & Coley, 2014). Whether adults generally reduce their reliance on essentialist construals relative to children, and under what conditions, remains an important question.

A tendency to resort to essentialist reasoning may also be true of our own science (Mahoney, 2023). Oyama (2000, 2002) proposed that theorists may treat statistical evidence of a trait as if it were "a hidden truth, rooted in the past and already there" (in other words, an essence). Siegler (1996) argued that the questions that developmental psychologists pose often treat age categories as uniform and static, rather than variable and continuously changing—as if 5-year-olds (for instance) can be understood as possessing a single, underlying, true nature. Similarly, Barrett (2017) suggested that scientists incorporate essentialism into their theories—erroneously implying that categories (such as emotion categories) are uniform and universal rather than variable and influenced by contextual factors. And in reporting the results of our research in scientific journals, there is an overwhelming tendency to use generic language, a tendency that glosses over exceptions (DeJesus et al., 2024). In all of these respects, there is a danger that we may lapse into essentializing assumptions about the very constructs we are trying to explain.

I end this chapter by reproducing (with some modifications) the final paragraph of my 2003 book, which I believe still holds today:

I conclude by considering why we essentialize in the teleological sense—in plain terms, what it buys us. There are, I think, two sorts of answers. First, each of the underpinnings of essentialism is useful for making sense of the world, each in its own distinct way. We attend to object history in order to recognize individuals through space and time. We defer to experts in order to benefit from cultural knowledge. We draw inductive inferences from property clusters, and distinguish appearance from reality, in order to make (generally) accurate predictions. We search for causes in order to create more useful tools and technologies. And so forth. In other words, each of the underpinnings of essentialism is motivated for independent purposes – not for essentialism per se. There is a second answer as well. Essentialism is a by-product, with unintended and unpredictable consequences. Each strand that underpins essentialism is beneficial in our interactions with the world. Yet the cumulative effect of these tendencies is more mixed, presenting deep problems alongside the benefits.

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CHAPTER SIX

Children's ability concepts: Their development, content, and consequences

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Contents

Abstract

In this chapter, we summarize theoretical and empirical work on the development of ability concepts in children. We first examine the form of children's basic concept of ability, asking whether it undergoes major differentiation during development or whether, instead, a near adult-like ability concept is available early on. We then ask when in development children's ability beliefs begin to exhibit coherence and motivational force. The body of evidence reviewed here points to previously unnoticed sophistication, depth, and coherence in young children's ability concepts. We close with a discussion of how theoretical and empirical attention to early ability concepts might advance our understanding of motivational processes in early childhood, as well as offer insights into how to optimize these processes.

Keywords: Ability, Concepts, Growth mindset, Intuitive theories

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Ability is central to who we are and what we do. Even in the course of an ordinary day, we may think about how much of it we have, how much of it others have, and what this might mean for our, and others', ability to master skills and succeed on important tasks. Whatever ability may actually be, or however it may contribute to success, as a concept it plays a central role in everyday thought and behavior. Given its centrality, it is critical to examine both how we come to understand ability and the potential implications of these understandings.

An examination of this sort is all the more pressing given that ability concepts figure prominently in nearly every contemporary psychological account of motivation (Bandura, 1982; Covington & Beery, 1976; Dweck, 1986; Weiner, 1985; Wigfield & Eccles, 2000). The extent to which we embrace challenges, persist after failure, and more generally try to achieve mastery, for instance, is intimately tied to our basic assumptions about what ability is like and what it does (Bian et al., 2018; Blackwell et al., 2007; Elliott & Dweck, 1988; Limeri et al., 2023; Rege et al., 2021; Yeager et al., 2019). In short, our concepts of ability are foundational to our motivation, to our decisions about our learning, and, ultimately, to what we achieve.

In light of their theoretical and practical significance, our goal in this chapter is to synthesize and evaluate evidence on early concepts of ability, focusing roughly on the grade school age range (5–11 years). Throughout, we consider classic and contemporary findings with the goal of arriving at a clear, comprehensive, and up-to-date overview of young children's thinking about ability. Such an endeavor, in our view, is needed to ensure that future research efforts are grounded in what is currently known about children's ability concepts. As we have foreshadowed, one question that readers might keep in mind in reading this chapter is: How similar is young children's ability-related thinking to that of older children's and adults'? To anticipate our conclusions, early research often pointed to dramatic differences, but newer methods have revealed fascinating similarities.

We discuss early concepts of ability in two major parts. We start by considering whether children have well-formed, adult-like ability concepts early on, or whether, instead, these concepts must be gradually acquired over early and middle childhood. After addressing the question of whether young children's ability concepts resemble adults', we then ask whether children's beliefs about ability are rudimentary and disconnected from their motivation and learning, or whether, instead, these beliefs are coherent and linked with achievement behaviors, such as challenge-seeking, early on. Against this backdrop, we conclude the chapter by outlining avenues for future research, highlighting open questions whose answers can yield insight into how best to sustain children's motivation.

In addressing these questions, we find it useful to regard children's ability concepts as embedded in domain-specific, theory-like knowledge structures called "intuitive theories" (e.g., Gopnik & Wellman, 2012; Kinlaw & Kurtz-Costes, 2003; Murphy & Medin, 1985; Wellman & Gelman, 1992). Like a scientific theory, an intuitive theory is used to interpret, explain, and predict everyday events within its scope (Keil, 2024). Here, it makes sense to assume that both children and adults have informal understandings of performance and the factors that contribute to it. This informal understanding can be thought of as an intuitive theory, comprising interconnected concepts such as ability, which can be used to explain what happened in the past (e.g., why did she fail the bar exam?) and make predictions about what might happen in the future (e.g., will she fail the bar exam again?). Importantly, with this perspective, and in keeping with cognitive development traditions, we can characterize developmental change, if any, in intuitive theories by asking how similar (vs. different) young children's ability concepts are relative to those of older children and adults (for a similar strategy, see Carey, 1985, 2000).

1. The development of children's ability concepts: Change or continuity?

Adults have intuitive understandings of ability as a cause, distinct from effort, that increases the likelihood of (but does not guarantee) good performance.[1] That is, adults' intuitive theory of performance has embedded in it *differentiated* concepts of ability and effort. When do children's intuitive theories begin to resemble this structure? Are the concepts embedded in this intuitive theory kept separate by children, or might they be conflated initially instead? These questions have implications not only for our basic understanding of conceptual development, but also for approaches to studying motivational processes in early childhood. If young children have an undifferentiated ability concept that includes effort, this might suggest that their motivation and learning behaviors (e.g., their responses to failure) look quite different from older children's and adults'. If, however,

[1] For our purposes here, we define ability as a personal resource that (a) increases the likelihood of, but does not guarantee, good performance and (b) is fairly consistent in its effects on performance across situations. One thing that distinguishes ability from other personal resources (e.g., effort) is that it is not under one's control at the time of performance. One's ability could change over longer temporal spans, but one cannot instantaneously increase their ability at will.

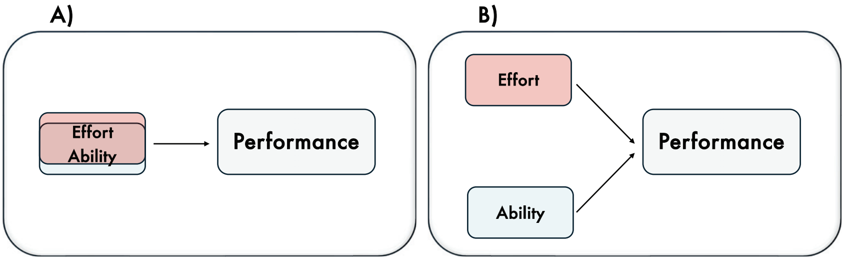

Fig. 1 Schematic of an intuitive theory of performance with ability and effort (a) as undifferentiated concepts versus (b) as differentiated concepts.

young children do have a distinct ability concept, this would suggest the early presence of a foundational aspect of motivation, and would motivate empirical examinations of how this concept is involved in motivational processes.

An influential theory put forth the idea that children's ability concepts develop quite gradually in a stage-like fashion, with early stages reflecting immature or misguided understandings and later stages reflecting mature, adult-like understandings. In particular, this canonical perspective suggested that children do not understand ability as a distinct source of performance until they reach 10 years of age (Nicholls, 1978; Nicholls & Miller, 1984a; 1984b). On this view, children start out with an undifferentiated concept of ability that is combined with something different: effort (see Fig. 1). The differentiated, mature concept of ability was thought to emerge only after a stage-like restructuring of children's intuitive theories (Nicholls & Miller, 1984a).

Accounts of this sort were not limited to children's understanding of ability. The idea that children's (initially jumbled) concepts become increasingly differentiated with age has been central to developmental psychology, and to cognitive development traditions especially (e.g., Marsh et al., 1984; Smith et al., 1985; Werner, 1957). This characterization has been applied to diverse sets of concepts: for example, to children's concepts of weight and size (Piaget & Inhelder, 1974), to their concepts of inanimate and animate, as well as inanimate and dead (Carey, 1985, 2000; Piaget, 1929), and to their concepts of numerosity and spatial arrangement (Piaget & Inhelder, 1969), to name a few. To foreshadow our conclusions, some of these characterizations were later challenged with evidence showing that young children's knowledge is often masked by (overlooked) features of the very procedures used to assess it (e.g., Gelman, 1990; Keil, 1979; McGarrigle & Donaldson, 1974; Nguyen & Gelman, 2002; Rose & Blank, 1974; Smith et al., 1985).

Evidence for the idea that children initially conflate ability and effort came mainly from a series of studies in which children were asked to make judgments about two students' abilities after being given information about how much effort they put into their work and about their performance outcomes (Chapman & Skinner, 1989; Miller, 1985; Nicholls, 1978; Nicholls & Miller, 1984b). In these studies, participating children were shown films of two students working side by side on schoolwork: One student is shown to work the entire time, while the other spends only some of the time working. It is eventually revealed that the students earned identical (and high) grades on their schoolwork, and at the film's conclusion, participating children are asked to respond to questions about the students' effort expenditure, ability, and so on.

Critically, when participants were asked to make judgments about the students' abilities, children younger than 10 usually said that the student who worked *harder* had more ability than the student who did not put in as much effort. The interpretation was that young children chose the student who put in more effort because effort and ability are not yet differentiated concepts: People who try harder are smarter. This account suggested that around age 10 is when these two concepts become neatly separated.

If young children really do think that a student who has to expend *more* effort for the same outcome has more ability, then their ability concepts are quite different from those of older children's and adults', who typically view the less-hardworking (but still high-achieving) student as having more ability. Further, this canonical finding was emerging alongside and fit well with results painting young children as characteristically lacking in sensitivity to ability-relevant information. For example, across a number of studies using a variety of tasks, young children's assessments of their own abilities and future performance were often notably and unreasonably high (Parsons & Ruble, 1977; Stipek & Tannatt, 1984; Yussen & Levy, 1975; see also Leonard & Sommerville, 2025; Xia et al., 2022). Claims from the canonical account could be used to explain this: If ability and effort are one conceptual unit, then securing a good performance outcome in the future is just a matter of trying harder.

Young children also seemed curiously resilient to failure—that is, failure did not seem to prompt the decrements in performance or persistence (Miller, 1985; Rholes et al., 1980) or the tempered expectancies for future success (Parsons & Ruble, 1977; Stipek & Hoffman, 1980) seen among older children. These findings were also interpreted against the backdrop of the canonical theory that children's concept of ability is not yet differentiated from effort: Without a distinct concept of ability, failure or otherwise subpar performance is not an indictment of one's abilities and therefore cannot amount to a motivation-thwarting event.

In short, this body of work constructed a picture of how young children's undifferentiated ability/effort concept functions in ordinary achievement situations. If young children are unable to distinguish between ability and effort, as the canonical account proposed, then one would not expect them to be demotivated by clear evidence of their inadequacy. Further, this picture of young children's optimism and resilience stood in sharp contrast to the image of older children and adults, who usually adjust evaluations of their own abilities following a failure experience or after learning that peers performed better than they did (e.g., Ruble et al., 1980). It seemed quite reasonable then, that the concepts underpinning these behaviors—namely, ability and effort—might undergo qualitative change and restructuring with age.

1.1 Problems for the canonical account of children's ability concepts

There was, at the same time, some evidence to suggest that this theory did not provide an adequate account of the development of children's ability concepts. It did not fit well with contemporaneous findings showing that in some cases, fairly young children could in fact make logical inferences about varying levels of ability, effort, and outcomes. In one study, 6-year-old children judged that a less-able student would have to try harder than a more-able student to complete a puzzle (Karabenick & Heller, 1976). Similarly, even younger children, 4-year-olds, seemed to understand that a character with high ability who obtained a poor outcome exerted little effort, and that a character who exerted little effort but obtained a good outcome had high ability (Wimmer et al., 1982).

There were also reasons to believe that the rosy canonical view of young children's reactions to failure was not entirely accurate either. Even quite young children—4- to 6-year-olds—seemed to show individual differences in how they responded to challenges and setbacks (Burhans & Dweck, 1995; Heyman et al., 1992; Smiley & Dweck, 1994; see also Butler, 2005; Kagan, 1981). Much like older children and adults, many young children displayed decreased motivation, including negative affective responses and lower persistence, after learning that they were unsuccessful in completing a task (Heyman et al., 1992). The canonical account also did not fit well with findings showing that children's behavior does, under some circumstances, seem to be responsive to ability-relevant information (Cimpian et al., 2007; Zhao et al., 2017, 2018; see also Rhodes & Brickman, 2008). In one study, for example, 4-year-old children's responses to failure were more negative when the failure was framed as resulting from lack of ability rather than lack of effort (Cimpian et al., 2007). Failure and setbacks could, for many young children, be demotivating experiences. So then why did young children seem to have a single, undifferentiated concept mixing ability and effort in some cases, but not others? Were their ability concepts truly undifferentiated, or could task demands have prevented them from accessing and deploying a differentiated concept of ability?

Today, there is considerable evidence that undermines the claim that young children have an undifferentiated ability/effort concept. Consider, for example, that by age 5 children distinguish ability from performance in surprisingly sophisticated ways. They judge that a more able runner who loses a race to a less able runner because of an intervening event (e.g., tripping over a stone) would, in the future, perform better than the less able runner (Yang & Frye, 2016). This suggests that young children have some notion of ability as a personal resource that—barring unusual intervening events—is consistent in its effects on performance across situations.

Children's understanding of ability is further revealed in their reasoning about how ability relates to others' mental states. When two puppets complete a low-effort task such as climbing a short structure to obtain a reward, but one of the puppets has previously expressed a preference for a reward requiring high effort (i.e., climbing a tall box), 5- and 6-year-old children infer that the puppet with the preference cannot climb (Jara-Ettinger et al., 2015). Further, even younger children understand that low (vs. high) ability corresponds with different mental states about, or subjective evaluations of, tasks: Three-year-olds understand that a person who found a task hard has less ability than a person who found the task easy (Heyman et al., 2003).

In these cases, it is clear that children's concepts of ability have some degree of nuance and sophistication. However, more direct evidence of young children decoupling ability from *effort* is really needed before concluding that they do in fact have a differentiated ability concept. To that end, recent work has reconsidered the most important empirical evidence for the canonical account of children's ability concepts by examining features of the two-student paradigm discussed earlier (e.g., Nicholls, 1978). Two features stand out. First, in these older paradigms the student who works less hard is always depicted with potential misbehaviors, such as playing with a textbook or pencil or looking around the room. Not only was this conduct less than ideal, but it was also in the context of the *classroom*. This specific presentation of low effort could have unduly influenced the responses of young children, who are greatly concerned with (and inundated with messages about the importance of) behavioral conduct and work habits (e.g., Stipek & Tannatt, 1984). This may have led children to view the student who works less hard—and their ability—more negatively than they would have in the absence of such behaviors. In addition to their being (viewed as) impermissible, these behaviors may not have been a good signal of a *lack of a need* to work hard—instead, they could have signaled, and been interpreted as, trouble with the schoolwork or disliking of it.

Second, after children viewed the two-student paradigm, they were always *first* asked by the experimenter if one of the students worked harder or if they worked the same (Nicholls, 1978; Nicholls & Miller, 1984b). It was always after children had responded to this question that they were then asked to evaluate the students' abilities. This is important because the question about effort may have incidentally set up a task demand that influenced subsequent evaluations of the students. After explicitly stating that the hardworking student put more effort into their schoolwork, young children may have had a globally positive view of this student, and thus assessed their abilities more favorably afterward. Overall, then, young children may have been anchored by *task demands* (here, the behaviors of the non-hardworking student and the order of the questions), such that when they later responded to questions about the students' respective abilities, they gave more positive evaluations about the hardworking student.

So perhaps young children do understand ability but had difficulty deploying these concepts in the presence of these two features. Maybe the question young children were answering was closer to "Who has more ability: Someone who performed well and behaved? Or someone who performed well and didn't behave?" In fact, later studies that disambiguated these two features by (a) including a clearer signal of a lack of a need to work hard (earlier completion of the schoolwork) and (b) not asking the question about effort first found that even very young children say that the less-hardworking student is smarter than the hardworking student (Heyman & Compton, 2006; Muradoglu & Cimpian, 2020). For example, in one study, 4- to 9-year-old children were shown vignettes of two students working side-by-side on schoolwork (Muradoglu & Cimpian, 2020). In one condition, children were shown the non-hardworking student working *intermittently* on their work, as in the canonical paradigm, and in another condition, children were shown the non-hardworking student working *continuously* (and finishing the task *before* engaging in off-task behavior). The latter condition, where low effort was depicted as continuous rather than intermittent, was designed to be a clearer signal of a lack of a need to work hard. Importantly, however, the amount of effort put in by this student, as operationalized by number of frames in a storybook, was equivalent across the two conditions. What varied was simply whether their effort was depicted as intermittent throughout the task or concentrated and continuous at the beginning of the task.

In this study, after children were shown the vignettes, they were then asked to assess each student's effort and ability. Children in one condition were asked the effort question immediately after viewing the vignettes, as in the canonical paradigm. Children in another condition were asked to make judgments about how hard the students found the schoolwork after viewing the vignettes. Here again, the latter condition was designed to prevent children from being unduly anchored by positive assessments of the hardworking student.

When children made judgments about the students' abilities *without* the two features of the canonical paradigm, they rated the less-hardworking student's abilities more highly than the hardworking student's, suggesting that ability and effort are differentiated by young children. Even the youngest children—4-year-olds—distinguished the two. Importantly, children still rated the non-hardworking student as working *less* hard than the hard-working student, suggesting that children can indeed successfully decouple ability from effort (Muradoglu & Cimpian, 2020; see also Leonard et al., 2019, for additional evidence that young children consider speed and performance information together). And, there was support for the idea that the intermittent display of effort on behalf of the less-hardworking student signaled difficulty with the task. Children's perceived difficulty ratings for the less-hardworking student were higher in the intermittent-low-effort condition than in the continuous-low-effort condition, and this shift in perceived difficulty mediated the effect of type of low effort on children's ability judgments.

These findings offer more compelling evidence against the claim that children do not acquire a mature, differentiated concept of ability until around age 10 by showing that much younger children—4- and 5-year-olds—have concepts of ability and effort that are differentiated (see also Cimpian, 2017). What appeared to be conceptual immaturity may have instead been the result of methodological features that masked children's conceptual sophistication.

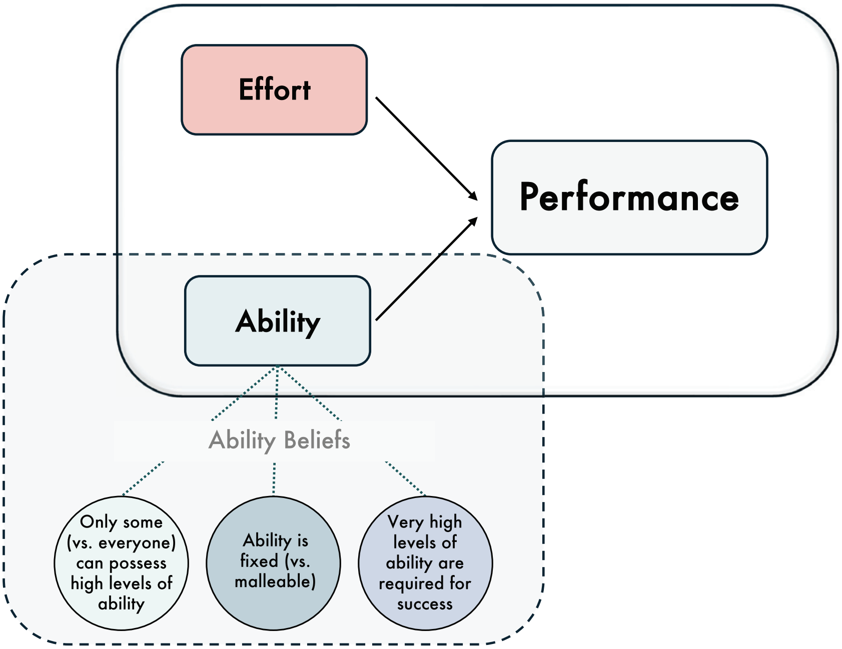

Fig. 2 Schematic of ability as a concept, distinct from other performance inputs, such as effort. Ability, like effort, bears on performance outcomes, and people can hold a number of nuanced beliefs about the nature of ability (in blue circles).

2. Beyond having a differentiated ability concept: The organization and impact of children's beliefs about the nature of ability

The aim of the prior section was to address whether young children have a unitary concept of ability and effort that increases in differentiation and clarity with age. The weight of the evidence suggests that this is not the case. Even very young children seem to initially have intuitive theories in which ability and effort are distinct concepts. However, it is one thing to have a notion of ability as a distinct cause, and yet another to have assumptions about what ability is like—that is, to have beliefs about its nature (see Fig. 2). These assumptions or beliefs involve content about what is true (and untrue) about ability. If, as we claim, young children do in fact have a clear concept of ability, then they might also build beliefs around this concept concerning what ability is like. And, if children have beliefs about what ability is like, these beliefs might be *motivationally relevant*—that is, they may be involved in

Table 1 Beliefs about ability, their conceptual meaning, and references.

Belief	Conceptual definition	References
Growth (vs. fixed) mindsets	Beliefs about the malleability of ability	Dweck, 1999; Dweck and Leggett, 1988; Dweck et al., 1995
Universal (vs. nonuniversal) mindsets	Beliefs about the distribution of potential for high ability in a population	Rattan et al., 2012; Rattan et al., 2018; Savani et al., 2017
Brilliance beliefs	Beliefs about the extent to which brilliance—that is, a very high level of ability—is required for success	Jenifer et al., 2024; Leslie, Cimpian et al., 2015

children's learning behaviors and achievement. The aim of this next section is to examine these possibilities.

We begin by summarizing theory and research on a set of beliefs with a long history in the field of achievement motivation: beliefs about how *malleable* ability is (Cain & Dweck, 1995; Dweck & Bempechat, 1983; Dweck & Leggett, 1988). This work suggests that such beliefs—now called "growth (vs. fixed) mindsets"—are foundational to motivation, learning, and achievement (Blackwell et al., 2007; Dweck & Leggett, 1988; Hong et al., 1999; Limeri et al., 2023; Rege et al., 2021; Yeager et al., 2019).

Yet, growth (vs. fixed) mindsets are only one part of a dense network of beliefs about ability's properties. As adults, our concept of ability is richly detailed with assumptions concerning not only how *malleable* (vs. static) it is, but also whether high levels of it are required for success in specific contexts, whether everyone (or just some people) can possess the highest levels of it, and so on (see Fig. 2). Over the years, researchers have expanded empirical inquiries beyond growth (vs. fixed) mindsets to consider these beliefs—called "brilliance beliefs" and "universal (vs. nonuniversal) mindsets," respectively (see Table 1).[2]

Are young children's ability concepts richly detailed with these kinds of ideas? Or might children's ideas about ability be more limited than those of adults? Further, if children *do* have a rich set of assumptions about the nature of ability, do these assumptions link up with each other in logically coherent ways, or are they somewhat scattered and independent? Here again

[2] Importantly, this list of beliefs—growth mindsets, universal mindsets, and brilliance beliefs—is not an exhaustive one.

it makes sense to first review the state of the evidence from investigations with older children (i.e., of middle- and high-school age) and adults before considering evidence from young children, which is relatively more limited. In what follows, we first describe these nuanced beliefs in more detail and then present evidence on their structure—that is, their dimensionality and interrelations—among adults. We then turn to emerging evidence that points to the presence of a structured, coherent, and motivationally active network of ability beliefs in early childhood.

2.1 Components in the network of beliefs about ability

2.1.1 Growth (vs. fixed) mindsets

A long tradition of scholarship has looked to a particular type of belief to understand the dynamics of motivation: beliefs about how *malleable* ability is (Dweck, 1986; Dweck & Leggett, 1988). Some people view ability as relatively dynamic and malleable (i.e., a growth mindset) and others view it as relatively stable and fixed (i.e., a fixed mindset; Cain & Dweck, 1995; Dweck & Bempechat, 1983; Dweck & Leggett, 1988). Assumptions about the malleability of ability have received a great deal of empirical attention because people's beliefs along this dimension have been shown to influence learning behaviors, and, as a result, achievement (Blackwell et al., 2007; Claro et al., 2016; Dweck, 1986; Dweck & Leggett, 1988; Dweck et al., 1995; Hong et al., 1999; OECD, 2019; Yeager et al., 2019). These links are present because ability beliefs can influence and structure key motivational processes, including people's goals (i.e., to increase mastery or showcase ability), understandings of effort (i.e., as useful or futile; as a way to increase ability or a sign of low ability), and attributions for and responses to failure (e.g., increase or withdraw effort) (Blackwell et al., 2007; Hong et al., 1999; Robins & Pals, 2002).

For instance, at the level of goals, people with growth mindsets tend to be more focused on increasing mastery of skill, in comparison with those with fixed mindsets, who are more focused on *showcasing* their abilities and preventing anything that might discredit them (e.g., Blackwell et al., 2007; Dweck & Leggett, 1988). Because beliefs about the malleability of ability predispose people to more (vs. less) adaptive approaches to and responses to achievement events, these beliefs can end up contributing to actual achievement outcomes as well. There is some evidence to suggest that, compared to people with stronger fixed views of ability, people with stronger growth views of ability tend to earn higher grades in school

(Blackwell et al., 2007; Romero et al., 2014; Yeager et al., 2016) and to perform better on standardized tests (Claro et al., 2016; Cury et al., 2008; McCutchen et al., 2016).

2.1.2 Universal (vs. nonuniversal) mindsets

One extension of work on growth (vs. fixed) mindsets concerns universal (vs. nonuniversal) mindsets. Whereas growth and fixed mindset beliefs provide answers to the question "*Can ability be developed?*" universal and nonuniversal beliefs provide answers to "*Can anyone (or only some people) possess the highest levels of ability?*" (Rattan et al., 2012, 2018; Savani et al., 2017). Universal beliefs involve assumptions that virtually everyone has the capacity to possess the highest levels of ability; nonuniversal beliefs involve assumptions that only some have the capacity to reach these levels of ability. To make the distinction with growth (vs. fixed) mindsets more concrete, consider the person who believes that ability can be increased (i.e., has a growth mindset), but simultaneously believes that the *highest* levels of ability can only be reached by a select few (i.e., has a nonuniversal mindset).

Prior work shows that these beliefs among instructors can play a role in students' motivational processes. In particular, students' *perceptions* of the extent to which universal (vs. nonuniversal) beliefs are endorsed by others in their learning environments seem to be important for students' sense of belonging. When students perceive that their instructors believe that *everyone* has the potential to possess the highest levels of ability (i.e., hold a universal mindset), students feel a stronger sense of belonging to their field. Further, students who had these perceptions of their instructors' beliefs outperformed students who perceived their instructors to hold nonuniversal mindsets, a difference that was explained in part by these differences in belonging (Rattan et al., 2018).

2.1.3 Brilliance beliefs

Another dimension that has been recently prominent in the study of ability beliefs involves the *importance of very high ability for success* in certain arenas. Some view very high levels of intellectual ability (i.e., "brilliance") to be a precondition for success in a particular domain or context, while others view this level of ability as less necessary for success, with effort and motivation being sufficient (Leslie, Cimpian et al., 2015; Muradoglu et al., 2023, 2025). Whereas growth (vs. fixed) mindsets and universal (vs. nonuniversal) mindsets concern the perceived *properties* of ability (that is, its malleability and its distribution in the population, respectively), brilliance beliefs instead concern ideas about ability's *importance*. These beliefs are often brought to bear when considering what it takes to excel in specific domains or areas of study. For instance, people tend to think that brilliance is necessary for success in domains such as mathematics, physics, and philosophy (Ito & McPherson, 2018; Leslie, Cimpian, et al., 2015; Rutten et al., 2024).

Importantly, endorsing such beliefs about brilliance is thought to thwart processes involved in motivation, particularly for some groups of people. In one study, for example, high school girls who believed brilliance to be central to success in the physical sciences, technology, engineering, and mathematics expressed weaker intentions to pursue such domains (Ito & McPherson, 2018). Similarly, environments whose members valued brilliance for success tended to prompt lower belonging and higher anxiety (Bian et al., 2018), as well as impostor feelings (Muradoglu et al., 2022), particularly among members of underrepresented groups.

2.2 The dimensionality, coherence, and correlates of ability beliefs among adults

Though conceptually distinct, these beliefs bear enough resemblance to each other that one might expect some amount of interdependence among them. For example, one can reasonably expect people with growth mindsets to more often hold universal (vs. nonuniversal) beliefs about ability, and to view brilliance as less important for success. Similarly, one might expect people with fixed mindsets to more often hold nonuniversal (vs. universal) beliefs about ability, and to view brilliance as more important for success. Empirically speaking, to what extent are these beliefs interconnected? And, if interconnected, *how* interconnected are they? Might they be so highly related that they are best understood as representing the same underlying construct?

The empirical evidence so far does not support the idea that these beliefs reflect a single underlying construct. Instead, growth (vs. fixed) mindset beliefs, universal (vs. nonuniversal) mindset beliefs, and brilliance beliefs seem to be distinguishable but related dimensions. First, correlations between growth (vs. fixed) mindsets and universal (vs. nonuniversal) mindsets range from .14 to .27, suggesting distinguishability but some interdependence (Rattan et al., 2012). Second, a more recent study that examined growth mindset beliefs, universal mindset beliefs, and brilliance beliefs together in college students found that these beliefs were empirically represented by separate underlying dimensions (Limeri et al., 2023). Though distinct, these beliefs showed a logical pattern of interdependence: Students who believed that ability was malleable tended to also believe that anyone could reach the highest levels of it ($r = .38$), and that brilliance was not a precondition for success (in STEM) ($r = -.24$). Students who believed that anyone could reach the highest levels of ability tended to also believe that brilliance was not a precondition for success (in STEM) ($r = -.42$) (Limeri et al., 2023; see also Bian et al., 2018; Porter & Cimpian, 2023). Thus, adults seem to carry with them an internally consistent belief system about the nature of ability. Importantly, however, the magnitude of the correlations between these beliefs point to interdependence but distinguishability.

This belief system also appeared to be bound up with motivational processes among students. Consistent with theory (Dweck & Leggett, 1988), students' growth mindsets were associated with their goals, such that students who viewed ability to be relatively malleable had goals for mastering course material. Also consistent with theory, these students were relatively less focused on simply avoiding performing worse than their peers. Further, compared to students who thought that ability was relatively unmalleable, students who thought that ability was relatively malleable had lower levels of evaluative concern—that is, they were less anxious about the prospect of their abilities being judged negatively by others. Students who viewed ability as relatively universal, as well as students who thought that success (in STEM) was not strongly dependent on brilliance reported weaker levels of evaluative concern as well (Limeri et al., 2023).

2.3 Evidence for distinct, coherent, and motivationally active ability beliefs in early childhood

An important developmental question concerns the extent to which these dimensions can be reasonably described as a coherent, motivationally active network of beliefs among children. This broad question can be broken down into three more concrete ones. First, are children's ideas about ability more limited than those of adults? Second, if children do have a rich set of assumptions about the nature of ability, do these assumptions link up with each other in logically coherent ways, or are they somewhat scattered and independent? Third, to the extent that these beliefs are distinguishable, are they motivationally *active* among young children? In other words, are key learning behaviors, such as embracing challenges, tied to children's assumptions about ability?

There is some reason to expect that children's ability beliefs might, like adults', be multifaceted and coherently interrelated. Indeed, at around age 5, children start to acquire culturally constructed ideas about ability, and their ideas show a notable degree of consensus with those of adults. For example, as early as first grade, children view high levels of intellectual ability to be more central to success in domains such as math than in domains such as reading and writing (Gunderson et al., 2017; Jenifer et al., 2024; see also Muradoglu et al., 2025). Also, at around age 6, children begin to associate high levels of intellectual ability with boys and men more than with girls and women (Bian et al., 2017; Jaxon et al., 2019; Kim et al., 2024; Okanda et al., 2022; Shu et al., 2022). In addition to ideas about who is *generally* more intellectually talented, young children seem to have adult-like stereotypical beliefs about who is more intellectually talented in certain domains. On both explicit and implicit measures, second-grade children associate mathematics with boys more than with girls (Cvencek et al., 2011; Miller et al., 2024). Similarly, around age 5, children begin to view men as more competent than women in science, technology, engineering, and mathematics (STEM; Shenouda et al., 2024).

Although these findings hint at detailed, adult-like beliefs, they do not speak to the organization or coherence of children's beliefs about the *properties* of ability. One recent study, however, offered more direct evidence on this matter by embracing the full nuance of these beliefs (Muradoglu et al., 2025). In this study, the authors examined 5- to 11-year-old children's beliefs about ability's malleability, its universality, and its importance for success.[3] To ensure that items tapping these beliefs were accessible for young children, the language concerned abilities in math and spelling (e.g., "good at math"), which represent concrete manifestations of ability, rather than general intellectual ability.

The authors were first interested in understanding the *structure* of these beliefs, asking whether they represent truly distinct dimensions of reasoning about ability (as seems to be the case among adults) or instead form a single belief amalgam. Though young children clearly seem to have some adult-like understandings of ability, there are still grounds for expecting their beliefs about its properties to be not neatly distinguishable. Consider, for instance, work on the development of self-concepts that has shown that young children can make self-assessments regarding their competencies

[3] The authors measured two other beliefs (innateness and responsiveness to intervention) for a total of five; however, for brevity and clarity, we do not discuss these additional two beliefs here.

in cognitive and physical domains (Harter, 1982; Harter & Pike, 1984). However, for children younger than 8, the structure of their self-assessments seemed to lack dimensionality. Rather than loading on two distinct factors, young children's self-assessments in cognitive and physical domains loaded onto a *single* factor. This was interpreted as evidence that the self-structure becomes more differentiated and dimensional with age (but see Marsh et al., 2002, for evidence that 4-year-olds distinguish between multiple dimensions of self-concept).

What were the dimensions underlying young children's ability beliefs? The results indicated that items tapping the beliefs of interest—ability's malleability, universality, and importance for success—loaded onto three separate factors. In addition, this structure appeared to be consistent both among the younger children in the sample (before about 8 years of age) and among the older children (Muradoglu et al., 2025). This suggests that although the three beliefs share a "family resemblance," even young children's mental representations differentiate between them. Because the structure of children's beliefs about ability does in fact reflect these nuances, it does not seem sensible to treat these dimensions as missing or merged.

The second main question addressed in this study concerned coherence: To what extent are these three beliefs correlated with one another? The results indicated that these beliefs were linked in a logically coherent manner: Children who thought that intellectual ability was relatively fixed also tended to endorse the idea that it was nonuniversal and that high levels of it were a precondition for success. Similarly, children who viewed ability to be relatively malleable tended to also view it as relatively universal, and they thought that high levels of it were not a strong requirement for success. These (disattenuated) correlations ranged from .32 to .64 in absolute value, suggesting small to moderate levels of concordance—and, importantly, non-redundancy (Muradoglu et al., 2025).

The third and final question in this study concerned the motivational significance of these beliefs: Are children's ability beliefs implicated in their motivation? Two findings suggested the answer is "yes" (Muradoglu et al., 2025). First, above and beyond age, children who viewed ability to be malleable (rather than fixed) opted for learning goals (rather than performance goals) and preferred more challenging tasks. Second, younger children who thought that very high levels of ability (i.e., brilliance) were a precondition for success in school reported higher levels of evaluative concern—that is, they were more sensitive about how their abilities might be perceived by others in academic contexts (see Fig. 3). These

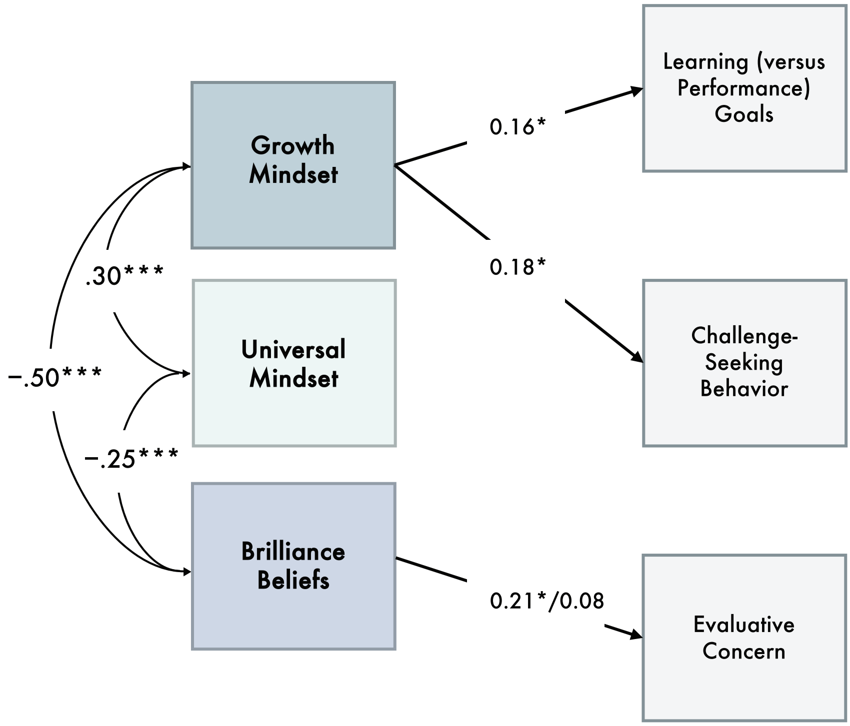

Fig. 3 Path model illustrating the relations among growth mindset, universal mindset, and brilliance beliefs, as well as relations between beliefs (growth mindset, universal mindset, and brilliance beliefs) and motivation (goals, challenge-seeking, and evaluative concern). Numerical entries are estimates of correlated residual variances (zero-order correlations) and standardized regression coefficients, respectively. Child age moderated the association between brilliance beliefs and evaluative concern; therefore, the path shows regression coefficients for younger and older children separately (i.e., younger/older). All other paths from beliefs to motivation adjust for child age. The data come from Muradoglu et al. (2025). *$p < .05$. **$p < .01$. ***$p < .001$.

relationships are both intuitive and aligned with theory (Dweck & Leggett, 1988; Dweck et al., 1995): An understanding of ability as a relatively fixed and highly important resource sets up a "world" in which the priority is to "prove and protect" one's abilities. It makes sense that in this "world," one would gravitate toward easier, lower-risk tasks and experience concerns about how others might perceive one's abilities.

Together, these findings suggest that even young children have a network of coherently organized beliefs about ability. Children's beliefs about ability's malleability, its universality, and its importance for success appear to be clearly distinct dimensions. Further, these beliefs are coherent, in that most of them correlated with one another in logically expected ways. Children with relatively stronger (vs. weaker) beliefs about ability's malleability also had relatively stronger (vs. weaker) beliefs about ability's universality, as well as relatively weaker (vs. stronger) beliefs about ability's importance for success. A final piece of evidence suggests that in addition to being distinct and coherently interrelated, these beliefs are motivationally active: Children's motivation appears to be, at least in part, organized around their beliefs about what ability is like. Children's beliefs about ability's malleability and importance for success were linked with the extent to which they embraced challenges and adopted goals for increasing their mastery, and with the extent to which they expressed concern over others negatively judging their abilities, respectively. Although it is not possible to draw causal conclusions here, the results nevertheless suggest that even quite young children might, much like adults, enter achievement situations with a network of ability beliefs that has some degree of sophistication, coherence, and motivational force.

3. Conclusions and avenues for future research

The literatures we have reviewed here yield three key conclusions about the psychology of ability in early childhood. First, children seem have a well-formed concept of ability, distinct from effort, by age 4 or 5. Second, by the age of 5, children also appear to have a multifaceted and coherently interrelated set of beliefs about what ability is like. Third, this set of beliefs is not only coherent but also already linked up with children's motivation. Before describing avenues for future research based on these conclusions, it is useful to briefly consider the impact of the older, "classic" arguments on the field's views of, and approach to studying, young children.

Because young children were painted as relatively unconcerned with ability-relevant information, they were regarded as both unshakably optimistic and highly resilient to failure (but see Heyman et al., 1992; Smiley & Dweck, 1994). This picture predictably led to efforts to understand (and ultimately improve) motivational processes among older age groups, which were assumed to be more vulnerable: namely, older children, adolescents, and adults. It is no wonder that our understanding of basic motivational concepts and processes in early and middle childhood is relatively limited as a result. And, admittedly, what we do know now may still be a somewhat incomplete account of young children's motivational concerns and problems. Our hope is that the revised picture we have proposed here, along with our suggestions for future research, will contribute to a deeper and more complete account of motivation in early childhood.

3.1 Avenues for future research

A key upshot of the work we have summarized here is that young children have sophisticated, motivationally active concepts of ability much earlier in development than previously theorized. One way of thinking about this is that we now have a much clearer picture of "the seeds"—that is, of the *beliefs* and *concerns*—that guide children's motivation. Yet these beliefs and concerns do not develop in a vacuum. A useful (and we would argue, valid) way of thinking about children's ability beliefs, and their motivational tendencies more generally, is to regard them as deeply embedded in (social) contexts (de Ruiter & Thomaes, 2023; Haimovitz & Dweck, 2017). A generative avenue for future research, then, will be to better understand these contexts—that is, to better understand "the soil" in which "the seeds" take root and grow (Walton & Yeager, 2020).

In the pursuit of better understanding children's contexts, one more general priority may be to make progress toward building a rich understanding of the ordinary, everyday moments that shape the beliefs and motivational tendencies children come to adopt. Our current understandings of young children's quotidian exposure to ability- and achievement-relevant cues, for example from teachers or parents, is somewhat limited. For instance, when a child makes a mistake in their schoolwork, how (if at all) do teachers respond? When children receive explicit evaluations of themselves in the form of tests or report cards, what do conversations between teachers (or parents) and children look like? When children are placed in ability groups, what are they told, if anything? We believe that research on children's everyday achievement-related experiences, including some of the events we have outlined here, could contribute substantially to an understanding of how academic contexts shape early motivational processes. In addition to its theoretical contribution, a clearer picture of the contextual factors that shape young children's beliefs about ability—that is, "the soil"—would lay the groundwork for intervention research to improve young children's beliefs and motivational tendencies.

3.1.1 Home contexts

Although much remains to be studied about these experiences, a few literatures offer useful starting points. One is work on feedback from adults. A branch of this work has focused on how so-called process praise (e.g., praise for one's effort or strategy) versus person praise (e.g., praise for one's ability) might shape the mindsets children adopt and their vulnerability to setbacks. This research has taken the form of carefully controlled experimental studies (Kamins & Dweck, 1999; Mueller & Dweck, 1998) as well as field studies whose focus was on praise from parents (Barger et al., 2022; Gunderson et al., 2013, 2018; Pomerantz & Kempner, 2013). The field studies were with children of very different ages and used different methods and measures. They, not surprisingly, yielded heterogeneous patterns of results, with some studies finding links between process (but not person) praise and children's mindsets (Gunderson et al., 2013, 2018), some finding links between person (but not process) praise and children's mindsets (Pomerantz & Kempner, 2013), and others finding no links between either type of praise and children's mindsets (Barger et al., 2022).

Rather than regarding this heterogeneity as a weakness, it can be viewed as a starting point and an opportunity for further investigation. Because field studies involve capturing processes in highly complex and multifaceted environments, perhaps one reason for their heterogeneous results is that some parents express process-oriented praise but enact other behaviors that contradict or undermine the message that it is effort or strategy that matters (see Walton & Yeager, 2020; Wang et al., 2024). In addition, children of different ages might be more or less sensitive to the different kinds of feedback. Finally, in some cases, the parent feedback might be overshadowed by strong feedback from teachers. Future research can (a) seek to resolve these inconsistencies by focusing on when and for whom adult feedback influences children's beliefs, and (b) try to broaden the scope of inquiry on this question to consider antecedents other than praise (e.g., Haimovitz & Dweck, 2016; Zarrinabadi et al., 2021).

3.1.2 School contexts

Any account of how contexts shape children's thinking about ability would be incomplete without attending to one of children's most important ecological "niches": school. Empirical attention to children's educational environments is all the more pressing in light of more general, secular trends toward increasingly ability-focused contexts. Over time, children with different academic abilities have become less likely to attend the same school (OECD, 2015). And, among 80 countries, the US ranks fifth in having one of the highest percentages of students attending schools that sort students by ability (OECD, 2023). Evaluation is very clearly a feature of every educational context, and recent evidence suggests that children as young as 3 care about how their abilities are evaluated by others (Asaba & Gweon, 2022).

Yet we know very little about the interplay between the features of young children's school environments and their thinking about ability (including their thinking about how their own abilities will be viewed by others) and their motivation. Children's views of ability could well be shaped by teachers' classroom practices and by teachers' expectations of their students (Brummelman & Sedikides, 2023; Darnon et al., 2023; Goudeau & Cimpian, 2021). Therefore, empirical attention to teachers' expectations for (and treatment of) higher- and lower-achieving students, as well as children's perceptions of these dynamics, could lead to valuable insights. Indeed, some researchers are now focusing on these very types of practices in the classroom and coaching teachers in the implementation of practices that shape students' active engagement with learning and pursuit of challenging work (e.g., Hecht et al., 2023a, 2023b).

Another useful endeavor may be to examine the course of ability beliefs and motivational processes across early and middle childhood as a function of the types of school environments children are in. Early theorizing on this topic suggested that as children progress through school, the character of their school environment changes to become more regimented, evaluative, and ability-focused (Eccles et al., 1984). This led many researchers to assume that children came to view ability as more fixed with age.

Although the idea that grade-related changes in school environments shape children's beliefs and motivational orientations is, at face value, compelling, it has not yet, to our knowledge, moved from theorizing to *testing*. Further, the empirical yield on whether and how elementary school-age children's beliefs change with age is quite inconclusive. Some studies have shown increasing growth-oriented views of ability with age (Gunderson et al., 2017; Jenifer et al., 2024; Muradoglu et al., 2024, 2025; Pomerantz & Saxon, 2001; Stipek & Gralisnki, 1996), while others have shown the opposite—stronger views that ability is fixed with age (Droege & Stipek, 1993; Heyman & Giles, 2004; Pomerantz & Ruble, 1997; Pomerantz & Saxon, 2001). Still others have found no changes with age (Heyman & Giles, 2004; Kinlaw & Kurtz-Costes, 2007; Muradoglu et al., 2024; Pomerantz & Ruble, 1997).

Part of these inconsistencies may stem from different approaches to measuring and operationalizing children's beliefs about ability (a point we return to later). Indeed, age-related changes in children's ability beliefs seem to depend, at least in part, on the specific question children are asked (e.g., Heyman & Giles, 2004; Muradoglu et al., 2024; Pomerantz & Ruble, 1997). Further, some of these investigations involved samples of older elementary-aged children (e.g., Pomerantz & Saxon, 2001), whereas others involved younger elementary-aged children (e.g., Jenifer et al., 2024; Muradoglu et al., 2024), making direct comparisons more challenging. Broadening the age range of focus, as well as the suite of questions in future work may clarify these patterns. Another interesting approach—and one that would speak to the influence of the school environment—would be to use longitudinal designs to compare the beliefs of children attending more- vs. less-academically focused elementary schools (see e.g., Stipek & Daniels, 1988).

3.1.3 Cultural contexts

At a broader level than the *school* context is the *cultural* context. A great deal of what we have discussed—ability concepts, beliefs, motivational tendencies, school contexts—can vary across cultural contexts (Holloway, 1988; Li 2003; Li, 2004; Li & Wang, 2004; Stevenson et al., 1990; Sun et al., 2021; Uttal, 1997). Consider, for instance, that children as young as 4 seem to have conceptions of learning that align with their culture's values. Whereas US American children's conceptions of learning seem to be bound up with ability and strategy use, Chinese children's conceptions appear to be bound up with personal virtues such as diligence and persistence (Li, 2004).

The beliefs of important socializers in children's lives exhibit cross-cultural variability as well. One study showed that although mothers in the US, Japan, and Taiwan all view effort to be a more important factor for children's school performance than ability, this difference is much larger among Taiwanese and Japanese mothers, compared to US American mothers (Stevenson et al., 1990). Similarly, compared to Japanese and Taiwanese mothers, US American mothers were less likely to endorse the idea that any student could be good at math or reading if they worked hard enough (Stevenson et al., 1990). It would be informative to explore potential visible manifestations of these beliefs among young children, with attention to how these manifestations may vary cross-culturally. One other important consideration is that, of the estimated 2 billion children in the world, a minority live in Western societies. It is therefore an exciting and important endeavor to study children's ability concepts, as well as their motivational concerns and problems, more globally.

3.1.4 Attention to measurement

We wish to end by acknowledging that a continuing challenge to these pursuits is a lack of concerted empirical attention to a key part of any psychological investigation: careful measurement of constructs (e.g., Hodson, 2021). One real problem is that measurement is a challenging undertaking in itself, and almost all of the constructs we have focused on here involve content that can be highly abstract. These challenges are only compounded when the target participants are young children, whose still-developing language skills may make it difficult to express their beliefs and thoughts about concepts such as ability. But this does not mean that children do not have abstract thoughts, nor does it mean that it is impossible to access and study such thoughts via careful and serious measurement pursuits (see Barroso et al., 2023; Muradoglu et al., 2024). Though there are likely a number of approaches to increasing comprehensibility, one used in some recent studies (Muradoglu et al., 2024, 2025) has involved using multiple concrete, domain-specific expressions of ability (e.g., "good at math," "good at spelling") to approximate children's beliefs about ability more generally. We look forward to future work that embraces the challenges—and opportunities—inherent in measuring young children's thinking.

3.2 Conclusion

Ability is a key concept in people's intuitive theories of performance. We reviewed evidence that calls into question canonical views of children's ability concepts, showing that core elements of adults' intuitive understanding of ability are present from early childhood. Our hope is that this effort to provide an up-to-date overview of young children's ability concepts can serve as a starting point for research that seeks to understand the formation and downstream consequences of young children's thinking about ability.

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CHAPTER THREE

Emotion understanding in infants and young children: How input shapes emotional development

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Contents

Abstract

Here, we will review the developmental literature on how infants and young children learn about emotions. We take a process-based perspective, highlighting how the protracted trajectory of emotional development unfolds concurrently with changes in children's cognitive abilities, and how variability based on context, culture, and experience shape this trajectory over time. We will also emphasize the role of *input* into this development, a factor that has often been ignored. Indeed, little research has systematically sought to understand how children's emotional development is shaped by the emotional input they receive. We focus on the development of three abilities that comprise emotion understanding. First, we will discuss the developmental trajectory of *emotional expression*, and how infants and young children come to express emotions and behave emotionally. Second, we will discuss how infants and young children learn to *regulate* those emotional responses, and how different environmental inputs affect that ability. Third, we will discuss the development of *emotion perception*, or how infants and young children learn how to perceive, categorize, and identify the emotions of others over time. Finally, we will discuss mechanisms for developmental change in all of these domains, and how other concurrent domains of development—including motor skill, language, theory of mind, and attachment—might have cascading effects on changes in emotion understanding.

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Over the course of a day, adults engage in an average of about 12 social interactions with various individuals, including friends, colleagues, teachers, children, and strangers (Zhaoyang et al., 2018). This suggests that over the course of our lifetimes, we will likely participate in hundreds of thousands of social interactions, spending a significant part of our lives engaged with others. Emotion understanding—or the set of abilities related to determining our own emotions, regulating them, and interpreting the emotions of others—is crucial for the development of healthy social interactions. Emotion understanding allows us to respond appropriately to others' needs, make predictions about social interactions, and even regulate our own emotions effectively. Research has shown that children who have better emotion understanding are rated as more socially skilled by their teachers, more likable by their peers, and are better able to navigate aggressive interactions. Likewise, children who have poor emotion understanding tend to be more aggressive, are more likely to develop behavioral problems and internalizing issues, and demonstrate lower academic achievement (Denham, 2019). But how do we develop emotion understanding? How do we learn to navigate our own emotions and the emotions of others?

There are various theories of what emotions are and how they come to be expressed and understood throughout human development. The dominant view in the social science literature has been that there are a set of basic discrete emotions—including anger, disgust, fear, happiness, sadness, and surprise—that are universally recognized and produced, regardless of individual experiences. According to this theory, basic emotions are innate, evolutionary adaptations to specific environmental challenges, and each has a distinct physiology and brain circuitry (Ekman & Cordaro, 2011; Izard, 2007; Panksepp, 2007). Evidence to support this view comes from classic research showing that individuals in an isolated, preliterate culture in New Guinea can perceive the emotional facial configurations of (White) American adults (Ekman et al., 1997). In this work, researchers showed participants photographs of White adults from the U.S. displaying various emotional facial configurations and asked them to match the configurations with emotional vignettes, such as "He is happy because his friends have arrived," or "He is sad because his mother has passed away." Despite having no exposure to U.S. culture, participants from New Guinea successfully identified each emotion category.

These findings were robust, and subsequent studies using similar methods confirmed that there is significant cross-cultural agreement in recognizing emotion categories across different regions, including Asia, Europe, and the United States (Krause, 2004). This consistency has been found to extend to more subtle emotions, like pride (Tracy & Robins, 2008). Additionally, others have shown that people *express* emotions similarly across cultures as well, producing comparable emotional facial configurations in response to a range of emotional stimuli (Ghiselin et al., 1974; Scherer et al., 1988, 1986).

Because of the popularity of this view and its intuitive nature, many social scientists have ignored the role of development in emotion understanding. The discrete emotions perspective assumes that emotion perception and expression are universal, and thus, implicit in this perspective is the idea that emotion knowledge should develop rapidly early in life, with the ability to express the facial configurations of basic emotions like anger, sadness, and fear early in infancy, potentially at birth. This view also predicts that emotion knowledge develops on a similar trajectory across all individuals, relatively independent of context, culture, or experience.

However, the developmental literature is inconsistent with these assumptions. In contrast to emotions being early-emerging and unchanging over much of development, research has revealed a protracted trajectory for the development of emotional expression and perception, with emotion understanding accumulating incrementally over the course of infancy, early childhood, and even into adulthood and old age (Woodard et al., 2022). Broadly, this literature has suggested that early emotion understanding is very general, with both the expression and perception of emotions mapping roughly onto global positive and negative affect, and then gradually becoming more specific over time, with the ability to express and differentiate between more discrete emotions like anger, sadness, and fear (Widen, 2013; Widen & Russell, 2008, 2010).

Further, many researchers have reported significant variation in emotion perception and expression based on context and experience (Barrett et al., 2011; LoBue & Adolph, 2019). For instance, a particular facial configuration can be interpreted as disgusted in one context, such as when presented on a person who is eating, but can be interpreted as anger in another, such as when presented on a man making a fist (Aviezer et al., 2008). Additionally, not all emotional facial configurations are "universal" in how they are interpreted. Research has shown that although emotions like happiness or sadness are cross-culturally recognized, other emotions—like achievement and relief—are only accurately recognized within specific cultural groups (Sauter et al., 2010). In fact, different cultures tend to focus on different cues for emotion recognition. Adults from Western countries, for example, rely heavily on facial configurations and verbal information, while adults from Japan prioritize contextual cues, and for example, are more likely to perceive facial configurations as happier when surrounded by other faces posing positive configurations (Ishii et al., 2003; Masuda et al., 2008; Tanaka et al., 2010).

Beyond variations in emotion perception, research has also highlighted cultural differences in emotional *expression*. For example, whereas adults from Western countries tend to express certain emotions through very distinct facial muscle configurations, individuals from East Asia show considerable overlap between categories of emotional facial configurations, especially for emotions like surprise, fear, disgust, and anger (Jack et al., 2014). Language also plays a crucial role in emotional expression, and the frequency with which people mention emotion-related words in response to different stimuli varies widely across cultures, even within Europe (Scherer et al., 1986). Furthermore, bilingual individuals express emotions differently depending on the language with which a stimulus is presented (Chen et al., 2012).

Differences in emotional expression between individuals have also been linked to societal norms. For instance, U.S. adults from low socio-economic backgrounds tend to express more anger than do individuals from higher socio-economic backgrounds, whereas the opposite pattern has been observed in Japan. This difference aligns with each culture's values: In Western cultures, independence and personal goal achievement are highly emphasized, and anger is often a response to blocked goals, particularly in lower socio-economic groups. In contrast, Japanese culture places a strong emphasis on social harmony and interdependence, discouraging anger in lower and middle socio-economic groups where it might disrupt group cohesion (Park et al., 2013).

As a result of these studies and many more showcasing the role of culture, context, language, and experience on emotion understanding, various alternative perspectives to the basic emotions approach have been proposed. We collectively call these views *process-based perspectives*. Although there are a variety of approaches that fit under this umbrella (including dimensional models, constructivist approaches, and emergent theories), what they all have in common is the idea that emotions are learned through experience and the process of appraisal, highlighting the role of cognition as fundamental to both emotion perception and expression (Barrett, 2006; Clore & Ortony, 2008; Coan, 2010; Cunningham et al., 2013; Lazarus, 1984; Zajonc, 1980). In support of this view, one recent study demonstrated that when researchers gave over 300 people artificial intelligence (AI) tools to generate facial configurations that represent discrete emotions like happiness, fear, sadness, and anger, no two faces looked alike (Binetti et al., 2022).

Here, we will review the developmental literature on how infants and young children learn about emotions. We take a process-based perspective, highlighting how the protracted trajectory of emotional development unfolds concurrently with changes in children's cognitive abilities, and how variability based on context, culture, and experience shape this trajectory over time. We will also emphasize the role of *input* into this development, a factor that has often been ignored. Indeed, little research has systematically sought to understand how children's emotional development is shaped by the emotional input they receive. This notion is central to a processed-based perspective.

We focus on the development of three abilities that comprise emotion understanding. First, we will discuss the developmental trajectory of *emotional expression*, and how infants and young children come to express emotions and behave emotionally. Second, we will discuss how infants and young children learn to *regulate* those emotional responses, and how different environmental inputs affect that ability. Third, we will discuss the development of *emotion perception*, or how infants and young children learn how to perceive, categorize, and identify the emotions of others over time. Finally, we will discuss mechanisms for developmental change in all of these domains, and how other concurrent domains of development—including motor skill, language, theory of mind, and attachment—might have cascading effects on changes in emotion understanding.

Altogether, taking a developmental perspective on how we learn to express, regulate, and perceive the emotions of others has implications for emotion theory, and the degree to which emotions are universally shared across individuals versus shaped by context and experience. Further, the work reviewed here has practical implications for how emotional input might shape emotion understanding in both adaptive and maladaptive ways. Thus, we will conclude by discussing how future research focusing on how variability in the environment—or in the emotional input that infants and young children receive—shapes our emotional life for better or for worse.

1. Emotional expression

One way that children begin to develop emotion understanding is by experiencing emotions themselves and expressing them. There is some debate, consistent with the various theories of emotion discussed in the previous section, about whether newborn infants experience and express a vast array of emotions. According to the discrete emotions perspective, emotional expressions should be present at birth, and early in life emotional expressions should be identical to adult emotional expressions (Izard et al., 1987). Indeed, the very first studies in this area suggested that by 1 or 2 months of age U.S. infants can express facial configurations that represent fear, disgust, anger, sadness, happiness, and surprise (Izard et al., 1980). However, these findings were met with criticism, because, as often has been observed with newborns, spontaneous emotional facial configurations in very young infants were only produced very quickly and at random times. In other words, these early facial configurations were fleeting and not produced in response to an appropriate eliciting event, suggesting that they may not actually correspond to any underlying emotional state at all (Camras & Shutter, 2010). Further, in US samples, 4-month-old infants exhibit a combination of interest and surprise when confronted with tickling, tasting a sour substance, arm restraint, and the approach of a masked stranger. It is not until 12 months of age that US infants express clearly differentiated facial configurations of happiness, disgust, anger, and fear (respectively) in response to these events (Bennett et al., 2005).

In contrast, process-based views predict a more protracted developmental trajectory for the expression of specific emotions, with emotional expression co-developing with cognitive abilities. For example, happiness is a response to something rewarding and should only occur after infants have associated particular people or objects with a specific reward. Likewise, sadness is a response to the loss of a rewarding person or object and thus emerges with the ability to remember a rewarding person or object that is absent. Anger is a response to a blocked goal, requiring the ability to perform actions based on goals. Finally, fear is a response to a perceived threat, dependent on the ability to perceive a stimulus as a threat. Thus, according to a process-based perspective, infants would first have to experience or learn about rewarding events (for happiness and sadness), engage in goal directed behavior (for anger), and recognize that a person or object is threatening (for fear) to experience and express each of these specific emotions.

In fact, infants do not express discrete happiness, or smiling, in response to a familiar person until 1 to 2 months of age, when they have developed the ability to recognize a familiar person. All prior smiling, or the neonatal smile, is thought to be a reflex, and is typically elicited during sleep, suggesting that even happiness is not reliably expressed without some cognitive prerequisite. Moreover, the emergence of the social smile, in contrast to being universal, appears to be experience dependent. Wörmann and colleagues, for example, demonstrated that the emergence of the social smile varies across cultures, with earlier emerging social smiles in cultures that practice more distal parenting characterized by face-to-face interactions (Kärtner et al., 2013; Wörmann et al., 2012, 2014).

Research has shown that the expression of other emotions emerges even later. For example, not until around 4 months of age do US infants exhibit discrete facial configurations of anger in response to an appropriate eliciting event—in most cases, in response to interrupting a learned contingency. In one classic study, researchers taught U.S. infants that whenever they pull on a string, an image of a smiling baby would appear on a screen. Infants were unable to learn this contingency until approximately 4 months. At that age, but not earlier, infants also showed evidence of a stereotypical angry facial configuration when the contingency was later interrupted (Lewis et al., 1990). This suggests that the ability to learn a contingency and then recognize when that contingency has been interrupted are cognitive prerequisites for the elicitation of discrete anger.

Similar findings have been reported on the development of the expression of fear. In the standard stranger approach paradigm, a stranger (typically a male adult) approaches an infant in one of many contexts, for example, while seated on the floor, in a highchair, or on their mothers' laps. Infants' behaviors—including facial configurations, crying, and escape behaviors—are measured. Importantly, although US 6-month-old infants can show that they differentiate between the faces of their mothers and strangers by looking longer at strangers, they do not respond differentially to the *approach* of a stranger until several months later (Bronson, 1972; Lewis et al., 1978; Lewis & Rosenblum, 1974; Sroufe, 1977). Thus, perceiving a face or individual as unfamiliar is not sufficient to elicit a fear response; infants do not show fear until they perceive the face of a stranger as a threat—an assessment that requires infants not only to recognize when someone is novel versus familiar, but decide that the approach of the novel person is threatening or undesirable. Moreover, even when infants show fear in response to a stranger, typically between 8 and 12 months for US infants, classic studies have shown that the expression of fear in infants varies by context. US infants exhibit almost no evidence of fear when tested at home or when seated on a mother's lap, but they exhibit more negative emotional responses in the absence of their mothers, when tested in the lab, or when a stranger approaches rapidly or attempts to pick the infant up (Bronson, 1972; Ricciuti, 1974; Smith, 1974; Sroufe, 1977). This suggests that fearful responses towards strangers do not develop until about 8–12 months of age, and that context predicts these early responses.

The expression of more complex emotions also develops alongside cognitive abilities. The self-conscious emotions like embarrassment, envy, and empathy—which all require infants to evaluate their behaviors with reference to others' expectations or societal norms—do not appear until about 18 months of age, around the same time children begin to engage in self-referencing behavior; pride, shame, and guilt appear even later, around 36 months (Lewis, 2019). Further, once infants are old enough to reason about others' expectations, they might experience a more complex emotional response to an event that months earlier would have elicited a different response altogether. For example, researchers have reported that when 2-year-old US children experience failure, they are most likely to express sadness, whereas the very same failure in 4-year-old children is more likely to result in an expression of guilt (Lewis, 2019).

The trajectory for the development of disgust is even longer, despite assumptions that it is one of our most basic emotions and should be present at birth. Newborn infants are certainly capable of responding negatively to things that taste bad, but emotion researchers often call this *distaste*, which is a physiological response that requires gustatory contact (Rottman, 2014). Disgust may cause a physiological response, such as nausea, but requires no actual gustatory contact. In other words, in order to experience distaste, you have to taste something; disgust is more complex, and it does not require actually tasting anything. Indeed, disgust can be evoked not just by food, but also by blood, insects, or even by moral transgressions. Accordingly, while distaste is present at birth, disgust has a much more protracted period of development. For example, one classic study showed that U.S. children do not consistently reject a glass of juice that has been stirred with a used comb or that has a dead grasshopper visibly floating inside until about 6 to 8 years of age (Rozin et al., 1985). Although insects are not commonly eaten in the U.S. or Europe, grasshoppers, ants, and beetles are commonly consumed in other parts of the world, such as in Mexico, Thailand, China, and parts of Africa and Australia. This suggests that disgust responses develop slowly, likely based on experience and cultural norms.

More recent work has highlighted that even within these emotion categories, there is also significant variation in emotional facial configurations across the lifespan. For example, in one study researchers coded 7- to 9-year-old children's facial configurations while they were having conversations with their mothers. Children consistently produced configurations of happiness during joyful conversations, but they produced anger, fear, and sadness more inconsistently; those configurations were not necessarily produced in conversations when they were discussing events that should have produced these emotions (Castro et al., 2018).

Furthermore, even when children do produce discrete emotional facial configurations in response to appropriate elicitors, these configurations do not necessarily look like the stereotypical configurations we often picture (and use in our research). According to the discrete emotions perspective, basic emotions should result in very specific facial configurations, such as smiling for happiness and scowling for anger. In fact, Ekman and colleagues have developed a system of coding emotional facial configurations called the Facial Affective Coding System (FACS) to reliably identify when a specific emotion is present via facial configuration. However, research with both adults (Durán & Fernández-Dols, 2021) and children (Camras et al., 2007a; Camras & Shutter, 2010; Castro et al., 2018) suggests that we express emotions with considerably more variability than the posed stereotyped configurations represent (Barrett et al., 2019; Durán & Fernández-Dols, 2021). For example, one study reported that US children between the ages of 3 and 6 rarely express the full stereotypical facial configurations for any emotion except happiness. When they express other emotions, some of the facial actions (e.g., shape of mouth, position of eyebrows) match the stereotypical configurations for those emotions, but rarely do all of the children's facial actions match the stereotypical configuration (Gaspar & Esteves, 2012).

The research discussed which has primarily been conducted with US children provides evidence that emotional expression develops slowly and reflects contextual factors. Comparison of developmental trajectories across cultures provides even stronger evidence of this conclusion. First, there is significant cultural variation in emotional expression across the lifespan. For instance, although Japanese and American adults tend to report feeling happiness, sadness, and anger, their experience of high arousal emotions (e.g., anger, fear, happiness) and low arousal emotions (e.g., sadness, calm) differ. In Western or individualist cultures, high-arousal emotions are more valued, and adults thus report experiencing more high-arousal than low-arousal emotions. In contrast, in Eastern or collectivist cultures where low-arousal emotions are more valued, adults report experiencing more low-arousal emotions than high-arousal emotions (Lim, 2016).

Cross-cultural differences in emotional expression emerge even in infancy. As described earlier, the emergence of the social smile varies between cultures that differ in their use of distal parenting, which includes more face-to-face interactions (Kärtner et al., 2013; Wörmann et al., 2012, 2014). Infants across cultures also differ in the *intensity* of their emotional facial configurations. For example, one study showed that 12-month-old infants in the US were more intense in their expressions of pleasure and discomfort than were infants in Chile (Muzard et al., 2017). In another study of US, Japanese, and Chinese 11-month-old infants' responses to various emotion-eliciting activities in the lab, Chinese infants were less expressive, both smiling and crying less than US and Japanese infants (Camras et al., 1998). Likewise, Gartstein and colleagues reported that at 12 months, Spanish female infants showed more intense fear responses than did US infants (Gartstein et al., 2016).

Moreover, the development of these emotional expressions depends on experience, and there is evidence that there is more cross-cultural variability in emotions that requires some understanding of cultural expectations. In a large cross-cultural study of adults in China, India, Korea, Japan, and the United States, participants read brief stories that represented 22 emotions, and their emotional responses were coded. Interestingly, there was cross-cultural overlap (between 67–89 percent) for simpler emotions like surprise, fear, and anger, but very little correspondence (18–44 percent) for emotions like disgust and shame that have a longer developmental trajectory (Cordaro et al., 2018). Altogether, this work suggests that the developmental trajectory of emotional expression is closely tied to concurrent developments in cognition, with emotions that involve simpler cognitive processes, like happiness, developing sooner, and more complicated emotions like disgust and the self-conscious emotions developing later once children have some understanding of social norms.

2. Emotion regulation

Emotion regulation—or the ability to control one's emotions—similarly, and not surprisingly, has a protracted developmental trajectory. At first, young infants are not able to regulate their emotional responses on their own and typically require caregivers to regulate their emotions for them. Techniques like rocking, swaddling, and most importantly touch, have been shown to be helpful in regulating infant negative affect, especially in newborns. For example, one group of researchers randomly assigned full term newborn infants in the Netherlands to receive either daily skin-to-skin contact or care-as-usual (which did not involve skin-to-skin contact) for 5 weeks. Infants who received the skin-to-skin contact cried less and slept more than infants in the control group. Further, there was a dose response effect, with more skin-to-skin contact leading to greater reductions in crying and longer total sleep duration (Cooijmans et al., 2022).

Further, skin-to-skin contact can have long-lasting effects on emotion regulation abilities, even into adulthood. For example, in another intervention study, a group of premature infants in Israel were assigned to either 2 weeks of skin-to-skin contact or standard incubator care. Infants who received the skin-to-skin contact demonstrated better emotion regulation 6 months later, and impressively, 10 years later. In fact, another study went on to report that US adults who were held and cuddled as infants were the most likely to be the most emotionally healthy and well-adjusted adults (Narvaez et al., 2016).

By 4 to 6 months of age, US infants begin to soothe themselves via touch, particularly at night, by stroking their own skin or petting a toy or stuffie. As they get older, these strategies become less physical and more cognitive, especially as children learn what emotions are acceptable and not acceptable to express. Importantly, there are different expectations about emotional expressivity within families, neighborhoods, and cultures, which emotion scientists have called *display rules*. As children develop, they learn the display rules specific to their environments and are tasked with regulating their emotions accordingly.

In a study of over 5000 people from 32 countries, researchers reported some consistency in display rules along with several cross-cultural differences. Across all the countries they tested, researchers reported that individuals thought it was more acceptable to be emotionally expressive with members of one's ingroup than one's outgroup. Further, certain negative emotions like contempt, disgust, and fear were the least acceptable emotions to express for all groups, and the expression of sadness was consistently associated with more acceptance in ingroups versus outgroups. However, there was cross-cultural variability as well. Western individualistic cultures accepted more expressiveness overall than Eastern collectivist cultures, especially for positive emotions. Individualist cultures also had less variability in their regulation across emotions overall, likely because emotional expressiveness is generally more acceptable in these cultures, requiring less regulation.

Cross-cultural variation in the ability to regulate emotions is also evident in childhood. Conclusions about the development of emotion regulation are typically drawn from studies using Walter Michel's classic marshmallow task (or some variation) to test a related ability—delay of gratification. In the task, preschool-aged children are offered a marshmallow, or some equally appealing snack, but then are told that the experimenter needs to leave the room. The child is told that if they wait until the experimenter comes back, the child would be given two marshmallows to eat instead of just one. However, if the child is unable to wait, they can ring a bell to bid the experimenter's return and eat the single marshmallow instead of waiting for two. In the original study conducted in the US, children typically had difficulty waiting, particularly the youngest children (Mischel & Ebbesen, 1970).

Cross-cultural studies have suggested that there are cultural differences in children's ability to wait in this task, suggesting cultural differences in delay of gratification. In one study, researchers compared the behavior of 4-year-old children in Germany—a Western industrialized country—to Nso children, who are from an indigenous tribe in the Northwestern region of Cameroon. Like the US children tested in the original study, German children had difficulty waiting to eat the marshmallow (only 30 percent waited the entire time), and needed to use strategies to distract themselves, like sitting on their hands, talking to themselves, or looking away from the marshmallow. In contrast, the Nso children had little difficulty waiting for the second reward, with 70 percent waiting until the end. The researchers speculated that because children in the Nso culture are expected to control their negative emotions at a very early age, even young children were better able to regulate their emotional responses when faced with a challenge than German or US children who are more likely to be encouraged to express their negative emotions (Lamm et al., 2018). This suggests that children can regulate their emotions at a much earlier age in countries where the outward display of negative emotions is discouraged.

Other work suggests that delay of gratification may develop within cultural practices in which children are required to wait, even within cultures where outward displays of negative emotion are discouraged. In one study, preschool-aged children were tested in Japan and the US either on the standard marshmallow task or on a second similar task that involved waiting to open a wrapped gift. In both cases, children could wait for two treats (two marshmallows or two gifts), or if they could ring the bell and have just one. Instead of global differences in the ability to wait, children from the two cultures could better regulate their emotions—and delay gratification—in different tasks. The US children had difficulty waiting in the marshmallow version of the task, but the Japanese children did not. However, the results were reversed for the wrapped gift—in this task, the Japanese children had difficulty waiting to open the gift, but the US children did not (Yanaoka et al., 2022). Importantly, these results reflect the norms of each country: In Japan, children are often expected to wait to eat lunch, for example, until everyone in their classroom is served—a norm that is not common in the US. In contrast, children in the US have experience waiting to open wrapped gifts, for example, by waiting until Christmas day to open presents that are visible under the Christmas tree or waiting until the end of a party to open a pile of wrapped presents on display. Children in Japan, in contrast, are often given gifts sporadically throughout the year instead of having specific events, and open those presents immediately. Altogether, this body of work suggests that like emotional expressiveness, emotion regulation is closely tied to concurrent developments in cognition, with significant cross-cultural variability especially with respect to culturally specific display rules.

3. Emotion perception and labeling

Finally, children's ability to perceive, understand, and talk about the emotions of others develops over time as well. Research examining US infants' perception of photographs of posed stereotypical emotional facial configurations suggest they have precocious abilities to discriminate such images, consistent with the discrete emotions perspective. Studies have provided evidence that even in the first few months of life infants discriminate between photographs of smiling (happy), frowning (sad), and wide-eyed (surprise) stereotypical facial configurations (Farroni et al., 2007; Field et al., 1983; Young-Browne et al., 1977). Further, around 5–6 months of age, they differentiate between wide-eyed gasping poses (fear), frowning poses (sad), and scowling poses (angry) (Schwartz et al., 1985; Serrano et al., 1992). By 6–7 months, infants group different models posing the same emotion category (C. A. Nelson et al., 1979) and detect category boundaries when images were morphed from one stereotypical configuration to another (Kotsoni et al., 2001). These findings suggest that infants' ability to differentiate between a wide variety of emotion categories develops rapidly in the first year of life.

However, other studies have suggested a more protracted developmental trajectory for emotion perception. In one study, US 3- to 6-year-old children and adults were asked to sort images of emotional facial configurations according to whether they were related. The faces differed in valence, with some positive and some negative, and they represented several distinct emotions (e.g., happiness, fear). In general, across age, children did not become more competent with this task, but rather younger and older children used different information to sort the faces. The younger children relied on valence, whereas it was not until 5 years of age that children consistently sorted the images based on discrete emotion categories (Woodard et al., 2022). Results like these suggest that although infants are sensitive to perceptual differences between facial configurations, children's understanding of the emotional expressions of others continues to develop into childhood.

Furthermore, although infants can discriminate between faces with different emotional facial configurations, it is impossible to know from most studies whether infants are simply discriminating between the perceptual features of various emotional facial configurations or whether they can discriminate between the emotional meaning that these configurations represent. It is not until infants can use emotional facial configurations to guide their action that we can know for sure that they are interpreting meaning from these configurations. The first evidence of such a process is the emergence of *social referencing* late in the first post-natal year in US infants (Sorce et al., 1985). When infants experience an ambiguous situation, they look to a caregiver or trusted adults' facial configuration and modify their behavior based on the configuration they see.

For example, at 12 months of age, infants look to their mothers' facial configurations to guide behavior when presented with an ambiguous visual drop-off (Sorce et al., 1985) or slope (Tamis-Lemonda et al., 2008). Infants are much more likely to traverse across the drop-off or down the slope if their mothers have a positive emotional facial configuration than if they have a negative configuration, for example conveying fear. Infants behave similarly when presented with novel objects: US 12-month-old infants avoid touching novel objects that are paired with a fearful facial configuration compared to novel objects paired with a happy or neutral configuration (Mumme & Fernald, 2003). Similarly, after mothers pose either a positive or negative facial configuration towards one of two novel toys, toddlers are more likely to avoid touching the toys that are paired with negative configurations than toys that are paired with positive configurations (Gerull & Rapee, 2002). Such results suggest that late in infancy, US children have begun to associate facial configurations with different meaning, and use those differences to determine how to behave, particularly when faced with novel or ambiguous situations.

As further evidence of developing emotion perception, between the ages of 2 and 3 children begin to produce the expected labels for various stereotyped facial configurations, first by correctly distinguishing smiling (positive) versus scowling or frowning (negative) poses, then later, by correctly producing the labels for various emotion categories (Widen, 2013; Widen & Russell, 2008). Importantly, in most of these studies the stimulus sets of emotional facial configurations are high in emotional intensity and represent only the most stereotypical of emotional facial configurations. Thus, the results may not generalize to the more varied and subtle facial configurations children encounter in their everyday lives. A handful of studies using facial configurations of more varied intensities suggest a very protracted development in children's ability to accurately identify and label low intensity configurations. Indeed, although children between the ages of 7 and 10 are highly accurate at identifying high intensity emotional facial configurations, it is not until much later that children accurately identify lower intensity configurations. Moreover, this developmental trajectory differs for different categories of emotion (e.g., happy versus disgusted). In general, this research suggests that the perception and labeling of emotional facial configurations does not reach maturity until adulthood (Gao & Maurer, 2010; Thomas et al., 2007).

4. Mechanisms and cascades

Thus far, we have discussed the trajectory for the development of emotion understanding, which includes the development of emotional behavior, emotion regulation, and emotion perception, over the first several years of life. But what are the mechanisms by which infants acquire emotion knowledge in these domains? Because much of the literature is based on the discrete emotions perspective, there is remarkably little research on identifying the mechanisms by which emotional understanding develops. That is, there is a large literature documenting when children of different ages perceive, identify, and understand something about different facial configurations, as well as how children express and regulate their own emotions, but there is less work attempting to understand how these abilities develop.

We can think about the development of emotion understanding by considering emotions as part of a system, with development reflecting cascading interactions across multiple domains. As part of this approach, it is important to recognize that emotion understanding is not just one thing. As described above, emotion understanding involves *emotional expression* and the *regulation* of that expression, often requiring the internalization of cultural norms. Emotion understanding also involves *perceiving* and *recognizing* the emotions of others, typically relying on emotional facial configurations, but there are many cues to a person's emotional state. Finally, emotion understanding involves verbalizing one's own and other's emotional state. Thus, it is too simplistic to think about the development of emotion understanding as one thing; instead, as described here, there are various developmental trajectories for different aspects of emotion understanding, and those trajectories differ for different emotions.

Furthermore, the development of emotion understanding does not happen in isolation but rather changes with development in other systems. As described here, emotion understanding is closely tied to developmental changes in cognitive processes, social interactions, and learning of cultural norms and expectations. It is impossible to understand the development of emotion understanding without recognizing how changes in the aspects of emotion reflect changes in other domains.

4.1 The role of input

There are several mechanisms by which infants and young children acquire emotion knowledge, from general purpose mechanisms like statistical learning, to the transmission of facial and verbal emotional information by observation or interacting with others. All these mechanisms involve the child processing and evaluating their experiences and input, however little is known about the kind of emotion input children actually receive. Focusing on the role of input has long been important in some areas of development, for example in how children acquire language (Hoff, 2003; Huttenlocher et al., 1991; Rowe & Snow, 2020) or how they form attachments to their caregivers (Bowlby, 1979). Increasingly, researchers have begun to consider how the nature of the input shapes other aspects of development, such as infants' emerging understanding of objects and object categories (Smith et al., 2018).

Input comes from interactions with others and what others model. Infants presumably learn about emotional expressions by observing the emotional expressions of others. This is evidenced by the fact that cultural differences in emotional expression are observed even in infancy (Camras et al., 2007b, 1998). Similarly, children receive input through *social referencing* or looking to others for emotion information about something or someone that is otherwise ambiguous. As mentioned above, by 12–18 months of age, US infants use the emotional facial configurations of others to guide their behavior, suggesting not only that infants recognize those facial configurations as emotion related, but also that they use them to learn about the relevant emotions in ambiguous situations.

Experimental studies have shown that exposure to specific emotional facial configurations leads to specific kinds of emotion learning. For example, children between the ages of 7 and 10 years are slower to approach novel animals that were previously paired with a fearful versus happy facial configuration. Most importantly, children also report a higher rate of fear about the animals that were paired with a fearful facial configuration when compared to those paired with a happy configuration (Askew & Field, 2007; Broeren et al., 2011). Further work has shown that these effects—namely increased fear—can last up to 6 months after initial exposure (Field et al., 2008; Muris et al., 2003), suggesting that information gleaned by social referencing can not only influence infants' behavior in the moment, but can also shape children's longer-term emotional responses (Dubi et al., 2008; Gerull & Rapee, 2002).

Children also learn from the emotional language they hear. For example, 6- to 9-year-old children show increased heart rate and slower approach responses to novel animals after being presented with negative versus positive or neutral verbal information (Field & Lawson, 2003; Field & Schorah, 2007). Likewise, US 5-year-old children report more fear of a stranger after hearing parents provide negative verbal information about the stranger (Aktar et al., 2022). At younger ages, exposure to labels can aid infants in forming categories of emotions (Price et al., 2022; Ruba et al., 2020), just as labels aid infants in forming categories of spatial relations (Casasola, 2005).

Children can also build their emotion knowledge through parent-child conversations. For example, mothers' use of language about desire predicts their 15-month-old toddlers emotion understanding when the children were 24 months (Taumoepeau & Ruffman, 2006, 2008). Similarly, in a study where US children and parents discussed a wordless storybook with images of actors experiencing various emotions, parents' questions and emotion talk were related to children's emotion talk over time (Reschke et al., 2024). This suggests that emotion language can serve as input to help children understand their own emotional states and how to express their emotions in culturally appropriate ways.

Children also develop emotion understanding from detecting statistical regularities in the facial configurations they observe. For example, children may detect statistical regularities in the input and use statistical learning to identify emotion boundaries. A large number of studies have shown that infants have sophisticated statistical learning abilities to identify and categorize sounds in speech (Saffran et al., 1996) and objects in visual stimuli (Needham et al., 2005). It is therefore plausible that infants also detect statistical regularities in facial configurations and use those regularities to identify emotion boundaries.

In fact, one study has shown that US 6- to 8-year-old children and adults categorize emotional facial configurations using statistical regularities. In this study, after learning how to categorize morphed faces as angry or neutral (in reality each face was created by morphing an image of an angry face and an image of a neutral face), participants shifted their category boundaries when they experienced a change in the distribution of angry and neutral faces (Plate et al., 2019). This finding demonstrates that at least by 6 years of age, children are sensitive to the distribution of facial configuration features in their experience, and their emotion boundaries reflect that distribution.

Much less is known about whether *infants* use statistical learning to recognize emotion boundaries. One problem is that despite a few studies examining infants' visual input for the presence of faces (Jayaraman et al., 2015, 2017; Sugden et al., 2014; Sugden & Moulson, 2017), we know very little about what facial configurations infants actually see. However, research with clinical populations suggests that infants do learn the statistical regularities in their input. For example, infants of depressed or anxious mothers show different perception of and reactions to emotional facial configurations in the lab. Depressed mothers express more negative affect than do non-depressed mothers (Murray et al., 1993), even in face-to-face interactions with their infants (Aktar et al., 2017; Feldman et al., 2010). Accordingly, infants of depressed mothers—who presumably had more negative emotion input than infants of non-depressed mothers—show less interest in negative facial configurations than infants of nondepressed mothers (Cohn et al., 1986; Field, 1992), and they take longer to disengage or look away from stereotypically happy configurations (Diego et al., 2004; Field et al., 1998). This result may reflect the same processes shown by Plate et al. for older children; these infants may have a shifted emotion boundary reflecting the distribution of negative and positive emotion input in their experience. As a result, positive facial configurations may be more novel to infants of depressed mothers, and these differences may reflect infants of depressed mothers showing a novelty response to positive configurations and a habituation response to negative configurations.

Similar patterns have been observed with infants of anxious mothers. Research suggests that anxious mothers are less sensitive to their infants' needs, less emotionally responsive, and more interfering when compared to non-anxious mothers (Kaitz & Maytal, 2005). Infants of anxious mothers have difficulty disengaging or looking away from negative facial configurations, especially angry or threatening stereotypes, when compared to infants of non-anxious mothers (Morales et al., 2017). Further, longitudinal work has shown that infants of anxious mothers engage less and become slower to detect emotional facial configurations overall over the course of the first two years of life (Vallorani et al., 2024). Similar to infants of depressed mothers, infants of anxious mothers may have formed different emotion boundaries reflecting the distribution of the positive and negative facial configurations in their experience.

4.2 Cascades across domains

As described earlier, emotion understanding does not develop in isolation. Instead, developments in other domains have cascading effects on emotion understanding over time, exposing children to new kinds of input and experiences. Consider, for example, developmental changes in infancy in basic cognitive abilities such as perception, attention, and memory. Changes in each of these abilities affect the developing ability to reason about emotions (Ruba & Pollak, 2020). For example, at birth, infants have extremely poor visual acuity and immature visual attention and perception. Nonetheless, some studies suggest that newborn infants distinguish between faces with different emotional facial configurations (Farroni et al., 2007; Field et al., 1983; Young-Browne et al., 1977). Given their limited visual abilities, any such discrimination between different emotional facial configurations cannot be due to attention to, perception of, and recognition of the details of the facial configurations associated with those emotional expressions. However, the discriminations young infants make based on their limited visual experience may be a critically important step in the development of emotion understanding. Indeed, research with individuals born with dense central cataracts, and therefore are essentially blind from birth, suggests that early visual experience is critically important for the development of typical face perception (Le Grand et al., 2001). This early visual experience may also be important in the development of visual perception of emotions.

This is just one example of how developmental changes in basic cognitive abilities might have a cascading effect on the development of emotional understanding. As infants' visual perception abilities develop, they can actually detect the features of facial configurations and differentiate them from one another, allowing categorization based on more sophisticated and subtle features and configurations. Infants must learn to attend to the most relevant features and combine information stored in memory from different encounters. Thus, the development of emotion understanding is closely linked to the development of core cognitive abilities.

Second, language development contributes to emotional development. For example, the ability to perceive and learn emotion labels may help young children recognize emotion boundaries. Research with US infants suggests that at the end of the first year and early in the second year of life, the presence of labels facilitates their attention to other kinds of categories (e.g., horses or purple things) (Forbes & Plunkett, 2019; Fulkerson & Waxman, 2007; Waxman & Booth, 2003). Thus, it is possible that young children's learning of emotion categories is influenced by their developing ability to recognize labels as markers of abstract categories (Hoemann et al., 2020). By the age of 2 or 3, US children do use emotion labels in this way. Studies have demonstrated that toddlers in the second year can associate some emotion labels with visual stimuli (Ruba et al., 2019) and at least by age 2 or 3, being given labels seems to help children categorize some emotional expressions (Price et al., 2022) Thus, young children's developing ability to use labels to organize visual stimuli into categories likely influences how they categorize facial configurations into emotion categories.

Further, as described earlier, language is one way parents might transmit emotion knowledge to their children. An analysis of parent-child interactions in the CHILDES database (for dyads speaking North American English or British English) showed that parents used emotionally valenced words in the utterances surrounding emotion words (Nencheva et al., 2023). For example, when using words with their child such as "happy" or "sad," parents' other utterances near those words were positive (for happy) or negative (for sad).

Motor development may also contribute to changes in emotion understanding. For example, before infants can sit independently, they likely spend a great deal of time laying on their backs, mostly looking at other people's faces. Indeed, there is significant exposure to facial information early in development, but this exposure gradually declines as infants begin to sit on their own and shift their focus to handling objects (Fausey et al., 2016; Sugden et al., 2014; Sugden & Moulson, 2019). The relatively high exposure to face perception early in infancy may similarly be important for infants' developing perception, recognition, and understanding of facial configurations. That is, early motor abilities (or general lack thereof) may result in more face exposure, providing infants with critical visual experience they need for not only developing face perception but also developing emotion understanding.

As infants get older and develop the abilities to sit, crawl, and walk, there are fewer faces in their field of view (Franchak et al., 2018) and they must shift to other sources of input (e.g., language, prosody, context) to learn about emotions. Furthermore, it has been suggested that once infants are able to locomote, either by crawling or walking independently, this independence might cause increased bouts of negative affect from caregivers, as infants can, for the first time, defy the rules of the house (Campos et al., 2000). Altogether, this suggests that changes in motor ability could readily cascade into important changes in the content and type of emotional input that infants receive.

Development of children's theory of mind, or the ability to reason about the mental states of others, between the ages of 2.5 and 5 years of age, is another system that contributes to emotional development. As mentioned above, the self-conscious emotions like embarrassment, envy, and empathy all require children to evaluate their behaviors with respect to others' expectations. Not surprisingly, there is research showing a causal link between experiencing the self-conscious emotions and training in theory of mind. In one study, US elementary school children were given training in acting, which requires children to put themselves in the mindset of another person. Children who were given the acting training showed enhanced theory of mind and empathy one year later when compared to a control group that received visual arts or music training (Goldstein & Winner, 2012). Thus, advances in children's ability to reason about the minds of others can cascade into changes in children's ability to experience self-conscious emotions, and to more accurately reason about the emotions of others.

Finally, attachment quality, or how secure an infant feels in the presence of their primary caregivers, can have cascading effects on emotion perception and regulation. Using event related potentials (ERP), researchers have shown that although children in institutionalized care have greater neural responses to fearful emotional faces than children in a non-institutionalized comparison group, this difference disappears when children establish a secure attachment relationship with a foster parent (Nelson et al., 2013; Young et al., 2017). This suggests that feeling securely attached to a caregiver may buffer infants' responses to certain negative or threatening facial configurations so that these configurations are perceived as less threatening.

5. Conclusions and future directions

Here, we reviewed the developmental literature on how infants and young children learn about emotions based on a process-based perspective, highlighting how the protracted trajectory of emotional development unfolds concurrently with changes in children's cognitive abilities, and how variability based on context, culture, and experience shape this trajectory over time. As such, the work reviewed here moves us beyond traditional emotion perspectives by highlighting how variation in input shapes the developmental trajectory of emotion knowledge over time. This trajectory may appear abrupt at first blush, suggesting that infants learn basic emotion categories in the first year of life, but the evidence we discussed points to a much more protracted developmental trajectory, with significant change across childhood.

However, additional future research that focuses on how specific input drives emotion understanding is still needed. Recent work suggests that even brief changes in social interactions can have a lasting effect on emotion and mental state knowledge. For example, one very recent study examined false belief understanding in two cohorts of US 3- to 5-year-old children who were either tested before or after the COVID-19 lockdown in 2020. They reported that children tested after lockdown had significantly worse false-belief knowledge than children tested before (Scott et al., 2024). This suggests that there could be significant cohort effects based on global events that encourage or suppress social interaction, which can potentially in turn, have an effect on emotion understanding.

This is important, as changes in emotion understanding can shape children's long-term mental health. For example, we discussed here how infants of anxious mothers have difficulty disengaging or looking away from negative facial configurations, especially angry or threatening stereotypes, when compared to infants of non-anxious mothers (Morales et al., 2017). Difficulty disengaging from emotional stimuli has been associated with the development of anxiety in infants and children. Thus, emotional input from depressed or anxious caregivers might set the stage for the development of these same emotional problems in the infants themselves (Burris et al., 2019). This suggests that the emotional input children receive can put them at risk for various emotional problems later in life, making further research on this topic important for potentially mitigating the development of maladaptive patterns of emotion reasoning.

Finally, we made the case in this chapter for the importance of input in building emotion knowledge. We provided examples from studies of cross-cultural and research from clinical populations. However, to date, we still know very little about the *natural* emotional input that infants typically receive in the first few years of life, at a crucial time when infants are first learning to recognize and differentiate between emotion categories. As discussed above, most of the research in this domain focuses on how infants and young children perceive and label photographs of posed, stereotypical emotional facial configurations. This is problematic for two reasons. First, as mentioned above, by the time infants reach 6 months of age, they experience a sharp decline in their attention to faces in general, likely because of the shift from predominantly laying on their backs to sitting independently. Thus, the reliance only on emotional information from static posed facial configurations is not necessarily representative of infants' daily experience.

Second, in the real world, adults rarely express instances of emotion using posed stereotypical configurations (Barrett et al., 2019), so it is likely that the input infants receive is also highly variable and context-dependent (Barrett et al., 2019; Hoemann et al., 2020; LoBue & Adolph, 2019). However, scientists have very little information about what emotional expressions infants actually see, and therefore what real-world input they rely on to develop emotion understanding. So while the existing developmental literature can tell us about infants' increasing ability to detect and classify stereotypical, static facial configurations, it might mischaracterize the real-world learning problem infants face altogether. In other words, rather than learning relatively consistent facial configurations that are centered around a stereotype, infants must learn to make inferences about dynamically changing facial and vocal expressions that are fully contextually situated. This is likely a good thing, as researchers have argued that the best way to improve learning and generalization is to expose the learners to more variable input (Raviv et al., 2022). But if this is indeed the case, we still have a great deal to learn about how infants build emotion knowledge in real world contexts. These data are vital for discovering how infants infer the emotions of others, build expectations about social interactions, plan their behavior adaptively, and how specific experiences and individual differences shape emotional development (Hoemann et al., 2020). Further, they are *absolutely critical* for discovering how atypical emotional environments, such as those associated with maltreatment, anxiety, or depression, might shape maladaptive trajectories of emotional behavior. We hope this review helps draw attention to the importance of input in the development of emotion knowledge, and in turn drives future research in the field with this important focus.

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ADVANCES IN CHILD DEVELOPMENT AND BEHAVIOR

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VOLUME SIXTY NINE

ADVANCES IN CHILD DEVELOPMENT AND BEHAVIOR

CHAPTER NINE

Preparing for equitable family and community engagement through home visiting: linking culturally responsive/sustaining pedagogy and developmental frameworks

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Contents

Abstract

This manuscript examines how home visiting by educators aligns with and informs culturally responsive and sustaining pedagogy (CR-SP) through the lens of developmental theory. Drawing on a synthesis of multiple qualitative studies, we analyze evidence of home visiting's influence on four dimensions of CR-SP: teachers' beliefs, dispositions, instructional practices, and integration of family and community knowledge. We find

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the strongest alignment between home visiting and teacher beliefs, including enhanced self-reflection, cultural competence, and asset-based thinking. While evidence of impact on responsive instructional practices is emerging, findings show limited influence on teachers' sociopolitical consciousness, curriculum adaptation, or integration of family knowledge into classroom environments. To guide our analysis, we apply asset-based educational frameworks alongside developmental theories, including Bronfenbrenner's ecological systems theory, Vygotsky's sociocultural theory, and developmental systems theory. This interdisciplinary approach highlights home visiting as a potentially powerful relational and developmental bridge between families and schools, while also underscoring the need for structural supports—such as targeted professional learning, reflection tools, and policy guidance—to realize its full promise. We conclude with implications for theory, research, and practice, calling for more robust and longitudinal inquiry into how home visiting can support equity-driven, developmentally aligned teaching across varied educational contexts.

Keywords: Classroom practice, Culturally responsive pedagogy, Family engagement, Home visiting, Teacher beliefs

Fostering equitable family engagement requires schools and teachers to implement a repertoire of responsive practices (Mapp & Bergman, 2021). Assets-framed, relationship-building home visiting (also called family-centered home visiting) is one such family engagement practice. This type of home visiting practice is intended to center families' stories and assets, build relationships between educators and families, and inform asset-based teaching (Moll et al., 1992; Park & Paulick, 2021; Paulick et al., 2024; Venkateswaran et al., 2018). There is evidence that family-centered home visiting is associated with families' increased involvement with school, children's improved academic performance, and shifts in teachers' beliefs and practices (McKnight et al., 2019; Meyer et al., 2011; Sheldon & Jung, 2018; Venkateswaran et al., 2018).

Home visiting programs more broadly have a long history in the United States. Such programs have taken a variety of forms over the years, but they have primarily centered practitioners' expertise, supplementing or supplanting families' caregiving practices. For example, home visitors were sent into low-income, immigrant homes at the turn of the 20th century with the goal of "Americanizing" mothers' caretaking and homemaking (Bhavnagri & Krolikowski, 2000; Roibal, 2016). Similarly, programs since the 1960s have involved public health workers and social workers visiting homes in order to provide services and education within the families' space (National Home Visiting Resource Center). These types of home visits that center practitioners' knowledge continue to occur, with an emphasis on the program goals and provider's expertise, rather than explicitly foregrounding the cultural and linguistic assets, knowledge, and experiences of families (e.g., Grindal, 2016;

Jacobs, et al., 2016; Nievar et al., 2018; Peacock, 2013; Walsh et al., 2019). Within the same era, family-centered home visiting began. These assets-framed, relationship-building teacher home visits have aims that are different from home visits conducted by public health workers or social workers. Specifically, assets-framed ethnographic home visits are intended to inform educators about families (e.g., Moll et al., 1992); relationship-building home visits are intended to build trusting collaborations between educators and families (e.g., Venkateswaran et al., 2018). Visits that combine those two aims center families' stories and, in so doing, build rapport (Paulick et al., 2024).

Over the past 25 years, family-centered home visiting by pK-12 teachers has gradually gained traction across the United States. Despite this growing interest, research specifically focused on teachers conducting home visits remains limited, particularly in terms of how assets-framed home visiting might impact teaching practice. Qualitative studies, including ethnographies and case studies of locally developed family-centered home visiting initiatives, have employed teacher reflections, surveys, and interviews as primary data sources (Johnson, 2014; Lin & Bates, 2010; McIntyre et al., 2001; Meyer et al., 2011; Peralta-Nash, 2003; Whyte & Karabon, 2016). These studies highlight that while teachers often express apprehension about entering families' homes and may initially possess narrow views of family assets (Whyte & Karabon, 2016), both educators and families tend to view home visits as beneficial (Lin & Bates, 2010; McIntyre et al., 2001; Peralta-Nash, 2003). For example, Meyer and Mann examined relationship-building home visits conducted by 26 rural early elementary teachers. Their pre- and post-survey analysis revealed that teachers reported improved communication and stronger relationships with families along with a deeper understanding of the children and their home environments. A replication study by Meyer et al. (2011), involving 17 original and 11 additional participants, confirmed these findings 5 years later.

To build on this emerging work and guide our analysis, we turn to a set of theoretical frameworks that foreground equity, cultural responsiveness, and developmental alignment between home and school. These frameworks help us examine not only *whether* home visiting supports equitable education, but *how* it does so, and where its potential remains unrealized.

1. Theoretical frameworks

Our study is grounded in a combination of asset-based pedagogical frameworks and developmental psychology theories that, together, support a holistic understanding of how family-centered home visiting can inform and enhance equitable, culturally responsive educational practices. The primary frameworks that inform asset-based teaching include Culturally Responsive Teaching (Gay, 2010), Culturally Relevant Pedagogy (Ladson-Billings, 1995, 2014; Powell et al., 2016), and Culturally Sustaining Pedagogy (Paris, 2012). These approaches center students', families', and communities' cultural identities, values, and knowledge systems as fundamental resources for learning. They outline core beliefs (e.g., understanding one's positionality and engaging in cultural self-reflection), dispositions (e.g., high expectations and sociopolitical consciousness), and practices (e.g., student-centered instruction, responsive discourse, and family collaboration) that educators can enact to disrupt inequities and promote historically marginalized students' success. We also attend to the dimension of integration, which reflects the alignment of beliefs, dispositions, and practices in ways that support students' community-based identities, heritage languages, and critical awareness of power and oppression. Taken together, they are Culturally Relevant/Responsive/Sustaining Pedagogy (CR-SP).

To complement this educational lens, we incorporate insights from developmental psychology. Bronfenbrenner's bioecological systems theory (1979) emphasizes the importance of interconnected environments in shaping children's development. In particular, the mesosystem—the interaction between home and school—is a critical site for fostering consistent, responsive developmental support. Home visits serve as a direct mechanism for strengthening this mesosystem, allowing teachers to engage with families within their own cultural contexts and deepen mutual understanding. We also view learning and development as culturally mediated and rooted in social interactions (Vygotsky, 1978). Teachers' participation in family-centered home visits allows them to better recognize and build on the cultural practices, languages, and values that shape students' development outside of school. This awareness supports instructional practices that are not only developmentally appropriate but culturally meaningful, helping students learn within their zone of proximal development through culturally congruent support. We also draw on developmental systems theory (Lerner, 2006), which emphasizes dynamic, reciprocal relations between individuals and their ecological contexts. From this perspective, optimal development occurs when children's individual strengths and characteristics are supported by environments that reflect and respond to their cultural and social realities. Family-centered home visiting can be seen as an intentional effort to improve the developmental alignment between school and home, creating conditions where families' values and practices are recognized, respected, and integrated into classroom learning. This alignment supports not only children's social and emotional growth but also fosters cognitive and language development, as children experience consistency between home and school that reinforces meaning-making, narrative structures, and concept development in culturally congruent ways. When teachers incorporate what they learn from home visits into instruction, they are better positioned to build on students' existing knowledge and linguistic repertoires, which is especially important for multilingual learners and children from culturally diverse backgrounds.

Finally, our work is informed by Mapp and Bergman (2021) "More Liberatory Approach to Family Engagement," which identifies equitable engagement as relational, collaborative, culturally responsive, and rooted in mutual respect. Their model emphasizes that achieving such engagement requires intentional, reflective work by educators to confront bias, build trust, and truly listen to families and communities. Together, these theoretical frameworks illuminate how family-centered home visiting can serve as both a developmental and pedagogical intervention—strengthening teacher beliefs and practices, improving family-school relationships, and ultimately supporting children's academic and social development in culturally sustaining ways.

2. Literature review

2.1 Conceptual intersection between home visiting and CR-SP

Most teachers in the U.S. are from dominant cultural and social identity groups, and most families are from historically-marginalized groups (NCES, 2024). This results in power differentials between schools and homes that can lead to mistrust between families and teachers (Hong, 2020; Ishimaru, 2019). Family-centered home visits are intended to build relational trust between educators and families. Relational trust across school communities, according to Bryk and Schneider (2002), requires mutual respect, a belief in the competence of others, personal regard, and integrity. All of those elements take time, proximity, and opportunities to connect safely, and family-centered home visits provide those opportunities (Paulick et al., 2023, 2024). In a study of family-centered home visits between kindergarten and second-grade teachers and their students' families, Paulick et al. (2024) found that teachers can employ rapport-building strategies that connect them with families as fellow human beings and as competent, child-centered professionals. That rapport-building led to the kinds of substantive discussions that are the basis for trust building. Ideally, family-centered home visits would translate into what Mapp and Bergman (2021) describe as liberatory family engagement, situating families and educators as equal collaborators with different expertise and positioning that collaboration as powerful enough to dismantle systemic inequities.

Research has also shown that family-focused home visits contribute positively to teachers' attitudes toward children and families, foster more effective disciplinary and communication practices (McKnight et al., 2019), and enhance students' academic performance, attendance, and family involvement (Sheldon & Jung, 2018; Wright et al., 2018). Furthermore, family-centered home visits have been shown to significantly influence teachers' beliefs, promoting greater cultural responsiveness, reduced bias, and increased recognition of family strengths and assets (McKnight et al., 2022). By focusing intentionally on building genuine relationships, home visits have the potential to transform educators' perceptions, moving them away from deficit-oriented views and toward recognizing and valuing the strengths within families and students, thereby fostering effective instructional approaches and enhancing meaningful collaboration between schools and families (Baeder, 2010; McKnight et al., 2022). Moreover, evidence suggests that students who experienced family-centered home visits demonstrated reduced chronic absenteeism and improved student performance in ELA and math (Sheldon & Jung, 2018). Notably, the potential benefits extended beyond students and families who directly experienced home visiting, as students attending schools with at least 10 percent of families participating in home visits were both less likely to be chronically absent and more likely to reach proficiency on standardized ELA tests (Sheldon & Jung, 2018). These findings were across elementary and secondary schools, indicating that it is developmentally appropriate to enact family-centered home visits across children's and adolescents' schooling.

Family-centered home visits can inform asset-based teaching practices. Asset-based teaching requires that teachers see and understand the Funds of Knowledge (Moll et al., 1992), Community Cultural Wealth (Yosso, 2005), and cultural models (e.g., (Gallimore & Goldenberg, 2001)) that families possess. Those assets are the historical and contemporary values and skills that are passed on to children, and values and skills vary widely across families. Furthermore, when teachers are exposed only to mainstream values and skills, it may be challenging for them to see and honor the values and skills that families with marginalized identities possess. For example, mainstream

American culture places a value on competition, while many families teach the value of collaboration to their children. Paulick et al. (2022) found that home visiting teachers can use artifacts in the home to elicit families' stories, skills, and cultural models. Knowing about those assets provides essential information for enacting culturally responsive/relevant/sustaining pedagogy (Gay, 2010; Ladson-Billings, 1995, 2014; Paris, 2012; Powell et al., 2016).

While modest, the research base for the impact of family-centered home visits on educators' attitudes and beliefs is considerably more robust than the research base on the impact of these visits on educators' teaching practices. There is empirical work indicating that family-centered home visits may fall short of realizing culturally sustaining tenets (Park & Paulick, 2021) and may lead to surface-level impacts on teaching (Szech, 2022), particularly without substantial professional learning opportunities before and after the visits. Without those professional learning opportunities, power imbalances persist, teachers may be blind to families' assets, and translating assets into pedagogical practice is difficult.

Overall, the research on the relationship between family-centered home visiting and teachers' beliefs, dispositions, and practices lags considerably behind the practice of family-centered home visiting. This conceptual study asked: *How does home visiting support and enable CR-SP? Which CR-SP beliefs, dispositions, and practices are not served by family-centered home visiting? What is left to know and do at the intersection of family-centered home visiting and CR-SP?*

3. Methods and data sources

With the aim of conceptually analyzing how family-centered home visiting aligns with and advances CR-SP principles, we first reviewed existing CR-SP frameworks across foundational literature (Gay, 2010; Ladson-Billings, 1995, 2014; Paris, 2012; Powell et al., 2016). This initial examination identified four critical dimensions of CR-SP:

1. Beliefs: Teachers' beliefs involve a deep awareness and understanding of their own identities, the development of cultural competence, and a commitment to ongoing reflective practices that examine culture and identity, thereby laying the essential foundation for culturally responsive and equitable teaching.
2. Dispositions: Teachers' dispositions, characterized by maintaining high expectations for student academic success while demonstrating a steadfast commitment to nurturing students' sociopolitical consciousness, empower learners to engage critically with social issues.

3. Practices: Teachers' practices that encompass the ability to design and adapt curricula and assessments in ways that are flexible, responsive, and student-centered, ensuring equitable access to dominant cultural norms and language, while also engaging in culturally responsive discourse and actively collaborating with families to support children's learning and development.
4. Integration: Teachers' integration involves the seamless interplay between teachers' dispositions and practices, actively supporting students in preserving community and heritage languages, centering families' and communities' narratives and traditions, and critically examining practices that may reinforce oppression or hegemonic norms, ultimately cultivating an inclusive and empowering educational environment.

Together, these four dimensions—beliefs, dispositions, practices, and integration—offer a comprehensive framework for understanding culturally responsive and sustaining pedagogy (CR-SP). They serve as an analytic lens through which we examine how home visiting may influence educators' orientations and practice toward equity in diverse educational contexts

We then employed qualitative content analysis techniques to systematically identify overarching categories that capture the key findings across our studies on home visiting. Through iterative coding, categorization, and synthesis of findings, we identified six overarching themes describing how home visiting aligns with and informs CR-SP principles:

1. Differentiation: Family-centered home visiting informs differentiation of curricula, instruction, and assessment by revealing the strengths, interests, and cultural contexts of children and families.
2. Understanding: Family-centered home visiting broadens teachers' understanding of children's and families' identities, values, and lived experiences, thus supporting culturally responsive relationships.
3. Professional Development: Effective implementation of home visiting to truly center families require targeted professional training for most teachers.
4. Efficacy: Teachers' self-efficacy in culturally responsive and sustaining practices increases significantly when training is combined with hands-on home visiting experiences and structured reflection opportunities.
5. Asset Integration: Teachers effectively leverage artifacts within families' homes during visits to elicit rich narratives, histories, and cultural practices, deepening culturally responsive engagement.
6. Dialogue: Ideally, school-based conferences serve as contexts primarily for educational progress discussions, while home visits uniquely offer

space for child- and family-focused dialogue centered on cultural assets and community contexts.

These six themes illustrate the multifaceted ways in which home visiting can support and extend the principles of CR-SP, from shaping instructional approaches to deepening relational and cultural understanding. Together, they provide a nuanced foundation for examining how home visiting operates as a practice that fosters equity-oriented teaching and meaningful school–family partnerships.

Finally, to facilitate a detailed comparison between CR-SP frameworks and family-centered home visiting practices, we developed a comparative matrix. This analytic tool enabled us to explicitly map home visiting practices onto the identified CR-SP components, clearly highlighting areas of alignment and identifying existing gaps or areas requiring further development or research attention.

4. Results and implications

We found evidence of alignment between the CR-SP framework and the themes representing the evidence from our studies of home visiting (Table 1). Overall, evidence from the research literature to date indicates that family-centered home visiting is associated with teachers' beliefs to a large extent and their dispositions somewhat. There is some evidence that home visiting supports teachers' responsive discourse, but otherwise there is not yet a body of evidence demonstrating the impact of home visiting on teaching practice.

4.1 Beliefs

Our findings suggest that family-centered home visiting supports teachers' understanding of their positionality, enhances their sense of self-efficacy for engaging culturally competently with students, and facilitates an ongoing process of self-reflection. Additionally, home visiting appears to promote an asset-based mindset by encouraging teachers to recognize the strengths and cultural wealth within families and communities while also challenging biases and deficit perspectives. Our analyses further highlight that targeted professional development explicitly addressing the goals, purposes, and effective implementation of home visits appears essential. Teachers require structured preparation and ongoing training to clearly communicate home-visiting purposes to families, effectively center family perspectives, and optimize culturally responsive interactions.

Table 1 Matrix exploring alignment between CR-SP and the themes from home visiting studies.

	Findings					
	Differentiation	**Understanding**	**Professional development**	**Efficacy**	**Asset integration**	**Dialogue**
Culturally Relevant/Responsive/ Sustaining Teaching Tenets	Home visiting informs differentiation of curricula, instruction, and assessment	Home visiting broadens/nuances teachers' understandings of who children and families are and what they value	Training necessary for teachers to center families; families also need to be made aware of the purpose	Training, hands-on practice and reflection is associated with increased self-efficacy for CR-SP	Teachers can leverage artifacts in families' homes to elicit families' stories	Conferences for school-focused discussions; home visits for child-focused and family-focused discussions
BELIEFS						
Teacher understanding of their own identities and positionality			✔	✔		
Teacher's cultural competence		✔	✔	✔	✔	✔
Teacher's ongoing self-reflection		✔	✔	✔	✔	
DISPOSITIONS						
High expectations for academic success of students	✔					
Commitment to developing students' sociopolitical consciousness						

PRACTICES						
Utilizing and modifying curricula and assessments that are flexible, responsive, and student-centered while still providing access to dominant culture and language	✔					
Engaging in responsive discourse				✔	✔	✔
Collaborating with families in service of children						✔
INTEGRATION						
Supporting students in maintaining their community/heritage languages and cultural models						
Centering children's, families', communities' stories, ways of knowing/being, practices		✔			✔	
Interrogating and critiquing youth and community practices that perpetuate oppression and hegemony						

4.2 Dispositions

We found limited evidence that family-centered home visiting helps teachers develop high expectations for students' academic success. However, there was no supporting evidence that home visiting promotes teachers' commitment to developing students' sociopolitical consciousness. These gaps indicate a need for further research to explore whether and how visiting might empower families to engage as equal partners in their educational process, thereby potentially influencing teachers' dispositions over time.

4.3 Practices

We found evidence that family-centered home visiting supports teachers in engaging in responsive discourse with families, which contributes to relationship building and trust. However, there was limited evidence that it aids in utilizing and modifying curricula and assessments to be flexible, responsive, and student-centered while providing equitable access to dominant culture and language. Similarly, although teachers sometimes leverage artifacts within families' homes to elicit rich narratives and cultural practices, there is limited evidence that home visiting consistently fosters broader collaboration with families outside of the home environment or leads to changes in classroom environments and pedagogy that fully reflect these insights. Additionally, our findings indicate that family-centered home visits serve a distinct role separate from school-based conferences. While school-based conferences primarily address students' educational progress, home visits uniquely facilitate deeper, culturally responsive dialogues, specifically oriented towards understanding family strengths, values, cultural assets, and community contexts.

4.4 Integration

We found limited evidence that family-centered home visiting leads teachers to center children's families' and communities' stories, ways of knowing, and practices. We also found no evidence that home visiting fosters teachers' integration of practices that support maintaining students' community/heritage languages or the critical examination of practices that perpetuate oppression and hegemony. Future studies should examine not only these integrative practices but also the potential long-term impact of assets-framed, relationship-building home visiting on sustained teacher growth, including the development of community partnerships and shared resource models that could enhance culturally responsive and sustaining education.

5. Significance

This study examined how family-centered home visiting practices align with and inform culturally responsive and sustaining pedagogy (CR-SP), with the goal of identifying where such visits support teachers' beliefs, dispositions, and instructional practices that promote equity in education. Despite the growing use of home visiting to strengthen family-school partnerships, relatively little empirical work has explored its connections to CR-SP or its role in shaping teacher development. Our analysis across multiple studies revealed strong evidence that family-centered home visiting positively influences teachers' beliefs—particularly their self-reflection, cultural competence, and asset-based mindsets. We found more limited evidence of impact on teachers' dispositions and classroom practices, and minimal evidence that home visiting fosters the deeper integration of culturally sustaining pedagogies in everyday instruction.

Our findings indicate that family-centered home visiting can significantly enhance teachers' understanding of their positionality, self-efficacy in cultural competence, and foster an ongoing process of self-reflection, all of which are crucial for implementing CR-SP. However, the limited evidence in promoting high expectations for academic success and the lack of evidence in developing sociopolitical consciousness, flexible curriculum adaptation, and deeper family collaboration highlight critical areas for further research and professional development. Addressing these gaps is essential for maximizing the impact of home visiting on educational equity and ensuring that teaching practices comprehensively support diverse student populations.

Our inquiry was guided by a combination of asset-based pedagogical frameworks and developmental psychology theories, each of which positions families' and communities' cultural knowledge and practices as central to children's learning and growth. We drew on Culturally Responsive Teaching (Gay, 2010), Culturally Relevant Pedagogy (Ladson-Billings, 1995, 2014), and Culturally Sustaining Pedagogy (Paris, 2012), which collectively emphasize educators' beliefs, dispositions, and practices that honor and incorporate students' lived experiences. This study contributes to these frameworks by illuminating how home visiting can serve as a relational entry point into this work—supporting the foundational beliefs and perspectives that underlie CR-SP, even when practice-level transformation is less evident. To complement these educational frameworks, we incorporated insights from developmental psychology, particularly Bronfenbrenner's bioecological systems theory (1979), Vygotsky's sociocultural theory (1978), and developmental systems theory (Lerner, 2006). These perspectives emphasize the critical role of social interactions, cultural contexts, and aligned systems in supporting children's holistic development. Home visiting offers a promising avenue for strengthening the mesosystem—the connections between home and school—and for creating more consistent, culturally aligned experiences that support children's social, cognitive, and linguistic development. Together, these frameworks provide a robust foundation for understanding how home visiting can function as both a developmental and pedagogical intervention aimed at advancing educational equity.

5.1 Beliefs: foundational shits toward culturally responsive practice

Our findings demonstrate that home visiting consistently supports shifts in teachers' beliefs, particularly in ways that align with culturally responsive and sustaining pedagogical frameworks. Across studies, teachers who participated in home visits described increased self-awareness, deeper cultural competence, and a clearer understanding of their own positionality in relation to the families they serve. These reflections often emerged through direct, relational encounters with families in their home environments, where teachers confronted and challenged their assumptions, recognized the strengths and cultural wealth present in students' lives, and began to view families as critical partners in education.

This evidence aligns strongly with the core belief components outlined in Culturally Responsive Teaching (Gay, 2010), Culturally Relevant Pedagogy (Ladson-Billings, 1995, 2014), and Culturally Sustaining Pedagogy (Paris, 2012), all of which position teacher self-reflection and cultural awareness as foundational to equity-oriented practice. By engaging directly with families in non-school settings, teachers are afforded opportunities to see students within the context of their homes, languages, traditions, and values—insights that are central to asset-based teaching.

From a developmental perspective, these relational experiences reflect the kind of culturally grounded, context-sensitive learning emphasized in both Vygotsky's sociocultural theory and Bronfenbrenner's ecological systems model. Vygotsky (1978) argued that learning is socially mediated and culturally situated; home visiting enables teachers to better understand the sociocultural tools children bring to the classroom, improving their capacity to scaffold instruction within students' zones of proximal development.

Similarly, Bronfenbrenner (1979) emphasized the importance of strong mesosystem connections—those between home and school—in supporting child development. Home visiting functions as a direct mechanism to strengthen this mesosystem, fostering developmental continuity and mutual understanding between educators and families.

The role of professional development is also crucial in amplifying the belief-level impacts of home visiting. Our findings suggest that when home visiting is paired with structured training—particularly training that helps teachers reflect on bias, define the purpose of visits, and learn strategies for culturally responsive interaction—teachers are more likely to engage in meaningful self-reflection and to recognize the cultural assets within families. In the absence of this intentional preparation, opportunities for learning may be limited or misdirected. Thus, effective professional development is not simply a supplement to home visiting but a necessary condition for ensuring that visits promote shifts in beliefs that lay the groundwork for deeper changes in practice.

5.2 Dispositions: building trust in the absence of critical engagement

While our findings indicate that family-centered home visiting can contribute to shifts in teachers' beliefs, there is more limited evidence that it influences teachers' dispositions—particularly their expectations for students' academic success and their commitment to developing students' sociopolitical consciousness. A few studies suggest that home visits may help educators recognize students' strengths in ways that could support higher expectations, but this shift was inconsistently documented and rarely articulated explicitly by teachers. Even more notably, we found no evidence that home visiting alone cultivates dispositions aligned with sociopolitical consciousness, such as a commitment to helping students recognize and challenge injustice in their lives and communities.

This absence is significant when considered in light of culturally relevant and sustaining pedagogical models, which emphasize sociopolitical consciousness as a critical dimension of equity-oriented teaching (Ladson-Billings, 1995, 2014; Paris, 2012). While home visiting creates conditions for educators to connect with families, it may not provide the structural or conceptual tools necessary to support critical reflection on systemic inequities, power dynamics, or educators' own roles in upholding or challenging oppressive systems. In other words, relationship-building alone does not automatically prompt educators to engage with questions of justice or to translate personal insight into a broader critical consciousness.

There are several potential reasons why home visiting, as currently implemented, may fall short of fostering these deeper dispositions. Most notably, the practice often lacks built-in reflection tools or prompts that center justice, equity, or power. Without guided opportunities to examine structural inequalities or to interrogate how cultural knowledge intersects with educational opportunity, teachers may form stronger personal connections with families without critically examining the larger systems shaping those families' experiences. In the absence of this critical framing, educators may unintentionally reinforce colorblind or individualistic interpretations of equity, rather than developing the sociopolitical awareness necessary to engage in transformative practice.

These findings point to a clear need for professional learning experiences that go beyond relational skill-building. While training in effective communication and cultural responsiveness is essential, it may require opportunities for critical reflection, structural analysis, and engagement with the historical and political contexts that shape educational inequity. Educators need time and support to explore how their positionality, school policies, and classroom practices relate to broader systems of power. Integrating this kind of justice-oriented professional development with home visiting could help ensure that teachers not only build trust with families but also develop the dispositions necessary to support students as agents of change in their own lives and communities.

5.3 Practice: building relationships but needing scaffolds

Our findings suggest that home visiting supports one particularly meaningful CR-SP-aligned practice: responsive discourse. Teachers who engaged in home visits often described improved communication with families, increased empathy, and a deeper capacity to listen and respond to families' perspectives. This form of culturally responsive dialogue fosters trust and affirms family voices, positioning teachers as learners and families as experts in their children's lives. These relational practices reflect a core tenet of culturally responsive pedagogy and represent a concrete way in which home visiting can shape daily educator–family interactions.

Beyond these relational gains, however, the evidence is more limited when it comes to the influence of home visiting on curriculum design, assessment strategies, and classroom pedagogy. While teachers often report gaining valuable insights into students' cultural contexts, interests, and strengths through home visits, few studies to date have systematically examined how, or to what extent, these insights are applied to instructional practice. It remains unclear whether this reflects a gap in teacher enactment or simply a gap in research focused on instructional transfer.

Developmental systems theory (Lerner, 2006) provides a helpful framework for interpreting this limitation. From this perspective, learning and development are optimized when there is alignment across a child's ecological contexts. In the case of home visiting, teachers gain contextually rich knowledge about children and families—but without institutional supports or pedagogical tools to guide application, that knowledge may not consistently transfer into the classroom. Even if educators are motivated to apply what they learn, structural barriers—such as rigid curricula, lack of planning time, or absence of collaborative support—may prevent that knowledge from shaping teaching practices in sustained ways.

These findings point to a need for caution in assuming that stronger relationships with families will naturally lead to more culturally responsive instruction. Home visits may create the foundation for asset-based teaching, but their full potential is unlikely to be realized without intentional scaffolding. Schools must invest in professional development, curricular flexibility, and planning structures that explicitly support the translation of relational insight into practice. At the same time, more research is needed to explore how and when this kind of transfer occurs, and what conditions best support it. Strengthening this alignment between relationship-building and classroom implementation is critical to ensuring that home visiting functions not only as a tool for connection, but also as a lever for equity-driven teaching.

5.4 Integration: toward deep alignment of home and school contexts

Among the four CR-SP dimensions examined in this study, integration emerged as the area with the least evidence of alignment. While teachers often reported gaining meaningful insights into students' families, cultures, and home environments through family-centered home visits, we found minimal documentation that these insights translated into sustained, systemic shifts in curriculum, instruction, or school policy. It is important to note, however, that this lack of evidence may reflect limitations in the current research base rather than a definitive absence of integrative practice. Few studies to date have explicitly examined how teachers incorporate family and community knowledge into classroom environments over time, leaving open important questions about what forms of integration may be occurring but remain unstudied or undocumented.

That said, our findings suggest that even when relational insight is achieved, teachers may lack the time, tools, or institutional support to meaningfully apply what they learn in structurally transformative ways. Integration—defined here as the alignment of beliefs, dispositions, and practices with families' cultural knowledge, community languages, and local epistemologies—likely requires more than relational exposure. It demands shifts in the broader systems that shape teaching and learning: curriculum and assessment frameworks, language policies, professional development opportunities, and teacher preparation programs. Without these structural supports, even the most well-intentioned educators may struggle to translate family knowledge into classroom practice.

From a developmental perspective, the absence of sustained integration may hinder alignment between children's home environments and formal learning contexts. Sociocultural theory (Vygotsky, 1978) emphasizes that development is shaped through culturally mediated interactions, while developmental systems theory underscores the importance of coherence across ecological settings. Without mechanisms for integrating family knowledge into instruction, the potential for developmental alignment across home and school remains limited. As we argue that integrating family and community knowledge into educational practice is not peripheral but essential to advancing equitable developmental outcomes—especially for students from historically marginalized communities (Palacios & Paulick, 2024).

These findings point to an important direction for both research and practice. Family-centered home visiting may be a powerful entry point for building relational trust and cultural understanding, but realizing its full potential will require systems-level support. Future studies should investigate how, when, and under what conditions integration occurs—and how schools can create environments that actively support the transfer of family and community knowledge into the heart of classroom learning.

5.5 Revisiting theory, policy, and practice: the promise and limits of home visiting

Our findings suggest that home visiting holds significant promise as both a developmental and pedagogical intervention, particularly in shaping teachers' beliefs and laying a foundation for culturally responsive practice. Across studies, home visits helped teachers reflect on their own positionality, engage more empathetically with families, and adopt a more asset-based view of students' cultural contexts and experiences. These shifts align with central tenets of CR-SP and reflect the kind of contextual sensitivity emphasized in developmental theories. When teachers gain a deeper understanding of the cultural tools, languages, and values children bring to school, they are better positioned to support learning within students' developmental zones and to strengthen coherence across home and school environments (Bronfenbrenner, 1979; Palacios & Paulick, 2024; Vygotsky, 1978).

Yet, while home visiting fosters meaningful relational insight, we found less evidence that it prompts transformation in teachers' dispositions or instructional practices—particularly around sociopolitical consciousness, curriculum design, and sustained integration of family knowledge into classroom environments. This limited impact may reflect both real implementation gaps and limitations in the current evidence base, as few studies have directly examined how insights from home visits are—or are not—carried into classroom teaching. Still, the pattern points to a core theoretical and practical insight: relational exposure is a key component, but alone it is not sufficient for structural change. For home visiting to fully support CR-SP and developmental alignment, it must be embedded within broader systems of support.

This finding surfaces a critical tension: family-centered home visiting is often framed as a strategy to build relational trust, yet it does not inherently cultivate critical consciousness or lead to systemic shifts in pedagogy or policy. While teachers may reflect on their own biases and build empathy–which we acknowledge as key components of family-centered, assets-based, relationship-building home visits–these insights often remain at the level of individual reflection rather than extending into curriculum, assessment, or instructional reform. To move beyond this limitation, we argue for the development of professional learning that intentionally links home visiting to equity-focused instructional planning. This includes embedding structured reflection protocols, prompts for culturally sustaining curriculum design, and opportunities for teachers to collectively make sense of what they learn during visits. Policy and institutional supports may be essential: teachers need time, guidance, and frameworks that help them apply relational knowledge in meaningful ways. Moreover, schools and districts should distinguish clearly between the purposes of home visits and school-based conferences—recognizing that the former creates space for understanding families' lives, assets, and values, while the latter often centers on academic performance. Clarifying these roles can ensure that home visits are not reduced to compliance activities, but are instead positioned as transformative tools for relational and instructional alignment.

In this way, our findings extend both CR-SP and developmental theory by identifying the conditions under which home visiting can serve as a lever for equity. When supported by intentional professional development and institutional infrastructure, home visiting can help create the alignment between ecological contexts that developmental theories call for, while also advancing the pedagogical commitments central to CR-SP. However, realizing this potential requires a deliberate shift—from treating family-centered home visiting as a relational practice alone to embedding it within a broader vision of culturally sustaining, developmentally aligned schooling.

5.6 Toward a more robust evidence base: limitations and areas for future research

While this study offers important insights into how home visiting aligns with and informs culturally responsive and sustaining pedagogy (CR-SP), several limitations of the current evidence base warrant careful consideration. First, much of the available research is cross-sectional or short-term, limiting our ability to assess the sustained impact of home visiting on teachers' beliefs, dispositions, and practices over time. Longitudinal studies that follow teachers across multiple years are needed to understand whether early shifts in mindset or relational practice lead to deeper instructional transformation and improved student outcomes. Additionally, there is a need for research at the intersection of developmental and educational theory to better understand how home visiting may function differently across age groups and stages of development, as the dynamics of home visiting may shift as children grow older. Adolescents, for example, may play a more active role in guiding the conversation and shaping what is shared during a home visit–similar to the way that parents typically guide interactions for younger children. These developmental differences warrant closer examination even as the overarching goals of home visiting remain consistent: enabling educators to learn from students and families in ways that deepen their understanding, interrupting inequitable power dynamics, and informing culturally responsive and sustaining teaching.

Notably, key areas of teacher development remain underexplored. While our synthesis found evidence of shifts in teacher beliefs, there is limited research examining how these beliefs are translated into classroom practices—particularly in terms of curricular adaptation, culturally sustaining pedagogy, or differentiated assessment. Likewise, there is a striking absence of research exploring the impact of home visiting on teachers' sociopolitical consciousness, a central component of CR-SP that relates to educators' understanding of power, justice, and structural inequity in education. Without attention to this domain, the transformative potential of home visiting remains only partially realized. Additional gaps include the long-term impacts of home visiting on integration—that is, how family and community knowledge is sustained and embedded in instructional practices over time. Few studies have tracked how teachers maintain and evolve their use of family insights in the months and years following a visit. Moreover, there is limited evidence regarding the role of home visiting in supporting multilingual learners or the maintenance of heritage languages, both of which are foundational to culturally sustaining pedagogies and critical for promoting language equity in schools.

To address these gaps, future research should leverage mixed-method designs that combine rich qualitative data with longitudinal quantitative tracking. Studies that document not only teacher reflection but also classroom practice—along with student and family perspectives—would provide a more comprehensive view of how home visiting functions across educational systems. Attending to equity-oriented outcomes, such as sociopolitical awareness, heritage language use, and culturally sustaining instruction, will also be key in evaluating the broader transformative potential of home visiting as both a developmental and pedagogical intervention.

6. Conclusion

Family-centered home visiting offers significant promise as a relationship-centered practice that connects families and schools in meaningful, developmentally supportive ways. It can help educators build trust, better understand the cultural and linguistic strengths of students and families, and begin to shift how they see their role in fostering equity. Yet, to fully realize the potential of the CR-SP framework, home visiting must be supported by more than good intentions. Teachers need ongoing professional learning, time for critical reflection, and school-wide systems that help them apply what they learn during visits to their classroom practices. Strengthening the alignment between what teachers come to understand through home visiting and how they teach, assess, and engage students in the classroom is essential for creating inclusive and excellent learning environments. These findings build on both culturally responsive and developmental theories by showing how home visiting can foster alignment between children's home and school environments while also revealing the systemic supports needed to translate that alignment into lasting educational practice.

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SIGNIFICANCE

Earlier Intervention Necessitates Earlier Identification
Autism spectrum disorder (ASD) is a lifelong condition that affects ~2.3% of the population and carries a lifelong financial cost of $1.4-2.4 million per individual, thus representing a pressing public health challenge[29,30]. Early intervention plays a pivotal role in the lives of children and families affected by ASD – improving functional outcomes for the individual, enhancing the quality of life for the family, and reducing a lifelong financial cost[31,32]. Unfortunately, with ASD diagnosis and therefore intervention onset occurring after age 4-6 years[33,34], we are missing out on critical periods of early intervention that could capitalize on peak neuroplasticity within the first 2 years of life[35–37]. **This is a critical barrier in the field**–earlier identification of ASD within the first year of life can *catalyze the onset of very early intervention,* leading to greater *improvements in long-term outcomes*. Addressing this critical barrier by identifying the earliest point of developmental divergence in ASD is a high priority.

The search for biological and behavioral markers of ASD has resulted in increased understanding of altered neurodevelopmental pathways, and **atypical attention** is among the most promising biomarkers that persists across the lifespan[38–42]. Research on infant attention suggests that infants with ASD exhibit reduced neonatal attention[6,43], diminished social monitoring[44,45], disrupted engagement of social sustained attention[46], and a developmental decline in eye-looking[7], all prior to diagnosis. **Still, this evidence is limited to <u>behavioral</u> studies of attention, and few extend to earlier than 2-3 months of age.** To advance our understanding of disrupted attention in ASD, there is a need to examine specific biological substrates associated with this developmentally persistent phenotypic feature. The <u>timing</u> and <u>context-specificity</u> of attention differences measured at <u>multiple levels</u> (e.g., behavioral, physiological) in <u>early development</u> may reveal key neurodevelopmental mechanisms in ASD. This project takes the critical next step in examining **biological underpinnings of disrupted attention in ASD** by testing whether reduced autonomic regulation of attention ("heart defined attention") is **a key early-emerging feature associated with ASD. Significantly,** this work will:

1. **Advance current etiological theories of ASD** that link, with heightened specificity, an established, lifelong cardinal feature of ASD – atypical attention – to autonomic regulation of attention in the first months of life.
2. **Pave the way for new developmental models of ASD** wherein disrupted attention has proximal effects on interactive behavior that interferes with learning and has distal, cascading effects on long-term outcomes.
3. **Lead to the development of new models for earlier intervention** that enrich early experiences and enhance learning opportunities *from the first months of life*, ultimately reducing lifelong costs and improving lifelong outcomes for individuals with ASD.

There is an urgent need to test specific mechanistic hypotheses about precisely *when* attention patterns diverge in ASD, the *neurobiological systems* implicated in this disruption, and the *in-the-moment* influence of attention on interactive behaviors in infancy, *prior to the emergence* of symptoms.

The First Three Months: A Critical Period of Neurodevelopment and Autonomic Maturation
The brain undergoes faster structural and functional maturation, as well as significant cross-domain integration, from birth to 3 months of age than at any other postnatal time[47]. From birth, the autonomic nervous system (ANS) plays a vital role in the adaptive competence of the newborn. ANS has two major component systems: sympathetic ("fight or flight") and parasympathetic ("rest and digest"). In response to ever-changing environmental contexts, these systems enact physiological changes that regulate basic functions, such as heart rate and respiration. In early prenatal development, physiological functions are driven primarily by sympathetic influence. Higher cortical processes begin to integrate autonomic control in the late prenatal and early postnatal period leading to increasing influence of the parasympathetic system. As ANS matures, parasympathetic tone serves as the foundation for several higher-level behaviors that support calm states of alertness and **sustained attention** to people and objects, ultimately providing countless opportunities for social and language learning. In this way, ANS is considered a "developmental resource" in the neonatal and early infant period that mediates infant-environment interactions[11]. While some neurodevelopmental models of ASD suggest atypical emergence of cortically-driven endogenous attention systems[48], **the developmental course of parasympathetic influence on attention in ASD has never been investigated,** representing a clear scientific gap to be filled[9,48,49,11]. Current evidence that forms the **scientific basis of this study** includes: 1a) *very early* disrupted attention, measured behaviorally, in ASD, 1b) an ANS-regulated marker of attention that is well-studied in TDs, but not ASD, 2) a developmental profile of PT infants who experience very early ANS dysregulation, but not ASD (in most cases), and 3) some evidence of a dynamic, temporally coordinated, ANS-attention-behavioral system in TDs, but less in ASD. This strong scientific foundation (described in further detail below) **calls for further research** on the "physiological phenomena" of attention[49] and presents **<u>three specific gaps</u> in prior research** that prevent a full understanding of the role of autonomic regulation in atypical attention in ASD: **(1)** The maturational course of

autonomically regulated attention and baseline autonomic functioning as it emerges *within the first three months of life* for infants at elevated genetic likelihood for ASD remains unknown. **(2)** The developmental course of (and relations between) autonomically regulated attention, baseline autonomic functioning, and ASD symptomology in infants who are *born with* baseline ANS dysfunction (PTs) has not been delineated. **(3)** The dynamic role of HDA in supporting *interactive behavior,* moment-by-moment, is entirely unexplored. These gaps have prevented our ability to test a full model of ASD wherein early autonomic dysregulation of attention *within the first months of life* has cascading effects on interactive behavior and the emergence of ASD symptoms. Examining this novel mechanistic hypothesis while capitalizing on an especially critical period of rapid neurodevelopment (1-3mos) will advance scientific knowledge of ASD pathogenesis, ultimately paving the way for novel paradigms of very early interventions.

1. Autonomic Regulation of Attention: A Critical Opportunity for ASD Research. Autonomic regulation and attention are critical components of an adaptive engagement system that selects relevant features of the environment and creates rich opportunities for learning and social interaction[10]. ANS serves a regulatory function in attention by modulating heart rate responses during periods of attention and information processing. In a healthy ANS-mediated attention system, a neonate overrides metabolic demands to generate a physiological response (heart rate deceleration) that facilitates and enhances attention and stimulus intake[50]. This parasympathetically-driven physiological response, termed here **heart defined attention** (**HDA**),[51] is characterized by 3 discrete attention phases (*Fig 1*) that are associated with cognitive and learning processes[52]. Some argue that HDA indices are more precise measures of information processing than looking time alone[53–55]. Limited studies of infants later diagnosed with ASD show atypical behavioral measures of attention[7,43,56] in very early infancy (see PI's work, *Fig 2*)[57] as well as atypical general autonomic functioning in later infancy[58,59], pointing to the possibility of autonomically-driven dysregulation of attention. **Our pilot work**, and that of others, show differences in HDA quantity (**%SA:** %time in heart-defined sustained attention) and quality (**HR-decel:** magnitude of heart rate deceleration**;** *Fig 1*) in infant siblings of children with autism (**ASIBs**)[24,49] (see *Fig 4* below).

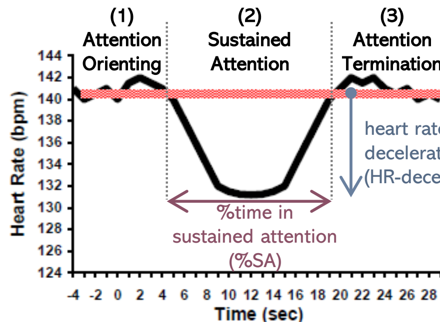


Figure 1. Heart defined attention phases and proposed DVs: %SA, HR-decel

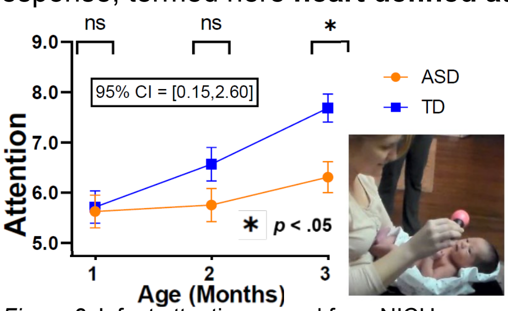


Figure 2. Infant attention scored from NICU Network Neurobehavioral (NNNS) exam. Higher score indiciates better performance.

2. Preterm (PT) Infants as a Model of Global ANS Dysfunction. Significant ANS maturation occurs in the second half of pregnancy, and preterm birth (born <37w gestation) significantly disrupts ANS development and the overall functioning of the PT newborn[60]. For PTs, reduced ANS activity fails to normalize at term age equivalent[25,61] and may alter long-term temporal programming of ANS maturation[62]. While the vast majority of PTs experience prolonged global ANS dysfunction, *only a small minority of PTs develop ASD* (7% in PTs vs. 20% in ASIBs)[16,17]. Thus, inclusion of **PTs as a model for global ANS dysfunction** in this study allows for heightened precision and specificity in delineating the link between *attention-specific* ANS dysfunction (i.e., HDA) and ASD symptoms. **Systematic, longitudinal examination of autonomic measures of attention (HDA) in infants with elevated liability for ASD (ASIBs) contrasted to PT and TD control groups** is necessary to test our mechanistic hypothesis that disrupted autonomic regulation <u>*specific to*</u> attention (vs. global ANS dysfunction observed in PTs and measured in this study) carries critical consequences for learning and development[63] and is associated with ASD symptoms. Measurement of general, baseline autonomic function in addition to attention-specific autonomic regulation, as proposed in this study, will clearly show how autonomic regulation of attention differs across **ASIBs, PTs,** and **TDs**.

3. Modeling Autonomic, Attentional, and Behavioral Dynamics in ASD. A healthy, regulated autonomic and attention system creates rich opportunities for learning and social interaction[10]. After the third month of life, a newly emerging, cortically driven organization of behavior results in anticipation and preparation for interactions with people and objects. The dynamic coordination of autonomic arousal and attention affords calm states of alertness, an optimal context for learning[64]. Quantification of the dynamics of this system[65] can reveal how autonomic regulation of attention is associated with, and may even predict, interactive behaviors as they unfold, moment-by-moment, in infancy. Autonomic arousal can index "anticipatory readiness" for dynamic changes in attention and behavior[66,67]. Emerging research in ASD suggests unique patterns of ongoing physiological activity and attention during social tasks[68,69], however, **there is a significant lack of data on temporal dynamics of HDA and ongoing behaviors[70] in ASIBs** in infancy as symptoms of ASD begin to unfold. Such ANS-driven dynamics could facilitate ongoing social engagement, which may help to guide ongoing interactive behavior, and ultimately increase opportunities for learning. **The proposed study will fill this gap** with an analysis of temporal dynamics of infant gaze, heart rate changes, and behavior during naturalistic interactions between age 6-12 months across ASIBs, PTs, and TDs. Critically, the direct link from ***physiology*** (heart-defined attention) to **in-the-moment interactive *behavior*** will enable translation of findings to pilot behavioral interventions and explore biofeedback tools[19,71,72] to directly target the quality and quantity of infant HDA during interactions.

Model of ANS-Regulated Attention and ASD

Based on current literature and our preliminary data, we propose a model of ASD wherein initial global ANS (primarily sympathetic) function supporting neonatal arousal and reactivity is intact at birth. Yet, subsequently emerging parasympathetic regulation of attention is disrupted. This disruption emerges over the first year of life and interferes with infant sustained attention and ensuing behavior, leading to decreased learning, especially social learning, opportunities, and increased ASD symptoms. **Expected Outcomes by Group** (see *Fig 3): ASIBs*, who have *no/low* global ANS dysfunction at birth (*Fig 3a*), will show emerging ANS dysregulation *of attention* over the first year (attenuated HDA from 1-3m, weakening HDA-behavior associations from 6-12m; *Fig 3b*) leading to elevated ASD symptoms at age 3 (*Fig 3c*). **PTs**, who *do* have global ANS dysfunction at birth (*Fig 3a*), should show gradual improvement in HDA (1-3m) and HDA-behavior associations (6-12m), ultimately resulting in low ASD symptoms at age 3. The rate of ASD in PTs is higher than that of the general population, and we hypothesize that a small subset of PTs with elevated ASD symptoms at outcome will show *both* disrupted global ANS *and* ANS-dysregulated attention in early infancy. **TDs** (*no* global ANS dysfunction) will show no/low ANS dysregulation of attention and low ASD symptoms at age 3.

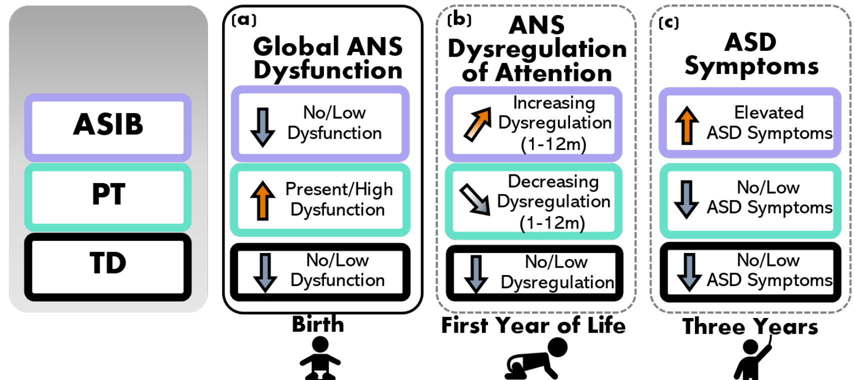
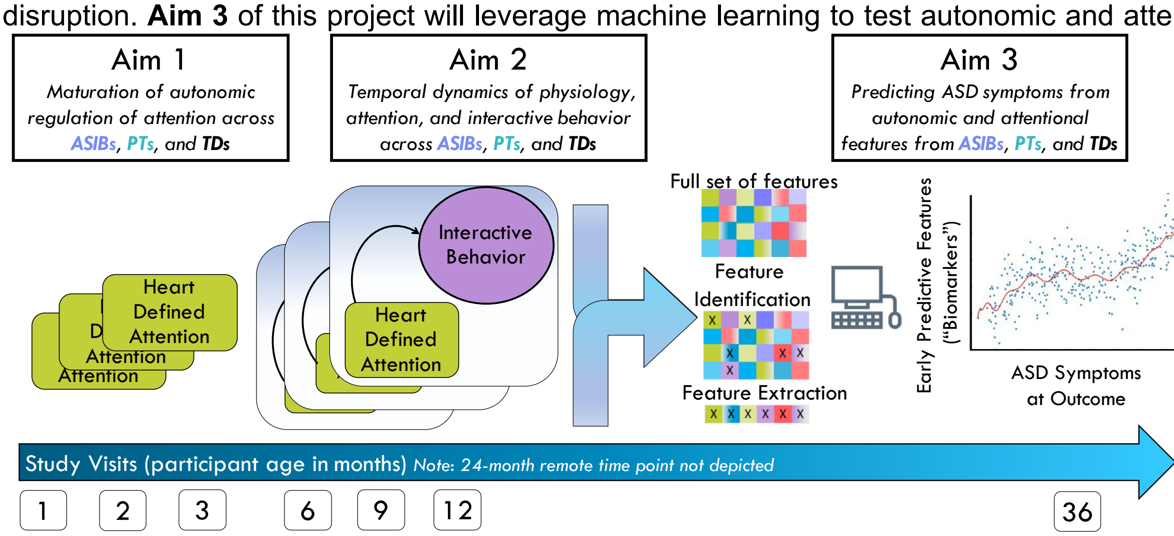


Figure 3. Driving framework for expected outcomes to be tested (b) and (c) given initial global ANS function/dysfunction at birth (a). ANS dysregulation of attention (b) will be measured measured early (1-3 months) via heart defined sustained attention (%time in sustained attention, HR deceleration). ANS dysregulation of attention will be measured later (6-12 months) via HDA-behavior associations.

Translation of Findings to Non-invasive Biomarkers of the ASD Phenotype

Early intervention plays a pivotal role in the lives of children and families affected by ASD – improving functional outcomes for the child, enhancing quality of life for the family, and reducing lifelong financial burden[73]. Very early onset of intervention enhances child gains, which has prompted the PI and others to develop and pilot interventions for at-risk infants, some *as young as 6 months*[31,74–77]. The nascent, but rapid, progress in the development of very early interventions demands innovative studies to identify early markers of social functioning. Offering intervention to all infants with genetic or congenital risk is likely not feasible. But a **valid, cost-effective, non-invasive biomarker** can help identify those who have the greatest likelihood of developing social-communication challenges and ASD, and thus are in greatest need. In a landmark study, Jones and Klin (2013)[7] identified altered trajectories of social attention (2-6 months) as the earliest known predictor of ASD. However, this exceptional study did not evaluate potential biological mechanisms underlying such an early disruption. **Aim 3** of this project will leverage machine learning to test autonomic and attentional features from 1-12 months (Aims 1-2) as viable biomarkers of ASD symptoms. Notably, **Aim 3 is not contingent** on significant results from Aims 1-2 as statistical significance is not a requisite for including these variables in feature selection



procedures. See **study overview** above.

The proposed research will have a significant impact on the broader understanding of ASD by utilizing an innovative, developmentally informed framework: autonomic regulation of attention[78]. We will generate linkages from ANS and attention to moment-by-moment infant behavior and utilize advanced analytic methods to predict ASD symptoms years later. The prior research and our preliminary data (see *Prelim. Data* below) provide support for the overall study premise to identify disrupted autonomic regulation of attention as an early indicator of ASD.

The potential for translation of findings to actionable interventions is high; promising studies of behavioral interventions for PT infants show positive maturational shifts of autonomic regulation[79], suggesting malleability of this early-developing system. Developmentally informed interventions that target this rapidly maturing autonomic system of attention have the potential to make significant, sustained changes in attention, cognitive, and social outcomes for infants with an elevated likelihood for ASD.

INNOVATION

This proposal seeks to shift current research paradigms by studying novel indices of attention across three unique infant groups using advanced modeling of dynamic, moment-by-moment, behavior in the first year of life.

Attention as an Autonomically-Regulated Process and Predictor of ASD. Atypical attention has been among the most promising early markers of ASD[38,80], but the biological underpinnings of disrupted infant attention in ASD is currently unknown. This proposal will utilize an innovative, physiological measure of attention (HDA) in addition to a baseline autonomic marker (respiratory sinus arrhythmia, RSA) to test the hypothesis that disrupted attention is linked specifically to altered ANS-regulated attention and is associated with later-emerging ASD symptoms. Repeated measurement of RSA and of HDA across interactive contexts from the first months of life is an innovative approach that will provide an enriched dataset more powerful than previous studies. This innovation will allow us to understand *mechanisms* and *roles* of ANS and attention in the unfolding of ASD.

Preterm Infants: Controlling for Global ANS Dysfunction. PT infants are born with an exceptionally immature autonomic regulatory system that may not fully develop until well after birth[13,81–83]. Yet, the vast majority of PTs ultimately experience typical outcomes (~80-92% of PTs do not have ASD or developmental delays by preschool age[84,85]). Thus, PT infants serve as a unique control group that experience early autonomic dysfunction *and* mostly typical, non-ASD outcomes. Measures of HDA and baseline RSA within this innovative three-group research design will allow us to delineate: 1) how global ANS dysfunction (in PTs) differs from autonomic dysfunction specific to attention (in ASIBs) and 2) how maturation and improvements in ANS over the first year of life for PTs may support interactive behavior and learning, potentially buffering against ASD-specific outcomes.

Moment-by-Moment Measures of a Dynamic, Interactive System. Dynamic biological rhythms are central to the development of the infant[9]. Physiological and social maturation are dynamic processes, and variability within this system during critical neurodevelopmental windows may predict later functioning[9,86]. This proposal examines novel questions of how physiological correlates of attention (i.e., HDA) relate to infant interactive behavior. By measuring HDA during live interactions with people and objects, we will explore the dynamics of attention, physiology, and behavior in an integrated way that has never been done before. Aim 2 will capture this integration, generating a rich dataset that can inform the development of interventions.

APPROACH

Key empirical support for the proposed project include the PI and investigative team's preliminary data and previous research on disrupted attention in infants who later develop ASD[6,44,45,48,87–89]. However, prior research has been largely limited to behavioral measurement of attention in contrived experimental contexts (e.g., eye tracking) in later infancy (after 3-6 months). Moreover, very few studies have directly compared ANS or attention in ASIBs and PTs. And no studies have examined the temporal dynamics of autonomic regulation of attention and interactive behavior in these groups. These issues limit advancement of etiological theories of ASD and implications for treatment. This proposal details specific plans to address the gaps in prior research.

Research Team. The project PI (Bradshaw) has >14 years of experience conducting research in largescale, NIH-funded, interdisciplinary, autism centers (Yale Child Study Center, UCSB, UCSD, Emory), resulting in 40 publications. As an early-stage investigator (ESI), the PI is currently **leading** two NIH-funded studies (K23, R21), one Foundation award (James S. McDonnell Foundation), and two internally funded pilot studies. In the proposed project, the PI will be supported by five Co-Is and one psychologist (below), four of whom are senior investigators with an established track record of grants management (R01s) and successful mentorship of and collaboration with ESIs. In addition to regular individual meetings between the PI and Co-Is, **regular meetings with the full investigative team** (PI+Co-Is) will occur at key points throughout the project (≥3x/year) to troubleshoot issues and think through larger project questions and plans from multiple perspectives. **Jane Roberts** (expertise in autonomic dysfunction in ASD/Fragile X; primary mentor for PI's K23) has significant grants management experience and publications with the PI[90]. **John Richards** (co-mentor on PI's K23, collaborator on internal pilot grant) developed the theoretical model of heart-defined attention[52,91] and has led an R01 for >40 years focused on this model of infant attention in TD and PT infants. **Robin Dail** (expert in ANS in PT infants, collaborator on internal pilot grant) is leading a multisite R01 with PT infants in NICUs across North/South Carolina. Her expertise is critical for understanding ANS maturation in the context of PT birth and heightened medical morbidities. The PI's collaborative conceptual work with **Jana Iverson**[8] (expertise in TD and ASIB infant motor/communication development) on developmental cascades in ASD has led to the novel questions proposed in this study. **Christian O'Reilly** (expertise in computational modeling, early biomarkers, and signal processing) has worked with the PI using machine learning to predict ASD likelihood from heart activity (preliminary data for Aim 3). Finally, a licensed **Clinical Psychologist** with expertise in developmental psychology and autism (Dr. Emily Neger) will assist the PI with clinical training, supervision, reliability monitoring, and expert clinical judgements. The unique, complementary areas of expertise among the investigative team, along with diversity across stage of career and discipline, will enhance diverse perspectives, enrich expertise, foster ingenuity, and result in successful achievement of scientific aims.

PRELIMINARY DATA

ASD-Specific Differences in Behavioral Measures of Early Infant Attention. In the neonatal period, attention to the environment occurs during calm states of alertness, facilitated by optimal levels of autonomic regulation. Our recent set of studies (P50MH100029)[6,57] investigated *behavioral* measures of **attention** in 1-3-month-old ASIBs who later developed ASD. We identified specific timepoints, between 2-3 months of age, when maturation of attention in ASD (n=18) and ASIBs (n=41) lagged compared to TDs (n=39; *Fig 2*, above). These preliminary data highlight **feasibility** in **recruitment** and **measurement** of attention in neonates and **quantifiable differences** in behavioral measures of attention across groups. The current proposal will expand on these data by increasing sample size (n=75 ASIBs), adding a second comparison group (n=75 PTs), and examining a specific biological substrate of disrupted attention (heart-defined attention; HDA) as well as general autonomic integrity (baseline RSA). In addition, all data collection will occur in the home, which will greatly increase participant accessibility and enrollment (especially for communities of color and rural populations).

Infant Autonomic Regulation of Attention is Associated with ASD Symptoms. Preliminary data related to Aim 1 include trajectories of HDA during naturalistic parent-infant interactions from 1-4 months of age across n=10 ASIBs, n=20 PTs, and n=30 TDs (K23MH120476; R21DC017252; PI: Bradshaw). These trajectories are characterized by a significant increase in heart rate (HR) deceleration during sustained attention (F=10.24, p < .001, *Fig 4a*) and point to a dampened HR deceleration that is developmentally transient for PTs, but persistent for ASIBs[23].[949260] Moreover, HR deceleration during sustained attention to people at 3 months is associated with early-emerging symptoms of ASD[26] (social-communication skills at 9mo[92]; r=0.50; p<.05; *Fig 4b*). Our team's past research also suggests dampened baseline RSA is present very early in PTs[93], but only later in ASD[90] (similar to other work[58]). Building on these preliminary findings, we propose to test the hypothesis that autonomic dysregulation of attention is present early in ASIBs (Aim 1), disrupts temporal dynamics of physiology, attention, and interactive behavior (Aim 2), and is associated with elevated ASD symptoms at age 3 (Aim 3).

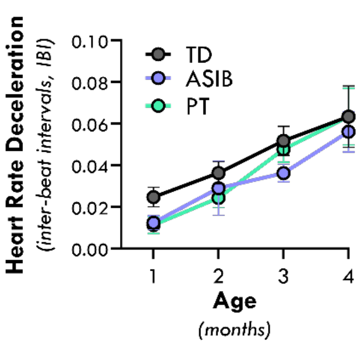


Figure 4a. Trajectories of HDA deceleration during social interactions.

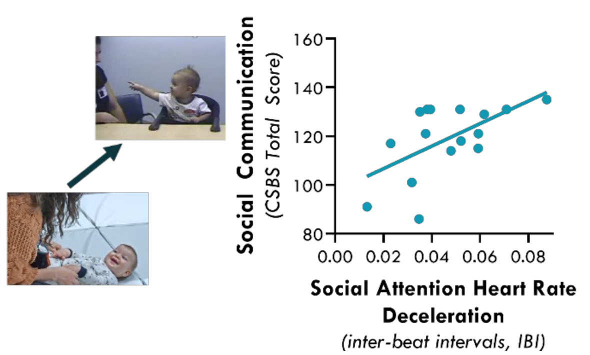


Figure 4b. Association between HDA deceleration during social interaction at 3-4mos and 9mo communication.

Feasibility of Time-Locking HDA Phases to Interactive Behavior. Our data show that HDA-related HR deceleration during a social interaction at 3 months is associated with social-communication outcomes at 9 months[92] (see above, *Fig 4b*). One interpretation of this preliminary finding is that more sustained attention during social interaction reflects increased social engagement and learning, leading to better social-communication skills 6 months later. Yet these data also point to questions about whether and how increased autonomic regulation of attention may facilitate interactive behavior (and therefore learning) ***in the moment***. Aim 2 will address this question. Utilizing full electrocardiogram (ECG) waveform recordings (1024Hz) and frame-by-frame video coding of gaze fixation, ongoing heart-defined phases of attention (HDA) are identified epoched based on the onset of interactive behavior (also coded frame-by-frame from video). Our preliminary data (K23MH120476) demonstrate **feasibility in capturing infant interactive behavior in 6-12-month-olds**[26,94] **and synchronizing with HDA time-series data**[23].

In a preliminary analysis of N=12 infants, we identified the onset of interactive behavior (object exploration) and calculated the proportion of time (i.e., relative frequency) of HDA phase in a time-series of 5 seconds prior to each behavior onset (*Fig 5)*. Findings point to between-group differences (TD vs ASIB) in autonomic regulation of attention immediately preceding behavior. As expected, TD infants showed a consistent pattern of HDA change within the 2 seconds immediately preceding the behavior**,** characterized by an increase in attention (increased **attention orienting**, green and **sustained attention**, red). By contrast, **ASIBs did not show a clear pattern of HDA change** immediately preceding the behavior. Instead, attention patterns remained stable up to 5 seconds prior to the behavior, characterized by less overall attention (more **inattention**, grey; less **attention orienting** and **sustained attention**). The proposed project will build upon this preliminary work in a rigorous examination of **moment-to-moment dynamics** of HDA and interactive behavior across the first year.

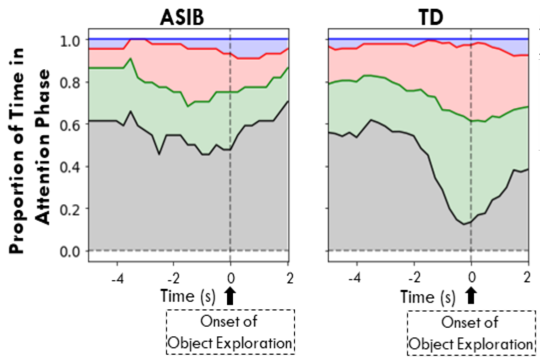


Figure 5. HDA phases (relative frequency of each attention phase) leading up to object exploration behavior for 9-month-olds.

Prediction of ASD Symptoms using Measures of ANS and Attention. In collaboration with Co-I (O'Reilly), ECG data from 3-6-month-old ASIBs (N=7) and TDs (N=16) during interactions were collected and cleaned. Several features were extracted (e.g., mean inter-beat-interval (IBI), standard deviation, root-mean-square differences between adjacent IBI, discrete functional analysis, sample entropy) and an ablation study was used for optimal feature selection. A multilayer perceptron was used to classify ASD likelihood (ASIB vs. TD) with 86% accuracy (precision: 65%, sensitivity: 65%, specificity: 94%). These preliminary results compare advantageously with a previous study reporting a 75.3% accuracy[95] and **demonstrate scientific and technical feasibility** in feature extraction and application of machine learning models to predict ASD likelihood. In this proposal, we will develop a more nuanced analysis predicting ASD symptoms on continuous scales using newly developed deep neural implementation of mixed-effect linear models[96]. The proposed project will use autonomic and attentional features collected for Aims 1 and 2 to better understand how HDA can be leveraged in the formulation of biomarkers in ASD. **Our preliminary data suggest sufficient power** for applying these methods to a sample size that is over ten times larger than that of this pilot study (N=200; n=75 ASIBs). Further, large effect sizes (detectable on relatively small samples) are critical for developing effective ASD biomarkers.

Regression Over Continuous Scales Rather than Binary Diagnoses Classification. ASD symptoms will be captured using Autism Diagnostic Observation Schedule (ADOS-2) calibrated severity scores (CSS)[97]. Our preliminary data (PI Bradshaw[57] and Co-I O'Reilly) suggest sufficient variability in CSS scores across groups of 3-year-olds with and without ASD to determine meaningful associations between early infant biomarkers and ASD symptoms. In the current proposal, we will further build on this direction using ANS-related predictors.

OVERALL EXPERIMENTAL DESIGN

Participants. A total of 200 infants will be included in this study: 75 ASIBs, 75 PTs, and 50 TDs. **Note that our analyses are not powered to detect associations between predictors and binary ASD classification. Rather, for Aim 3, our sample size is powered to detect associations with a continuous measure of ASD *symptoms* at age 3** (see *Power Analysis* below). Our power analyses, published studies,[57] and preliminary data suggest that this proposal is sufficiently powered to detect associations of infant biomarkers and ASD symptoms[6,57,92]. All participants will be followed longitudinally from 1 month to 3 years, at which point **stable** ASD symptoms can be quantified using ADOS CSS[98,99]. In accordance with NIH guidelines for consideration of **relevant biological variables,** we acknowledge that ASD prevalence is higher in males; however, males and females will be recruited equally, and sex included as a covariate in all analyses.

Inclusion Criteria: 1) **ASIB:** born fullterm ($\geq$37w), have a full biological sibling with ASD (confirmation of ASD in the older sibling: record review, direct clinical measures (SCQ[100], CBCL[101], SRS-2[102]), expert clinical judgement by licensed psychologist with expertise in ASD); 2) **TD**: born fullterm, have a full biological TD sibling (confirmed with SCQ, CBCL, SRS, expert clinical judgment), no history of ASD in 1st-3rd degree relatives; 3) **PT**: gestational age (GA) at birth between $24_{0/7}$–$32_{6/7}$ weeks, have a sibling, no family history of ASD in 1st-3rd degree relatives. **GA Considerations:** PTs will be evenly stratified into 3 GA bins to ensure relatively even distribution of risk for ANS dysfunction: (24-26w; 27-29w; 30-32w). GA at birth will be included as a covariate in all analyses, similar to our previous work[57,103]. **Exclusion criteria** for all participants: failed newborn hearing screen, known major hearing or visual impairments, non-febrile seizure disorders, known genetic/chromosomal syndromes (e.g., Down Syndrome), congenital heart condition known to effect heart rate, severe brain hemorrhage and periventricular leukomalacia (PVL). PTs will provide medical records for accurate eligibility assessment. To maximize recruitment efforts of PTs, we are not excluding twins (but multiple birth will be considered in analyses). TDs or PTs with an ASD diagnosis at outcome will be included in analyses (while considering GA and genetic likelihood in models) to increase generalizability.

Feasibility of Recruitment/Enrollment. The current project capitalizes on significant, established recruitment infrastructure, which has led to successful recruitment of ASIBs and PTs at USC. Resources include: the UofSC Neurodevelopment Registry (N>1,100; PIs: Roberts and Bradshaw), the USC Infants and Children Registry (N>150; PI: Richards, Bradshaw), multiple community partnerships (NICUs at the two largest hospital systems in the state, SC SPARK, CAN Center at USC, DDSN, Pediatric practices, autism services providers). USC investigators have **<u>direct outreach access to all children with ASD in the state</u> of South Carolina (total N>8,000**; n=6,700<8yrs) through the Department of Disabilities and Special Needs (DDSN). This outreach has been exceedingly successful for recruitment in the past (see *Letters of Support*). Our longitudinal studies (K23MH120476, R21DC017252) reflect consistent, successful recruitment and retention of ASIBs, PTs, TDs, **1-3 months of age** with **>95% retention**. The PI's lab consistently receives **30-50 referrals *per week*** from interested research participants in the local community and the average enrollment rate for ASIB studies at USC is **12-18 participants/month.** For 1-3-month-old ASIBs specifically, active recruitment in the PI's lab and at USC results in **enrollment of 4-6/month**. This track record demonstrates feasibility regarding enrollment for the proposed project (target average:10-15 enrollments/month across all groups). Our outstanding recruitment records are bolstered by established connections to supportive networks in South Carolina (ASD service providers, schools, pediatricians, hospital NICUs) and state organizations serving ASD and PT populations have pledged tremendous recruitment support (see *Letters of Support*). **To maximize enrollment, we will leverage in-home data collection** for study visits**.** The investigative team has ~20 years of experience with in-home data collection (TDs, PTs, ASIBs age 1wk-3yrs; N>450) and our data evidence that 99% of families (446/450+) prefer in-home visits when given the choice of home vs. lab. This approach will significantly increase participant recruitment/retention rates and **support access for families with limited resources or located in rural areas.**

Attrition. Resources for participant incentives and study visits include over-enrollment to account for expected attrition (rate of 5% based on current longitudinal studies of ASIBs, PTs, and TDs enrolling at 1-4 months). This includes an additional N=16 participants to account for attrition (~5% lost to follow-up, n=10). Thus, power analyses (described below) are based on the final expected sample of N=200.

Operational Considerations. The current project will capitalize on existing infrastructure and resources from the PIs ongoing NIH awards (K23MH120476, R21DC017252), foundation grants (James S. McDonnell Foundation), institutional awards, and institutional funds. The PI's lab is staffed with 1 research assistant professor, 1 postdoc, 4.5 research specialists, 1 psychologist, 1 fulltime recruitment coordinator, and 4 graduate and 20 undergraduate students. Two current RAs are trained in most proposed data collection and processing procedures and will be dedicated to this project. Two more RAs will be hired specifically for this project (see *Budget Justification*). The PI and lab psychologist are licensed clinical psychologists and at least two research staff are trained and reliable in all clinical measures.

EXPERIMENTAL PARADIGMS AND DATA COLLECTION

Data Collection Overview*.* A key component of this longitudinal study is dense sampling of infant behavior to capture change during a rapid period of neurodevelopment. Accordingly, study visits will occur at months 1, 2, 3, 6, 9, and 12, with an additional 24-month questionnaire-only check-in (which will also support retention), and a 36-month evaluation of ASD symptoms and overall development. <u>Adjusted age will be used for all **PT** visits</u>. At visit months 1-3, autonomic regulation of attention will be measured with simultaneous video and heart rate recording during administration of the well-established standardized neonatal measure NNNS-II (NICU Network Neurobehavioral Scales-II; see *Early Infant Attention Task* below)[104]. At visit months 6-12, trained research staff will video record two standardized infant interactions. <u>Research staff will be masked to participant group; parents will be reminded at each visit to refrain from discussing any family or medical history with staff</u>. All research staff will be trained to reliability (>80% agreement) in administration and scoring (where applicable) of the clinical and experimental measures; reliability will be monitored monthly by the project psychologist. A comprehensive evaluation of ASD symptoms and overall development will be administered at the 36-month time point. While study aims are not designed (nor powered) to use ASD diagnosis as a binary outcome, a clinical best estimate (CBE) procedure will be used to assign an overall diagnosis at 36 months in order to comprehensively characterize our sample. **ADOS calibrated severity scores** (RRB, Social Affect, Total) will be used as a metric of ASD symptoms for Aim 3, similar to our previous work[57]. In addition, each study visit will begin with a 5-min baseline ECG recording while the infant is laying or sitting still watching a video of fractals with classical music (to encourage restfulness/ reduce movement). As in previous work[58,59], **baseline autonomic function** will be indexed with respiratory sinus arrhythmia (RSA; a measure of cardiorespiratory control that reflects general autonomic integrity[105]; CardioBatch[106].

	CONSTRUCT	EXPERIMENTAL CONTEXT	MEASURES	TIME POINT (MONTHS) 1	2	3	6	9	12	24	36
Aim 1	Autonomic Regulation of Attention	Baseline Heart Activity (5 min); Attention Items (NNNS) (10-20 min)	RSA, HDA	X	X	X					
Aim 2	Moment-by-Moment Dynamics of Physiology, Attention, and Interactive Behavior	Baseline Heart Activity (5 min); Social/Object Interaction (10min/ea)	RSA, HDA, Interactive Behavior				X	X	X		
Aim 3	ASD Symptom Prediction	ADOS (45-60 minutes)	Autism Symptoms (Calibrated Severity Score)		Features from Aim 1 data			Features from Aim 2 data			X
Clinical Care	Clinical Care for Referrals	ASQ-3 (10 min)	Developmental Milestones						X	X	X

Figure 6. Data collection timeline and primary dependent variables. Note that some important measures are not depicted, including covariates (e.g., NICU-related morbidities, GA, motor skills).

Early Infant Attention Task from the NNNS-II (Aim 1). Attention will be measured from 1-3m during a standardized, readily translatable measure (NNNS-II) that is grounded in a developmental, neurobehavioral framework and aligns with study aims and hypotheses. The NNNS-II is a 15-20-minute direct evaluation of neurological functioning and behavioral adaptation of at-risk (e.g., medically compromised, preterm) and healthy neonates[107]. The PI has >6 years experience administering the NNNS, is a certified NNNS-II examiner and trainer, and has published on attention using the NNNS in TDs and ASIBs[6,57]. Of interest to the current study are the NNNS-II Attention items, which examine the infant's ability to block out external (and internal) stimuli and attend to visual and auditory stimuli. Six attention items include inanimate (ball and rattle) and animate (face and voice) stimuli that are presented visually with sound (rattle; face/voice), visually without sound (ball; face), and auditory only (rattle/voice out of sight). Administration is done with the examiner holding the infant in their lap in a flat or semi-elevated position. To obtain optimal performance, examiners implement handling strategies to keep the infant in a calm, alert state (handling strategies are recorded). Attention item administration = 3-6min.

Infant Interaction Task (Aim 2). Infant HDA and interactive behavior will be recorded during two semi-structured, table-based infant interactions at 6, 9, and 12 months (see *Fig 7* and our work using these protocols in 6-12-month-olds[26,94]). To capture both social and object interactive behaviors, two interactions will be completed. The **Social** Interaction occurs with an examiner and caregiver (ecologically valid context for infants) and is designed to elicit social interaction behaviors, similar to common social-communication measures for infants (e.g., CSBS[108]). The PI is a close collaborator of the CSBS author (Amy Wetherby)[26,75,109] and our research staff are trained and supervised in the CSBS-BS by Dr. Wetherby's research staff. The **Object** Interaction (no caregivers) is designed to elicit independently initiated object exploration behavior. Our research team has successfully administered this experimental task to quantify object behaviors[94]. Social and Object interactions are equated on duration and number of objects presented to the infant (n=5; *Fig 7*).

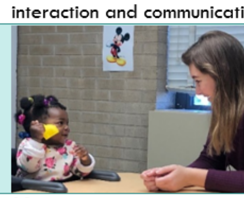
		Social Interaction	Nonsocial Interaction
Protocol Description		Semi-structured interaction between infant, examiner, parent, and one object at a time. Examiner provides presses to elicit infant social interaction and communication.	Semi-structured interaction between infant and one object at a time. Distinct object shapes and functions provide various opportunities to elicit infant exploratory behavior.
Experimental Setup			
Task Length		10 min	10 min
Activities Presented		wind-up toy · squeaky toy · balloon · koosh ball · bubbles	rattle/sensory teether · soft blocks · ring stacker · noise maker · pop-up
Action Examples	Earlier Actions	- Eye contact, Vocalization	- Grasp, Mouth
	Later Actions	- Point, Word Approximations	- Stack, Press

Figure 7. Infant Interaction Protocol

Autism Symptoms (Aim 3). ASD symptoms at 36m will be captured using ADOS-2 calibrated severity scores (CSS): Social Affect, Restricted and Repetitive Behaviors (RRBs), and Total[97]. ADOS examiners will be trained to research reliability and supervised by the PI and project psychologist. When the administering examiner is unintentionally unmasked on a study visit, a masked examiner will score the ADOS from video to ensure scientific rigor. CSSs range from 1-10 and represent an excellent measure of ASD symptoms, relatively independent from IQ and language abilities[97,110].

Data Processing

Electrocardiogram (ECG) Recordings. ECG recordings will be collected at each time point between 1-12 months via Actiheart[111] wireless heart rate monitors, sampled at 1024Hz. The research team has extensive experience using these monitors for successful, continuous recording of infant ECG. Video recordings of all experimental paradigms will be collected for synchronization of ECG recordings with behavioral events (behaviorally coded later from video). Specialized software programs, CardioPeakSegmenter and CardioEditPlus[106,112], will be used for R peak detection, segmentation, and artifact removal.

Baseline Autonomic Function will be indexed using overall RSA from a 5-min baseline recording[106].

Synchronization of ECG and Video Recordings. Actiheart wireless heart rate monitors are incredibly user-friendly and allow unrestricted movement for infants and caregivers. The start time of the ECG recording (based on a remote computer time) is used as the synchronization signal and is captured (video recorded) at the beginning of the experiment (i.e., the start time and run time of the ECG recording is captured on the video that will later be coded for infant behavior). Our custom data processing pipeline then crops the ECG recording to align ECG and Video start times ($time_0$(ECG)= $time_0$(Video)), allowing precise alignment of behavioral coding (described below) and ECG. This method has been used successfully in the PI's lab for the past 4 years[23,92].

Behavioral Coding of Infant Gaze. Coding of infant looking is needed to identify heart defined phases of attention. Thus, frame-by-frame video coding of infant looking using DataVyu software will be completed by trained and reliable undergraduate RAs masked to group. The PI has 16 years of experience in behavioral coding and the PI's lab has established procedures and manuals for coding infant looking during interactions[23,92]. Coders will identify the onset and offset of infant looks to pre-identified target stimuli (object, caregiver, examiner) during NNNS attention items (1-3m) and infant interactions (6-12m). For all interactions, only one object at a time will be presented. Our established training/reliability procedures ensure high research rigor.

Behavioral Coding of Infant Interactive Behavior. The occurrence of infant interactive behaviors will be coded from video by masked RAs (Aim 2). Dr. Iverson (Co-I) is a renowned expert in capturing infant social[113–116] and object[117–119] interactive behavior in PTs[120,121] and ASIBs[114,117] and will assist with this process. Due to rapid development in social and object interaction skills from 6-12 months, behaviors will be categorized as "earlier" and "later" emerging (*Fig 7* above). This will allow our measures to account for maturation in interactive behaviors and avoid floor and ceiling effects at each time point. For example, we expect no infants show "later" social acts at 6m (words and gestures) but many infants show "earlier" social acts at 6m and 12m (social smiles).

HDA. The PI and Co-Is O'Reilly and Richards developed customized Matlab and Python scripts to synchronize heart rate (HR) with infant looking to calculate onsets and offset of HDA phases (sustained attention defined as 5 successive beats with HR slower than pre-look HR median[22,23,92]). Summary variables calculated for analysis will include duration of sustained attention (**%SA**) and heart rate deceleration during sustained attention; these have been associated with enhanced learning[122] and will serve as the primary DVs (Aims 1-2).

Covariates and Demographic Data. Sociodemographic data (race, ethnicity, income, education, home address) and medical history (pregnancy, birth, and early postnatal period) will be collected at the first visit. For PTs, morbidity scores[123] will be derived from medical records and in-depth NICU questionnaire (APGAR scores, overall time in NICU, presence/duration of medical supports (e.g., ventilator), neurological damage (IVH), infections, and surgeries). Due to the potential impact of NICU-related stimulation and stress on cardiorespiratory functions, these variables will be considered as covariates in analyses. Co-I Dail has expertise in evaluating and selecting relevant variables. Annual questionnaires about parental concerns, developmental milestones (ASQ-3[124]), and any early intervention services will be collected. Given the impact of maternal depression on early social development, mothers will complete a depression measure (EPDS[125]) at months 1, 3, and 6. The PI and project psychologist have the clinical expertise to evaluate and address clinical depression and will refer mothers for appropriate care. Motor questionnaires[126] will be completed at each visit to control for impact of motor delays on attention and interactive behavior. Outcome cognitive and motor skills will serve as covariates in all analyses.

Data Management. Management of ECG, video, questionnaires, and behavioral assessments will be led by the Project Manager. Undergraduates will be trained in cleaning and editing of raw ECG files (CardioEdit, CardioBatch), video files will be recorded to SD cards and immediately transferred to our secure server. REDCap will be used for parent questionnaires and to store behavioral data (using double data entry/validation). Original forms will be scanned, stored on a secure server, and shredded to protect participant confidentiality. Experimental data will be processed and linked to demographic/behavioral data for statistical analysis.

SPECIFIC AIM MEASURES AND ANALYSES

Aim 1. Compare the maturation of autonomic regulation of attention in the early infant period (1-3m) across ASIBs, PTs, and TDs.

Overview and Hypotheses. Behavioral measures of attention at 2-3 months reveal significant associations with ASD/ASD symptoms [6,7,57], but biological underpinnings have not been identified. Behavioral measures of looking coordinated with heart rate deceleration reflect focused attention, brain arousal system activation, and enhanced learning[78,122,127]. Thus, HDA in early infancy reflects physiological regulation required to attend to, and learn from, the environment. **Aim 1 will directly test the hypothesis that the maturation of autonomic regulation of attention in young infants from 1-3 months is delayed for PTs and disrupted for ASIBs**.

Measures. Baseline autonomic function will be measured with RSA extracted from a 5-min baseline recording. Autonomic regulation of attention will be measured during the six Attention items of the NNNS-II (see *Data Collection Overview* above). Attention to each stimulus will be coded from video and temporally synchronized with ECG data to calculate: %time in heart-defined sustained attention (**%SA** = time in SA / task duration) and heart rate deceleration in sustained attention (**HR-decel** = maxHR–minHR during each SA phase).

Analytic Plan. We will utilize latent growth curve models (LGCM) to characterize trajectories of HDA (%SA; HRdec) as well as baseline autonomic function (RSA) from 1-3 months. LGCMs can be considered comparable to multilevel models and, in many cases, they are numerically identical, however there are several specific advantages to LGCM approaches[128]. For a single manifest variable measured at a minimum of three time points, an LGCM can be fit to estimate two latent factors, i.e., intercept and slope. This approach is established within a structural equation model (SEM) framework, which allows us to take advantage of the flexibility and broadness of SEM models with the following added benefits: 1) ability to assess global model fit[128,129], 2) easy handling of missing data within and between visits using the full information maximum likelihood strategy[130], 3) inclusion of latent predictor(s)/outcome(s) in model[128,129,131]. Recent research has confirmed that this technique can be applied to small sample sizes with missing data[130]. We will first plot individual trajectories of HDA (%SA, HRdec) to explore differential patterns. The model fit will be determined using chi square tests and fit indices, such as the root mean square error of approximation and the comparative fit index. Mean values of growth factors (intercept, slope) will be estimated and compared across groups (TD, PT, ASIB). We will also examine the variance of the growth factors to test whether individuals share the same starting points and rate of change over time. Covariance between growth factors will provide information about individuals' growth rates related to their starting point. <u>Effects of baseline or outcome characteristics (e.g., gestational age at birth, NICU-related morbidity, biological sex, SES, cognitive outcomes) will be controlled for in all analyses</u>. This is critical as a substantial minority of preterm infants also show cognitive delays[132]. **Power Analysis.** Estimated power for growth curve models was calculated following a simulation-based approach (1000 simulation replicates)[34]. Power rates were computed by assuming 3 measurement occasions and linear growth. Population values used for the simulation were obtained by fitting the linear latent class models (LCM) to our pilot data. Given 200 subjects, power rates to detect individual variability in intercepts and slopes were 0.99 and 0.99, respectively. To detect group differences in trajectories, we computed the power to detect a significant difference in terms of mean slopes between two groups (n=75, n=50) and estimated power was 0.81. Finally, given N=200, we can detect a medium effect (R^2=0.25) when predicting continuous outcomes (ASD severity) with 86% power.

Expected Outcomes and Interpretation. Due to global ANS dysfunction at birth, **PTs** will show reduced baseline RSA and autonomic regulation of attention (less **%SA** and less **HRdec**) from 1-3 months compared to **TDs** ($PT_{1\text{-}3mo} < TD_{1\text{-}3mo}$). Due to *intact* global ANS functioning for **ASIBs** at birth, baseline RSA for **ASIBs** will be comparable to **TDs** from 1-3 months. **ASIB** HDA, on the other hand, will be comparable to **TDs** at 1 month ($ASIB_{1mo} = TD_{1mo}$), but as attention and heart rate become increasingly coordinated over time, **ASIBs** will diverge from **TDs** by 3 months ($ASIB_{3mo} < TD_{3mo}$), similar to our behavioral work[6,57]. Due to the hypothesized attention-specific ANS deficit for **ASIBs**, HDA will be reduced for **ASIBs** by 3 months ($ASIB_{3mo} < PT_{3mo} < TD_{3mo}$).

Aim 2. Determine the moment-to-moment influence of autonomic regulation of attention on interactive behavior across the first year (6m, 9m, 12m).

Overview and Hypotheses. **ASIBs**[26,117,133,134] and **PTs**[135–137] follow unique developmental trajectories of interactive behavior. We have three hypotheses regarding the predictive association between HDA and in-the-moment interactive behaviors: 1) In alignment with our pilot data, **TDs** will show strong, consistent patterns of HDA that predict interactive behaviors, 2) **PTs'** initially immature ANS will cause initially weak predictive associations that improve over time as ANS matures, 3) **ASIBs** will show disrupted associations that weaken (worsen) over time as ASD symptoms emerge. In addition, baseline ANS function trajectories will be compared.

Measures. Interactive behavior will be recorded at 6, 9, and 12 months using a standardized protocol (see above, *Infant Interaction Task*, *Fig 7*). The onset of each interactive behavior will be recorded and synchronized with time series HDA to examine, with fine temporal resolution, the influence of HDA on interactive behavior. Because motor

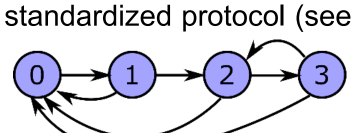


Figure 8. Possible transitions between HDA phases.

differences/delays may influence the frequency/function of interactive behavior[138], visit-specific motor questionnaires[126] will be included as a covariate.

Analytic Plan. We will use HDA times series locked to behavior onsets to study the dynamics of the autonomic attentional system leading up to interactive behavior. We will compare modeling capability of the following alternative approaches: **(1) Continuous-time Markov chain (CTMC):** Our previous work has employed Markov chains to understand dynamics of social attention in toddlers with ASD[139]. A CTMC describes a continuous stochastic process in which the process changes state according to an exponential random variable, moving to a different state with probabilities specified for a Markovian process. We will use this approach to model the process of changing from one HDA phase to the next when moving backward in time, starting from a given action. The structure of the model is fixed given four HDA phases (not looking, orienting, sustained attention, termination) and includes one time constant per state and seven transition probabilities (see *Fig 8*), for a total of 11 parameters. These parameters will be estimated per subject and the distribution of these parameters will be compared between groups. Subject-specific parameters will be estimated in a Bayesian framework using the Markov Chain Monte Carlo (MCMC) approach. **(2) Simple historical mean:** We will compute the inter-beat-interval (IBI) historical mean and corresponding confidence intervals leading up to a behavior. These trajectories will be used to assess group differences in the historical mean and in its time derivatives (speed, acceleration, jerk) to assess group differences in autonomic-attention-action system dynamics. Cluster-based permutation tests will be used to determine periods of time showing statistical differences between groups while accounting for multiple comparisons. This approach is considered as a simpler approach to offer a benchmark for the other, more complex, approach, and previous studies show that simpler approaches often yield better outcomes[140]. **Model Comparison.** We will evaluate the effect size obtained with these approaches to determine which one best captures the dynamics of HDA leading to marked behaviors and extract relevant features (e.g., fitted modeling parameters, historical means over temporal windows, rate of decrease in predictability) as potential biomarkers for ASD symptoms (see Aim 3). **Power Analysis.** Between-group comparisons for the different parameters estimated (i.e., CTMC parameters, IBI means/derivatives over time windows) will be entered into growth models to evaluate this longitudinal data. We therefore expect the same statistical power as Aim 1: range of .81-.99 to detect a medium effect for N=200.

Expected Outcomes and Interpretation: We hypothesize that: (1) baseline autonomic function (RSA) from 6-9 months will be reduced in **PTs** compared to **ASIBs** and **TDs,** but show gradual improvement over time**;** (2) **ASIB** RSA will be largely similar to **TDs** (similar to previous work showing later-emerging RSA disruption in ASD[58]). In terms of HDA-behavior coordination, **PTs** (global ANS dysfunction at birth) will show initial reduced HDA-behavior coordination, and therefore weak HDA-behavior associations, compared to **TDs**. But the effect size of this group difference will decrease over time ($PT_{6mo}<TD_{6mo} \rightarrow PT_{12mo}\sim=TD_{12mo}$). **ASIBs** (hypothesized developmental cascade wherein attention-specific ANS deficits are present early, by 3 months, and contribute to disrupted emergence of interactive behavior[8]) will show increasingly weak associations of HDA with interactive behaviors compared to **TDs** ($ASIB_{6mo}<TD_{6mo} \rightarrow ASIB_{12mo}<<<TD_{12mo}$). **ASIBs'** autonomic-attention-behavior system dynamics may be initially comparable to **PTs** ($ASIB_{6mo} = PT_{6mo} < TD_{6mo}$) due to prolonged ANS dysfunction for **PTs**. But by 12 months, ASIBs will significantly diverge ($ASIB_{12mo} < PT_{12mo} \sim= TD_{12mo}$).

Aim 3. Predict ASD symptoms at age 3 years from infant autonomic and attentional features.

Overview and Hypotheses. Univariate statistical methods for biomarker discovery may be inadequate for capturing the complexity of biological processes involved in a multifaceted, heterogeneous neurodevelopmental disorder[27]. Machine learning (ML) provides unique tools for examining relationships binding multiple interacting variables[28]. Many biomarker studies of ASD (blood, metabolic) have implications for elucidating pathophysiology and improving diagnostic accuracy[141–144], but the lack of conceptual or functional links from **biomarker to behavior** prevents direct translation to behavioral interventions. This aim will incorporate biobehavioral measures of baseline autonomic function (RSA) and autonomic regulation of attention (HDA) to predict ASD symptoms. We hypothesize features collected from 1-12 months will predict ASD symptoms at age 3 years.

Measures. We will use a ML approach considering as features the variables and parameters estimated for Aims 1-2 (regardless of whether the impact of the variables has reach statistical significance for the specific analyses proposed in those aims), as well as simple additional statistics (e.g., overall time in HDA phases) and features established in our preliminary studies (i.e., statistics of heart rate variability, discrete functional analysis, sample entropy from pre-processed ECG with empirical mode decomposition and discrete wavelet transform). As in our previous work[57], ASD symptoms (36m) will be measured using the ADOS-2 Calibrated Severity Scores.

Analytic Plan. To optimize the use of our sample while precluding overfitting and ensuring reproducibility on new data, performance regarding ASD symptom prediction will be cross-validated using a leave-one-subject-out approach. These cross-validated predictions will be compared against the traditional regression approach (i.e., with no test-train split) to assess the degree of overfitting that would result from not splitting the dataset. If overfitting is negligible, this model will be considered final. Else, we will identify the subjects for which the cross-validation prediction is significantly different from the non-cross-validated model, reject the subjects, and re-test. Previous studies using ML in ASD with children and infants concentrated on diagnosis classification. They used various classification methods (support vector machine, k-nearest neighbors, naive Bayes, random forest) and obtained accuracy roughly in the 0.75-0.85 range[145–147]. These studies vary widely with respect to the properties of their sample (e.g., large differences in sample size), their constructs (analysis of electroencephalogram versus spoken language), their objectives (diagnosis in infants versus in children), and their methods in general. As preliminary results, we showed that ANS properties assessed through ECG can be used for prediction by obtaining an 86% accuracy for the classification of ASD risk (ASIB vs TD) using a multilayer perceptron. In this proposal, we will use similar **deep learning models in the context of regression**[96] **to predict continuous phenotypic outcomes** (autism symptomology; ADOS CSS scores). Continuous outcomes will be assessed through the goodness of fit of regression models (e.g., coefficient of determination, R^2). **Power Analysis.** Sufficient power in our sample is supported by our preliminary demonstration of high predictivity of ECG-derived features for capturing ASD likelihood (classification accuracy=.86 with only N=23). Clinical significance, rather than mere statistical significance, requires large effect sizes (e.g., large R^2). As opposed to p-values, effect sizes do not depend on power and sample size. Reliability of R^2 estimates will be determined using confidence intervals computed with bootstrapping. For a sample size of 200 infants, a single predictor (derived from the combination of many features), and an R^2=0.5, the expected 95% confidence interval (CI) goes from 0.40 to 0.60 (for R^2=0.6, 95% CI = .51-.68). Thus, this study is well-sampled for reasonably narrow CIs on at least medium effect sizes.

Aim 3 Expected Outcomes and Interpretation: We hypothesize that early autonomic-attentional features will predict ASD symptoms that do not emerge until age 2-3 years. Further, through ablation study, we will examine which features from which time points (e.g., 1-3mos vs. 6-12mos) contribute most to predictability. We expect predictability to be highest for attention-specific features and to increase with age.

RIGOR AND REPRODUCIBILITY

Rigor and reproducibility of this project is apparent in our prospective longitudinal design with masked examiners, monitoring of inter-rater reliability/agreement (>80%) for experimental and clinical measures, consideration of potential confounds (GA, medical morbidities, sex), sample size selected based on a priori power analyses, and an investigative team with expertise in grant management/physiology (Roberts, Richards), ANS/prematurity (Dail), infant behavior (Iverson), and signal processing, machine learning, computational neuroscience (O'Reilly).

Potential Problems, Alternative Strategies, & Benchmarks for Success. This project will be initiated nearly upon funding as we have existing recruitment infrastructure and a team trained in most experimental and clinical measures. Activities will occur as follows: recruitment Y1-2, data collection/processing Y1-5, data analyses Y2-5, publications Y3-5 (see *Human Subjects-Study Timeline*). **Recruitment and attrition** are potential concerns for any longitudinal study. Considering past success of the PI and Co-Is in neonatal PT and ASIB recruitment, a plan to see participants in their home statewide, a large research staff, and significant resources dedicated to recruitment, we do not anticipate any difficulty with meeting enrollment targets. We will access institutional funds to support recruitment/retention, dedicate a fulltime recruitment coordinator solely to this project, utilize multiple state-specific autism databases, and employ retention strategies (frequent contact with families through social media and clinical feedback). It is highly possible that **examiners may be unmasked to participant group** during home visits. We mitigate this by: attempting to schedule visits when siblings are out of the home, reminding families to keep examiners masked to family history, collecting data on when examiners are unmasked (to explore in analytic models), process data using masked coders/psychologists (ECG data, attention data, behavioral data, ADOS scores). Our statistical design is **robust to within and between-visit missing data**, we will use state-of-the-art ECG artifact corrections using wavelet and empirical mode decomposition, and we rely on refined longitudinal modeling (i.e., growth curves). In order to ensure robust findings, we will cross-validate our results by employing additional analytic strategies (e.g., mixed effects regression, general linear models) when appropriate. **GA in PTs:** While recent evidence suggests that ANS dysfunction relates more to morbidity than GA[123], we will stratify PT recruitment by three evenly divided birth GA bins (all <32w) to ensure sufficient ANS immaturity. **Significant phenotypic heterogeneity** is expected in ASIB and PT groups. Therefore, comprehensive characterization of ASD and developmental profiles at 36m for all participants will allow for analyses to extend beyond between-group comparisons (Aims 1-2) to include key covariates or stratifiers that can parse out heterogeneity and reveal complex, meaningful interactions. **Aim 3 is not dependent on Aims 1-2** as significant results are not a requisite for including these variables in feature selection procedures (i.e., failure for a test to reach statistical significance does not prove the absence of useful predictive information) and, in fact, may elucidate novel findings above and beyond Aims 1-2.

RESEARCH STRATEGY

SIGNIFICANCE

Early Identification Catalyzes Early Intervention, *Especially for Vulnerable Infants*

Autism spectrum disorder (ASD) is a neurodevelopmental disorder that affects ~2.3% of the population and carries a lifelong financial cost of $1.4-2.4 million per individual, thus representing a pressing public health challenge.[30,31] Early intervention plays a pivotal role in the lives of children and families affected by ASD – improving functional outcomes for the individual, enhancing the quality of life for the family, and reducing the lifelong financial cost.[32,33] Unfortunately, with an ASD diagnosis and intervention onset occurring after age 4-6 years,[34,35] we are missing out on a critical period for early intervention that could capitalize on peak neuroplasticity in the first 2 years of life.[36,37] Later identification of ASD occurs disproportionately in more vulnerable populations,[30,31,38] such as those with medical comorbidities, neurodevelopmental impairments (NDIs),[32] as well as those from marginalized racial/ethnic groups and lower socioeconomic status (SES), all of which are highly prevalent in infants born preterm (PT).

Across the world, 15 million PT infants are born each year.[39] Prematurity is *the* leading cause of short and long-term morbidities[40,41] and PT infants are at a significantly increased likelihood for ASD and NDI. The risk for NDI outcomes increases exponentially with lower gestational age (GA)[15] and the occurrence of neonatal morbidity in the Neonatal Intensive Care Unit (NICU).[21,22,42] Socioeconomic disparities play a substantial role in PT birth outcomes, with evidence for multiplicative effects of SES and GA on NDI outcomes.[26,43,44] The burden of PT birth equates to at least $26 billion in the United States (US), which includes medical, educational, and lost productivity expenses.[45] The American Academy of Pediatrics emphasizes the importance of early intervention and coordinated efforts for follow-up care to maximize childhood health and development outcomes.[46]

The rate of ASD in PT infants is more than triple that of the general population,[3] with recent reports of a 10-fold increase for very PT infants (VPT; born <32 weeks GA)[4] and a 20-fold increase for extremely PT infants (<28 weeks GA).[5] Thus, PT infants are at an elevated likelihood for ASD while also experiencing medical, developmental, and socioeconomic risk factors that put them at a very high risk of *later identification* of ASD and *even later* intervention. **This is a critical barrier in the field** – earlier identification of ASD within the first year of life of PT infants can help address this health disparity by *catalyzing earlier intervention,* leading to greater *improvements in long-term outcomes*. Evidence that forms the **scientific basis of this study** includes: 1) the importance of autonomic nervous system (ANS) in environmental adaptation, social interaction, and learning, 2) the presence of ANS dysfunction at birth in VPT infants and throughout the lifespan of individuals with ASD, 3) the exacerbation of ANS dysfunction by PT morbidities, and 4) the elevated prevalence of ASD and NDI phenotypes among VPT infants. This strong scientific foundation and the absence of clear mechanistic explanations for distinct developmental pathways present three gaps:

(1) **Multimodal measurement** has not yet been leveraged to analyze the specific and combined contributions of ANS subcomponents (i.e., sympathetic and parasympathetic) to PT autonomic dysfunction and developmental outcomes.

(2) **Prematurity-associated morbidities** have a compounding effect on ANS dysfunction, but their impact on the association between ANS and developmental outcomes is not well understood.

(3) **Prospective, comprehensive characterization of trajectories** of ASD-specific features and neurodevelopmental impairment throughout infancy in VPTs is lacking.

These gaps **call for further research** on ANS dysfunction as a mechanism for ASD in VPT infants.

ANS Dysfunction: A Mechanistic Hypothesis for Increased Incidence of ASD in PT Children

Existing evidence suggests that ANS dysfunction is experienced by both PT infants and children with ASD. In this study, we will evaluate **ANS dysfunction** as a novel mechanism for the increased incidence of ASD in VPT infants by surveilling NICU-based indicators of ANS functioning and comprehensively phenotyping the emergence of ASD symptoms and NDI across the first 3 years of life. Determining the association between PT ANS dysfunction and ASD/NDI outcomes by age 3 years will significantly improve scientific knowledge of ASD pathogenesis. Ultimately, it will pave the way for translatable neonatal indicators of risk that will lead directly to tailored developmental follow-up and **reduce disparities in timely diagnosis and intervention** for this vulnerable population. This study is well aligned to address PAR-21-201 which calls for studies designed to elucidate the etiology, epidemiology, and diagnosis in relation to ASD.

Significant ANS maturation occurs in the second half of pregnancy, and PT birth (born <37w GA) significantly disrupts ANS development and the overall functioning of the PT newborn.[47,48] For PTs, reduced ANS activity fails to normalize at term age equivalent,[49] and may alter long-term temporal programming of ANS maturation.[50] Following birth, ANS plays a vital role in the adaptive competence of the newborn through balanced adaptive responses of two major subcomponents: the sympathetic nervous system (“fight or flight”; SNS) and the parasympathetic nervous system ("rest and digest"; PNS). In response to ever-changing environmental contexts, these systems enact physiological changes that regulate basic functions, such as heart rate and body temperature. In early prenatal development, physiological functions are driven primarily by sympathetic influence. Higher cortical processes begin to integrate autonomic control in the late prenatal and early postnatal period leading to increasing influence of the PNS. As ANS matures, parasympathetic tone serves as the foundation for several higher-level behaviors that support calm states of alertness and sustained attention, ultimately providing countless opportunities for social and language learning. In this way, ANS is considered a "developmental resource" in the neonatal and early infant period that mediates infant-environment interactions.[51]

In the case of PT birth, a drastic transition to the extrauterine environment occurs amidst ongoing ANS maturation and is a clash that is compounded by exposure to medical stressors. As a result, autonomic maturation is impaired throughout the PT period, initiating a developmental cascade that may disrupt the physiological adaptation to the environment and integration of autonomic control into later-developing, higher-level cortical processes important for social learning and cognitive development. These early disruptions may be at the root of the emergence of ASD symptoms (social-communication delays, social interaction impairments, repetitive behaviors, atypical sensory reactivity). Emerging evidence for this developmental pathway from our group links early infant ANS/attention regulation in full-term infants to later social-communication and ASD.[18,52,53]

1. Multimodal Measurement of PT ANS Dysfunction

Current studies of PT ANS dysfunction are largely limited to unimodal measurement of ANS function, most often heart rate indices (HRIs), such as heart rate variability, derived from electrocardiographic (ECG) recordings[54]. The current study overcomes this limitation by leveraging multimodal measurement of ANS function that includes recordings of **peripheral and central body temperature** and **ECG**.

<u>Temperature-Derived Measures of ANS Dysfunction</u>. Body temperature is regulated in part by vasomotor tone, or constriction and dilation of blood vessels, which may become altered with autonomic instability.[55] In an infant with a healthy, stable ANS, the temperature is higher for the central body than the periphery, with a <u>C</u>entral (abdominal) - <u>P</u>eripheral (foot) <u>T</u>emperature <u>d</u>ifference (CPTd) of 1-2º C.[56] Thus, CPTd is a measure of the **thermal gradient**, or blood perfusion across the infants' body. The SNS plays a critical role in preventing abnormal thermal gradients through peripheral vasoconstriction.[57] Under normal conditions, the vasomotor center in the brain continuously transmit signals to the sympathetic vasoconstrictor nerve fibers throughout the body.[58] This continual firing is called vasoconstrictor tone and is responsible for maintaining vasomotor tone, which allows constriction of peripheral vessels to conserve heat. Peripheral temperature correlates with peripheral blood flow.[59] As blood flow to the extremities increases, higher peripheral temperatures occur. <u>Our research has found that assessing vasomotor tone through surveillance of the CPTd offers a novel approach to evaluating ANS dysfunction</u>.[60,61] Longitudinal abnormal thermal gradients (CPTd <0° or CPTd≥2°C) are a sign of microcirculatory dysfunction and can be used as a sign of ANS dysfunction (likely driven by abnormal SNS activation), which may be an indicator of neurodevelopmental outcome and ASD.[7] As in our previous work,[61,62] this study will use continuous measurement of PT abnormal thermal gradients as an index of ANS dysfunction.

<u>ECG-Derived Measures of ANS Dysfunction.</u> ECG recordings of heart activity yield several **HRIs** that can characterize ANS function, including beat-to-beat heart rate, transient heart rate accelerations/decelerations (i.e., tachycardias and bradycardias), and heart rate variability (HRV), and hear rate variability at the frequency of respiration (i.e., respiratory sinus arrhythmia; RSA). Each HRI represents unique and combined effects of sympathetic and parasympathetic activity, where sympathetic activity generally increases heart rate and parasympathetic activity reduces heart rate. The magnitude of HRV and RSA measures increase over the first years of life, and higher magnitudes indicate a more adaptive, balanced ANS system.[63] This process is disrupted in PT infants, resulting in an increase in RSA magnitude that is 2-3 times slower than that of term infants by 6 months age.[64] Time and frequency domain analyses of HRV show that ANS function is adversely affected by PT-related morbidities such as mechanical ventilation, infections, and patent ductus arteriosus (PDA).[20] This detrimental influence of PT birth on RSA and HRV has been found to persist into childhood and even adulthood.[65],[66] Existing research, including our own, link abnormal HRIs (e.g., transient heart rate accelerations) in the PR period with the later emergence of ASD, with atypical RSA emerging in later infancy (age 2-3 years) for those with ASD, suggesting a potential decline in parasympathetic regulatory mechanisms.[7,67] While RSA could be affected by external influences on respiration (e.g., ventilatory rate, pressure support, and continuous positive airway pressure), previous studies have documented success in measuring RSA in PT infants on mechanical ventilation and we will include respiratory support and severity of lung disease as covariates in all analyses of ANS dysfunction based on RSA.[68,69] HRIs can be obtained through standard cardiopulmonary monitoring systems and, for this study, will be obtained via HeRO bedside monitors,[70] as we have done

previously.[7,62] Continuous heart activity recording offers a novel way to evaluate ANS function, including both sympathetic and parasympathetic activation and balance, in VPT infants in the NICU.

The multiplicative effects of ANS dysfunction on several biological functions make mechanistic interpretations of unimodal data insufficient. For example, abnormal thermal gradients may be caused by vasoconstriction resulting from sympathetic overactivation or by reduced vasodilation caused by parasympathetic under activation. Imbalance of the SNS and PNS has been suggested as an ASD-specific biomarker,[10] but a debate remains on what precisely causes this imbalance (e.g., overactive sympathetic or underactive parasympathetic). Multimodal assessment (temperature, ECG) that includes **multiple** biological indicators will allow for a clear, specific description of ANS function and a nuanced understanding of adaptive and maladaptive sympathetic and parasympathetic responses. This will help parse contributions of SNS and PNS in the overall evaluation of ANS dysfunction and test their relative predictive contributions to ASD and NDI.

2. Prematurity-Associated Morbidities, ANS Dysfunction, and Neurodevelopmental Outcome

Infants born PT experience multiple morbidities (see Table 1) and higher major morbidity rates are associated with worse neurodevelopmental outcomes.[71]

Table 1. Preterm Infants <32 weeks GA Morbidities and Incidence

Morbidity	Definition	Incidence
Severe Brain Hemorrhage	Grade III-IV, Periventricular Leukomalacia	6-19%
Sepsis	Late onset blood infection > 7 days of age	9-50%
Necrotizing Enterocolitis (NEC)	Infection of the GI tract, requiring medical and/or surgical treatment	5-15%
Bronchopulmonary Dysplasia (BPD)	Chronic Lung Disease, requiring oxygen use at 36 weeks postnatal age	32-44%
Patent Ductus Arteriosus (PDA)	The Ductus Arteriosus fails to close post birth and requires medical or surgical treatment for closure if hemodynamically significant	5-60%
Retinopathy of Prematurity (ROP)	Disorder of the developing retina of preterm infants that can lead to blindness	3-42%

Preterm Morbidities and ANS Dysfunction. The risk for morbidities increases with earlier GA[72,73] and recent research suggests that PT morbidities may mediate the adverse effect of GA on PT ANS dysfunction.[20,71,74] All PT morbidities are associated with periods of ANS dysfunction due to the morbidity-related stress on infants' already fragile homeostasis – many directly attacking the brain, heart, or lungs. Some researchers posit that ANS dysfunction underlies the need for oxygen in PTs and, in turn, this increased oxygen exposure can cause hyperoxia and induce oxygen radical disease, which links ANS dysfunction to BPD, ROP and NEC morbidities in VPT infants.[72,75] Thus, early-life ANS dysfunction may both cause and exacerbate PT morbidities.

Preterm Morbidities and Developmental Outcome. As ANS plays a large regulatory role in mitigating stressors of extrauterine life in the VPT infant, the combined effects of ANS dysfunction and PT morbidities can greatly decrease optimal health outcomes. The risk for ASD and NDI increases with decreasing GA and limited research has shown an association between PT morbidities and ASD.[76] For example, the occurrence of 2-4 PT-related morbidities has been associated with positive ASD screening (M-CHAT-R/F)[24] and PT sepsis has been associated with a later ASD diagnosis.[77] Some medical treatments for morbidities (e.g., steroids for BPD) have also been associated with ASD.[78] This already limited research largely comes from retrospective analysis of medical records and has not addressed how PT morbidities *moderate* associations between ANS dysfunction and *dimensional* ASD *and* NDI outcomes. Aim 2 will evaluate these PT morbidities as potential moderators of the association between ANS dysfunction in the PT period and ASD/NDI outcomes at 36 months.

Social Drivers of Health (**SDOH**). It is also critical to consider the role of **SDOH** in child development and mental health research.[79] Demographic factors and SDOH, such as race/ethnicity, maternal education level, insurance, and housing stability, have major effects on access to maternal prenatal and postnatal healthcare as well as infant and child healthcare, developmental supports, and medical and mental health diagnoses.[80,81] These barriers to care can heavily influence preterm morbidities and neurodevelopmental outcomes, as our work has suggested.[82-86] Likewise, early infant experience, including maternal stress, postnatal hospitalizations, and early interventions, can influence outcomes. As such, demographic variables, SDOH,[87,88] maternal stress,[89] and infant context (intervention, hospitalizations) will be assessed and accounted for in analytic models.

3. Comprehensive Characterization of Emerging ASD-Specific Symptoms and NDI

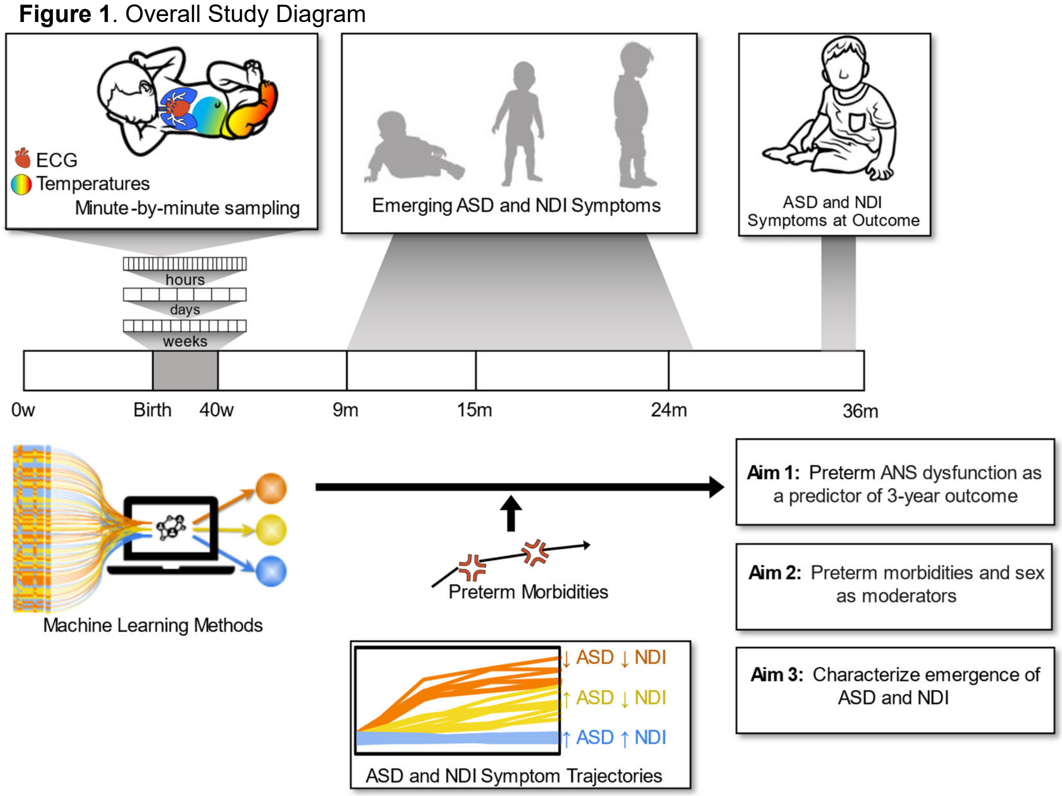
Rates of both ASD and NDI in VPT infants are significantly greater than that of the general population,[3] with many overlapping symptoms common to VPTs both with and without ASD including: motor dysfunction, language deficits, social impairment, atypical sensory processing, and executive function challenges.[26] This is further complicated by health disparities wherein PT birth disproportionately effects minoritized and lower SES families and, in turn, lower SES adversely effects developmental outcomes for individuals with and without ASD,[90,91] as our work has also shown.[83] These commonalities complicate differential diagnosis of ASD and NDI. Yet another complication is characterizing development and change during a critical period of rapid brain development and skill acquisition. Our preliminary data, along with existing research,[27,92] suggest differential trajectories of development across domains for VPT infants where strengths and weaknesses are not stable. Moreover, direct observation of ASD-specific symptoms in the first year of life using validated assessment tools[29] have yet to be applied to a cohort of VPT infants. Understanding individual differences and shifts in phenotypic profiles of both ASD-specific characteristics and NDI (cognitive, motor, language impairments) from as early as 12 months of age, over the first 3 years of life, is critical for informed and specific diagnostic and intervention.

INNOVATION

This proposal seeks to shift current research paradigms related to PT birth and ASD by prospectively studying a large cohort of VPT infants for ANS dysfunction during their entire NICU hospitalization and monitoring them for the emergence of ASD/NDI symptoms over their first 3 years of life. Through this investigation, we aim to identify abnormal thermal gradients and HRIs that can be used as a biomarker in infancy to forecast atypical neurodevelopmental outcomes and, specifically, ASD. Study results will inform the novel mechanistic hypothesis that ANS dysfunction is an underlying link between PT birth and ASD outcome. Although our extensive experimental data collection protocol is not meant for direct translation into a revised standard of care, it will inform the relevance of considering ANS monitoring, potentially via contactless, more scalable strategies.

Continuous Measurement of ANS Dysfunction in the NICU. PT infants are born with an exceptionally immature autonomic regulatory system that may not fully develop until well after birth.[65,93-95] We propose to measure ANS dysfunction through continuous recording of biological measures (temperature, ECG) from birth to discharge. Sampling for an average of 9 weeks per infant (average NICU stay = 9 weeks) will yield an average

Figure 1. Overall Study Diagram



of 90,720 minute-by-minute temperature samples and 1512 hours of continuous recordings of ECG *per infant*. This innovative approach will generate an unprecedented volume of longitudinal, multimodal indicators of ANS.

Machine Learning Approach. A machine learning (ML) approach is necessary for the high volume of biological signals that will be collected and processed throughout this study. Our analytic approach leverages ML and artificial intelligence (AI) techniques that support nonlinearity and investigation of multiple ANS measures, while also retaining interpretability, clinical significance, and clinical relevance, of ML algorithm results. Specifically, we will use support vector machine regression, a supervised ML tool for nonlinear regression, in addition to permutation feature importance techniques developed in the context of explainable AI, to identify the most meaningful ANS features in our prediction models. We will also use DBSCAN, an unsupervised ML algorithm to identify clinically meaningful longitudinal phenotypic profiles. These innovative methods represent powerful tools that provide this study with enormous potential for understanding the impact of PT birth on the emergence of ASD symptoms.

APPROACH

Preliminary Data

Thermal Gradients as a Measure of ANS Dysfunction. Our series of studies have shown that ANS dysfunction in VPT infants is observable via abnormal thermal gradients, defined by a condition of warmer, better perfused extremities with simultaneous colder and less perfused core body parts intermittently over their first month of life.[60,61,96-98] CPTd is a measure of microcirculatory dysfunction that can capture this phenomenon.[99] We, and others, have shown that abnormal thermal gradients, defined as a CPTd < 0º or CPTd > 2º C, is correlated with the occurrence of infection in VPT infants. [61,99,100] The state of circulatory dysfunction and centralization of blood flow in relationship to the presence of bacteria, will cause the CPTd to become increased with researchers measuring CPTd values on average of 2º C, but as high as 9º C[99,101] in PT infants with sepsis. In severely ill infants, with decreased GA and immature ANS, the CPTd can be quite small (0-1º C) or negative, indicating warmer periphery than core (see **Fig. 2**).[101,102] We used Grant dataloggers and thermistors in our study (R01NR017872) to measure CPTd to predict infection,[62] a preliminary analysis showed that when an infant had CPTd >2ºC for more than > 50% of the time, it had an 86% higher odds of infection compared to infants without temperature differentials >2ºC.[103] These data demonstrate that continuous CPTd, derived from readily measurable dual temperature vital signs, can be continuously recorded in VPT infants over their NICU hospitalization and effectively utilized as one of two multimodal measures of ANS function.

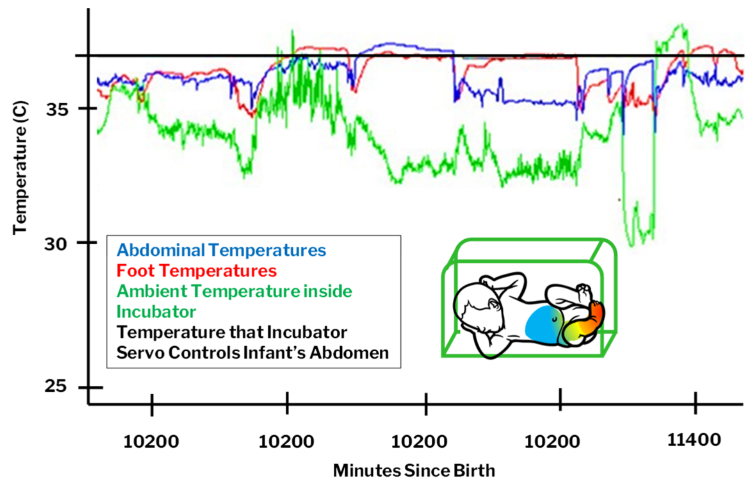


Figure 2. VPT infant body temperatures with intermittent hypothermia, negative central-peripheral temperature difference and normal thermal gradient over 12 hours.

Heart Rate Indices (HRI): A Gold Standard Measure of ANS Dysfunction in the NICU. Heart rate is regulated by the ANS, with sympathetic activation causing accelerations and parasympathetic activation causing decelerations.[1] ANS dysfunction can cause arrhythmias, vascular irritability, and hypo and hypertension.[1] HRIs, including beat-to-beat heart rate, HRV, mean, standard deviation, and skewness of IBI can be recorded through bedside cardio-pulmonary monitors as we and others have done.[16,20,104] For example, we have used simultaneous, continuous measurement of temperature and HRIs in the NICU to find the body temperature where infants were most thermally stable.[104] Researchers have used HRV through continuous ECG recordings to measure autonomic maturation in VPT infants hospitalized in a NICU.[105] We recently completed enrollment of 352 infants (R01NR017872) employing HRIs[70] as an index of ANS dysfunction and predictor of infection.[62] In an interim analysis of 55 VPT infants, we found that more abnormal HRIs were associated with significantly increased odds of infection on that day.[103]

ANS Dysfunction Measured with Thermal Gradients and HRIs: Associations with Later ASD Symptoms. Preliminary findings[7] from a funded pilot study (PIs: Dail, Bradshaw) significantly contribute to the scientific basis for the proposed work. In this prospective, longitudinal trial of N=20 VPT infants (a subsample of participants enrolled in R01017872, PI: Dail), continuous measures of minute-by-minute thermal gradients and hour-by-hour abnormal heart rate indices, measured using Heart Rate Characteristic (HRC) scores derived from HeRO Monitors,[70] were collected for 28 days after birth (>40,000 samples/infant). Following NICU discharge, comprehensive neurodevelopmental follow-up consisting of standardized measures of cognition, language, and motor skills were collected at adjusted ages 6, 9, and 12 months for N=12 infants. At 12 months, assessments of social communication and early ASD symptoms were administered. Across all infants in the first 28 days of life, ANS dysfunction was indicated by an average HRC score >1 and abnormal negative thermal gradients for about 20% of the time (6 of 28 days). Neonatal abnormal HRC scores were strongly associated with ASD symptoms at 12 months ($r=0.81$, $p<.01$; **Fig. 3**), as was birth GA, birth weight (BW), and abnormal negative thermal gradients. A regression model including GA, BW, and HRC scores revealed a significant, unique predictive effect of abnormal HRI (HRC score) on ASD outcomes (model $r=0.86$, $p=.017$; HRC Score: $r=.72$, $p=.029$).[7] **This strong predictive model stands as compelling preliminary data for Aims 1-2.** This R01 is critical in delineating precise HRIs (e.g., tachycardia, bradycardia, HRV, etc.) and other indices (thermal gradients) that are associated with ASD outcomes. Each index may be sensitive to different underlying neural processes and parasympathetic-sympathetic dynamics. We will therefore go beyond the HRC scores to evaluate how different indices differentially map to ASD versus global ANS dysfunction.[106-109]

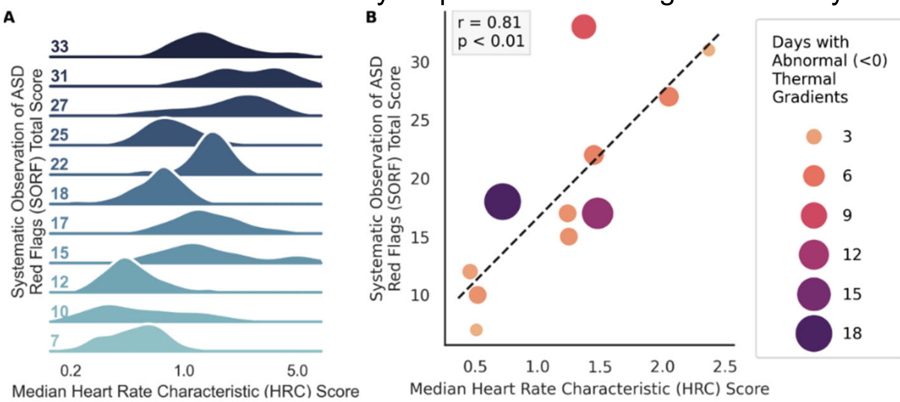


Figure 3. Association between neonatal abnormal heart rate characteristics (HRCs) and 1-year ASD symptoms. Higher scores indicate more heart rate abnormalities and more ASD symptoms. A) Ridgeline plot displaying the distribution of neonatal abnormal HRC scores across participant ASD symptoms (SORF total composite score) at 1 year. B) Scatterplot and associated regression between neonatal abnormal HRC scores across 1-year ASD symptoms (SORF total composite score).

Precise timing and dynamics of multiple ANS measures. HRC scores used in our preliminary data are coarse, hourly estimates of a single composite score. In this study, we will derive **more temporally resolved HRIs** from ECG recordings. Collecting temperatures continuously, with high temporal resolution, will allow for sophisticated analyses of the complex interplay of neonatal autonomic regulation across the preterm period.

Disentangling ASD from NDI in VPT Infants Across the First Three Years. Our preliminary data suggest that significant neurodevelopmental impairments did not emerge in VPTs until 12 months of age when over 50% of the sample exhibited social-communication delays and exceeded the clinical cutoff for ASD risk (Fig 4).[7] At 12 months, infants were characterized by ASD and NDI as follows: ASD features (high = 33%; moderate = 25%; little-to-none = 42%); NDI manifested by cognitive and/or motor delays (NDI = 67%; no NDI = 33%). 25% of the sample had no evidence of ASD or NDI. Of interest, 50% of the infants with high ASD features evidenced no motor or cognitive NDI. These preliminary descriptive statistics illustrate the **significant variability of developmental outcomes** for VPTs using direct clinical observation and ASD-specific measures. Minimal change in developmental skills was observed prior to 12 months, but significant developmental divergence was observed at 12 months. These **preliminary data** suggest **feasibility** for Aim 3: 1) participant retention and **completion of longitudinal follow-up measures** with VPT infants at evidence-informed time points (based on preliminary data); 2) obtention of **sufficient variability in developmental trajectories and ASD/NDI outcomes** with significant, distinct ASD and NDI symptoms already emerging at 12 months; and 3) use of measures with **sufficient sensitivity** to uniquely quantify ASD and NDI symptoms in infancy. Our data (Fig. 4) suggest significant variability and complexity in development and outcomes and Aim 3 is crucial for unveiling developmental nuances and advancing etiological frameworks of ASD in VPTs.

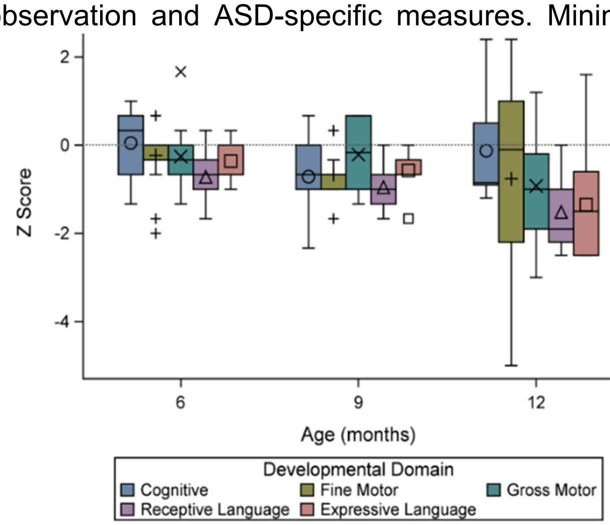


Figure 4. Bayley-4 subscale scores over time represented in z-scores where 0 is the average, -1 is one standard deviation (SD) below the mean, -2 is two SDs below the mean, etc.

Research Team

This study is a natural collaboration between **University of South Carolina's (USC)** MPI Robin B. (Knobel) Dail, Health Sciences Endowed Professor, College of Nursing (CON) and MPI Jessica Bradshaw, Associate Professor of Psychology, with the mutual interest of ANS dysfunction in infancy. The current project is an extension of the MPIs' individual research programs and will capitalize on existing infrastructure, resources, and ongoing/completed awards (Dail: R01NR017872;[62,82] Bradshaw: R01MH1364619, K23MH120476, foundation).

Neonatal Expertise: MPI Robin Dail is a nurse scientist with expertise in morbidity and mortality in PT infants. She has collaborated with **Prisma Richland NICU** neonatologists, on studies of ANS in VPTs over the last 6 years. **Co-I Ravi Rao**, Neonatologist and Medical Director will oversee a 50% research coordinator hired for the proposed study (see Budget Justification). **Co-I Kayla Everhart** is a CON neonatal researcher and has worked with MPI Dail since 2020. She is currently running a funded RCT in Prisma Richland's NICU with MPI Dail. Drs. Dail, Everhart, Rao, and research nurses/coordinators are all trained in collecting thermal gradient and

HRI data from Prisma NICU patients and maintaining a REDCap database with demographic and morbidity data. Dail's Data Manager will oversee data transfer/storage.

ASD, Neurodevelopment, and Longitudinal Data Expertise: MPI Jessica Bradshaw (Psychology) has expertise ASD/NDI in infants/toddlers and in prospective, longitudinal designs and longitudinal modeling.[28,85,110,111] Bradshaw maintains a lab staffed with 1 postdoc, 5 research coordinators, 1 psychologist, 2 recruitment coordinators, 5 graduate and 20 undergraduate students. Staff/students are trained in proposed data collection and processing procedures, 1 of whom will be dedicated to this project. 3 additional RAs will be hired specifically for this project (see *Budget Justification*) and Bradshaw's clinical staff (clinical psychology postdoc and social worker) will be available for RA training, case consultation, and family feedback when needed.

Physiological Data Science and Statistical Expertise: Co-I Christian O'Reilly (College of Engineering & Computing) is an electrical and biomedical engineer and neuroscientist with expertise in scientific computing, advanced statistical analyses, biosignal processing, and computational neuroscience with applications to neurodevelopment. He has established collaboration with the other investigators on this proposal (R01MH132925; Institutional grants).[7,108,112,113] Co-I O'Reilly will supervise 1 graduate student to support the analyses proposed in this project. **Co-I John Richards** (Psychology) is a renowned developmental cognitive neuroscientist with methods expertise in heart rate indices, EEG, and MRI as well as content expertise in nervous system development in PT infants.[17,114,115] Richards and Bradshaw have collaborated closely on studies of infant attention and physiology for the past 5 years (Co-I, R01MH132925; mentor K23MH120476) and has collaborated with Co-I O'Reilly on cortical source analyses in infancy.[113]

Study Design

This prospective, longitudinal study will enroll and follow 260 VPT infants. Minute-by-minute, continuous ANS monitoring will be provided during their entire NICU stay (admission duration range: 5-16 weeks). Following NICU discharge, comprehensive assessments will be administered at adjusted ages 12, 24, and 36 months.

Setting and Equipment. Participant enrollment will occur at Prisma Richland NICU, which is 2 miles away from USC campus where all study investigators are located. Standard-of-care electronic medical records (EMR), SpaceLabs cardiopulmonary monitors, and HeRO bedside monitors (equipped to store continuous R-R intervals, see *Letter of Support*), will be used for data acquisition. Central and peripheral temperatures will be recorded and saved using Grant Squirrel Dataloggers,[62] (n=26 available for use). This setting and available equipment allows for enrollment of 12 infants per month, with measures collected for an average of 3 months per infant. Prisma NICU capacity (70 beds) and admission rate (~250/yr) far exceeds this planned enrollment rate, ensuring study feasibility. **A NICU Research Nurse (RN) will dedicate 50% effort to this project** and will: obtain consent from mothers within 24 hours of the infant's birth, educate clinical staff on study protocols throughout the study, place and monitor infant instrumentation and skin condition (described below), monitor data acquisition, and download/enter/save all collected data (weekly for EMR data, 2x/month for physiological data). Upon NICU discharge, the RN will be responsible for ensuring all medical and demographic data from EMR is entered into a REDCap and all physiological data is transferred to USC Project Manager. All data will be housed at USC on a secure server. Developmental assessments (adjusted age 12, 24 and 36 months) will be completed in the participants' homes by MPI Bradshaw's research staff (team of 2 per visit). In-home data collection has resulted in adequate retention, particularly for our underrepresented populations, but families may come to the lab if they prefer. Research Specialists will have clinical experience and training in assessments to be administered. MPI Bradshaw's lab also has 5 clinical psychology graduate students, 4 of whom are supported by training grants (NIH T32, F31, NSF GRFP), and who can contribute to clinical data collection and training as needed. Our team has conducted in-home data collection for >5 years and has designed custom hardware and software solutions. Our in-home data collection kits include 5 A/V recording systems, assessment kits (2 Bayley-4s, 3 CSBSs, 3 ADOSs), and 3 Surface Pros for completing REDCap surveys. Due to different home sizes and resources, we bring a variety of infant highchairs, adult chairs, and tables to ensure a standardized setup in any context. Data will be entered into REDCap and videos stored on secure servers.

Participants. Over the first two years of the study, we will enroll a total of 260 VPT infants born at Prisma Richland NICU between **24 and 31 6/7 weeks GA and at least 500 grams birthweight.** Infants born less than 24 weeks GA are fragile and frequently unstable. We consistently have used these enrollment GAs and weights for our studies with little attrition due to instability or mortality.[61,62,116] <u>Exclusion criteria</u> include: major anatomical or cardiac defect known at birth, anatomical brain damage or known in utero infant asphyxia, and known genetic or chromosomal abnormality that is known to affect heart function or neurodevelopment (e.g., Down Syndrome, Fragile X). <u>To account for attrition</u>, we are over-enrolling by 30%. Infants may be withdrawn from the study by parents or attending physician during NICU hospitalization, or a very few may die due to complications of prematurity. Based on our pilot data and previous studies, enrolling 260 VPT infants in the NICU should yield at least 200 retained infants who will complete the final 3-year assessment visit and enable sufficiently powered analyses for our aims. We expect to enroll n=12/month. Prisma NICU admits 200-250 VPTs annually (~50% males; 56% Black/African American, 35% White, 9% Hispanic). We will monitor enrollment demographics to ensure our sample is representative of NICU hospital admission and our analyses will account for sex as a biological variable. Given the NICU capacity and admissions rate, our enrollment success rate (90% for NICU families), and our conservative plan to over-enroll by 30%, we will easily reach our target enrollment within 2yrs.

Procedures

All study procedures are outlined in PHS Human Subjects and Clinical Trials Information form. Consent will be obtained from parents before their infant is born or after birth up to 24 hours of age. Data will be collected on the primary variables of abdominal and foot skin temperature, and ECG for each infant included in the study until NICU discharge, transfer, or death. Demographic data, maternal/prenatal history, family history of ASD, and PT-related morbidity data will be collected from the EMR and follow-up questionnaires to be used as covariates (race, maternal education, GA) and moderators (sex, morbidities) in analyses. Family history of ASD will be assessed via parent interview and ASD screeners for any biological siblings. Developmental assessments will begin at 12 months, following a virtual touch point at 6 months with parents. This virtual touch point will take place over the phone and will include a brief medical history since NICU discharge, re-explanation of the study and study procedures, demographic update (contact information, maternal education, income), and visit scheduling. Follow-up data will be collected in participant homes at adjusted ages 12, 24 and 36 months.

Primary Variables/Measures

Neonatal ANS Function: Thermal Gradients. Abdominal (central) and foot (peripheral) temperatures will be measured continuously with a thermistor (disposable skin sensor, 499B, Cincinnati Sub Zero, Cincinnati, OH) covered by Mepitac Silicone adhesive tape.[62] These thermistors and tape are used for temperature monitoring in standard care. Thermistors record temperatures every minute using portable, high-precision dataloggers (Squirrel SQ data logger, Grant Instruments), resulting in a CPTd value every minute, as done in our previous studies.[61,62] Abnormal thermal gradients are either negative (CPTd < 0º C; foot temperature warmer than abdomen) or high (CpTd > 2º C; abdominal temperature much warmer than foot). Per each 24-hour period, days with < 80% of valid temperature data will be dropped. Our previous studies suggest this method results in < 10% data loss. **Heart Rate Indices: Pre-Processing.** HRIs will be derived from continuous ECG recordings from standard of care cardiopulmonary and HeRO monitors within the NICU (see *Letter of Support*). Continuous, beat-by-beat R-R intervals will be recorded from birth to discharge via the FDA-approved device HeRO Solo from Medical Predictive Science Corporation (MPSC). This same system was used for preliminary data reported here.[7] MPSC utilizes an FDA-approved, automated QRS detector to remove noise/artifacts and to detect R-waves; it has been validated against hand-annotated ECG waveforms with precision >99% and sensitivity >90%.[117,118] Further validation is evident in its use in the largest randomized controlled trial conducted among VPTs, where displaying HRIs based on R-R intervals generated by the MPSC QRS detector to clinicians resulted in a 20% reduction in all-cause mortality and a 40% reduction in mortality after infection.[119,120] In addition, our team has developed custom pre-processing pipelines that will further validate the MPSC-generated R-R intervals, as we have done in previous work.[108,110,121] Specifically, isolated missed beats will be detected using standard non-parametric outlier analysis and corrected through interpolation. Segments with >5 contiguous missing/edited beats will be rejected from analysis. Windows of 1h with >10% discarded data will be inspected and dropped as needed. This procedure will be validated and adjusted using a large dataset of manually cleaned ECG available from previous and ongoing projects run in Bradshaw's lab. **HRIs**. The following HRIs will be extracted: HRV, RSA, NN sample entropy, mean NN, NN standard deviation, and NN skewness.[67] To support comparison, we will also include four ECG-derived features that we used in a previous study,[108] namely, the root mean square of the difference between successive NN (RMSSD), the ratio of the two main axis of the ellipse fitting the scatter plot of NN vs preceding NN (i.e., Poincaré plot; SD1SD2), the coefficient of variation (std/mean) of NN (CVNN), and the HRV triangular index (HTI) calculated as the number of NN values in the modal bin of the normalized 8-ms-bin NN histogram. Continuous wavelet transforms (CWT) will be used to estimate RSA (see Table 2b below). We will complement our analysis of the relationship between ASD and ANS by refining our assessment of sympathetic and parasympathetic branches of this system. Toward this goal, we will investigate the relationship between ASD and the Sympathetic and Parasympathetic Activity Indices (SAI and PAI),[106] the Cardiac Sympathetic Index (CSI), and the Cardiovagal Index (CVI).[107,108] This extensive set of indices will be used specifically for ML prediction of ASD and NDI (Aim 1).

PT-associated Morbidities (birth to NICU discharge). Analyses for Aim 2 will use an established Morbidity Score previously adopted to study associations between ANS development and PT-related morbidities in the NICU.[20] This summative morbidity score quantifies the number of morbidities experienced at any time and includes the following factors: incidence of culture positive infections, NEC diagnosis, presence of PDA requiring medical or surgical intervention, severe brain injury (ultrasound readings of Grade III or IV intraventricular hemorrhage and/or periventricular leukomalacia), diagnosis of BPD or chronic lung disease, need for mechanical ventilation, and ROP at ≥ stage I (Table 1).[20] Morbidities will be extracted from the EMR weekly.

Social Drivers of Health (SDOH) and Early Infant Experience. Given the significant effect of SDOH, and the prenatal and postnatal environment on infant/maternal care and the developing brain,[122,123] the following measures will be accounted for in analyses. At consent (birth) and outcome (3 years), mothers will complete the 15-item, evidence-based, patient-centered tool PRAPARE (Protocol for Responding to and Assessing Patients' Assets, Risks, and Experiences).[87,88] PRAPARE measures SDOH across four domains: 1) personal characteristics (e.g., race, ethnicity), 2) family and home, 3) money and resources, and 4) social-emotional .health, and provides an overall Risk Tally Score that represents the number of social drivers of health risk (range: 0-22). PRAPARE is a recommended for assessing maternal unmet social needs as potential barriers to healthy prenatal and postnatal outcomes.[124] In addition, the following four measures (variables) will be collected on an ongoing basis to evaluate infant caregiving context and early life experience[125]: maternal perceived stress (any moderate to high score > 20),[126,127] depression (any score > 12)[128,129], infant health (number of days of infant post-discharge hospitalization), and infant developmental support (months of early intervention participation). Note that clinically significant/elevated levels of maternal stress/depression, as defined by the instruments, will be flagged immediately and caregiver will be contacted by the lab's clinical psychologist and/or MPI Bradshaw (licensed clinical psychologist) who will provide resources/referrals as needed (same protocol as ongoing R01).

ASD and NDI Outcome (36 Months): ASD Symptoms. The ADOS-2 will be administered at age 36 months adjusted and Calibrated Severity Scores (CSS)[130] for Social Affect (SA CSS) and Restricted and Repetitive Behavior (RRB CSS) will be used as dimensional measures of ASD outcomes in Aim 1-2. ADOSs will be administered by clinicians trained to research reliability and supervised by MPI Bradshaw and lab psychologist, with co-scoring to consensus occurring for at least 20% of the sample. CSS ranges from 1-10 and represents an excellent measure of ASD symptoms, relatively independent from IQ and language abilities,[130,131] that has been shown by our group and others to have sufficient variability for biomarker prediction.[132,133] For Aim 3, the Total CSS score will be used to compare derived clusters. **Neurodevelopmental Impairment**. Consensus is lacking on the definition of severe NDI outcome for PT populations.[134] But when assessing NDI, studies tend to use normative thresholds to define impairment (e.g., 1.5 or 2 standard deviations below the mean) for the following areas: motor, cognitive, language.[15,134] However, this dichotomous approach fails to capture subtle challenges and nuanced profiles in VPT infants.[135] This study will use a dimensional measure of neurodevelopmental impairment as the NDI outcome for Aims 1-2. In line with other studies of neurodevelopmental outcome in VPTs,[136] the Bayley Scales of Infant and Toddler Development, 4th Edition (Bayley-4)[137] will be administered at adjusted age 36 months, and the Cognitive, Motor, and Language standard scores (mean=100, SD=15) will be used as continuous measures of impairment. These scales explain a sizable proportion of later cognitive and academic outcome variability and are invaluable for characterizing early development and guiding intervention efforts.[138]

Emerging ASD and NDI (12-24 Months): ASD Symptomatology. Emerging ASD symptom profiles will use a direct measure of very early ASD symptoms, as has been used in our previous studies,[7,18,28,29,52,85] at 12 and 24 months. Specifically, we will use the Systematic Observation Rating of Flags for ASD (SORF), an observational measure of ASD red flags based on the DSM-5 framework that will be scored from the Communication and Symbolic Behavior Scales-Behavior Sample (CSBS).[139] The SORF has been found to distinguish children with ASD from children with NDI (but no ASD) and typically developing children, and our study team has documented the sensitivity and specificity of this measure for differentiating 12-month-old infants later diagnosed with and without ASD.[140] In infants 16-24 months of age, the SORF has large effects and good discrimination (area under the curve=.84-.87).[140] For Aim 3, the SORF Social Communication Domain score and the SORF RRB Domain score will be used to characterize emerging ASD symptoms. **Neurodevelopmental Impairment.** NDI will be measured dimensionally using standard scores from all 5 subscales of the Bayley-4: Cognitive, Fine Motor, Gross Motor, Receptive Communication, and Expressive Language. Scores will be adjusted for infants' GA. These subscales will be used in cluster analyses described in Aim 3.

Data Management

A HIPAA-compliant, USC-based, REDCap database will be created to collect all data in this study. The NICU RN will review and enter neonatal demographic and clinical data. Direct import from EMR into REDCap is not feasible as an RN is required to harmonize EMR data across individual patients to be suitable for analysis. Per our recent 5-site R01, this protocol involves clinical reading and interpretation of multiple tests results with varying response options. Co-I Everhart will train the RN to reliability in this process and will oversee all data harmonization procedures, as she has done for over 350 VPT infants in our multisite R01. The RN will download and store temperature data from dataloggers and ECG data from HeRO monitors every two weeks until the infant is discharged from the NICU. All data will be accessible over an encrypted and password-protected connection. The Data Manager at USC will clean temperature data and transfer data into SAS (9.4) datasets. Co-I O'Reilly's team will pre-process ECG data and compute HRIs (see above). Clinical data collected after NICU discharge will be stored on our secure server (videos) and entered in REDCap (assessment scores) following each study visit. Double data entry will be used to ensure data validity.

Data Analysis

Aim 1: Examine PT ANS dysfunction as a specific predictor of ASD symptoms at outcome. The analysis for Aim 1 will leverage ML by using a **support vector regression** from Scikit-Learn[141] to implement non-linear regression models that predict symptoms of ASD (Social and RRB CSS from ADOS-2) and NDI (Cognitive, Motor, Language standard scores from Bayley-4) at outcome from a set of features describing the preterm ANS and its dysfunction. **Table 2a** lists the measures that will be considered for this analysis. For each measure, four primary features will be extracted: 1st, 50th, and 99th percentiles, and the inter-quartile range for the measured values across the NICU stay. The slope of the relationship between the median value of these measures (averaged per 24-h period) and GA will also be considered. We will also investigate cross-modal temporal associations by calculating the correlation across time between all pairs of median measures (elevated per 1h-hour period) and Fisher Z-Transforming these correlations. We will add these correlations to the feature set if their intra/interparticipant variance ratio (estimated with bootstrapping) is statistically significant compared to chance-level assessed on a surrogate dataset obtained by randomizing the temporal order of the median measure time series. The correlation matrix between these features will be hierarchically clustered using the Seaborn *clustermap* function, and highly correlated features will be pruned to remove multicollinearity. Linearity of this relationship will be tested, and a power transform used if necessary. We will further evaluate the association of these features with demographic covariates (race, sex, maternal education, GA) and, if statistically significant, the effect of these covariates will be regressed out of the features.

Table 2a. ANS Measure Descriptions for Aim 1 Machine Learning

Measures	Modality	Interpretation
% time CPTd < 0^0 C	Temperature	Peripheral body warmer than core body, indicative of ANS dysfunction or immaturity[59,123]
% time CPTd > 2^0 C	Temperature	High core to peripheral temperature indicative of Sympathetic activity, seen with sepsis and stressor activity in PT infants[89,124]
SAI	Heart Rate	Index of sympathetic activity
PAI	Heart Rate	Index of parasympathetic activity
CSI	Heart Rate	Index of sympathetic activity
CVI	Heart Rate	Index of parasympathetic activity
HRV	Heart Rate	Measured as the natural logarithm of RMSSD. Decreased values indicate ANS dysfunction and are characteristic among individuals with ASD[125]
Mean NN	Heart Rate	Increased with overactive sympathetic and decreased parasympathetic or vagal tone.[16] Highly predictive features for likelihood of ASD according to previous study.[99]
NN standard deviation	Heart Rate	Spread of the NN distribution. Index both sympathetic and parasympathetic activity.[126]
NN skewness	Heart Rate	Transient HR decelerations (bradycardia) result in negative skewness (parasympathetic). Transient HR accelerations (tachycardia) result in positive skewness (sympathetic).
NN sample entropy	Heart Rate	Decreased values are thought to be associated with a decoupling of the ANS from external inputs (i.e., non-responsiveness, non-adaptability).[127,128]
RMSSD	Heart Rate	Parasympathetic activity; vagally mediated variation in heart rate.[129]
SD1SD2	Heart Rate	Reflect the balance between short-term vagal activity and sympathetic modulation.[129]
CVNN	Heart Rate	STD(NN) captures both low and high-frequency components of the heart-rate signals. High-frequencies variations are due to RSA (parasympathetic). The normalization of STD(NN) by MEAN(NN) makes CVNN more stable when comparing individuals or conditions with different baselines.[129]
HTI	Heart rate	Depressed HTI is a robust index of sympathovagal imbalance.[130]
RSA	Respiration & Heart Rate	Periodic slowing and speeding up of heart rate with the respiratory cycle. Higher RSA indexes more vagal flexibility, which promotes balanced parasympathetic and sympathetic influences to the heart.

Table 2b. RSA Calculation

RSA Calculation Method: Continuous Wavelet Transform (CWT)	
Step	**Relevant References**
1. Transform R-R (beat-by-beat) series into a sample-rate (10Hz) time series, then use "cwt" function of MATLAB. The output from cwt is all the **wavelet-transformed IBI series,** that is the IBI time series transformed by the wavelet filterbank.	*Parameters used*: "analytic Morlet", "amor", and 48 "voices per octave" (# of wavelets). *CWT Analysis and RSA*: Grossmann & Morlet, 1984; Cnockaert et al., 2008; Wachowiak et al., 2016
2. Use inverse CWT ("icwt") with frequency bands set to the min and max of respiration, determined by age-appropriate norms. The output of icwt is the **RSA time series** for the respiration frequency chosen for this age.	*Respiration rate in infants*: Gagliardi & Rusconi, 1997; O'Leary et al., 2015; Rusconi et al., 1994; Tveiten et al., 2016
3. To be consistent with established RSA calculation methods (PB-MPF), use log(var(icwt time series)) to **calculate RSA**.	*Porges-Bohrer Moving Polynomial Filter*: Porges, 1982; Porges & Bohrer, 1990
Advantages. Compared to other filters, the CWT approach results in higher resolution for frequencies, better drop-off, and is more modifiable.	

Social Drivers of Health (SDOH) and Early Infant Experience. Further, we will test for significant associations between these features and relevant SDOH (PRAPARE Risk Tally Score), and early infant experience (maternal stress, maternal depression, infant health, infant intervention) variables. If these associations are statistically significant, the effect of these covariates on the features will be regressed out. We recognize the complex, interconnected, and multifaceted nature of demographics, SDOH, and early infant experience, all of which cannot be fully and meaningfully captured in a single score, while also acknowledging the possibly high number of influential variables and the collinearity therein, which raises significant analytic challenges.[142] Thus, our analytic plan for Aims 1-2 will account for these summative variables, while also allowing us to further probe individual items in subsequent analyses.

We will then use permutation feature importance techniques developed in the context of explainable artificial intelligence and implemented in Scikit-Learn[141] to identify which features contributed the most to this prediction. Since most of our features can be interpreted in a mechanistic way (e.g., see Table 2a), this analysis will allow us to confirm and refine our hypotheses on the role of ANS dysfunction in the emergence of ASD and NDI. All ML analyses will be conducted using nested subject-level-cross-validation (train: 70%; test: 15%; validation: 15%) to allow reliable and reproducible model fitting and hyperparameter tuning. The effect size for ANS as a predictor of outcomes at 3 years of age will be evaluated as model's R^2 values (i.e., coefficients of determination) and their statistical significance will be assessed by looking at whether their bootstrapped 95% confidence intervals overlap with 0. Since model assessment is cross-validated, negative R^2 values and confidence intervals overlapping with 0 are expected in the absence of a true relationship.

Aim 2: Evaluate preterm morbidities and sex as potential moderators in the association between PT ANS dysfunction and symptoms of ASD and NDI at outcome. While Aim 1 (above) leverages machine learning, the statistical analysis for this Aim 2 relies on a **classical moderation modeling** where the association between PT ANS dysfunction (independent variables; listed in the next paragraph) and symptoms of ASD and NDI at outcome (dependent variables) will be tested for **moderation by sex and a previously validated preterm morbidity index,**[20] as defined in the PT Morbidities section. The linearity of this relationship will be tested, and a power transform will be used if necessary. The correlation matrix between independent features will be analyzed and highly correlated features will be discarded to avoid multicollinearity issues. If a significant moderation effect is found for the morbidity index, we will conduct post hoc moderation analyses to determine which of its constituting factors (severe brain injury,[143] infection, or NEC[145], BPD[75,146] or need for mechanical ventilation,[147] ROP,[149] and/or PDA) contribute to the moderation. Python (StatsModels Mediation class), which supports both mediation and moderation analyses, will be used.

We will use the same dependent variables as those used in Aim 1 (i.e., ADOS-2 CSS and Bayley-4 Scales). Our analyses will align with our preliminary investigations[7] and will characterize ANS function during PT periods using 1) the proportion of days during which negative thermal gradients are present > 30% of the time, 2) the proportion of days during which CPTd > 2º C are present > 10% of the time, 3) IBI skewness (based on its key role in HRI predictors used in our previous work[53] and the work of others[67]), and 4) a small set ($\leq$ 3) of features that will be shown to have the highest predictive power in Aim 1. By using a combination of ANS measures that are determined based on preliminary data and that are highly predictive according to the analyses of Aim 1, we ensure that Aim 2 does not significantly depend on Aim 1. We will correct for multiple independent tests using the Bonferroni method. We will first perform these analyses using average values (e.g., rates and frequencies of ANS measures; no repeated measures). Then, we will add the effect of GA by considering 24-h averages for the independent variables and replace the fixed effect linear moderation by a mixed-effect one, including both the intercept and slope of the effect of GA. For this second analysis, we will test if GA needs to be transformed (e.g., power transform) prior to its use for linear modeling and evaluate the effects of potential covariates, including SDOH and early infant experience, as listed above.

Aim 3: Characterize and differentiate the emergence of ASD symptoms and NDI in VPT infants. We will characterize the development of VPT infants using direct measures of ASD symptoms and NDI administered at ages 12 and 24 months. Given the developmental heterogeneity observed in both infants with ASD and VPT infants, we will use both SORF subdomains (Social Communication and RRB Domain scores) to characterize ASD symptoms and all subtests of the Bayley-4 to characterize NDI (Cognitive, Fine Motor, Gross Motor, Receptive Communication, Expressive Communication). We will use a supervised clustering approach to identify subgroups of developmental trajectories that differ in terms of the emergence of ASD symptoms and NDI and that are predictive of the outcome at 36 months. To better capture the trajectories of autonomic ASD characteristics, and considering that we have two time points, we will consider the mean value across time points and the slope between time points for the two ASD and five NDI subscales, for a total of 2 x 7=14 features. The supervised approach will compute SHAP (Shapely Additive exPlanation) values associated with the features used for an XGBoost regression predicting the ADOS CSS Total score at 36 months (5-fold cross validation).[150] The SHAP values will then be used as input to the unsupervised The DBSCAN algorithm for clustering.[151,152] This approach will be implemented using Scikit -Learn[141] and the SHAP Python package.[153] Although the nonlinear XGBOOST and DBSCAN algorithms are not vulnerable to multicollinearity issues, we will use a Principal Component Analysis to evaluate the proportion of variance captured by every component and reduce redundancy in the feature set, if necessary. The Silhouette Coefficient[154] will be computed to assess the qualify of the clustering, the t-distributed Stochastic Neighbor Embedding approach[155] will be used to visualize the separability of the clusters, and the stability of the clustering will be assessed through bootstrapping.[156,157]

By using a supervised clustering approach based on the outcome, we maximize our chances of obtaining stable clusters that group trajectories predictive of different outcomes. These clusters will, therefore, support the possibility of using early symptom trajectories to predict ASD diagnosis. Clusters will be visualized as seven subplots (one per ASD and NDI subscales) with the time points along the x-axis and the cluster mean values in y-axis with bootstrapped 95% confidence intervals as obtained from Python Seaborn *lineplot* function. We will test whether extracted clusters form statistically significant subgroups by testing differences in centroids using a Hotelling T-squared test. We will validate the clinical significance of extracted clusters by assessing whether subgroups differ based on 36-month ASD outcome (ADOS CSS) using t-tests, adjusted for multiple comparisons.

Sample Size and Statistical Power Considerations. Aim 1 uses a ML-based regression approach to establish the statistical and clinical significance of the relationship between PT ANS abnormality and ASD/NDI outcomes. For this aim, the R^2 values for regression model fitting can be seen as the squared values of Pearson's coefficient of correlation which can be used for power analysis. With N=200, we can expect to detect a correlation of at least 0.20 with a 20% false negative rate (i.e., β=0.2) and a 5% false positive rate (i.e., α=0.05). This coefficient of correlation corresponds to a 0.04 R^2 value, which can be interpreted as the percentage of variance explained. Once corrected for 7 independent tests (i.e., 7 subscales, Bonferroni correction), this R^2 increases to about 0.0625. Therefore, we consider this sample size sufficient, as effects that cannot explain reliably more than 6.25% of the total variance in the final diagnosis are probably of low clinical significance. For **Aim 2**, we used G*Power[158] to determine the minimal sample size required to have a 0.80 power (i.e., β=0.2) to test for moderation (two moderators: medical morbidity index and sex) in a model including six predictor main effects and their centered interaction with moderators, and using an α=0.01 (i.e., 0.05 Bonferroni-corrected for testing 5 dependent variables). For a f^2=0.25 (considered a medium effect size), we will need N=130, which demonstrates that we have a large enough sample to run our main analysis and expect to obtain significant results if a clinically relevant effect exists. Our larger sample size (N=200) affords us extra maneuvering margin in case of smaller effects and for integrating co-variates (race, GA, and maternal education). For **Aim 3**, we will be able to detect significant differences in multivariate means between pairs of clusters each grouping 50 participants with β=0.2, α=0.05, and 14 response variables if the effect size has at least a Mahalanobis distance (a multivariate generalization of the Cohen's d) of 0.92, implying that we will be able to establish that these clusters are different only for relatively large effect sizes. If significant effects cannot be detected with our sample, we will increase the statistical power of this test by reducing the number of clusters (hence increasing the number of participants per cluster) and reducing the number of response variables (i.e., feature selection). Similarly, we will be able to detect effects having a Cohen's d of at least 0.57 for differences in mean ASD symptoms (Total CSS) between pairs of clusters with N=50 each (power=0.8; α=0.05; two-tailed).

Potential Challenges and Alternative Solutions. Many of the potential problems and alternative solutions have been uncovered in our previous studies, and effective solutions have always been implemented. As in the past, all potential issues will be discussed with study team members and immediately addressed as needed and as outlined in the MPI plan. All-investigator meetings will occur biannually to review progress, identify challenges, and generate solutions. **Enrollment**: Due to the ebb and flow of Prisma Richland NICU enrollment, we have adopted conservative enrollment quotas with a minimum of 12 infants per month (~50% of their annual VPT admissions), which allows us to over-enroll by 30% to account for attrition and loss to follow-up. For further backup, we have another NICU in the Prisma system in Greenville, SC that we can use as a second site for enrollment. Thus, we are confident we will meet our enrollment quotas for this study. **Follow-up Assessments**. We have built in a 6-month virtual visit (and incentive) to retain infants from discharge to the first follow-up ~1 year later. All follow-up assessments will be completed in the families' homes at their convenience to increase research accessibility. This strategy has greatly increased participation in our current studies, but in-lab visits are still available if desired. We will also access our Community Advisory Board as needed to overcome any barriers. All neonatal morbidities and severity will be recorded and could be used for Aim 2 in case the morbidity score shows limited variability. Later morbidities that may impede diagnostic determination (e.g., severe motor delays) will be evaluated in all-investigator meetings to determine course of action for inclusion in analyses.

ADVANCES IN CHILD DEVELOPMENT AND BEHAVIOR

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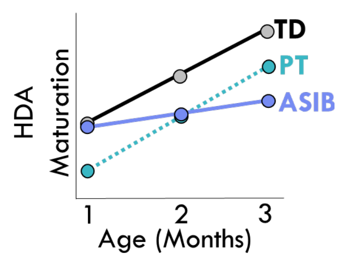
SPECIFIC AIMS

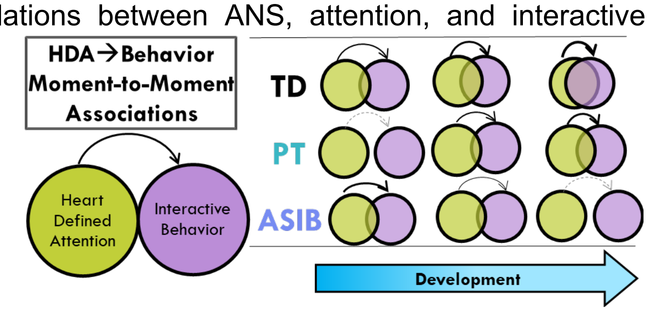
Atypical attention is among the earliest-emerging, lifelong features of autism spectrum disorder (ASD) and a particularly compelling candidate for an ASD biomarker[1–4]. Our work, and that of others, suggests that disrupted attention may be detectable and developmentally predictive in early infancy[5–8], yet the biological underpinnings of this disruption remain unknown. The autonomic nervous system (ANS) is an early-emerging bioregulatory system critical for supporting neonatal arousal and infant attention[9–11]. Early disruptions in ANS's regulation of attention ("autonomic regulation of attention") may have cascading consequences for interaction, learning, and development[8,9,11]. Broad ANS dysfunction is observed in older children already diagnosed with ASD[12], but there remains a tremendous gap in knowledge regarding **whether** and **when** ANS dysfunction contributes to attention abnormalities and how infant autonomic regulation of attention predicts the emergence of ASD symptoms.

The current study fills this gap by examining autonomic regulation of attention in the neonatal period and the developmental consequences therein for the emergence of ASD symptoms in infant siblings of children with ASD (**ASIBs**), infants born preterm (**PTs**), and typically developing (**TD**) infants. A critical innovation of this study is leveraging a comparison group of PTs who, intriguingly, experience global ANS dysfunction from birth[9,13–15]. While PTs show slightly elevated rates of ASD (~7%) compared to the general population (~2%)[16], the vast majority of PTs do *not* develop ASD. In this study, **inclusion of PT infants as a model for global ANS dysfunction** will help investigate our hypothesis that early and broad ANS dysfunction (not specific to attention) *does not* lead to ASD-specific symptoms. This developmental process (broad ANS dysfunction → non-ASD specific outcomes) will be contrasted to **ASIBs** who experience significantly higher rates of ASD (~20%)[17]. Contrasting **PTs** to **ASIBs** will help delineate how *attention-specific* ANS dysfunction predicts the emergence of *ASD-specific* symptoms. Thus, our innovative **three-group research design** will compare **ASIBs** to two control groups (**PTs** and **TD** infants) to determine the specificity of autonomic regulation of attention as a key, early emerging feature associated with ASD, with incredible translational potential for a cost-effective biomarker.

Aim 1. Compare the maturation of autonomic regulation of attention in the early infant period (1-3m) across ASIBs, PTs, and TDs. Attention in the first months of life is supported by flexible modulation of arousal and alertness, regulated by the ANS[18–20]. Autonomic regulation allows for attention to salient features of the environment and inhibition of responses to irrelevant (internal or external) stimuli. ANS regulation of attention is vital for learning and can be measured using **heart-defined attention (HDA)** (i.e., coordination of heart rate changes within moments of visual attention[21,22]). Maturation of ANS and attention systems across the first months of life lead to increasing coordination of heart rate with attention. Our pilot work with ASIBs shows differences in HDA quantity (%time in HDA) and quality (magnitude of heart rate deceleration)[23,24]. Building upon this work, Aim 1 will test the hypothesis that **ASIBs** show **typical HDA initially** compared to **TDs**. Then from 1-3m, as ANS commands an increasingly larger role in regulating attention, the development of HDA for **ASIBs** will attenuate. In contrast, **PTs** will show **delayed HDA maturation** from 1-3m (adjusted age) due to broad ANS dysfunction[25].



Aim 2. Determine the moment-to-moment influence of autonomic regulation of attention on interactive behavior across the first year (6m, 9m, 12m). Autonomic regulation of attention, measured via heart activity and contextualized within moment-by-moment behavior, has extraordinary potential to transform early identification of ASD by revealing specific mechanistic relations between ANS, attention, and interactive behavior. Our preliminary data suggests: (1) temporal coordination between HDA and interactive behavior in **TDs** and (2) reduced HDA-behavior coordination in **ASIBs**. Aim 2 will test the hypothesis that **ASIBs** show weakening predictive associations between HDA and interactive behavior from 6-12 months, as early symptoms of ASD emerge[26]. In contrast, we expect **PTs** to show initially weak associations that strengthen over time (with ANS and behavioral maturation)[25].



Aim 3. Predict ASD symptoms at age 3 years from infant autonomic and attentional features. ASD biomarker discovery requires quantitative modeling techniques that adequately capture the complexity of biological processes involved in such a multifaceted, heterogeneous neurodevelopmental disorder[27,28]. Aim 3 will leverage machine learning techniques to develop a biomarker predicting the later emergence of ASD symptoms based on early (1-12 months) features of the autonomic and attentional systems (variables from Aims 1-2). We hypothesize that features related to infant autonomic regulation of attention will predict ASD symptoms.

CHAPTER SEVEN

The role of parenting in children's math learning: A dual-pathway model

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Contents

Abstract

Parents' involvement in children's math learning is a critical, yet underutilized, resource. Drawing on cognitive and motivational perspectives, this chapter introduces a dual-pathway model highlighting the importance of parents' cognitive and motivational math parenting practices in fostering children's math learning. Cognitive practices include the content, level, and structure of parents' math talk and gestures. Through these practices, parents introduce numeracy concepts and scaffolded support within children's zone of proximal development, thereby providing cognitive resources for the development of children's math skills. Motivational practices include parents' autonomy-support (vs. control), positive (vs. negative) affect, and process (vs. person)-orientation around math. These practices fulfill children's basic psychological needs, which develops children's math beliefs, motivation, and engagement. Empirical evidence demonstrates that parents' cognitive and motivational practices each contribute uniquely—and in an additive fashion—to children's math learning, with the possibility of interactive effects. Proximal antecedents (e.g., attributes of children, parents, and the learning context) can shape both types of practices. By delineating distinct yet complementary mechanisms of parents' influence, the dual-pathway model offers a unified framework to inform future

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research, educational practices, and family-focused interventions aimed at enhancing children's math learning.

Keywords: Learning, Math, Motivation, Parent involvement, Parenting

1. The role of parenting in children's math learning: a dual-pathway model

There is much evidence that parents play a significant role in children's learning (for a recent review, see Grolnick & Pomerantz, 2022). In fact, parents have been deemed an underutilized resource in fostering children's math learning (e.g., Harackiewicz et al., 2012), which is declining in the United States (National Science Board & National Science Foundation, 2024). Much of the theory and research on the role of parenting in children's math learning has focused on the idea that parents can advance children's development of math skills prior to formal schooling by providing cognitive input (e.g., Gunderson & Levine, 2011). In this work, parents' provision of cognitive resources (e.g., numerical concepts) is the central pathway by which parents can support children's math learning, with much evidence that indeed parents' cognitive practices (e.g., relatively advanced math talk) matter (e.g., Eason et al. 2021; Ramani et al., 2015). A large body of theory and research, however, indicates that parents can provide children with motivational resources (e.g., feelings of autonomy) that foster their learning (for a review, see Pomerantz, Kim, & Cheung, 2012). Significantly, parents' motivational practices (e.g., supporting children's autonomy in solving math problems) have recently been identified as contributing to children's math learning (e.g., Oh et al., 2022; Wu et al., 2022). Thus, the cognitive pathway may not be the sole pathway by which parents shape children's math learning (Wu et al., 2024).

In this chapter, we introduce a dual-pathway model of the role of parenting in children's math learning by integrating the theory and research on parents' practices assumed to afford cognitive resources with that on their practices assumed to afford motivational resources. In brief, as shown in Fig. 1, although a central pathway by which parents can contribute to children's math learning is via their cognitive practices, they can also contribute via their motivational practices. The cognitive and motivational pathways provide distinct resources to children that can together optimize children's math learning over the course of their development. In line with ecological theories of development in which children, parents, and

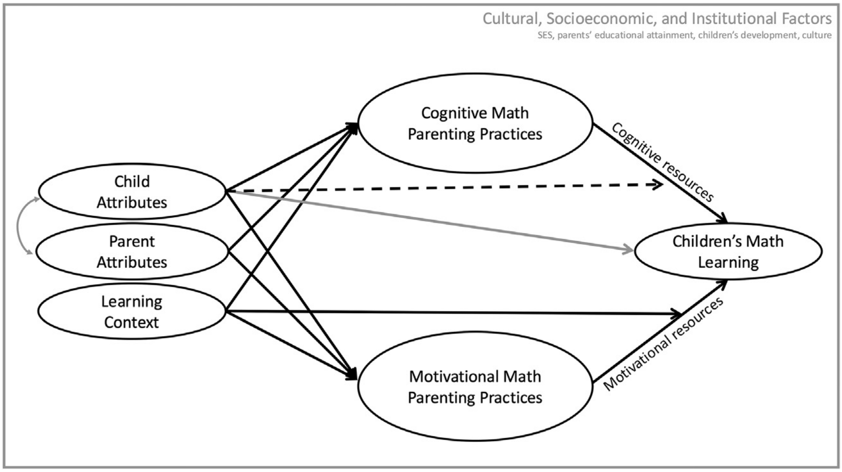

Fig. 1 The dual-pathway model of the role of parenting in children's math learning. *Note.* The dashed arrow indicates that there is no empirical evidence for the link. The gray arrows, lines, and text reflect processes that are present in the socialization process but not discussed here due to significance or a lack of theoretical and empirical work.

the multiple social environments in which they reside contribute to the socialization process (e.g., Belsky, 1984; Brofenbenner & Morris, 2006), the model includes forces at multiple levels that may shape parents' cognitive and motivational practices and thus children's math learning. In delineating the model, we begin by discussing the cognitive and motivational resource parenting pathways. We then turn to the most proximal forces (i.e., attributes of children, parents, and the learning context) that may shape the two pathways. We end with a discussion of key directions for future research and implications for educators.

2. Cognitive and motivational resource parenting pathways

The central idea behind the dual-pathways model is that both parents' cognitive and motivational practices contribute to children's math learning by providing distinct resources to children that build distinct strengths. In brief, parents' cognitive practices in the math context (for types and definitions, see Table 1) convey or support engagement with math concepts among children. Consequently, such practices may directly develop children's math skills. Parents' motivational practices in the math context (for types

Table 1 Cognitive math parenting practices: types, definitions, and evidence.

Type	Definition	Evidence Summary
Math talk	Verbal references to math concepts, such as counting, number recognition, and problem-solving.	There is a small positive association between parents' math talk and children's math learning (for reviews, see Daucourt et al., 2021; Silver et al., 2024), but findings are limited and vary across studies (e.g., Levine et al., 2010; Ren et al., 2022; Thippana et al., 2020).
Level of math talk	The complexity of the math content or strategies discussed with children, ranging from basic concepts to more advanced concepts relative to what children are learning.	Advanced math talk, within children's range of learning, predicts enhanced math abilities and engagement (e.g., Casey et al., 2018; Gunderson & Levine, 2011; Ramani et al., 2015; Wu et al., 2024).
Structure of math talk	Prompts (i.e., questions, directions, and hints that encourage math engagement and problem-solving) or statements (i.e., providing information, facts, and instructions related to math concepts) about math.	Parents' prompts, particularly advanced prompts, predicts more advanced math talk, engagement, and skills among children. Parents' statements have less positive, and event negative, effects (e.g., Eason et al., 2021; ; Kurkul & Corriveau, 2018; Wu, Oh, Hyde, & Pomerantz, 2024)
Math gestures	Use of hand movements to illustrate or represent math concepts during interactions with children.	Gestures used in combination with math talk is associated with children's enhanced math engagement with key math concepts (i.e., cardinality principle, counting strategy; e.g., Congdon et al., 2017; (Goldin-Meadow et al., 2012); (Gibson, Gunderson, & Levine, 2020)).

and definitions, see Table 2) allow children to meet basic psychological needs, such as autonomy and relatedness, which develop children's math beliefs, motivation, and engagement. Importantly both sets of strengths—that is math skills and math beliefs, motivation, and engagement—appear to contribute to children's math achievement, at least by the end of the

Table 2 Motivational math parenting practices: types, definitions, and evidence.

Type	Definition	Evidence summary
Autonomy support (vs. control)	Autonomy support includes offering choices, encouraging independent problem-solving, and providing guidance only when needed. Control entails pressure, excessive direct instruction, problem-solving on behalf of children, or coercive tactics to ensure compliance.	Parents' autonomy support (vs. control) predicts higher math achievement in children (e.g., Cimon-Paquet et al., 2020; Oh et al., 2022; Silinskas & Kikas, 2019a).
Affect	Emotional tone ranging from positive (e.g., happiness, joy, and love) to negative (e.g., irritation, anxiety, and sadness).	Parents' positive (vs. negative) affect predicts heightened math motivation,engagement, and achievement (e.g., Silinskas & Kikas, 2019b; Wu, Barger, Oh, & Pomerantz, 2022).
Responses to children's perfor-mance	Process-oriented responses focus on emphasizing internal, controllable factors such as effort and strategy use. Person-oriented responses highlight innate ability or intelligence, implying that performance reflects stable, inherent traits.	Process-oriented responses predict improved math achievement (e.g., Gunderson et al., 2018). Person-oriented responses predict children's heightened math anxiety, dampened preference for math challenge, and lower math achievement (e.g., Barger et al., 2022).

elementary school years and continuing into middle and high school years (e.g., Jordan et al., 2009).

2.1 Cognitive parenting practices in the math context

The sociocultural theory, originally proposed by Vygotsky (1978), posits that social interactions with more knowledgeable others, such as parents, play a critical role in children's cognitive development (Gauvain, 2001; Rogoff, 1990). Parents' use of language has been emphasized in these interactions as the "cultural tools" that help children make meaning and develop abstract concepts (Vygotsky & Cole, 1978, p. 123). As such, parents' math talk (i.e., verbal reference to a variety of concepts involved in math) features prominently in the sociocultural perspective. It has been argued that through such talk parents provide information about key math concepts that facilitate children's math learning (e.g., DePascale et al. 2021). In addition, borrowing from the bootstrapping account in the language acquisition area (e.g., MacWhinney, 2014), Gunderson & Levine (2011) suggest that parents' cognitive input, often via their math talk, matters for the development of children's math skills (e.g., numerical knowledge and vocabulary). This aligns with broader cognitive theories of learning, which postulate that more frequent exposure to numerical concepts helps children build mental representations that gradually refine their math understanding over time (e.g., Sweller, 1988).

Despite the theoretical importance of parents' math talk, research on its impact has been mixed. Some studies support the notion that frequent exposure to early math talk enhances children's math skills by the time they enter school (e.g., Levine et al., 2010; Susperreguy & Davis-Kean, 2016), whereas others do not (e.g., Ren et al., 2022; Thippana et al., 2020). Evidence from meta-analyses also reveals limited support for this association. Daucourt et al. (2021) meta-analysis of nine studies with children prior to entering formal schooling reported a near-zero association ($r = 0.03$) between parents' math talk and children's math achievement. A more recent and comprehensive meta-analysis (Silver et al., 2024) including 22 studies of the effects of parents' math talk with children from two to seven years old, found a small, but statistically significant, effect ($b = 0.10$, $SE = 0.03$, $p = 0.002$). When children's general cognitive abilities or prior math performance were taken into account, however, this effect was reduced in size becoming non-significant. These findings suggest that simply engaging in math talk frequently may not be enough.

Indeed, consistent with Vygotsky's (1978, 1986) ideas on scaffolding, the content of parents' math talk matters. More advanced content within the range of children's developing skills pushes children toward what Vygotsky (1986) called the "zone of proximal development" (p. 86). The cognitive perspective also suggests that exposure to relatively advanced concepts can motivate children to understand more cognitively demanding numeracy concepts (e.g., Gunderson & Levine, 2011). In line with this view, it appears that parents' more advanced numeracy talk, within the range of children's developing skills, benefits children's math learning (e.g., Casey et al., 2018; Ramani et al., 2015). For example, Gunderson and Levine (2011) found that during observations of natural parent-child interactions when children were one to two-and-a-half years old, the amount of numeracy talk parents devoted to sets (i.e., counting or labeling sets of present objects) consisting of higher (i.e., from 4 to10), but not lower (i.e.

from 1 to 3), numbers of items, was associated with four-year-old children's cardinal-number knowledge, even after adjusting for parents' socioeconomic status. Similarly, parents' focus on more advanced numerical concepts—such as cardinality, ordinal relations, and arithmetic—is associated with stronger numerical reasoning and magnitude comparison skills among preschool children, over and above parents' other teaching practices (Ramani et al., 2015).

Although the level of parents' math talk is important, Vygotsky's (1978, 1986) ideas on scaffolding suggest the structure of such talk also matters. In this regard, recent research guided by the sociocultural perspective has examined parents' prompts (i.e., questions, directions, and hints, often to encourage children's engagement) and statements (e.g., provision of information, facts, and instructions, often to convey relevant concepts) in the context of their math talk with children (e.g., Eason et al. 2021; Wu et al., 2024). Drawing from Vygotsky (1986) and Rogoff (1998), Eason et al. (2021) make the case that prompts may be more likely than statements to push children into the zone of proximal development by providing the opportunity for children's more advanced engagement. Parents' prompts may nudge children toward active engagement in the activity at hand as they often require a response that relies on exploration and operations (e.g., Vandermaas-Peeler et al., 2016). Parents' statements, in contrast, may provide useful input on numerical concepts, but children's reception may be relatively passive with minimal exploration and operations on the concepts (Wu et al., 2024). Similarly, Sigel et al., (2002) "distancing strategies" framework highlights that cognitively demanding prompts—such as those requiring children to predict, deduce, or reason through problems—are particularly effective in advancing learning.

The empirical evidence is consistent with these ideas in that it shows that parents' prompts, but not necessarily their statements, are linked to advanced math engagement. For example, studying mothers and their two- to four-year-olds, Eason et al., (2021) found that during activities such as pretend cooking, brick building, and math story reading, the more parents used both prompts and statements, the more children engaged in math talk, but only prompts were linked to greater diversity in children's numerical vocabulary. A similar pattern has been observed among parents and their children in early elementary school. Using a minute-to-minute within-dyad analytic approach, Wu et al., (2024) showed that when both the level and structure of parents' math talk (i.e., prompts and statements) were examined simultaneously, parents' advanced prompts, but not statements, were associated with children's subsequent engagement in math strategies, which were often advanced as well, even after taking into account children's engagement the prior minute.

Parents' provision of cognitive resources often extends beyond verbal input to include non-verbal elements such as gestures. From a cognitive perspective, gestures play a fundamental role in shaping children's math understanding by providing an additional modality for processing math concepts (e.g., Butterworth, 1999; Fuson, 1988). For example, Fuson (1988) makes the case that gestures can function as a cognitive bridge between verbal number knowledge and abstract numerical representations, reinforcing key counting principles such as one-to-one correspondence and stable order. Pointing gestures, for instance, can help children track number words during counting, ensuring accurate item-to-number mapping and supporting early math development (e.g., Wiese, 2003). In parent-child interactions around math, gestures frequently accompany math talk and act as a supplementary channel of cognitive input, enhancing the communication of both foundational and advanced math concepts (e.g., Flevares & Perry, 2001; Goldin-Meadow, 2009). Although research on parents' gestures is still emerging, early findings suggest they provide symbolic cognitive support, improving eight- to nine-year-old children's comprehension of math concepts (e.g., Congdon et al., 2017; Wakefield et al., 2019).

2.2 Motivational parenting practices in the math context

Parents' involvement in children's math learning can also confer benefits on children by providing them with motivational resources that foster their math learning (e.g., Oh et al., 2022; Wu et al., 2022, 2024). Self Determination theory (SDT; Deci & Ryan, 1985; 2000) posits that people have three fundamental psychological needs from an early age: autonomy, competence, and relatedness. Parents' involvement in children's learning process is an important context for meeting these needs and can provide children with the motivational resources that facilitate math learning (e.g., Grolnick et al., 1991; Grolnick & Pomerantz, 2022). Parents' involvement is especially beneficial when parents adopt practices that are autonomy supportive (vs. controlling), affectively positive (vs. negative), and process (vs. person) oriented (for a review, see Pomerantz, Kim, & Cheung, 2012).

Parents' involvement can be characterized as autonomy supportive, in that parents provide children with choices, allow them to take initiative, and adopt children's perspectives. The more parents' involvement is autonomy supportive, the more children feel agentic and competent, feelings that ultimately contribute to their intrinsic motivation (e.g., Lerner et al., 2022; Ryan & Deci, 2000). In the math context, parents can be autonomy supportive by letting children lead the way when solving problems and only providing hints when children ask for help, allowing children to feel autonomous and capable of doing math on their own. Parents' involvement can also be characterized as controlling in that parents pressure, direct, and restrict children (e.g., Grolnick & Pomerantz, 2022; Grolnick et al., 1991). Controlling parenting has been argued to undermine children's academic adjustment, as it dampens children's feelings of autonomy and competence (e.g., Grolnick et al., 1991; Grolnick & Pomerantz, 2009). Parents' control in the math context may manifest as directing children on how to solve problems or intervening early in the problem-solving process, which may lead children to feel helpless and incapable when it comes to math, thereby fostering their disengagement from math.

Although there is much evidence that parents' autonomy support (vs. control), contributes to children's motivation, engagement, and learning in general (e.g., Cheung et al., 2016; Grolnick et al., 2002), only recently has research examined the effects of such parenting in the math domain specifically. The evidence to date indicates that the more parents are autonomy supportive and the less they are controlling in children's math learning, the higher elementary and middle school children's perceptions of their math competence, persistence with math homework, and math achievement (e.g., Silinskas & Kikas, 2019a; Cimon-Paquet et al., 2020). For example, Wu et al., (2024) found that the more autonomy supportive (vs. controlling) parents were in working on math activity with their early elementary school children one minute predicted enhanced engagement in the activity among children the following minute adjusting for their engagement the prior minute. This may be especially true for children struggling with math as Oh et al., (2022) found that parents' autonomy support predicted increases over time in children's math achievement largely among children who were initially low achieving in math.

Parenting is thought to be imbued with affect (Dix, 1991), as parents' involvement in the learning process (e.g., helping with homework) can be enjoyable as well as frustrating for both children and parents (e.g., Pomerantz et al., 2005a; Wu et al., 2022). On the one hand, parents' involvement can be affectively positive, in that parents display positive emotions such as happiness and joy. This affect may help children experience and view math learning positively, which may foster beliefs, motivation, and engagement that support math achievement (e.g., Else-Quest et al., 2008;

Pomerantz et al., 2005a). On the other hand, parents' involvement can be affectively negative, as characterized by their display of negative emotions such as irritation and frustration, which may communicate to children that math is unpleasant and should be avoided, potentially diminishing children's beliefs, motivation, and engagement (e.g., Nolen-Hoeksema et al., 1995; Pomerantz et al., 2005a).

When parents' involvement is more affectively negative (vs. positive), elementary school children are more likely to exhibit helplessness and a weaker inclination to take on challenges (e.g., Estrada et al., 1987; Nolen-Hoeksema et al., 1995). For example, mothers' negative affect when assisting with eight- to 12-year-old children's homework predicts children's dampened motivation six months later; this effect, however, was reduced when mothers also reported greater positive affect when assisting children with homework (Pomerantz et al., 2005a). Research in the math domain is limited, with evidence that parents' negative, but not positive, affect contributes to children's math learning. Namely, Wu et al., (2022) found that parents' negative affect when helping first and second graders with math homework predicted children's dampened math liking, preference for math challenge, and achievement one year later, above and beyond children's prior math adjustment and parents' educational attainment.

Parents' responses to children's performance also can play a role in children's math learning. Some parents focus their responses on the process of learning, in that they emphasize effort and strategy, whereas others rely on person-oriented responses that focus on innate attributes of children (e.g., their math ability; for reviews, see Henderlong & Lepper, 2002; Pomerantz, Kim, & Cheung, 2012). Process-oriented responses encourage children to persevere through challenges and build their skills, thereby enhancing children's motivation and engagement. In contrast, person-oriented responses can have the opposite effect on children's learning. Given that math is often viewed as requiring innate talent (Leslie et al., 2015), parents may view children as either having math ability or not, leading them to adopt more person-oriented responses to their children's success and failures in math. These responses may signal to children that their performance reflects their ability, so that children come to equate low performance with low competence, undermining their motivation and engagement (e.g., Mueller & Dweck, 1998).

In line with this perspective, research suggests that parents' process-responses facilitate children's math learning, whereas their person-responses undermine it (e.g., Gunderson et al., 2013). Gunderson et al., (2018) found that parents' process-oriented, but not person-oriented, responses to toddlers' successes during everyday activities predicted stronger entity theories and greater preference for challenge in second and third grade, which predicted higher math achievement one year later, controlling for children's prior achievement. Among older children, parents' person-oriented, but not process-oriented, praise of children's day-to-day academic successes predicts 10-year-old children's stronger entity theories of academic ability and lower preference for challenge in school six months later, accounting for children's earlier theories and preferences (e.g., Pomerantz & Kempner, 2013). Similarly, parents' person, but not process, responses to first and second graders' math successes and failures predict children's heightened math anxiety, dampened preference for math challenge, and lower math achievement one year later, adjusting for a variety of potential confounds (Barger et al., 2022). Variation in the effects of process- and person-oriented praise may stem from differences in the contexts in which these responses were recorded.

Taken together, these findings highlight the role of parents' motivational practices in supporting children's math learning. However, drawing from Parent × Child socialization models (e.g., Bates et al., 1998; Kochanska, 1993), Pomerantz et al. (2005b) have made the case that children who have a heightened need for motivational resources may be particularly sensitive to their parents' motivational practices such that they are particularly likely to benefit from practices such as autonomy support that afford such resources and particularly likely to suffer from practices that do not. In line with this idea, elementary school children who struggle in school and thus may have relatively depleted motivational resources are particularly at risk in the face of controlling or affectively negative parenting (e.g., Pomerantz et al., 2005a; Ng et al., 2004). In the math domain, Oh et al. (2022) found that the more controlling parents were when interacting with their six- and seven-year old children around math, the poorer children's math achievement one year later but only among children who were already struggling with math.

2.3 Additive and interactive effects of cognitive and motivational parenting practices in the math context

The dual-pathway model we present here portrays parents' cognitive and motivational practices and the pathways by which they influence children's math learning as generally distinct with independent effects. It is possible, however, that the two types of practices could overlap and thus not have independent effects. This is most likely in regard to the structure of parents' math talk (i.e., prompts and statements) and parents' autonomy support (vs. control). For example, operationalizations of autonomy-supportive parenting often include prompts (e.g., Grolnick et al., 2002; Oh et al., 2022) as they can be instrumental in scaffolding children's exploration and treating children as experts. Prompts, however, are only one component of autonomy-supportive parenting and only sometimes reflect such parenting—for example, when they are used in a manner providing children with agency. Similarly, operationalizations of controlling parenting often include statements (e.g., Grolnick et al., 2002; Oh et al., 2022) given that statements can be directive; but statements are also only one component of parents' control, and when they are part of responsivity and support (e.g., used in response to children's request for assistance), are not controlling. As Wu et al. (2024) point out, autonomy support and control are also broader (e.g., they include parents' general child- vs. parent-centered practices) than prompts and statements, which in the context of math talk refer specifically to math concepts.

In line with the idea that parents' cognitive and motivational practices are generally distinct, Wu et al. (2024) found only modest associations between prompts and autonomy support ($r = 0.23, p < 0.001$) and control ($r = -0.18, ps < 0.001$) and statements and autonomy support ($r = 0.00$, *ns*) and control ($r = 0.24, p < 0.001$) among parents of early elementary school children during a math activity. Importantly, these cognitive and motivational practices also made distinct contributions to children's engagement during the activity. For example, suggestive of the transmission of cognitive resources (e.g., advanced math concepts), parents' advanced prompts one minute predicted children's engagement, particularly of advanced math strategies, the next minute over above not only children's engagement the prior minute, as described above, but also parents' autonomy support and control. In contrast, parents' autonomy support (vs. control) appeared to provide motivational resources that set the foundation for sustained persistence as such parenting foreshadowed children's engagement, but not necessarily of advanced strategies, the next minute adjusting for children's engagement the prior minute as well as parents' prompts and statements. Other overlaps among parents' cognitive and motivational practices have not been examined but are possible—for example, prompts may be more likely in the context of process praise as parents encourage children to think through their effort and strategies.

Although the evidence to date suggests parents' cognitive and motivational practices are likely to provide distinct resources and thus have additive effects, it is also possible the two types of practices and pathways have interactive effects (i.e., one modulates the other's contribution). For example, when parents combine advanced statements with autonomy support, they may provide useful cognitive resources that children can experiment with and apply with agency. It could also be that children are more open to parents' statements when parents establish feelings of relatedness via their positive affect. The one study that has examined the interactive effects of the structure and level of parents' math talk with their autonomy support (vs. control), however, did not find evidence for interactive effects for early elementary school children (Wu et al., 2024). Similarly, Ren et al. (2022) did not find interactive effects of parents' spatial talk and the frequency of their praise for early elementary school children.

3. Antecedents of cognitive and motivational resource parenting pathways

Given that both parents' cognitive and motivational practices can be important to children's math learning, a key issue is identifying what undergirds the two so as to understand how schools can best support parenting that facilitates children's learning. Although ecological theories of development (e.g., Belsky, 1984; Brofenbenner & Morris, 2006) suggest parents' practices may be shaped by a set of complex multi-level forces, we focus here on the three most proximal forces—attributes of children, parents, and the math learning context—as there has been the most theoretical and empirical attention to these forces. Moreover, such forces may be particularly easy for educators to target to facilitate the most constructive parenting.

3.1 Children's attributes

Although much of the theory and research on parents' involvement in children's learning has assumed a model in which parents set the course of children's learning via the resources they provide (e.g., Gunderson & Levine, 2011; Grolnick & Pomerantz, 2022), children are rarely passive recipients in the socialization process (e.g., Collins et al., 2000; Sameroff, 2009). Instead, they can play an active role in shaping their own learning through interactions with their environment (e.g., Oh et al., 2022; Pomerantz et al., 2005b). Piaget (1976), in his theory of cognitive development, similarly emphasized that children construct knowledge by engaging with the world around them. One of the most significant ways children influence their own learning is via the attributes they bring to interactions with parents as these attributes can shape the ways in which parents are involved in children's learning (e.g., Belsky, 1984; Pomerantz et al., 2005b).

Although a variety of children's attributes (e.g., temperament) have been deemed important drivers of the socialization process (e.g., Kiff et al., 2011; Sanson et al., 2004), in the learning context, children's competence-related attributes (i.e., capacity to understand, interpret, and engage with the academic subject at hand) are key given that parents' involvement in children's learning often revolves around such attributes—for example, children's capacities may determine the extent to which they can work independently (e.g., Pomerantz et al., 2005b). Children's competence-related attributes in the math context may be important drivers of both parents' cognitive and motivational practices but may operate differently for the two. The more children lack the capacity for math, the more parents may adjust their cognitive practices so that they provide children with the appropriate cognitive resources—an essential part of Vygotsky's (1986) concept of scaffolding. At the same time, however, parents' motivational practices may become less constructive such that they provide fewer motivational resources to children who lack capacity and thus may be most in need of such resources (e.g., Moorman & Pomerantz, 2008; Pomerantz et al., 2005b).

Cognitive parenting practices. Vygotsky (1986) highlights how parents scaffold children's learning by providing support that is appropriately challenging for children's current skill-level. In practice, children's math competence serves as a cue for parents to adjust their cognitive practices, tailoring their support to match children's developing needs (e.g., DePascale et al., 2021; Susperreguy et al., 2020). Parents provide cognitive input that aligns with their child's developmental stage, gradually increasing the complexity of the math concepts they introduce. For example, research on children aged two to four years, who are in the early stages of math learning, shows that parents primarily focus on foundational skills, such as counting and number recognition, with minimal introduction of more complex concepts like magnitude comparison or calculations (Vandermaas-Peeler et al., 2012). However, as children grow older and become proficient counters (around ages six to seven years), parents appear to naturally reduce their emphasis on these basic skills and shift toward more advanced numerical discussions (e.g., Wu et al., 2024; Zippert et al., 2019).

Beyond developmental trends, parents also respond dynamically to their individual children's competence, adjusting their cognitive input in real-time. Although direct evidence on this process remains limited, some studies have found concurrent associations in preschool-aged children, suggesting that the more competent children are, the more parents provide more advanced math input, while referencing foundational concepts for those who are less competent (e.g., DePascale et al., 2021; Vandermaas-Peeler et al., 2012). Two recent studies are suggestive of this pattern. Using vignettes in which children provided correct and incorrect responses to math problems, Çarkoğlu and Eason (2025) found that parents said they would provide more elaborative cognitive support—such as explanations and follow-up questions—after errors than after correct responses. Wu et al. (2025) applied a minute-to-minute within-dyad analytic approach to examine how first- and second-grade children's correct and incorrect use of strategies predicted parents' math talk from one minute to the next, while accounting for parents' prior talk. Parents provided more elaborative cognitive input when children made mistakes, offering both basic statements and prompts to support children's skill development. When children used strategies correctly, however, parents increased their use of advanced prompts to encourage higher-level math thinking.

Motivational parenting practices. Although parents may adjust their cognitive practices to children's capacity such that they provide children with optimal cognitive resources, this is not the case when it comes to parents' motivational practices. When children are having difficulty, parents may be most likely to use motivational practices with children that detract from the children's already sparse motivational resources. When children struggle with math, parents may feel pressure to intervene to ensure children meet performance standards, which can affect their motivational practices (e.g., Pomerantz et al., 2005b). Drawing on the SDT framework, Grolnick (2003) argues that "pressure from below", which can manifest in a lack of capacity among children, can undermine parents' motivational practices such that they are, for example, less autonomy supportive (vs. controlling) and affectively positive (vs. negative) in their interactions with children around learning.

Consistent with this idea, several studies find that when children struggle academically, parents respond by increasing their control (vs. autonomy support) and negative (vs. positive) affect. For example, in a minute-to-minute observational study conducted by Moorman and Pomerantz (2008), when four-year-olds displayed helpless (vs. mastery) behavior during a challenging cognitive task, mothers responded with heightened control the following minute over and above their practices the prior minute. Experimental research further supports this pattern, showing that parents become more controlling following fifth and sixth grade children's induced failure (vs. success) (Wuyts et al., 2017). Within the domain of math learning, longitudinal research finds that the lower children's math achievement in third grade the greater the increase in parents' controlling homework assistance by sixth grade (Silinskas & Kikas, 2019b). In addition, Silinskas et al., (2015a) found that children's lower math achievement in first grade predicted increased negative affect and decreased positive affect among parents when assisting with children's math homework between second and fourth grade. Unfortunately, to our knowledge, the role of children in parents' process (vs. person) praise has not been examined.

3.2 Parents' attributes

Parents join children in bringing their own attributes to their interactions with children in the learning context. Parents' cognitions such as their attitudes, beliefs, goals, and expectations have been identified as contributing to a variety of parenting practices (for reviews, see Bornstein, 2015; Darling and Steinberg, 1993; Darling and Steinberg, 2017). When it comes to learning, parents' competence-related cognitions have been considered as particularly important (e.g., Carkoglu et al., 2023; Eccles et al., 1983). Parents' competence-related cognitions can be focused on children's competence (e.g., as is the case for parents' perceptions of children's competence) as well as on their own competence (e.g., as is the case for their anxiety about their competence).

Particularly central in the theory and research on parents' socialization of children's learning has been Eccles' (1983) Expectancy Value model which posits that parents' perceptions of children's competence play a key role in shaping children's learning by acting as self-fulfilling prophecies with parents' treating children in line with their perceptions, regardless of their accuracy. Indeed, a large body of evidence indicates that the more parents view their children as possessing competence, including in the math domain, the more positive children's perceptions of competence and the higher their achievement over time, over and above children's prior perceptions and achievement (e.g., Simpkins et al., 2012; for a review, see Pinquart & Ebeling, 2020). This may be due in part to the influence parents' perceptions of children's competence have on their cognitive and motivational practices in the learning context (e.g., Skwarchuk et al., 2014; Susperreguy et al., 2020).

Beyond parents' perceptions of children's competence, parents' beliefs about their own competence as manifest in their math anxiety (i.e., fear or tension around math) which also includes an affective component, have gained increasing attention for their role in shaping children's math learning. Mounting evidence suggests that parents' math anxiety is predictive of children's heightened math anxiety and dampened math achievement (e.g., Maloney et al., 2015; Szczygiel, 2020), even after controlling for children's prior math adjustment (e.g., Oh et al., 2022). This is likely because math anxious parents may engage less frequently in math-related activities, and when they do, they may struggle to provide the cognitive and motivational resources necessary for fostering children's math learning (for a review, see Carkoglu et al., 2023).

Cognitive parenting practices. Because parents' perceptions of children's competence are often based, at least partially in reality, they may act much like children's actual competence. Indeed, parents' perceptions may serve as a cue for parents to adjust their cognitive practices, tailoring their support to match children's developing needs. A growing body of research examines how parents' perceptions of children's math competence are associated with their cognitive practices (for a review, see Douglas et al., 2021). The more positively parents perceive children's math competence and the more positive their expectations, the more frequent and advanced their math interactions with children (e.g., Skwarchuk et al., 2014; Susperreguy et al., 2020; Zippert & Ramani, 2017). For example, Uscianowski et al. (2020) found that parents with more positive perceptions of preschool children's math competence were more likely to indicate that they would use more complex math talk when reading a math book to children.

Parents' math anxiety may also play a role in their cognitive parenting practices in the math context. Individuals with math anxiety often avoid engaging in math-related activities in an effort to preserve their feelings of self-worth and control (e.g., Ashcraft & Moore, 2009; Hembree, 1990). Thus, parents with heightened math anxiety may engage in fewer math-related interactions, such as math talk, thereby limiting opportunities for children to develop essential math skills. Indeed, parents' heightened math anxiety is associated with less frequent math engagement and reduced math talk (e.g., Berkowitz et al., 2021; Kiss and Vukovic, 2021). For example, del Río et al. (2017) found that parents' heightened math anxiety was negatively associated with the frequency of their engagement with advanced home numeracy practices with kindergarten children, with such practices associated with lower math achievement among children. Similarly, parents' math anxiety has been associated with less frequent cardinal number talk with children under the age of three, particularly among higher socioeconomic families (Berkowitz et al., 2021).

Motivational parenting practices. It is also possible that parents' perceptions of their children's math competence may contribute to motivational parenting practices. However, despite theoretical support, research on the role of parents' perceptions of children's competence in shaping motivational parenting practices is practically nonexistent. In contrast, the role of parents' math anxiety in parents' motivational practices in the math learning context has received a fair amount of attention. Math-anxious parents may often feel incapable when working with children on math, even when it involves only rudimentary problems such as adding one- or two-digit numbers. Math-anxious parents are prone to frustration and stress when assisting children with math (DiStefano et al., 2020), which may lead to heightened negative (vs. positive) affect and control (vs. autonomy support). To avoid having to be involved in math, math-anxious parents may also be controlling to ensure children finish quickly (Oh et al., 2022) and because they are inflexible when it comes to math (Maloney et al., 2015; Retanal et al., 2021).

In line with these ideas, a concurrent study found that the more parents of sixth and eighth grade children were math anxious, the more they reported their assistance with children's math homework as being controlling and affectively negative (Retanal et al., 2021). Studying first and second grade children, Oh et al., (2022) found that math anxious parents were less autonomy supportive and were more controlling in assisting with math—both in the homework context and in a math game, particularly when children struggled with math, presumably because assisting children who struggle with math can be particularly challenging for parents who themselves feel anxious about their math competence and would prefer to avoid math altogether.

3.3 Learning context attributes

Parents' involvement can also be shaped by the attributes of the contexts in which learning takes place. One of the most widely studied context attributes in the math domain is the distinction between formal and informal learning contexts. Research has long categorized home numeracy activities along this dimension, drawing from similar classifications in the literacy domain (e.g., Skwarchuk et al., 2014). According to Skwarchuk et al., (2014), formal math activities involve structured, goal-directed interactions in which parents deliberately teach children numerical concepts, such as numbers, quantity, or arithmetic, with the explicit purpose of enhancing numeracy skills. These activities often take the form of completing math homework, worksheets, or flashcards.

In contrast, informal math activities occur in everyday experiences where math-related learning happens incidentally rather than as a primary objective. These activities include such things as playing number board games, engaging in measurement tasks during cooking or crafting, making quantity comparisons, and spatial reasoning games. In these contexts, parents may introduce math concepts more spontaneously and in ways that are naturally embedded in the activity at hand (e.g., Elliott & Bachman, 2018; Skwarchuk et al., 2014). These differences are not trivial as they have implications for both parents' cognitive and motivational practices.

Cognitive parenting practices. The math learning context plays a role in shaping the cognitive resources parents provide, for example, via the structure and level of their math talk (e.g., Bjorklund et al., 2004; Eason & Ramani, 2020; Zippert et al., 2019). In structured, instruction-focused formal activities such as homework, parents have clear opportunities to observe children's math skills in action and provide targeted scaffolding. The structured nature of these activities may guide parents in aligning their cognitive input with the specific math content their child is learning. Moreover, instructional materials for these activities, such as worksheets, can serve as a reference, helping parents provide developmentally appropriate support through math prompts and statements.

In contrast, because math is not the explicit focus of informal activities, parents may have fewer opportunities to engage in structured math talk leading their involvement to be more spontaneous and context-dependent. Even when math-related moments arise, parents may lack the necessary guidance or control to tailor their cognitive support to the level of the material or children's skills. As a result, informal settings may lead parents' math talk to be less instructional, with the level of the concepts introduced typically depending on the activity itself, often centering on basic numerical concepts, rather than the more advanced topics children encounter in school (e.g., Çarkoğlu & Eason, 2025; Zippert et al., 2019).

Research is in line with this perspective. For example, Bjorklund et al., (2004) found that when engaging five-year-old children in structured formal math learning contexts, parents provided significantly more prompts and cognitive guidance compared to when they played informal board games with children. Similarly, Eason and Ramani (2020) found that parents of four- and five-year-olds engaged in more math talk in structured, guided learning contexts than in unguided contexts, even when provided with the same materials. Among this same group of children, Çarkoğlu and Eason (2025) presented vignettes to parents in which children provided incorrect responses to math problems. They found that parents were more likely to correct errors directly in formal learning contexts, reinforcing accurate numerical understanding, whereas in informal settings, they often ignored or reframed mistakes in a playful manner. These findings suggest that structured learning environments may naturally elicit more cognitively supportive practices from parents.

Beyond the formal–informal distinction, other context attributes can also shape parents' cognitive involvement. Just as structured tasks like homework can guide parents in delivering targeted cognitive input, other math-rich environments can similarly support parents in integrating math talk into their interactions with children. Specifically, the nature of the materials used in math activities can influence how parents and children interact. For example, Masek et al. (2024) found that certain materials, such as grocery shopping sets and picture books, encouraged richer math talk among parents and their 24- to 36-month-old children compared to manipulatives like shape sorters or magnet board games. The richness of math talk in these contexts was positively associated with children's understanding of number words. Additionally, Zippert et al. (2019) found that when parents were given additional guidance on how to interact with a numerical tablet game, they engaged in significantly more math-related talk with their children.

Motivational parenting practices. Although formal learning contexts can provide rich opportunities for parents to offer cognitive input, they may not necessarily foster constructive motivational practices given the pressure parents may experience in formal contexts which are often evaluative. SDT (Deci & Ryan, 1985) suggests that contexts that emphasize evaluation and performance over learning can lead parents to feel pressured, ultimately shaping how they get involved with their children. Among formal learning activities, math homework stands out as a unique and highly structured context that becomes increasingly important as children progress through school. Unlike informal learning experiences that often emerge naturally through play, homework imposes external constraints such as time pressure, performance expectations, and rigid structures, often placing parents in a position where they feel responsible for ensuring children complete assignments correctly and on time (e.g., Silinskas et al., 2015b; Wu et al., 2022).

According to Grolnick et al. (e.g., Grolnick, 2015; Grolnick et al., 1997; Lerner et al., 2022), when parents experience pressure—either externally (e.g., from teachers or school policies) or internally (e.g., from their own anxieties about their child's success)—they are less likely to use constructive motivational practices. Instead, such pressure can undermine their own sense of autonomy and competence, leading them to adopt more controlling practices to quickly alleviate the immediate stressor, rather than focusing on long-term developmental goals, such as fostering autonomy and competence in children (e.g., Deci & Ryan, 1985; Grolnick et al., 2002). Beyond increasing control, pressure may also heighten parents' negative affect, particularly in evaluative contexts like homework, where children's performance may feel like a reflection of their own competence as a parent (e.g., Pomerantz et al., 2005), which could elicit heightened stress and frustration (e.g., DiStefano et al., 2020).

As a result, the structured and evaluative nature of formal activities, such as homework, may lead parents to become more controlling, prioritizing correct answers over exploration and independent problem-solving, while also displaying more negative (vs. positive) affect (e.g., DiStefano et al., 2020; Else-Quest et al., 2008; Wu et al., 2022). Experimental evidence supports this idea. Grolnick et al., (2002) found that when parents were placed in a high-pressure situation regarding children's performance, they were significantly more likely to exhibit controlling practices, limiting children's autonomy. Over time, this pressure-driven involvement can create a cycle of negative reinforcement, in which parents' heightened control diminishes children's sense of autonomy and competence, ultimately leading to reduced motivation and disengagement from math.

In contrast, it is possible that more informal and playful math activities allow parents to adopt less directive, more autonomy-supportive and affectively positive approaches. Using daily reports from parents of young elementary school children, Wu et al., (2022) compared parents' motivational practices across math homework and activities and found that parents' involvement in math activities (vs. homework) was characterized by less negative affect and more positive affect. Notably, parents' heightened negative affect in the homework context predicted children's reduced liking for math, lower preference for challenging math tasks, and lower math achievement over a year, even after controlling for their prior math adjustment.

4. Implications for educational recommendations and interventions

Although the dual-pathway model on the role of parenting in children's math learning requires additional empirical validation, it, along with the empirical evidence to date, highlights the importance of both cognitive and motivational resources provided by parents. The model has important implications for educational recommendations and interventions aimed at enhancing children's math learning. Most notably, efforts to harness the power of parents should target both parents' cognitive and motivational practices. Although educating parents about these two types of practices and how to use them will be important, efforts should also address attributes of children, parents, and the learning context that can support parents in optimizing both the cognitive and motivational resources they confer on children. For example, math homework—as well as other formal math learning activities—should be designed with attention to the pressure that homework can induce in parents. This may involve giving parents choice in the types of problems on which they work with children or moving away from grading homework. While this approach may support parents' motivational practices, it does not necessarily support their cognitive practices. To this end, homework design should also include the attributes (e.g., structure and guidance) that can enhance parents' cognitive practices.

5. Conclusion

In conclusion, the dual-pathway model underscores the critical role of parents' provision of both cognitive and motivational resources in children's math learning. Parents' cognitive practices, such as advanced math talk and using prompts, directly contribute to the development of children's math skills. Simultaneously, motivational practices, including autonomy support and positive affect, foster children's math beliefs, motivation, and engagement. The importance of both the cognitive and motivational pathways highlights the need for educational interventions that address both cognitive and motivational aspects of parents' involvement. Future research should continue to explore the distinct and interactive effects of cognitive and motivational parenting practices in the math context, as well as the antecedents that shape these practices. Of particular importance is to understand the influence of the broader cultural, socioeconomic, and institutional contexts, which have been relatively under-examined in regard to parents' role in children's math learning but are likely of much significance. By understanding the multifaceted nature of parents' involvement in children's learning, educators can develop targeted strategies to support parents in fostering their children's math learning, ultimately contributing to improved motivation and achievement in math as children navigate a variety of learning environments over the school years and as adults into their careers.

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CHAPTER FIVE

The sound-of-words model: A developmental perspective of phonolexical acquisition

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Contents

Abstract

The current chapter reviews 25 years of research on the so-called consonant bias in lexical processing, according to which consonants, rather than vowels, are most relevant to build the lexicon. The evidence so far suggests the C-bias might be prevalent in adulthood, though more work is needed on tone languages that might change this view. The findings from developmental studies offer a more nuanced approach, showing important crosslinguistic differences in the timing of acquisition of asymmetrical processing of consonants and vowels in lexically-related processes, and in the direction of the bias observed. This suggests that the bias is acquired on the basis of phonological and lexical cues. We propose a new model for the acquisition of these biases, the Sound-of-Words model, that could be used to guide further research on this issue.

Keywords: C-bias, Crosslinguistic variation, Early language acquisition, Infants, Language acquisition, Lexical acquisition, Lexicon, Phonological acquisition, Phonology

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Language acquisition is a crucial aspect of human development, being the single best predictor of school achievements in all subjects in primary school (third and fifth grades; Pace et al., 2018). Language is a complex construct, which can be described at different levels organized around fundamental building blocks, such as: the phonological level around phonemes and prosodic units; the lexical level around words and morphemes; the syntactic level around grammatical categories and rules. This general organization is shared by all human languages. At the same time, languages differ in crucial ways on all these levels. The existence of these commonalities and variabilities across languages has consequences for language processing and acquisition, and the aim of developmental psycholinguistics is to determine the mechanisms that allow infants to learn the different linguistic levels of their native language(s). This raises several central theoretical issues for infant development, such as the contribution of nature and nurture to acquisition, the links between the acquisition of the different linguistic levels, and the cascading developmental effects and predictive effects of early acquisitions on later developmental outcomes. The current chapter explores how the investigation of the emergence of processing biases in the acquisition of the phono-lexicon, namely the interface between the phonological system and the lexicon, brings new, related knowledge on these issues, and we propose a new model of the emergence of this interface, the Sound-of-Words model. According to this model, the emergence of a Sound-of-Words bias constitutes a major developmental milestone, attesting the binding of the phonological and lexical levels, and bootstrapping subsequent vocabulary development.

1. The "division of labor" hypothesis

This research takes as its premise the "division of labor" hypothesis (Nespor et al., 2003) by which two complementary phonological biases impact language processing and acquisition: "[C]onsonants, rather than vowels, are most relevant to build the lexicon, and vowels, rather than consonants, are most relevant for grammatical [and prosodic] information." This proposal was based on several linguistic observations, and emerging experimental data suggesting different contributions of consonants and vowels to lexical access in adulthood. For example, it underlined the fact that consonants are more numerous than vowels in most languages (Ladefoged & Maddieson, 1996), that lexical roots are mostly made up of consonants in Semitic languages, or that consonants tend to disharmonize within words (contrary to vowels). All these factors would make consonants more informative than vowels at the lexical level.

At the processing level, there was evidence that consonants and vowels tend to be processed differently, with more categorical perception of consonants than vowels in both adults (Fry et al., 1962; Liberman et al., 1957) and infants (Eimas et al., 1971; Kuhl et al., 1992). There is also evidence for differences in the timing of acquisition of consonants and vowels, with an earlier acquisition of native vowels – starting by 6 months- compared to native consonants –starting by 10 months (e.g., Best et al., 1988; Kuhl et al., 1992; Werker & Tees, 1984). Neuropsychological evidence also described two aphasic patients who presented contrasted error patterns when producing consonants and vowels, suggesting the involvement of different brain regions in the processing of the two types of phonemes (Caramazza et al., 2000). Moreover, the first experimental evidence in support of the division-of-labor proposal, revealing a consonant bias in lexical processing, was a series of word reconstruction studies conducted on Dutch-, English- and Spanish-speaking adults (Cutler et al., 2000; van Ooijen, 1996–). In these studies, adults were presented with pseudowords and asked to change a phoneme of each pseudoword to make it a word. Pseudowords (e.g. /kebra/) could be modified by changing either a consonant (/zebra/) or a vowel (/cobra/). In all three languages, they observed that adults retrieved words by changing more often and more rapidly a vowel into another vowel than a consonant into another consonant, establishing that consonants are stronger cues to word identity.

The importance of this "division of labor" proposal lies in the fact that if these two processing biases are present early in life, then they might bootstrap lexical acquisition, helping infants learn the words of their native language on the one hand, its grammatical and prosodic properties on the other hand. In the following, we evaluate the evidence that has been gathered since this proposal was made, first reviewing adult data, then developmental data. Since little work has been done on the grammatical/prosodic side of the "division of labor" proposal (see, however, Toro et al., 2008 for adult work; Hochmann et al., 2011, and Pons & Toro, 2010, for infant work), the present chapter focuses on lexical processing and the so-called "C-bias," according to which it is easier to learn and retrieve words on the basis of their consonants than on the basis of their vowels.

2. Evaluating the C-bias in lexical processing in adulthood

Further adult studies (see review by Nazzi & Cutler, 2019), using speech perception tasks tapping into word learning and lexical access, tested whether the C-bias in lexical processing is present cross-linguistically as predicted if it is an innate, universal bias. Various tasks were used, three of them tapping into processes of accessing known words: word reconstruction (in which adults are presented with a pseudoword and have to transform it into a word by changing either a consonant or a vowel), word spotting (in which they have to spot a word embedded in a longer pseudoword), and primed lexical decision (in which they have to decide whether a target is a word, the target being preceded by a prime that is either unrelated, or related to the target by its consonants or its vowels). Two other tasks are related to the learning of new word forms or new words: word segmentation (in which adults hear a continuous speech stream made up of a few pseudowords, and have to use distributional properties to find the words that make up the language), and word learning (in which they are taught pairs of new words differing by a consonant or a vowel).

At present, studies have been conducted in 16 languages, most of them Indo-European languages: English, Dutch, German, Danish, Norwegian, French, Italian, Spanish, Slovak and Russian. A few languages from other linguistic families have also been investigated: Sesotho, Berber, Japanese, Mandarin, Cantonese and Hebrew. A summary of the results found is presented in Table 1. For the Indo-European languages, on which most studies were conducted, the evidence overwhelmingly establishes a C-bias in all of these lexically-related processes, with just a few studies failing to show a bias (note that there is only one study on both Danish and Norwegian, failing to provide conclusive evidence). A C-bias was also found in Sesotho, Japanese and Hebrew, suggesting this bias extends beyond Indo-European languages, although no evidence was found in Berber. In contrast, results from Mandarin and Cantonese, two tone languages, fail to show evidence of a C-bias, and provide emerging evidence that a V-bias might be present in some languages (Gómez et al., 2018; Poltrock et al., 2018; Wiener, 2020; Wiener & Turnbull, 2016).

Hence, if a C-bias has largely been found, which would support the idea of a language-general C-bias in lexical processing, two limits have to be pointed out. First, these findings mostly come from Indo-European languages, with only a few of the around 7000 other languages investigated.

Table 1 Crosslinguistic consistency in adult auditory lexical processing (C: consonant bias; V: vowel bias; $V+T$: advantage of tone-carrying vowel; C-V n.d., no difference).

Languages	Tasks	Lexical bias	Reference
English	Word reconstruction	C	Van Ooijen (1996); Sharp et al. (2005); Moates et al. (2002)
	Word spotting	C	Norris et al. (1997, 2001); Newman et al. (2011)
	Syllable detection	C–V n.d.	Kearns et al. (2002)
	Primed lexical decision	C	Delle Luche et al. (2014)
	Word segmentation	C–V n.d.	Newport & Aslin (2004)
	Word learning	C	Creel et al. (2006); Escudero et al. (2016)
Dutch	Word reconstruction	C	Cutler et al. (2000)
	Word spotting	C	McQueen et al. (1998)
	Primed lexical decision	C–V n.d.	Cutler et al. (1999)
German	Word spotting	C	Hanuliková et al. (2011)
Danish	Word segmentation	C–V n.d.	Trecca et al. (2019)
Norwegian	Word segmentation	C–V n.d.	Trecca et al. (2019)
Spanish	Word reconstruction	C	Cutler et al. (2000), Marks et al. (2002)
French	Word spotting	C	Dumay et al. (2002)
	Primed lexical decision	C	New et al. (2008); New & Nazzi (2014); Delle Luche et al. (2014); Schmandt et al. (2022)
	Word segmentation	C	Bonatti et al. (2005), Mehler et al. (2006)
	Word learning	C	Havy et al. (2014)

Continued

Table 1 Crosslinguistic consistency in adult auditory lexical processing (C: consonant bias; V: vowel bias; $V + T$: advantage of tone-carrying vowel; C-V n.d., no difference).—cont'd

Languages	Tasks	Lexical bias	Reference
Italian	Word segmentation	C	Toro et al. (2008)
Slovak	Word spotting	C	Hanuliková et al. (2011)
Russian	Word spotting	C-V n.d.	Alexeeva et al. (2017)
	Word segmentation	C	Gomez et al. (2018)
Sesotho	Word spotting	C	Cutler et al. (2002a)
Berber	Word spotting	C-V n.d.	El Aissati et al. (2012)
Japanese	Word spotting	C	McQueen et al. (2001); Cutler et al. (2009)
	Word reconstruction	C	Cutler and Otake (2002)
Hebrew	Primed lexical decision	C	Lador-Weizman and Deutsch (2022)
	Word segmentation	C	Lador-Weizman and Deutsch (2024)
Mandarin	Word reconstruction	V	Wiener and Turnbull (2016); Wiener (2020)
	Word segmentation	$V + T$	Gomez et al. (2018)
	Word learning (Cantonese)	C-V n.d.	Poltrock et al. (2018)
Cantonese	Word segmentation	$V + T$	Gomez et al. (2018)
	Morpheme-spotting	C	Yip (2004a,b)
	Word learning	C-V n.d.	Poltrock et al. (2018)

It is thus crucial to further extend this kind of research to a larger range of languages and language families, in order to further probe the validity of such language-general claim. Second, if a V-bias in lexical processing is a feature that extends beyond Mandarin and Cantonese to all or most tone languages, this would challenge the view that the C-bias is the predominant pattern in lexically-related processing, given that tone languages constitute the majority of the languages of the world. Further studies on tone languages are thus key to confirm or refute the predominance of the C-bias. While waiting for further cross-linguistic extensions (that would allow us to specify the factors that modulate the expression of the C-bias in a given language), let us review whether a C-bias is found in development, how it might be set into place, and whether it is modulated cross-linguistically.

3. Hypotheses regarding the origin of the C-bias

Within the framework of phonological bootstrapping (Morgan & Demuth, 1996), the C-bias in lexical processing, if functional early in development, is viewed as a powerful tool for infants to bootstrap into lexical and syntactic acquisition. In terms of the origins of the C-bias, different hypotheses have been considered in the literature, making different predictions about the developmental timing at which the C-bias would be functional, and the existence of crosslinguistic modulation in its expression. According to the first one, or initial bias hypothesis, the C-bias would be innate, "human beings com[ing] into the world knowing that languages are structured in such a way" (Nespor et al., 2003, p. 224; see also Bonatti et al., 2007; Pons & Toro, 2010). This predicts that infants would potentially process consonants and vowels as distinctive linguistic categories from birth. It further predicts that a C-bias in "lexically-related" tasks should be observed across languages, irrespective of the distributional/acoustic characteristics of each language. A weaker version of this position is that the C-bias should be the predominant bias found cross-linguistically.

In addition, the C-bias might also result from the properties of the linguistic input infants are exposed to, and thus constitute a case of early attunement to the native language(s). Floccia et al. (2014) discussed acoustic/phonetic and lexical factors potentially involved in the emergence of the C-bias, and three acquisition hypotheses can be found in the literature: the acoustic/phonetic bias hypothesis (Bouchon et al., 2015; Floccia et al., 2014), the lexical bias hypothesis (based on Keidel et al., 2007), and the phono-lexical hypothesis (Nishibayashi & Nazzi, 2016; Poltrock & Nazzi, 2015).

According to the acoustic/phonetic bias hypothesis, the C-bias emerges as a result of differences in acoustic properties, as consonants tend to be shorter, less periodic and steady but tend to be perceived more categorically than vowels, leading to the construction of two distinct categories (Bouchon et al., 2015; Floccia et al., 2014). Since phonological acquisition starts during the first year of life (Best et al., 1988; Kuhl et al., 1992; Werker & Tees, 1984), the C-bias could emerge before the first birthday, independently of whether or not infants have acquired a sizeable lexicon. Moreover, it should be modulated over the course of development by the acoustic/phonetic characteristics of the ambient language, so that differences in the age at which the bias emerges and its strength should be found cross-linguistically.

According to the lexical bias hypothesis, the emergence of the C-bias would be related to lexical factors, linked to the phonological structure of the lexicon in acquisition. The C-bias would emerge as a result of the fact that, compared to vowels, consonants would be more informative cues to lexical identity in the native language. Work on English and French suggests that this higher informativeness of consonants could be related to density of phonological neighborhoods linked to C- and V-skeletons (Delle Luche et al., 2014; Keidel et al., 2007; New & Nazzi, 2014; Schmandt et al., 2022). Since calculating informativeness requires to have a sufficiently large lexicon, it was proposed that such a critical mass would not be reached before some point during the second year of life at the earliest, when lexical acquisition appears to accelerate. Hence, the lexical bias hypothesis predicts a later acquisition of the C-bias than the acoustic/phonetic bias hypothesis. Moreover, the lexical hypothesis also predicts crosslinguistic differences, since differences in the structure of lexicons across languages should impact the relative informativeness of consonants and vowels, and the ability and speed at which these properties can be acquired.

More recently, Poltrock and Nazzi (2015) proposed that rather than considering phonological and lexical properties as mutually exclusive sources from which the C-bias could be acquired, both should in fact contribute to the emergence of the C-bias. This proposal was based on adult work showing that both phonological and lexically-related properties contribute to lexical access in adulthood (New & Nazzi, 2014; Schmandt et al., 2022). It was also taking into account increasing evidence that lexical acquisition starts in the first year of life, as attested by studies showing that infants start recognizing the sound pattern of their name by 5 months

Table 2 Crosslinguistic and developmental predictions according to the different hypotheses explaining the origin of the C-bias.

Hypotheses	Origin	Cross-linguistic differences?	Timing of expression?
Initial	Innate	No	At birth
Acoustic/phonetic	Learned through acoustic/phonetic acquisition	Yes	Before 12 months
Lexical	Learned through experience with the native lexicon	Yes	After 12 months (sizeable lexicon)
Phono-lexical	Learned through phonological and lexical experience	Yes	Before 12 months

(Bouchon et al., 2015; Mandel et al., 1995) and of familiar words (Hallé & de Boysson-Bardies, 1994; Swingley, 2005; Vihman et al., 2004) by 11 months, and start comprehending some words by 6 months (Bergelson & Swingley, 2012; Tincoff & Jusczyk, 1999, 2012). It further considered findings establishing that phonological and lexical acquisitions are intertwined (Feldman et al., 2013; Stager & Werker, 1997; Yeung & Nazzi, 2014; Yeung & Werker, 2009). Lastly, it considered that acquisition of lexically-related properties could take place at the level of the pre-lexicon (including all word forms known by an infant) rather than the lexicon per se. Since infants' ability to segment word forms from fluent speech is found by 4 to 6 months in several languages (Berdasco-Muñoz et al., 2018; Bosch et al., 2013; Goyet et al., 2013; Höhle & Weissenborn, 2003; Jusczyk & Aslin, 1995; Kooijman et al., 2013), this pre-lexicon might become large enough to sustain the acquisition of lexically-related properties, and the emergence of the C-bias, before the end of the first year of life. As the other hypotheses stating that the C-bias is acquired, this phono-lexical hypothesis also predicts crosslinguistic differences in the emergence of the C-bias. Table 2 presents the different hypotheses proposed and their crosslinguistic and timing predictions.

4. Evaluating the C-bias in lexical processing in infancy and toddlerhood

To explore the innate and/or acquired determinants of the C-bias, a range of studies, starting from Nazzi (2005) with French-learning toddlers,

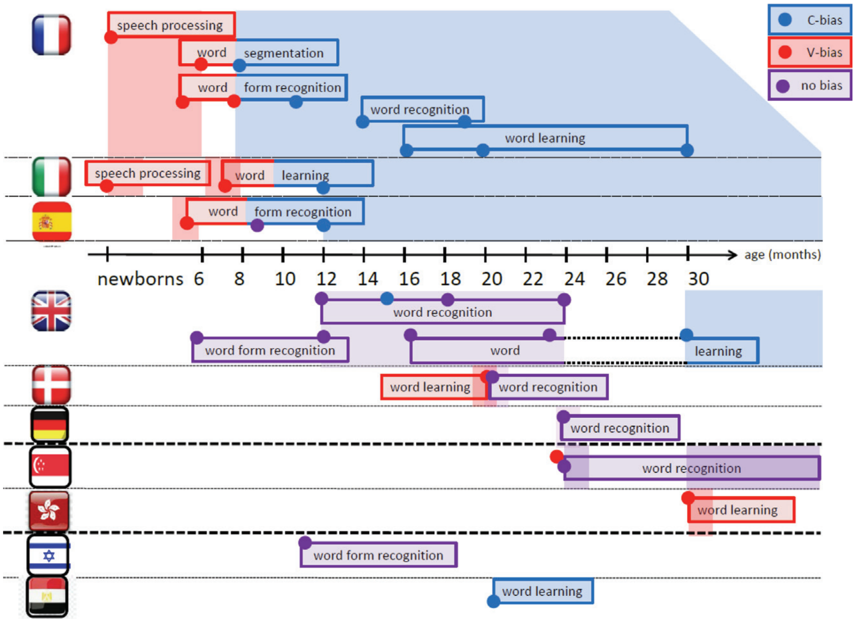

Fig. 1 Crosslinguistic data on C- and V-biases in infant/toddler lexically-related processing across languages. From top to bottom: Romance (French, Italian, Spanish), Germanic (English, Danish, German), Sinitic (Mandarin, Cantonese) and Semitic (Hebrew, Arabic) languages.

have been conducted in infancy and early childhood, in different linguistic populations. Most of these studies examined either infants' ability to learn pairs of new words that differ by one consonant or vowel, or their sensitivity to consonant or vowel mispronunciations of previously heard words or word forms. At this stage, the results found in languages from three different linguistic families suggest three different developmental trajectories (see review by Nazzi & Cutler, 2019, and visual summary in Fig. 1).

5. Romance languages

The first developmental trajectory was found for three Romance languages, and corresponds to an early acquisition of a C-bias. In French, the C-bias was first established by Nazzi (2005), in which 20-month-old French-learning toddlers were presented with triads of novel objects, comprising two objects given the same name and a third object given a name that differed by one phonetic feature. Following this learning phase, they were asked to group together the two objects with the same name. Infants successfully learned the object names when they differed by one of their consonants (e.g., /pize/ vs. /tize/) but not when they differed by one of their vowels (/pize/ vs. /pyze/). Further word-learning studies with French-learning toddlers established that a C-bias is found from 16 months to 6 years of age (Havy et al., 2011; Havy & Nazzi, 2009, 2014; Nazzi et al., 2009). A C-bias was also found in conflict studies in which, following a phase in which children learn the names of two objects (e.g., a /tide/ or a /pyde/), they were then asked to match a third name (e.g., a /pide/) to one of the preceding objects, which required neglecting either the consonantal or the vocalic mismatch. Although there is some variability in performance depending on the specific consonant and vowel contrasts tested, the C-bias in French appears to be a robust phenomenon. Indeed, it was found across different types of consonants (e.g., /nuk/ vs. /muk/, /rize/ vs. /lize/; Nazzi & New, 2007), different lexical structures (mono- and disyllables; e.g., Nazzi, 2005), and different positions within words (word-initial or not) and syllables (onset versus coda; e.g., Havy et al., 2014; Nazzi & Bertoncini, 2009; Nazzi & Polka, 2018; Nazzi et al., 2016). The C-bias also appears when toddlers are tested on their recognition of known words, for which they appear more sensitive to consonant than vowel mispronunciations (at 14 months: Zesiger & Jöhr, 2011; at 19 months: Von Holzen et al., 2023).

Additional studies on French explored whether the C-bias in lexically-related processes extends to the first year of life. To do so, infants were tested on their sensitivity to mispronunciations of the word forms of either early acquired words, or of recently segmented words (following a first experimental phase in which infants were presented with unfamiliar words embedded in sentential contexts). These studies revealed that a C-bias is present by 8 months when evaluated in a word form segmentation paradigm (Nishibayashi & Nazzi, 2016) and by 11 months when evaluated in a word form recognition paradigm (Poltrock & Nazzi, 2015; Von Holzen & Nazzi, 2020). However, prior to that, a V-bias is found, at 6 months for word form segmentation (Nishibayashi & Nazzi, 2016), at 5 and 8 months for word form recognition (Bouchon et al., 2015; Von Holzen & Nazzi, 2020). This early V-bias is consistent with earlier studies on syllable discrimination in newborns having found higher sensitivity to vowels than consonants (Bertoncini et al., 1988). Taken together, the findings on French establish that the C-bias emerges in development, as attested by a reversal from a V- to a C-bias, and that this emergence takes place at a young age, around 8 to 11 months of age. Crucially, a very similar trajectory was found with Italian-and Spanish-learning infants. In Italian, a study explored newborn's memorization of new word forms, and their sensitivity to a consonant or vowel mispronunciation of this word form (Benavides-Varela et al., 2012). The authors interpreted the pattern of brain responses measured with fNIRS (functional Near Infra-Red Spectroscopy) as a V-bias. This early V-bias is further corroborated by results from word learning experiments (based on anticipatory looking responses) showing that 6-month-olds give more weight to vowel information while 12-month-olds give more weight to consonant information, establishing a shift from a V- to a C-bias between these two ages in Italian (Hochmann et al., 2011, 2018). In Spanish, a study investigated whether infants are more sensitive to consonant or vowel mispronunciations of familiar word forms; it found a V-bias in 5-month-olds, no bias in 8½-month-olds and a C-bias in 12-month-olds (Bouchon et al., 2022). Taken together, the studies on French, Italian and Spanish establish that the C-bias emerges early in development in these three Romance languages, at some point in the second half of the first year of life.

6. Germanic languages

A second developmental trajectory was found for three Germanic languages, and would correspond to a late acquisition of a C-bias. In (British) English, studies exploring toddlers' ability to learn new words found a C-bias at 30 months of age (Nazzi et al., 2009), but equal sensitivity to consonant and vowel contrasts, hence no bias, at both 16 and 23 months (Floccia et al., 2014). Studies on word recognition exploring toddler's sensitivity to mispronunciations found no bias at 12, 18 and 24 months, with some weak evidence of a C-bias at 15 months (Mani & Plunkett, 2007, 2010). Studies investigating the specificity of early word forms either failed to find evidence of specificity for the infant's own name at 5 months (Delle Luche et al., 2017), or found equal sensitivity to consonant and vowel mispronunciations when recognizing familiar word forms at 11 months (Ratnage et al., 2023). Taken together, these findings on British English establish the late emergence of a C-bias, between 24 and 30 months, following an equal sensitivity to consonants and vowels at younger ages.

Results from two other Germanic languages are consistent with the late emergence found in British English. In Danish, two studies explored this topic at 20 months of age. Studying Danish was motivated by the fact that it has more vowels than consonants, and is characterized by extensive consonant lenition. Both of these phenomena could disfavor a consonant bias. In a first study in which the toddlers had to learn new words differing by either a consonant or a vowel, they succeeded only in the vowel-contrasted condition, attesting a V-bias (Højen & Nazzi, 2016). In a second study that was testing consonant and vowel specificity in word recognition, pitting consonant- and vowel-mispronunciations against each other, the toddlers failed to show a bias for either type of phoneme (Højen et al., 2023). In German, a study based on Højen et al. (2023) also failed to find a bias at 24 months of age (Piot et al., 2025). However, that study also tested French-learning 24-month-olds, who also failed to show a bias, when a C-bias was expected given the evidence presented above. Findings using this method should thus be interpreted with caution for the time being. That being said, it remains that no evidence of a C-bias could be found in Germanic languages before 30 months of age, contrary to the early emergence of the C-bias in Romance languages before the first birthday.

7. Sinitic languages

A third developmental trajectory was found for two Chinese languages, which are both tone languages, failing to find a C-bias in toddlerhood and childhood. The motivation for testing tone languages was that tonal information is mostly carried by the vowels, a phenomenon that could increase the relative processing weight given to vocalic information, and should thus reduce the C-bias. In Mandarin, studies on sensitivity to mispronunciations when recognizing familiar words provide some evidence of a V-bias in both Mandarin- monolingual and Mandarin-English-bilingual 24-month-olds (Wewalaarachchi et al., 2017). A similar study conducted on Mandarin-English-bilingual toddlers (aged 2.5–3.5 years) or preschoolers (aged 4–5 years) showed equal sensitivity to consonant and vowel mispronunciations (Singh et al., 2015). No bias was also found in both Mandarin-monolingual and Mandarin-English-bilingual 6-year-olds (Wewalaarachchi & Singh, 2020). In Cantonese, 30-month-olds tested in a word learning paradigm were found to learn pairs of words that differed by a vowel, but failed when the words differed by a consonant or a tone, establishing a V-bias (Chen et al., 2021). Taken together, the findings on these two Chinese languages suggest that a C-bias does not emerge in these languages, which is in line with the findings on adult speakers of these languages (Gomez et al., 2018; Poltrock et al., 2018; Wiener, 2019; Wiener & Turnbull, 2016).

8. Semitic languages

Last, work is starting on Semitic languages. A study investigating the sensitivity of Hebrew-learning 11-month-olds to consonant and vowel mispronunciations of familiar wordforms found no bias, infants being equally sensitive to consonant and vowel mispronunciations in stressed syllables and equally insensitive to consonant and vowel mispronunciations in unstressed syllables (Segal et al., 2020). This finding differs from the C-bias found in adult Hebrew speakers (Lador-Weizman & Deutsch, 2022, 2024). A study on Egyptian Arabic tested 21-month-old toddlers in a word learning conflict task where children are asked to pair objects whose labels share a consonant or a vowel (as done in Nazzi et al., 2009). Results show a clear consonant bias (Floccia et al., 2025).

In summary, the developmental findings reported above establish that the C-bias is not present early in life, given that in the first days or months of life, a V-bias is found. This V-bias might result from the greater salience (longer duration and higher energy) of vowels compared to consonants. They further reveal an early emergence of a C-bias, during the first year of life, for 3 Romance languages (French, Italian and Spanish), a late emergence of a C-bias for English in the third year of life (with no bias found in the second year for Danish and German), no early bias in Hebrew, while data on Sinitic languages establish either a V-bias extending during the third year of life, or no bias even in childhood. These findings are in large part consistent with the adult findings, in which Indo-European languages show a C-bias while Sinitic languages show either a V-bias or no bias (note however that whether these linguistic families are the right level to distinguish the different trajectories needs to be further tested). The increased weight given to vowels in Sinitic languages was attributed in both the adult and developmental studies to the presence of lexical tones in these languages, which are mostly carried by the vowels. As previously discussed for the adult findings, if this increased weight of vowels extends to the majority of tone languages, then the C-bias might not be the predominant pattern in lexically-related processing.

At any rate, the crosslinguistic modulation in bias found in the developmental literature constitutes, at best, only partial evidence for the C-bias proposal as stated by Nespor et al. (2003), that is, for the strong claim that it is an in-built property of the language processing system. Rather, it supports the proposal that the processing biases that emerge with the acquisition of the native language are acquired. While the early acquisition

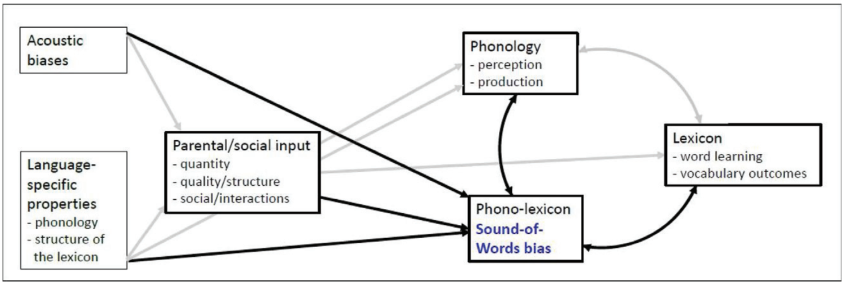

Fig. 2 The Sound-of-Words model, an integrative model of the acquisition of the phono-lexicon (black arrows = direct links with Sound-of-Words bias; grey arrows = other likely indirect links).

of the C-bias in Romance languages was initially taken as evidence for the acoustic-phonetic over the lexical hypothesis, it has been argued more recently that both levels likely contribute to the acquisition of lexically-related processing biases (phono-lexical hypothesis; Poltrock & Nazzi, 2015). This is also supported by the late emergence found in English (and possibly other Germanic languages), since toddlers would have sufficiently large vocabularies to perform such analyses at the time the bias emerges. Moreover, when a C-bias was not found, it was attributed to phonological properties (consonant reduction in Danish; presence of lexical tones in Mandarin and Cantonese), and possibly to lexical factors (higher informativeness of vowels in distinguishing words in the lexicon of these languages).

In the future, further studies will need to extend the work already conducted to the exploration of even more numerous, and more varied, languages, and provide more detailed developmental trajectories than what is currently available, given that, for most languages, very few age points have been tested. Such studies will contribute to the determination of the relative prevalence of C- and V-biases, and to the specification of the phonological and lexical factors that guide and modulate their acquisition. In the following, we propose a new framework intended both to take into account what we currently know of the biases found in lexically-related processes during development, and to guide further research on this topic: the Sound-of-Words model.

9. The sound-of-words model

According to the Sound-of-Words model (see Fig. 2), the C-bias would only be one option in the binding of phonological and lexical abilities. Developmentally, infants would first give more weight to vowels in language processing in general (due to their higher salience in the signal). The model then postulates how, for lexical processing, they would acquire a language-specific Sound-of-Words bias, be it a C- or a V-bias, as a result of parental/social speech input. Some language-general biases might influence that acquisition. For example, while the higher acoustic prominence of vowels over consonants might favor an initial V-bias, differences in the processing of consonants and vowels (more categorical for consonants, more continuous for vowels) might on the contrary favor the later emergence of a C-bias. Acoustic distance (such as differences in duration, intensity or spectral information) might also play a role in allowing discrimination and use of consonantal and vocalic contrasts to differentiate between words (see Bouchon et al., 2015).

Phonological and lexical properties of the language in acquisition would also impact the acquisition of the Sound-of-Words bias, which is reflected in the crosslinguistic variability in developmental trajectories revealed by the studies reported earlier. These properties still need to be better specified, and extending the repertoire of languages studied, varying their phonological properties, will allow this specification. Among the phonological factors that have been considered in the literature are the number of consonants and vowels in the inventory of the native language, and the ratio between them. The presence of lexical stress and vocalic reduction (in Germanic languages), consonantal reduction (in Danish), and lexical tones (in Sinitic tone languages) has been invoked. Syllabic structure (some languages allowing only consonant-vowel syllables, others allowing complex onset and coda consonants) might also modulate the bias. Among the lexical factors considered are cues that might signal the relative informativity of consonants and vowels to distinguish between words in the native lexicon, such as the number of consonantal and vocalic neighbors, or the number of unique consonant and vowel tiers. Their impact on the acquisition of the Sound-of-Words bias would be mediated by the quantity of input/words present in parental speech, but also by the quality of that input, in particular the variety/richness of the words spoken in the infants' environment. Their impact might also be related to the actual lexicon of each individual child. These effects of quantity and quality of input have been found to modulate other aspects of early language acquisition, such as word-form segmentation abilities and lexical acquisition (e.g., Hurtado et al., 2008; Weisleder & Fernald, 2013; Newman et al., 2016; Hoareau et al., 2019).

The Sound-of-Words model further postulates that the emergence of the Sound-of- Words bias, at the interface of phonological and lexical acquisition, would be linked to the concurrent development of related phonological and word learning abilities. It predicts that, at a given age, infants with better phoneme discrimination abilities and larger vocabularies would either acquire the Sound-of-Words bias earlier, or have a stronger Sound-of-Words bias. The model also states that the acquisition of the Sound-of-Words bias constitutes a major milestone in typical development corresponding to the binding between phonological and lexical acquisitions. And that this emergence, by allowing infants to focus on the sounds most relevant for lexical identity in their native language, would accelerate subsequent vocabulary development. Hence, in C-bias languages, infants with a stronger C-bias would be better word learners; similarly, in V-bias languages, infants with a stronger V-bias would be better word learners. This predicts that infants with a stronger bias at a given age would later have larger vocabularies than infants with a weaker bias. The age at which these predictive effects would be set into place likely depends on the timing of the acquisition trajectory for a specific language.

At present, very few studies have started to explore the links formulated by the Sound- of-Words model, and the evidence remains very scarce. For example, Bouchon et al. (2015) established that French-learning 5-month-olds' sensitivity to vowel-mispronunciations of their own name was stronger when the difference in spectral properties between their name and its mispronunciation was larger. However, this effect was not replicated in Von Holzen and Nazzi (2020). Von Holzen and Nazzi (2020) also investigated whether there was a link between infants' preference for their own name and the number of consonant or vowel tiers of the words they knew (based on parental vocabulary reports) at 5, 8 or 11 months. No links were found. With respect to the links between the C-bias and vocabulary development**,** Poltrock and Nazzi (2015) failed to find a link between the size of the C-bias in wordform recognition by French-learning 11-month-olds and their concurrent vocabulary size. Investigating the link between the C-bias and subsequent vocabulary size, no link was found in two studies, one investigating word-form segmentation at 8 months (Von Holzen et al., 2018), one investigating own name recognition at 5 and 8 months, with a reversed effect found at 11 months, infants with larger V-biases showing a faster vocabulary growth (Von Holzen & Nazzi, 2020). The study by Von Holzen et al. (2018) however established that higher sensitivity to all (consonant and vowel) mispronunciations of segmented word-forms was linked to faster vocabulary growth, which could support the proposal that the emergence of the C-bias is linked to overall better phoneme discrimination abilities. Future studies will be needed, investigating these postulated links on larger cohorts of infants, tested on multiple abilities at several age points, and exploring how individual variability in Sound-of-Words bias relates to input, phonological and lexical abilities.

Future research will also have to specify how the acquisition of the Sound-of-Words bias relates to the acquisition of other aspects of the phonological properties of native words, such as that of prosodic properties starting around 6 months (e.g., Höhle et al., 2009; Jusczyk et al., 1993a; Skoruppa et al., 2009), or that of phonotactic properties starting around 9/10 months (e.g., Gonzalez-Gomez & Nazzi, 2012a; Jusczyk et al., 1994). Given the crucial bootstrapping role given to the acquisition of the C-bias in the division-of-labor proposal, the acquisition of the Sound-of-Words bias could be thought of as a prerequisite for other acquisitions, or at least as playing a more important role in further acquisitions. At present, only one series of studies explored this question, investigating whether the acquisition of phonotactic properties by French-learning infants is modulated by whether they relate to consonants or vowels (Gonzalez-Gomez & Nazzi, 2012a, 2016). These studies established that acquisition of a consonant-based property (the Labial-Coronal bias) emerges between 7 and 10 months, while acquisition of a vowel-based property (the Posterior-Anterior bias) emerges between 10 and 13 months, suggesting that having acquired a C-bias, French-learning infants learn consonant-related properties faster. Future studies will have to extend this line of research to more properties and more languages. In particular, it would be key to determine whether the opposite acquisition lag is found in V-bias languages (that is, earlier acquisition of vowel- over consonant-related phonotactic properties).

Given the importance of language-specific properties and linguistic input in determining the Sound-of-Word bias in the Sound-of-Words model, future studies will also need to address how this bias is acquired in infants who are learning several languages in parallel. Indeed, given the cross-linguistic variability in acquisition of the Sound-of-Words bias found in the literature reviewed above, some bilinguals will learn two languages with the same bias (for example, French and Italian), while others will learn two languages with opposite biases (for example, French and Cantonese). Moreover, even if they are learning two languages with the same bias, acquisition of the bias might not follow the same timing (for example, French and English). Hence, bilingualism or multilingualism might differently affect the acquisition of the Sound-of-Words(s) biases depending on the specific languages infants are acquiring, and potentially on the language dominance pattern. The only current evidence on this issue comes from Mandarin-English bilinguals, tested only in Mandarin, showing a V-bias or most of the time no bias in lexical processing, as found for Mandarin-learning monolinguals (Singh et al., 2015; Wewalaarachchi et al., 2017; Wewalaarachchi & Singh, 2020).

10. Concluding remarks

In the present chapter, we reviewed evidence testing in both adults and infants the proposal that consonants play a larger role in lexically-related hypotheses than vowels. We presented evidence in support of this proposal, but also evidence of crosslinguistic variation suggesting that the C-bias might in fact only be one possible option. Accordingly, we proposed a new Sound-of-Words model of the acquisition of phonological biases in lexical processing that takes contribution from both nature and nurture, specifies links between the acquisition of the different linguistic levels and cascading developmental and predictive effects of early acquisitions on later developmental outcomes. This model could be seen as a guide to future research, and is likely to be later reformulated in light of the findings emerging from this future research. We would also want to encourage studying the acquisition of this bias in atypically developing populations (see Havy et al., 2010, on children with Williams syndrome; Havy et al., 2013, on deaf infants with cochlear implants; Gonzalez-Gomez & Nazzi, 2012b, on preterms). This would allow for the testing of the architecture of the Sound-of-Words model, but also for a better understanding of the specificity of language acquisition in neurologically-diverse populations, and for the setting up of efficient support interventions.

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